

DESIGN AND BEHAVIOUR GUIDE

Airwave Style Guide

Airwave, Airwave Hub and Airwave Companion, with the Android app and the Discord bot: one Aqua system, its rules, every screen, and how to keep it true.

PROFILE
snow-leopard-itunes-9

SOURCE FINGERPRINT
41724ca7c225392f

COVERAGE
90 surfaces · 17 journeys · 138 rules

INTERACTIVE VERSION
docs/design/styleguide.html

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Airwave Style Guide

snow-leopard-itunes-9 · three apps · 249 tokens checked · 90 surfaces · source 41724ca7c225392f

One guide for the three apps of Airwave — Airwave, the player (a page, also inside the Android app); Airwave Hub, the admin window the container serves; and Airwave Companion, the Windows app — and for the words the Discord bot uses. All three are a reconstruction of Apple's 2009–2010 interface, and each is built from a design file in `design/frontends/origin/`. Every value here is read from `tokens.json` as this page is built, the windows are drawn with the stylesheets and kits the products ship, and every rule, screen and journey is read from `design/` — so the page cannot describe one thing and show another.

The rules that outrank the visuals

A styleguide that opens with colour teaches the wrong lesson. These four decide whether a control is drawn at all, and each has overruled the aesthetics here at least once.

UX-PRIN-001

A control that cannot act is not drawn

No button that looks live and does nothing, no toggle whose state is decoration. If an action is impossible in this context the control is omitted or replaced by an explained state. The equalizer window ships without traffic lights: three circles that close nothing would be decoration.

UX-PRIN-002

An unavailable capability says so where the choice is made

Not in a tooltip, not in a log, not by disappearing. The control stays visible and, when chosen, reports the reason instead of acting. The Shared segment of the mode switch stays visible with no hub paired and reports why when pressed.

UX-PRIN-003

Say what the code does, not what the feature is called

Labels describe the implemented behaviour. When a label and the DSP (or the network, or the provider) disagree, the label changes. Solfeggio presets are described as filters with no claimed physical effect; retuning states that the fallback changes tempo.

UX-PRIN-004

A disabled control is still legible about why

Greyed, not gone. The reason is part of the accessible name, not only a tooltip. “500 Hz band, not used by 528 Hz (MI)”.

Brand and voice

Airwave is a self-hosted music suite: a player that works on the files you already have, a hub you run yourself, and a companion that brings a Windows library to it. Its look is a careful reconstruction of Apple’s 2009–2010 interface — the Snow Leopard window, the iTunes 9 list, the iPod’s Now Playing screen — chosen for its density and its honesty about state. The reference is `docs/design/APPLE_AQUA_2009_2010_UI_DESIGN_SPEC.md`.



Airwave

`music-player/public/icon.svg`

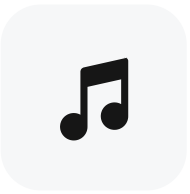
Also the Android launcher (brand colour #3b74c8) and the maskable PWA icon.



Airwave Companion

`windows-companion/resources/icon.svg`

The same tile with a folder behind the note: the one app that reaches the filesystem.



The note

`packages/aqua-ui/src/icons/glyphs.tsx`

Both icons draw this exact path; a unit test fails if they drift apart again.

USING THE MARK

- The tile is a rounded square (radius 112 on a 512 grid), a three-stop blue gradient and a soft gloss over the top half. Keep all three; the gloss is the period detail.
- Never recolour the note or redraw it by hand. Use the SVG or the `note` glyph.
- The blue tile is the product's mark. Inside the interface blue is never decoration — it means selected, active or default (see Colour).
- ● PROPOSED Clear space and minimum size have not been specified. Until they are, keep a quarter of the tile's width clear around it and do not use it below 16px; the tray icon is its own simplified drawing.

VOICE

Plain, specific and a little dry. The product explains what it is doing and what it cannot do, in the place you are looking. It does not apologise, cheer or use exclamation marks, and it never claims more than the code does (`UX-PRIN-003`).

`VOICE-001`

Empty states say what to do next ● OBSERVED

An empty list says what is missing and what to do next, in a few plain words: one sentence inside the list in the Airwave windows, a title and a line in the component library ("Nothing queued"). No apologies, no exclamation marks. "No folders yet — add the one this PC keeps these in." · "No guides yet — the guide will list channels without programmes."

`VOICE-002`

Errors explain the problem and the way out ● ADOPTED

Name what failed in the person's terms and what fixes it ("Add a bot token first (Admin → Discord → Token)"). Never show a stack trace or an internal identifier; a 500 carries a correlation id instead.

`VOICE-003`

One product name, set in one place ● ADOPTED

The suite is Airwave and its apps are Airwave, Airwave Hub and Airwave Companion; the names come from BRANDING in TypeScript, and "Now Playing" is the name of the player's home page only (DEC-025). Native resources that cannot import BRANDING (Android strings.xml, the SVG icons' aria-labels, the Electron builder config) repeat it and must be changed together. Machine identifiers keep their spelling.

WORDS WE USE

SAY	MEANING	NOT
Airwave	The suite, and the player's own name. The other two apps are Airwave Hub and Airwave Companion; in running text, "the player", "the hub" and "the companion".	Now Playing as a product name; names invented per screen
Now Playing	The player's home page: what is playing now. Nothing else.	The suite, the hub or the companion

SAY	MEANING	NOT
Solo / Shared listening	The two modes on the status bar.	Party mode, sync mode
Group	People listening to one shared queue on a hub.	Room, session
Pair	Connecting a device to a hub, confirmed by a fingerprint.	Log in, register a device
Shared link	A public link to a track or playlist the hub serves.	Share URL, public page
Fetch	Running yt-dlp or spotDL on this machine.	Rip, grab
Hub library	Files that live on the hub's data volume.	Cloud, server music

● OBSERVED The vocabulary above is how the code and the docs already talk; it has not been reviewed as a glossary.

REAL COPY

No folders yet
— add the one this PC keeps these in.
Companion · Library ▸ Saved Music

Nothing needs you.
The whole of an empty "Needs attention" list.
Hub · Overview

A link that doesn't answer
keeps what it last held, and is tried again by itself.
Companion · Live TV

A stream never implies a download.
Where a provider doesn't allow saving, there's no button — not one that fails.
Hub · Music ▸ Downloads

Three apps, one system

One period, one set of rules, three apps — and each app's interface comes from its own design file in `design/frontends/origin/`. The design file is the authority for how its app looks; this guide is the authority for what the three share: the rules, the words, the ledger of screens, and the checks that keep them in step.

What each app looks like now is its living mockup, `design/frontends/airwave-*.html`: one standalone file per app, generated from the app by `pnpm mockups:build` — every screen and state as the app drew it on sample data, as readable markup beside the app's own stylesheets. The owner may edit one by hand; `pnpm mockups:diff` shows what the edit changes, and it reaches the app through the workflow in AGENTS.md (DEC-038).

APP	DESIGN	WHAT IT WEARS	WHERE TO LOOK HERE
Airwave music-player · android · local-helper	`airwave-now-playing.html`	The design file itself, served as the shell: `music-player/index.html` is generated from it by `make-shell.py`, with its own custom properties, its own script for the drawn controls, and `src/shell/bridge.ts` behind its seams (DEC-019). It imports nothing from the component library.	The served shell in Elements (screenshots of the production build), the `player-shell-*` rows in Every screen and the `NP-*` rules. The page skin is the component library's version of the same look, kept in `music-player/src` but not served.
Airwave Hub docker-container/src/web	`airwave-hub.html`	`airwave-window.css` and `airwave-hub.css` (generated from the designs by `pnpm build:window-css`), AquaArt's drawings, its own kit `ui.tsx`, and `styles.css` for what the design left to its script.	The Airwave window in Elements, and its screens in the mockups.
Airwave Companion windows-companion/src/renderer	`airwave-companion.html`	`airwave-window.css`, the same AquaArt, its own kit `ui.tsx`, and `styles.css` for what a real window needs (it fills its frame; its chrome is the title bar).	The Airwave window in Elements, and its screens in the mockups.

The Airwave window
`--win-*, --chrome-*, --ink*, --aq-*` · `airwave-window.css, airwave-hub.css` · Lucida Grande · light only (DEC-003) · Airwave Hub, Airwave Companion

The component library
`--aqua-*, --np-*, --lib-*` · `aqua.css, now-playing.css` · packages/aqua-ui React components · imported by the player's React source, the gallery and this guide; no served product (DEC-026)

Colour

Chosen by role, never by picking a hex. Blue means *selected*, *active*, or *the default action* — it is not a brand colour and it is never decoration. Three palettes follow, straight from the token file: the Airwave window's (what the hub and the companion wear), then the component library's two.

THE AIRWAVE WINDOW — 56 COLOURS

Owned by the design files: `airwave-window.css` and `airwave-hub.css` are generated from them, and `tokens.json` records the values so `pnpm styleguide:check` notices when a design changes one. The last group is the hub's own: the design's `--ink-3`, `--success` and `--danger` fail 4.5:1 as text on white, so the hub reads sentences in darker inks of the same hues (DEC-022).

WINDOW AND CONTENT

desktop #6D737A	winOutline #747474	winBody #ECECEC	content #FFFFFF
contentMuted #F7F7F7	rowStripe #F1F6FB	rowDivider #D8DEE5	wellEdge #9A9A9A

CHROME AND HEADERS

chromeTop #F6F6F6	chromeUpper #E5E5E5	chromeLower #C4C4C4	chromeBottom #AFAFAF
chromeSep #858585	chromeHi #FFFFFF	headTop #F8F8F8	headBot #DADADA
headRule #ADADAD	headInk #3A3D41		

SELECTION AND FOCUS

selTop #69B5F3	selMid #378DDA	selBot #1764B2	selBorder #0F4E8B
selInk #FFFFFF	focus #3F9FE8	aqRing #7EA8E0	

AQUA GEL

aquaSpec #EAFBFF	aquaTop #A9E5FF	aquaMid #38B6F7	aquaLower #0D83D6

aquaBot
#075FA6

aquaRim
#07558F

INK

ink
#161616

ink2
#565B60

ink3
#96999C

aqInk
#000000

aqInkOff
#9A9A9A

STATUS AND THE FIRST-RUN GATE

success
#4E9D47

warning
#D99A1E

danger
#D64A44

warnBg
#FBF3DC

warnEdge
#D8B75E

LCD PLATE

lcdTop
#E1EAEB

lcdBot
#AEBCC0

lcdInk
#172326

FACES AND TRACKS

btnTop
#FEFEFE

btnMid
#F0F0F0

btnBot
#DCDCDC

btnEdge
#9C9C9C

trackEdge
#B9B9B9

trackLo
#DEDEDE

trackHi
#FBFBFB

CONTRAST INKS (HUB WEB, DEC-022)

inkQuiet
#6B6F73

successInk
#2F7A28

dangerInk
#B3322C

link
#1764B2

capOffInk
#6B6B6B

COMPONENT LIBRARY, WINDOW SKIN — 91 TOKENS

WINDOW AND CONTENT

desktop #6D737A	windowOutline #747474	windowBody #ECECEC	content #FFFFFF
contentMuted #F7F7F7	rowStripe #F1F6FB	rowDivider #D8DEE5	

CHROME

chromeTop #F6F6F6	chromeUpper #E5E5E5	chromeLower #C4C4C4	chromeBottom #AFAFAF
chromeSeparator #858585	chromeHighlight #FFFFFF		

SOURCE LIST

sidebarTop #DCE8F4	sidebarBottom #BECEE0	sidebarDivider #93A3B2	sidebarGroupText #536473

SELECTION AND FOCUS

selectionTop #69B5F3	selectionMid #378DDA	selectionBottom #1764B2	selectionBorder #0F4E8B
selectionText #FFFFFF	focus #3F9FE8		

AQUA GEL

aquaSpecular #EAFBFF	aquaTop #A9E5FF	aquaMid #38B6F7	aquaLower #0D83D6
aquaBottom #075FA6	aquaRim #07558F		

GRAPHITE

graphiteTop #E5E7E9	graphiteMid #AEB4B9	graphiteBottom #737A80

TEXT

text #161616	textSecondary #565B60	textDisabled #96999C

LCD

lcdTop #E1EAEB	lcdBottom #AEBCC0	lcdText #172326

STATUS

danger #D64A44	warning #D9A431	success #4E9D47

TRAFFIC LIGHTS

trafficClose #E9635B	trafficMinimize #E3B64B	trafficZoom #58A953

OTHER

headerTop #F8F8F8	headerBottom #DADADA	headerRule #ADADAD	headerText #3A3D41
warningBg #FBF3DC	warningEdge #D8B75E	ctlInk #151515	ctlFace1 #FDFDFD
ctlFace2 #F4F4F4	ctlFace3 #E6E6E6	ctlFace4 #DCDCDC	ctlFaceHover1 #FFF
ctlFaceHover2 #F8F8F8	ctlFaceHover3 #ECECEC	ctlFaceHover4 #E3E3E3	ctlFaceActive1 #C9C9C9

ctlFaceActive2
#D6D6D6

ctlFaceActive3
#DEDEDE

ctlFaceDisabled1
#FBFBFB

ctlFaceDisabled2
#F1F1F1

ctlBorder
#A3A3A3

ctlBorderBottom
#8D8D8D

ctlBorderDisabled
#C9C9C9

ctlFocusRing
#0D61A7

ctlDefaultInk
#06203A

ctlDefault1
#CFEAFF

ctlDefault2
#A3D4FB

ctlDefault3
#74B9F5

ctlDefault4
#5AA8EF

ctlDefaultBorder
#4684C8

ctlDefaultBorderBottom
#356DB0

ctlBoxInk
#1A2733

ctlBox1
#FDFDFD

ctlBox2
#F0F0F0

ctlBox3
#E4E4E4

ctlBoxBorder
#A0A0A0

ctlBoxOn1
#7DB4E6

ctlBoxOn2
#4F92D6

ctlBoxOn3
#3F81C6

ctlBoxOnBorder
#3D76AD

ctlFieldBg
#FFF

ctlFieldBorder
#9A9A9A

ctlFieldBorderTop
#7D7D7D

ctlFieldFocusBorder
#1768A7

ctlFieldInvalidBorder
#B8433D

ctlPlaceholder
#8B9095

dangerInk
#7A1712

COMPONENT LIBRARY, PAGE SKIN — 25 TOKENS

STATUS BAR

barTop
#F5F6F7

barUpper
#E8EAEA

barLower
#DCE2DE

barBottom
#D9DFE1

barEdge
RGB(31 41 45 / 22%)

FIELD

fieldTop
#F2F2F2

fieldBottom
#FFFFFF

MODE SWITCH

modeFaceTop #FDFDFD	modeFaceBottom #DCDEE0	modeOnTop #767C85	modeOnBottom #A9B0B8

SCRUBBER

railTop #E9ECEE	railBottom #FEFFFF	railFillTop #8FB9E4	railFillMid #2B9BEC
railFillBottom #A5E9FD			

LIST

listHeaderTop #F8F8F8	listHeaderBottom #DADADA	listSortTop #E8EEF7	listSortBottom #CFDBEB
listStripe #F2F5F9	listPlaying #DDE9F8	listSelectTop #5B8DD9	listSelectBottom #3A6CC4

MARKS

live #CC1027	star #F0A422

DARK

The page skin has a full dark palette and it is opt-out: every dark rule is guarded `@media (prefers-color-scheme: dark) { :root:not([data-np-theme='light']) { … } }`, so the system preference wins unless a page pins itself light. Redefine *only* tokens inside that block. Neither the Airwave window nor the library's window skin has a dark palette, because Snow Leopard had none (DEC-003); the player's shell carries its own dark values (NPD-002).

The page skin’s colours that change in the dark scheme, read from the shipped stylesheet when this page was built (88 properties). Light on the left, dark on the right.

`--ctx-bg` `rgb(247 247 250 / 88%) → rgb(40 42 48 / 90%)`	`--lib-bg` `#fff → #16181c`
`--ctx-ink` `#1b1c1f → #eef0f3`	`--lib-border` `#ababab → #08090b`
`--ctx-sep` `rgb(0 0 0 / 12%) → rgb(255 255 255 / 12%)`	`--lib-head-bot` `#dadada → #26292e`
`--lib-accent` `#2b6fd6 → #6aa3f2`	`--lib-head-emboss` `rgb(255 255 255 / 80%) → rgb(0 0 0 / 55%)`
`--lib-alt` `#f2f5f9 → #1b1f26`	`--lib-head-hot-bot` `#cfcfcf → #303338`

--lib-head-hot-top

#efefef → #45484f

--lib-head-ink

#3a3d41 → #c6cbd3

--lib-head-rule

#adadad → #0f1114

--lib-head-sep

#ccc → #17191d

--lib-head-top

#f8f8f8 → #3a3d44

--lib-icon

#8a8f98 → #8d939c

--lib-ink

#000 → #f2f3f5

--lib-np

#dde9f8 → #22354f

--lib-soft

#222326 → #d4d8df

--lib-sort-bot

#cfdbeb → #233043

--lib-sort-hot-bot

#c2d2e6 → #2b3b52

--lib-sort-hot-top

#dde7f4 → #36485f

--lib-sort-top

#e8eef7 → #2c3a4e

--lib-star

#f0a422 → #f4b241

--np-accent

#2b6fd6 → #6aa3f2

--np-avatar-face

#f3f3f3 → #2b2e34

--np-avatar-lip

rgb(255 255 255 / 70%) → rgb(255 255 255 / 10%)

--np-avatar-ring

rgb(0 0 0 / 26%) → rgb(0 0 0 / 55%)

--np-bar-1

#eef0f1 → #2f3238

--np-bar-2

#e8eaea → #2a2d33

--np-bar-3

#e2e6e5 → #25282e

--np-bar-4

#dfe4e1 → #212429

--np-bar-5

#dce2de → #1e2126

--np-bar-bottom

#d9dfe1 → #1a1d22

--np-bar-edge

rgb(31 41 45 / 22%) → rgb(0 0 0 / 60%)

--np-bar-emboss

rgb(255 255 255 / 55%) → transparent

--np-bar-top

#f5f6f7 → #34373d

--np-field-bottom

#fff → #24272d

--np-field-hint

#ababab → #7c8087

--np-field-icon

#8e8e91 → #85898f

--np-field-lip

rgb(255 255 255 / 70%) → rgb(255 255 255 / 8%)

--np-field-mid

#fbfbfb → #1b1e23

--np-field-ring

rgb(0 0 0 / 20%) → rgb(0 0 0 / 70%)

--np-field-shadow

rgb(0 0 0 / 32%) → rgb(0 0 0 / 75%)

--np-field-top

#f2f2f2 → #141619

--np-ink

#1b1c1f → #f1f2f4

--np-key-ink

#1b1c1f → #ccd1d8

--np-key-muted

#8b9099 → #7f858e

--np-key-on

#2f6ec0 → #5fa0ea

--np-list-alt

#f2f5f9 → #1b1f26

--np-list-badge

#dfe3e9 → #2f333b

--np-list-badge-ink

#3b3f47 → #d6dae1

`--np-list-border` `#ababab → #08090b`	`--np-vol-b` `#b7bdc4 → #878e96`
`--np-list-icon` `#8a8f98 → #8d939c`	`--np-vol-face` `#fff → #1b1e23`
`--np-live` `#cc1027 → #ff4d62`	`--srch-art-bot` `#e3e3e3 → #24262b`
`--np-live-halo` `rgb(204 16 39 / 32%) → rgb(255 77 98 / 36%)`	`--srch-art-ink` `#a9a9a9 → #8d939c`
`--np-meta` `#575757 → #9ba0a8`	`--srch-art-ring` `rgb(0 0 0 / 14%) → rgb(255 255 255 / 10%)`
`--np-page` `#fff → #0e0f12`	`--srch-art-top` `#fbfbfb → #33363d`
`--np-rail-line` `rgb(0 0 0 / 16%) → rgb(0 0 0 / 72%)`	`--srch-bg` `#f9f9f9 → #26282e`
`--np-seg-border` `rgb(0 0 0 / 36%) → rgb(0 0 0 / 70%)`	`--srch-border` `rgb(0 0 0 / 25%) → rgb(0 0 0 / 65%)`
`--np-seg-divider` `rgb(0 0 0 / 30%) → rgb(0 0 0 / 55%)`	`--srch-chev` `#8f8f8f → #9aa0a9`
`--np-seg-face-bottom` `#dcdee0 → #292c31`	`--srch-dot` `#c9c9c9 → #4a4e56`
`--np-seg-face-mid` `#f0f1f2 → #33363c`	`--srch-dot-on` `#7c7c7c → #b8bdc5`
`--np-seg-face-top` `#fdfdfd → #3a3d44`	`--srch-head-bot` `#eee → #25272d`
`--np-seg-ink` `#43464b → #c3c7cf`	`--srch-head-top` `#f6f6f6 → #2e3138`
`--np-seg-lip` `rgb(255 255 255 / 65%) → rgb(255 255 255 / 8%)`	`--srch-hot` `#ececec → #32353c`
`--np-seg-on-bottom` `#a9b0b8 → #23262c`	`--srch-ink` `#2c2c2c → #eef0f3`
`--np-seg-on-mid` `#939aa3 → #1c1f24`	`--srch-lip` `rgb(255 255 255 / 60%) → rgb(255 255 255 / 7%)`
`--np-seg-on-top` `#767c85 → #14161a`	`--srch-rule` `#dedede → rgb(0 0 0 / 50%)`
`--np-vol-a` `#9aa1a9 → #6e747c`	`--srch-soft` `#9b9b9b → #8d939c`

Aqua highlights — where the blue is allowed to shine

The specimens from here to Motion are the component library's controls. The principles are the Airwave window's too — the same light source, the same restraint about blue — but its controls are AquaArt's drawings, shown under The Airwave window.

Snow Leopard had flattened most of Aqua's gel by 2009, but not all of it: selection, the default action, the scroller and a few controls kept the glass. Those are exactly the places this system keeps it, and this is each one, live.

THE DEFAULT ACTION

Cancel Save Graphite ● SNOW LEOPARD · CORRECT

Save ● AQUA 10.2 · A DECADE EARLY — DRAWN ONCE HERE, AS THE THING NOT TO DO

The one button on a form that carries the Aqua blue: a pale cyan cap over a saturated body, a specular hairline at the top, a dark rim, and a slow pulse. The shape stays the Snow Leopard rounded rect — no pill, no glass lozenge, no gloss step across the middle.

CONTROLS THAT KEEP THE GEL

Checked Week Day Week Month Syncing artwork…

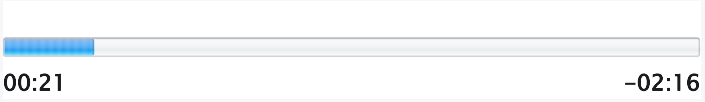
The checked box, the pop-up's end cap, the aqua-tinted segmented control and the indeterminate progress bar's barber pole are the last four places a form is blue.

THE OVERLAY SCROLLER



The full five-stop Aqua gel survives at strength in exactly one place: the list's overlay scroller, which fades in while you scroll and out 900 ms after you stop.

THE IPOD RAIL



Not gel, but the other blue on the page: the iPod Classic's progress bar, 12 px, near-square ends, and a fill that brightens downward — sky at the top through a saturated dip at 62% to a cyan glow along the lower lip. No knob: the device had none.

And the focus ring — `--aqua-focus` #3F9FE8 with a soft outer glow — is blue everywhere, because the thing that has the keyboard is the one thing on screen that must never be ambiguous.

Type

Two families, one scale. The window skin — the Airwave window and the library's alike — sets Lucida Grande, as OS X did; the Airwave window runs 12px body text with 11px secondary lines and 10px chips. The page skin sets **Helvetica** — the face the iPod classic drew its Now Playing

screen in, which is what the hero is modelled on — with Arial as its metric-compatible stand-in where Helvetica is absent. Each specimen is at its true size.

PAGE SKIN · HELVETICA

13px	**Sundress — A$AP Rocky**	`--aqua-font-system`
12px	Marlow & the Tidewater — Quiet Arithmetic	`--aqua-font-view`
11px	Searches this device only until a hub is paired	`--aqua-font-small`
10px	PREAMP · 32 · 64 · 125 · 250 · 500 · 1K · 2K	`--aqua-font-label`
9px	Mini controls only	`--aqua-font-mini`

WINDOW SKIN · LUCIDA GRANDE

13px	Airwave Hub	`--aqua-font-system`
12px	Copper Meridian — Orbital Cartographers	`--aqua-font-view`
11px	1,240 songs, 2.6 hours, 245 MB	`--aqua-font-small`
10px	LIBRARY · PLAYLISTS · CONNECTED	`--aqua-font-label`
9px	Mini controls only	`--aqua-font-mini`

Weights are **400** and **700**. There is no 500 and no 600: the profile predates variable UI fonts, and a half-weight reads as a rendering fault beside a true bold. Numbers that change in place — timers, dB values, tempo, track times — take `font-variant-numeric: tabular-nums`, so a changing digit does not shift the ones beside it.

Space

One scale, no arbitrary values. Each bar is drawn at its true width.

`--aqua-space-micro`	2px	
`--aqua-space-tight`	4px	
`--aqua-space-inline`	6px	
`--aqua-space-control`	8px	
`--aqua-space-small-group`	10px	
`--aqua-space-group`	12px	
`--aqua-space-frame-top`	14px	
`--aqua-space-strong-group`	16px	
`--aqua-space-window-edge`	20px	
`--aqua-space-major`	24px	
`--aqua-space-bottom-bar-regular`	32px	

SIZES

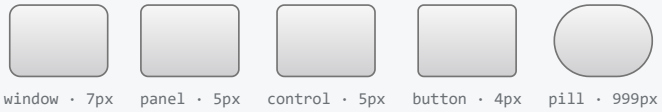
visualControlRegular 22PX	visualControlSmall 19PX	searchRegular 22PX	tableRow 20PX
listRow 18PX	railHeight 12PX	transportKey 48PX	playKey 64PX
barHeight 40PX	sidebarDefault 196PX		

32 px (`--aqua-hit`) is the floor for a pointer target, and **44 px** on a coarse pointer, which is what both platform guidelines ask for. A 12 px rail and an 18 px row are period-correct and stay that size — they get their hit area from an invisible `::after` that expands past the visual. Never shrink a hit target to match a visual.

Shape and material

One light source, above and slightly forward. Raised things are light at the top and dark at the bottom; recessed things are the reverse. Every shadow follows from that, and a rule that contradicts it looks broken even when the colours are right.

RADII



THE PUSH BUTTON



A rounded rect, one quiet vertical gradient, a hairline rim that darkens along the bottom, one pixel of white on top. The default action is the same shape with the Aqua blue on it — see the highlights above.

```
background: linear-gradient(to bottom, #fdfdfd 0%, #f4f4f4 47%, #e6e6e6 53%, #dcdcdc 100%);
border: 1px solid #a3a3a3;
border-bottom-color: #8d8d8d;
border-radius: 4px;
box-shadow:
  inset 0 1px 0 rgb(255 255 255 / 90%),   /* the pixel of light on the top edge */
  0 1px 0 rgb(255 255 255 / 55%);         /* the surface beneath catching it */
```

RECESSED FIELDS



Wells run the other way: shadowed at the top inside, with the lower lip catching light. Same source, opposite surface. The page skin's search pill is one, 26 px high.

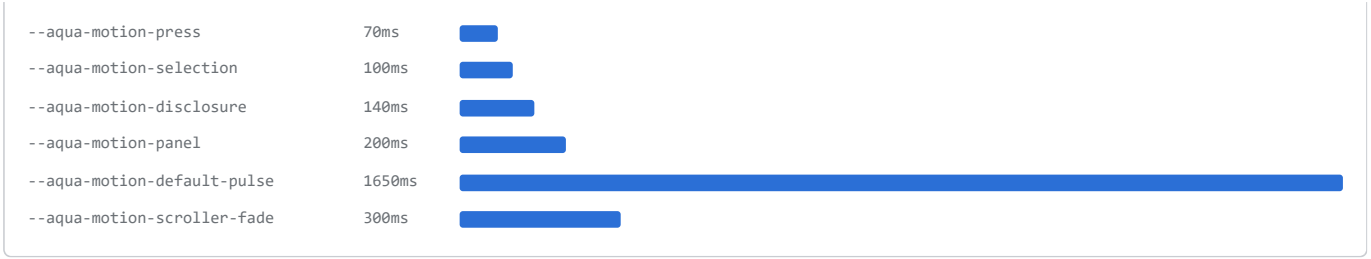
FADERS



The slider's knob is a small lozenge with a pointed edge, the equalizer's fader is the same control rotated; both ride a recessed groove.

Motion

Confirmation, never decoration: nothing animates that a person did not just cause. Everything eases on `cubic-bezier(0.2, 0.7, 0.2, 1)`. Bars are drawn in proportion.



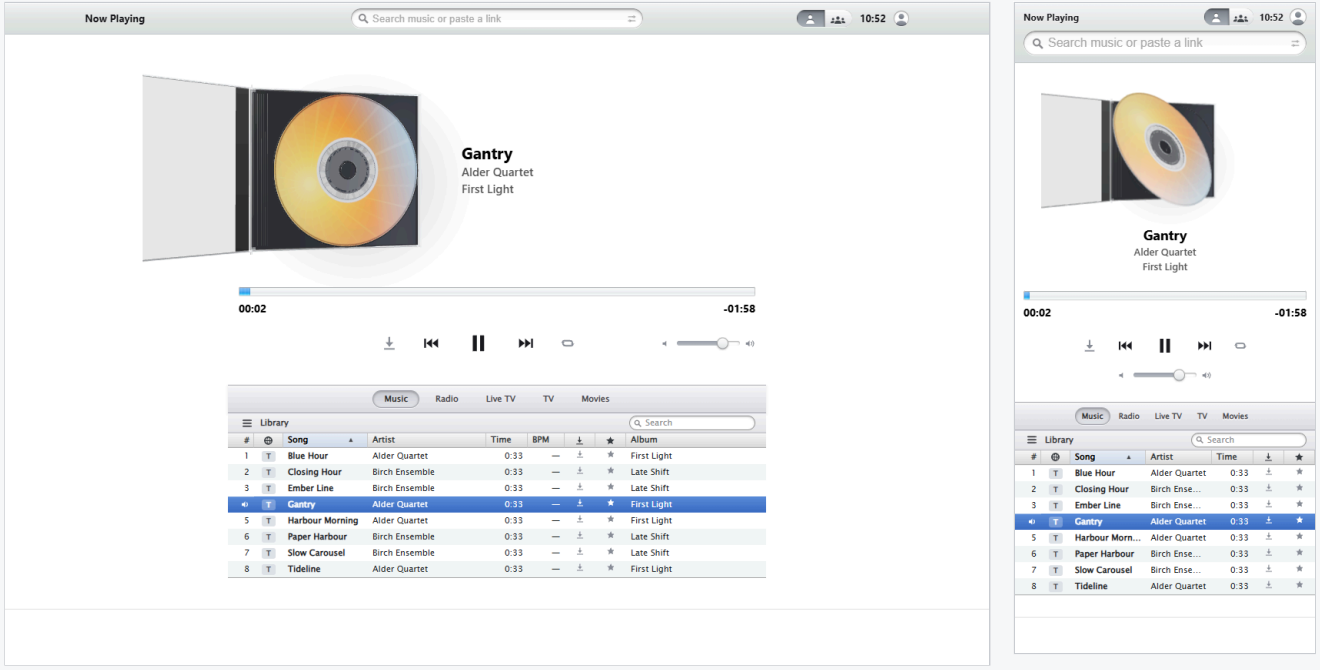
Reduced motion is a contract, not a courtesy. Under `prefers-reduced-motion: reduce` the live dot stops pulsing, the marquee does not scroll (the cell keeps its ellipsis), the sheet's travel drops to zero and it fades only, the constellation opens as a table rather than an animated field, and the 3D stage stops its idle drift. Two variables carry most of it — `--aqua-anim-state: paused` and `--aqua-sheet-travel: 0` — so a new animation should read one of them rather than inventing a third.

The served shell — the player as it is served

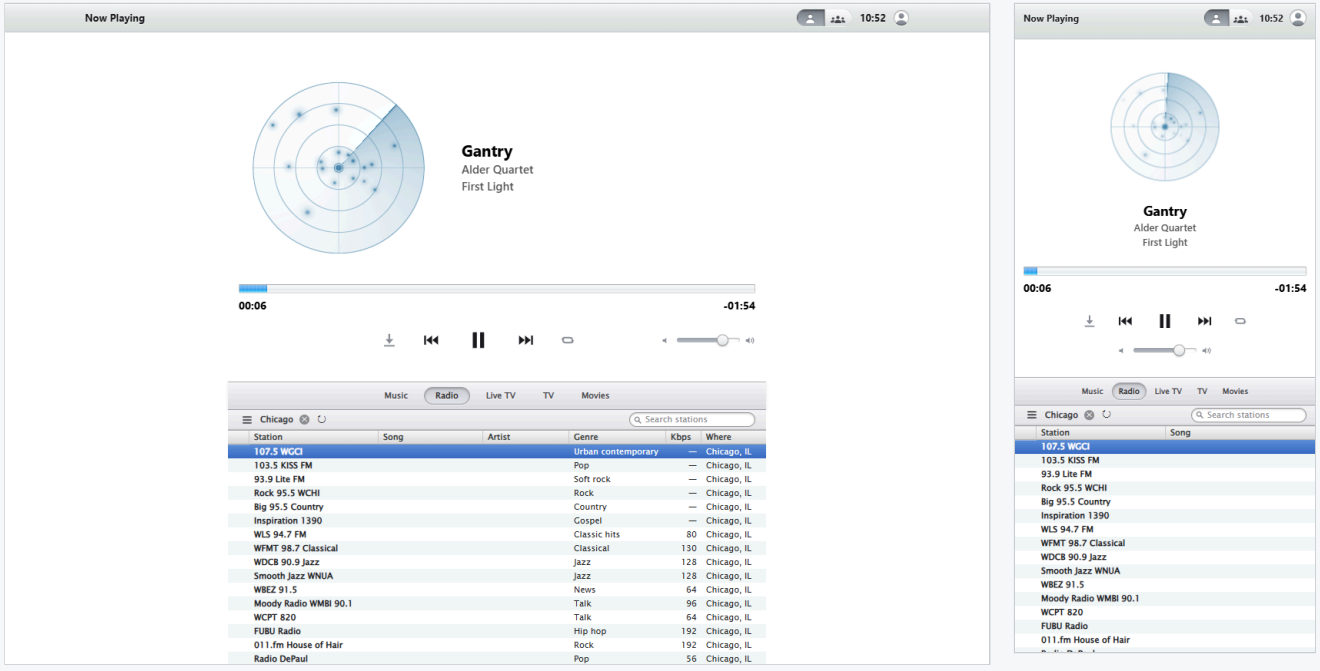
What a listener opens is `music-player/index.html`, the Airwave design served as the shell (DEC-019), and what Android and the local helper ship too. It is one document with its own stylesheet and scripts, so it cannot be put inside one of this guide's frames; it is shown here as it is, photographed from the production build. The page skin in the next chapter is the component library's version of the same look — the React interface kept in `music-player/src` — and is not what is served.

Taken 2026-10-20 from `music-player/index.html` (shell `296be7dbdad187d5`) by `pnpm styleguide:shell`, at 1280 × 860 and 390 × 844. Nothing reached the network: outside hosts were refused, and the companion's helper was answered by the script.

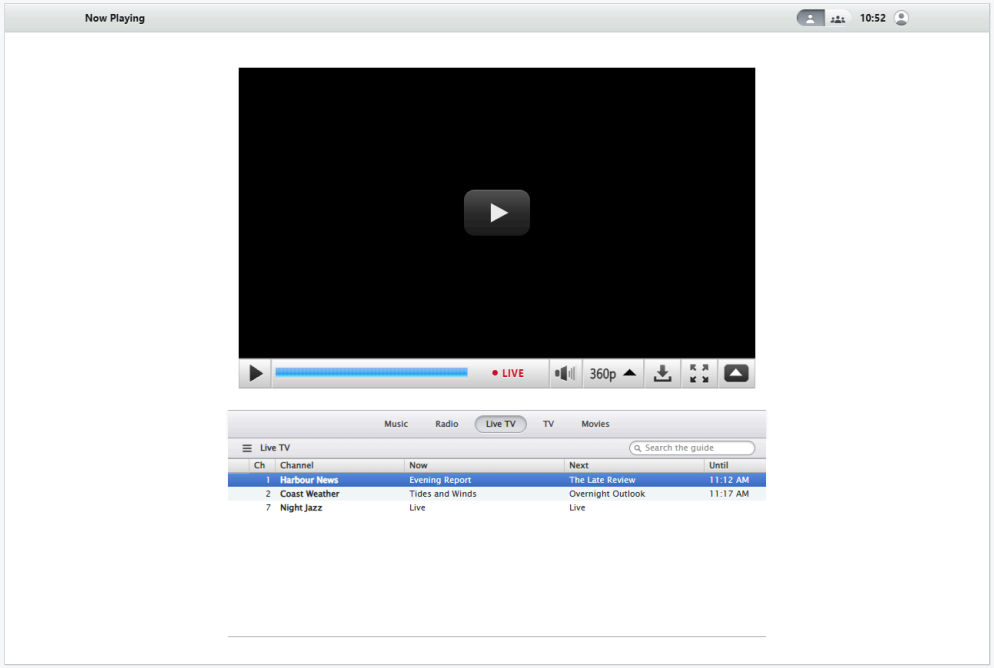
Music, playing Eight songs added through the shell's own import path, Gantry playing: the jewel case on the stage, the iPod rail, the transport, and the iTunes 10 list with the playing row selected.

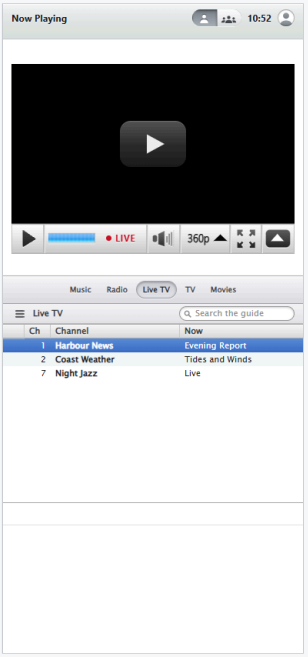


Radio The station directory unreachable, so the bundled stations for the nearest city show; the hero becomes the radar while the song keeps playing.

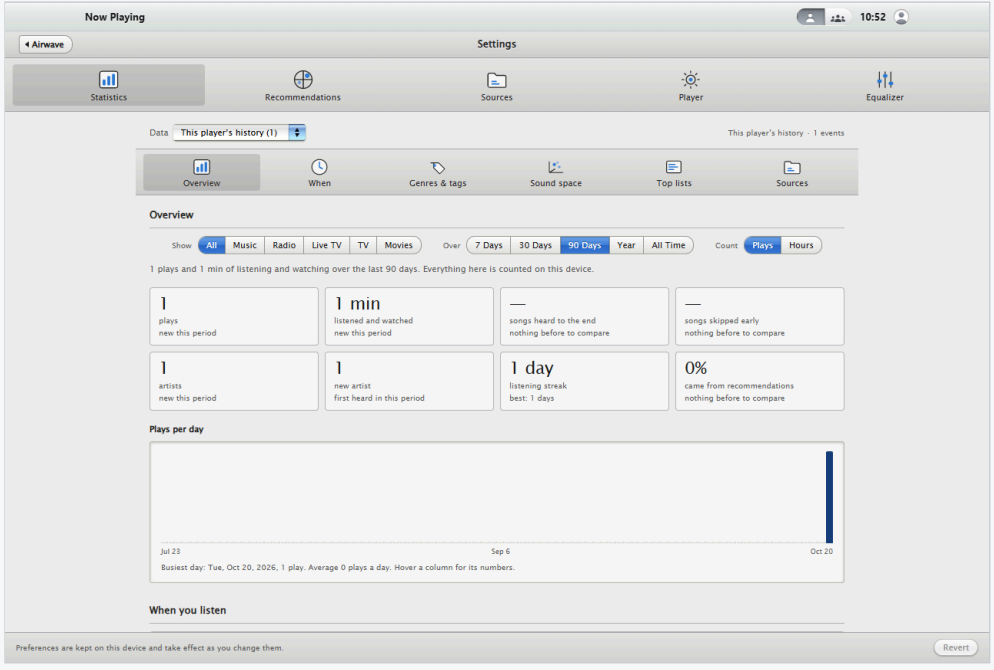


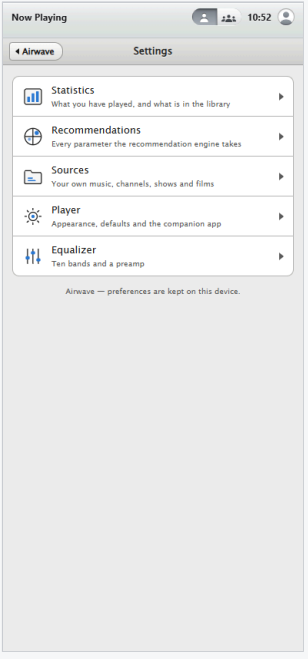
Live TV Channels and guide from the companion's helper on this machine (with no companion, a paired hub's copy fills the same guide): the QuickTime bar over the picture, Now and Next from the guide, and "Live" where a channel has none.





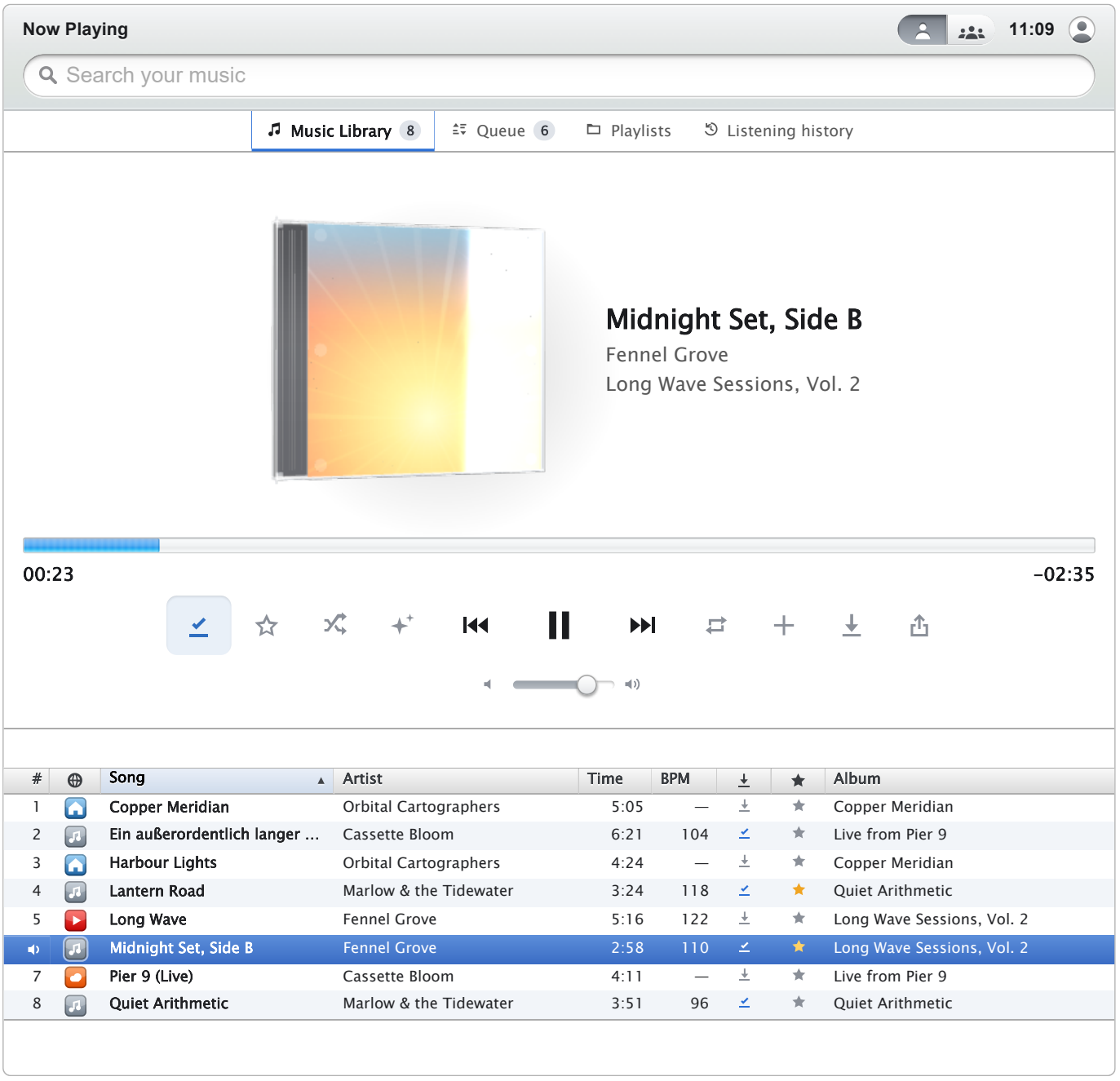
Settings Settings is a page with its own address. On a desktop it opens on Statistics with the section tabs across the top; on a phone it opens on the list of sections.





The page — what the player is made of

The music player's skin: a sticky status bar, a section strip, the hero with the jewel case on its stage, the iPod rail and a transport of eleven keys, and the iTunes 10 list under it. Everything below is live — press play and the case opens; pick a row and the stage follows it.



#	Song	Artist	Time	BPM	Album
1	Copper Meridian	Orbital Cartographers	5:05	—	Copper Meridian
2	Ein außerordentlich langer …	Cassette Bloom	6:21	104	Live from Pier 9
3	Harbour Lights	Orbital Cartographers	4:24	—	Copper Meridian
4	Lantern Road	Marlow & the Tidewater	3:24	118	Quiet Arithmetic
5	Long Wave	Fennel Grove	5:16	122	Long Wave Sessions, Vol. 2
6	Midnight Set, Side B	Fennel Grove	2:58	110	Long Wave Sessions, Vol. 2
7	Pier 9 (Live)	Cassette Bloom	4:11	—	Live from Pier 9
8	Quiet Arithmetic	Marlow & the Tidewater	3:51	96	Quiet Arithmetic

The transport is centred on the rail. Play sits on the progress rail's centre line at every width, with previous and next either side of it on the same line, and the keys never run past the rail's ends. The volume line sits beside the keys only when it fits on both sides; otherwise it goes beneath, centred, as it does here. When the keys do not fit on one line, previous, play and next take the first line on their own and the other eight keys share the line below. The row measures itself to decide this, because the room it needs depends on how many keys a product carries (DEC-016).

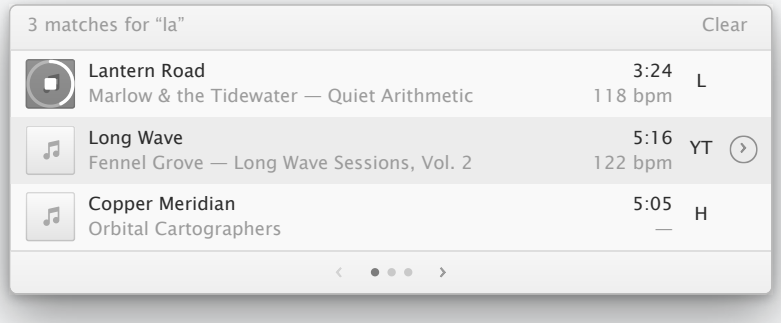
THE MUSIC LIST

#	Song	Artist	Time	BPM	Album
1	Copper Meridian	Orbital Cartographers	5:05	—	Copper Meridian
2	Ein außerordentlich langer…	Cassette Bloom	6:21	104	Live from Pier 9
3	Harbour Lights	Orbital Cartographers	4:24	—	Copper Meridian

#	Song	Artist	Time	BPM	Album
4	Lantern Road	Marlow & the Tidewater	3:24	118	Quiet Arithmetic
5	Long Wave	Fennel Grove	5:16	122	Long Wave Sessions, Vol. 2
6	Midnight Set, Side B	Fennel Grove	2:58	110	Long Wave Sessions, Vol. 2
7	Pier 9 (Live)	Cassette Bloom	4:11	—	Live from Pier 9
▶	Quiet Arithmetic	Marlow & the Tidewater	3:51	96	Quiet Arithmetic

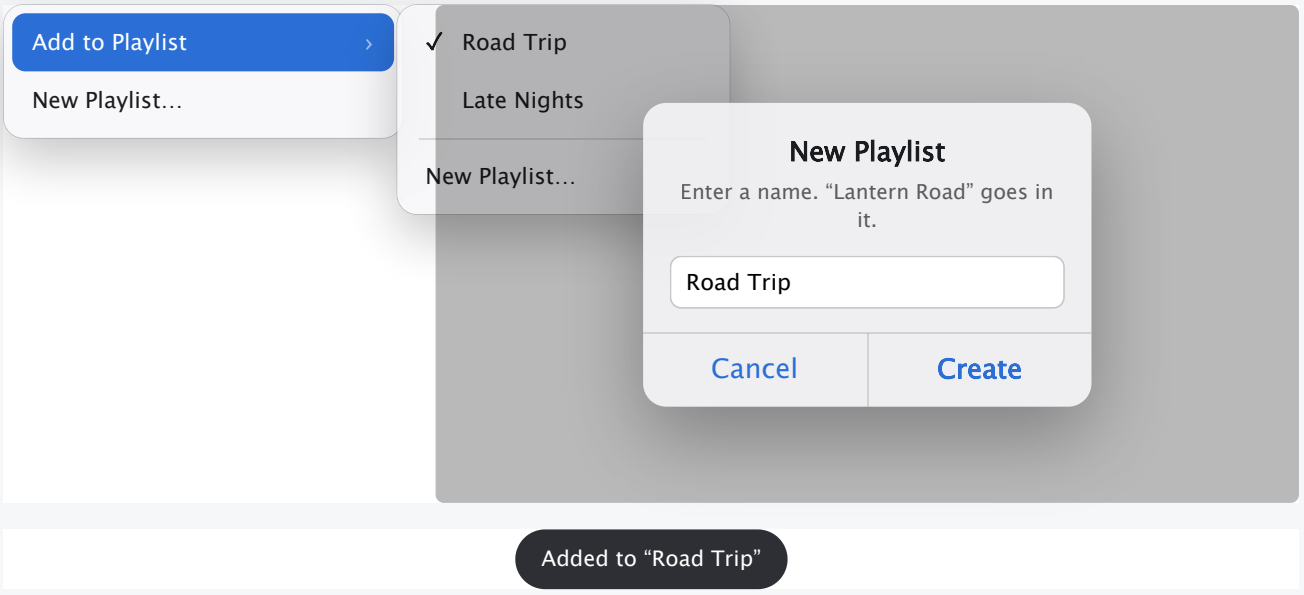
The same component the player renders, with the reference's stylesheet block: nine columns, 18 px rows, the stripe, no rules, an embossed header with the sorted column tinted, monochrome source badges read from each track's locator, icon columns for offline and star, the playing row in bold with a speaker glyph and a marquee, a right-click menu, and the gel scroller that fades in while you scroll.

THE SEARCH POPOVER



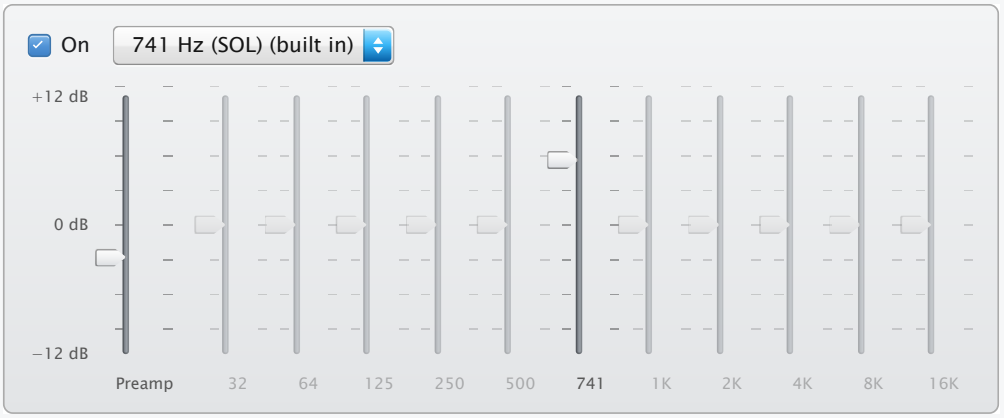
The iTunes 11 results popover: a gradient header with the count and Clear, 40 px rows whose artwork tile is a fifteen-second audition button with a countdown ring, a circled chevron on the hot row, and a pager. Arrow keys walk the rows and page off the end.

CONTEXT MENU, SHEET AND TOAST



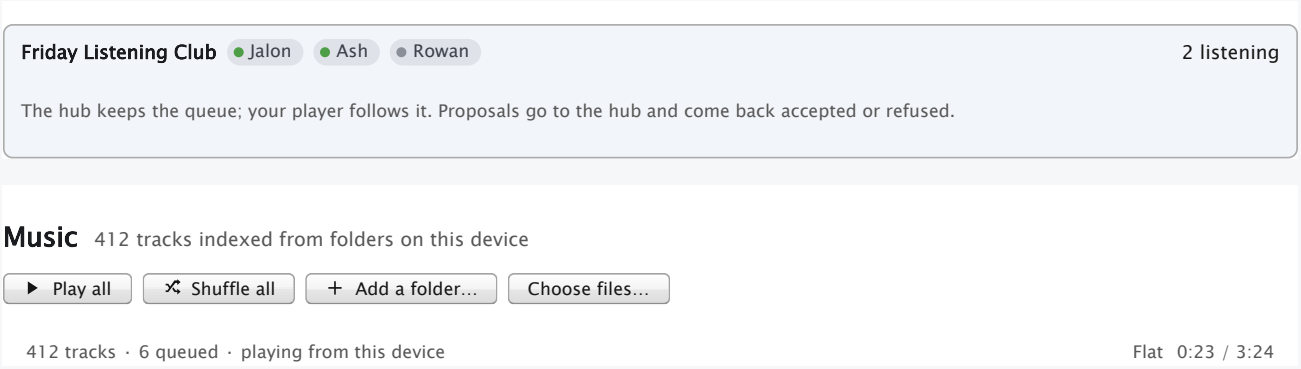
The desktop context menu with its playlist submenu; the New Playlist sheet; and the toast that answers a change. All three are the reference's, classes and all.

THE EQUALIZER WINDOW



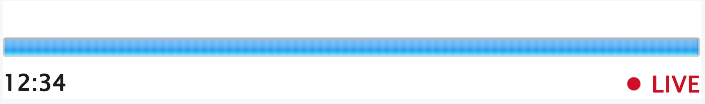
The iTunes equalizer: On beside the preset pop-up, then a preamp and the ten graphic centres on a ±12 dB scale with tick dashes flanking each rail. The window hugs its rails at 600 px (DEC-014) rather than stretching to the page. A one-band preset is chosen here so the greyed rail is visible — those faders name the preset that is not using them.

SHARED MODE, AND THE PAGE'S FURNITURE



The strip under the hero when you are in a group; a section's head and toolbar row; and the status line the bottom bar used to carry.

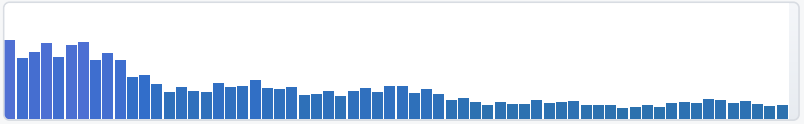
A LIVE BROADCAST



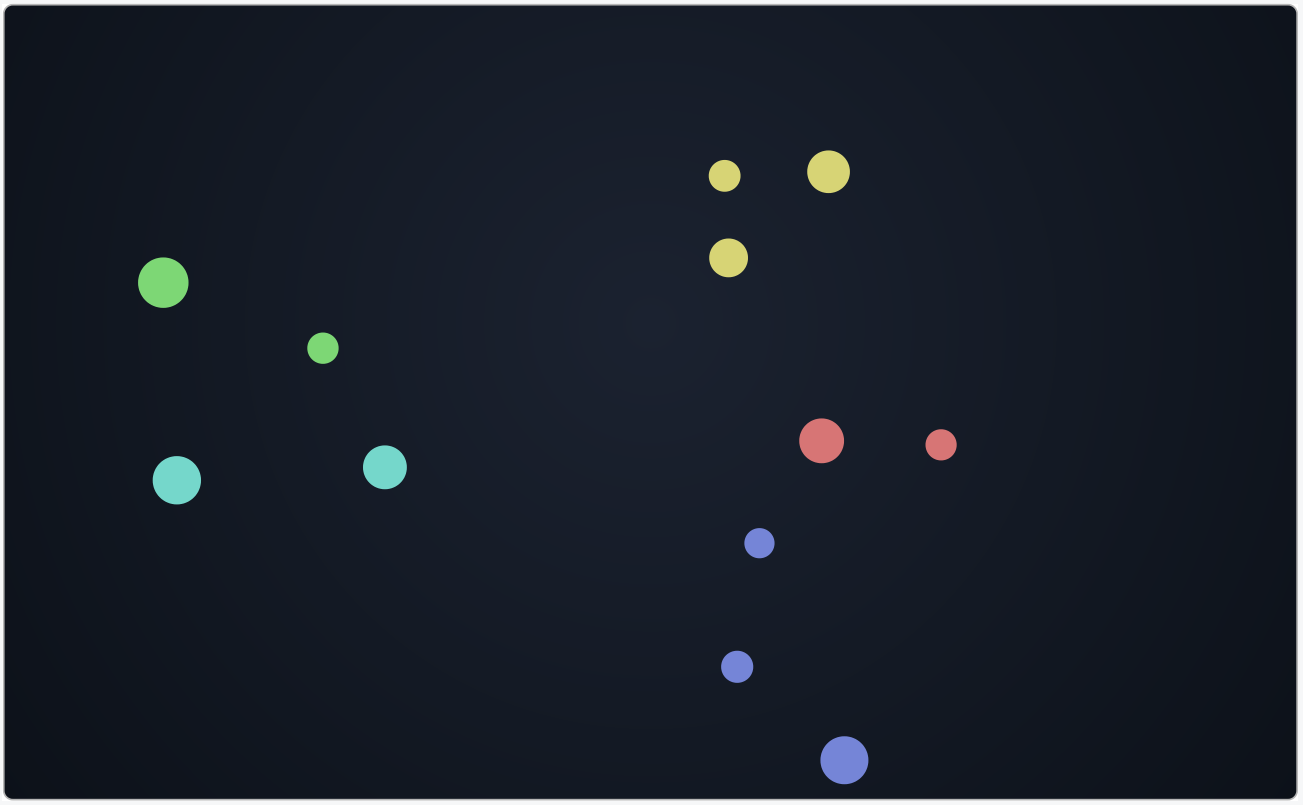
In shared mode the rail is not yours to drag: the duration stamp becomes the LIVE marker, the fill sits at the live edge, and the rail drops its slider role rather than announcing a control that refuses every change.

THE VISUALISERS — SPECTRUM, STAGE AND CONSTELLATION

The player draws three visualisers, and all three follow one rule: a visualiser must carry the data, not hide it. The two below are drawn here by the player's own code — `drawSpectrum` and `layoutStars` in `music-player/src/lib/`, the functions the views call — in the player's own `.player-spectrum` and `.player-constellation` styles. The third, the **jewel case**, is live on the hero at the top of this chapter.



The spectrum, under “How this is being played” in the Now playing section. Each analyser frame is averaged into 64 bars; a bar's hue and lightness rise with its level, so a loud band reads brighter as well as taller. It draws only while a song plays — paused, the loop stops rather than repaint a still picture — and under reduced motion it redraws four times a second instead of every frame. There is no audio here, so the specimen is fed a generated frame; print shows one fixed frame.



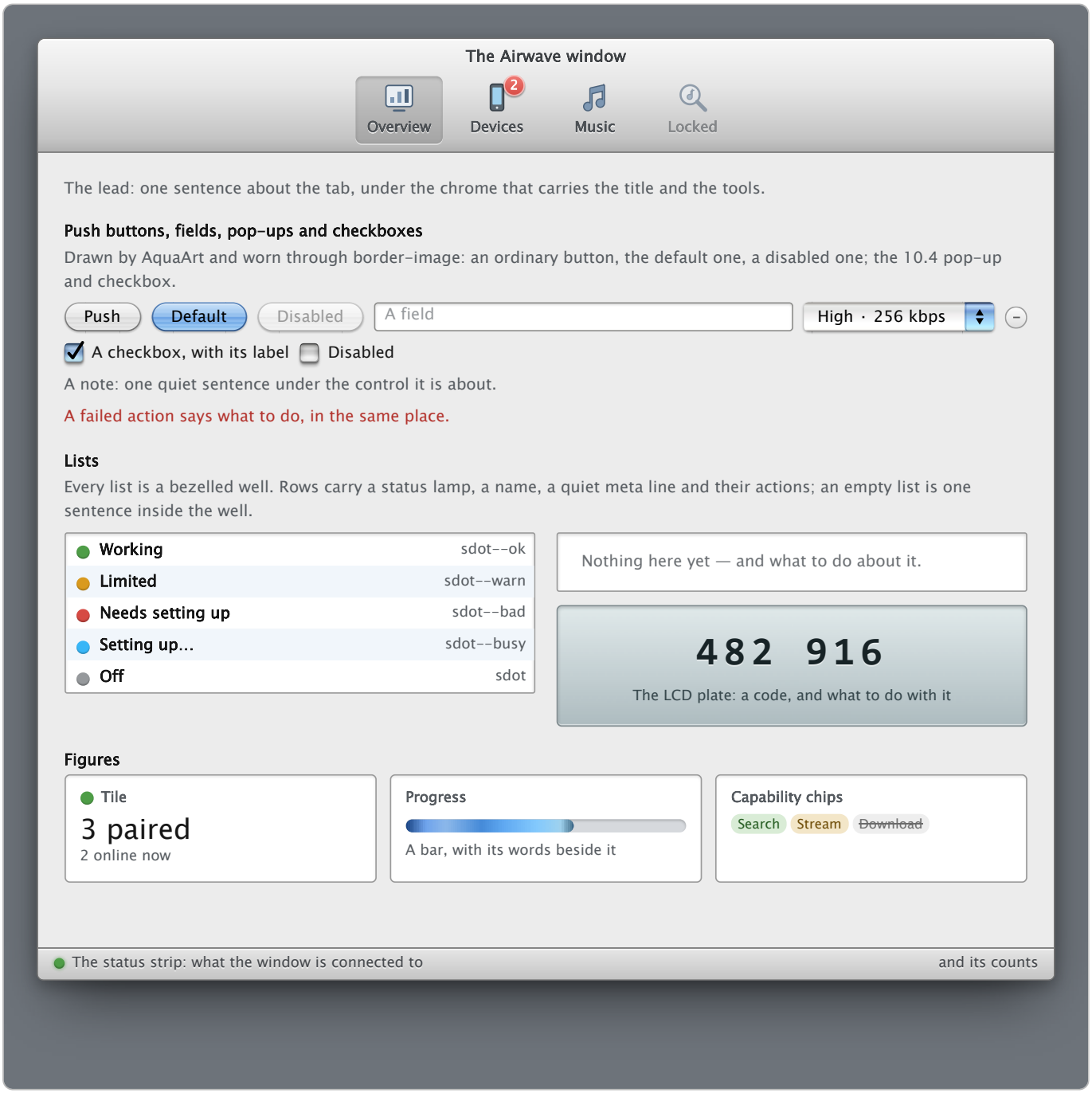
The constellation: the library as a star field, one star per album. Albums by one artist share an angular sector, a star's size is the album's length, and its colour is the artist's hue, so the layout is information, never scatter. The app renders it with WebGL and a slow rotation; this is the same placement projected through the same camera (55°, z = 42), as SVG so it also prints. Hover a star for its album.

The **jewel case** is `packages/aqua-ui/src/stage/jewel-case.ts`: the case and disc built from primitives, sleeve and tray-card textures generated from the track's real cover and running order, a drag with momentum, and a pose remembered per device. It stands in for Cover Flow (DEC-006). The stage and the constellation load Three.js lazily and share the chunk, so a listener who opens neither never downloads it. All three keep their meaning without motion: the spectrum slows to a level, the stage stops its idle drift and keeps the CSS cover, and the constellation opens as the same albums in a table, with identical selection and keyboard behaviour (`UX-MOTION-001`, `UX-STATE-001`). Both sections also appear whole in the mockups: *Now playing (audio chain)* and *Constellation*.

The Airwave window — what the hub and the companion are made of

A centred Snow Leopard window: one sheet of chrome carrying the title and a row of icon tools, a pane that scrolls beneath it, and a status strip at the foot. Airwave Hub has six tools (Overview, Devices, Music, Groups, Sharing, System) and Airwave Companion four (Library, Live TV, Remote, Settings). There is no source list, no traffic lights and no media toolbar: the tools are the navigation, a roving-tabindex tab strip (`UX-KEY-001`).

The frame below is not a drawing of it. It holds exactly what the hub's page loads — `airwave-window.css`, `airwave-hub.css` and the hub's `styles.css`, with none of the component library's stylesheets — AquaArt is installed in it as the products' `main.tsx` installs it, and the controls are the hub's own kit.



WHERE EACH PART COMES FROM

- **The stylesheets are generated.** `packages/aqua-ui/scripts/make-window-css.py` copies the `<style>` of `airwave-companion.html` into `airwave-window.css` and every rule the hub design adds into `airwave-hub.css`. Change the design file and run `pnpm build:window-css`; never edit the two by hand.
- **The controls are drawings.** AquaArt (`packages/aqua-ui/src/airwave/aqua-art.ts`, imported as `@now-playing/aqua-ui/airwave-art`) draws the Snow Leopard push button and the 10.4 pop-up and checkbox as SVG and publishes them as custom properties (`--aq-btn-22`, `--aq-def-22`, `--aq-pop-22`, `--aq-cb-14` and their pressed, disabled and checked forms). One implementation, shared by both products (NPD-021, DEC-026).

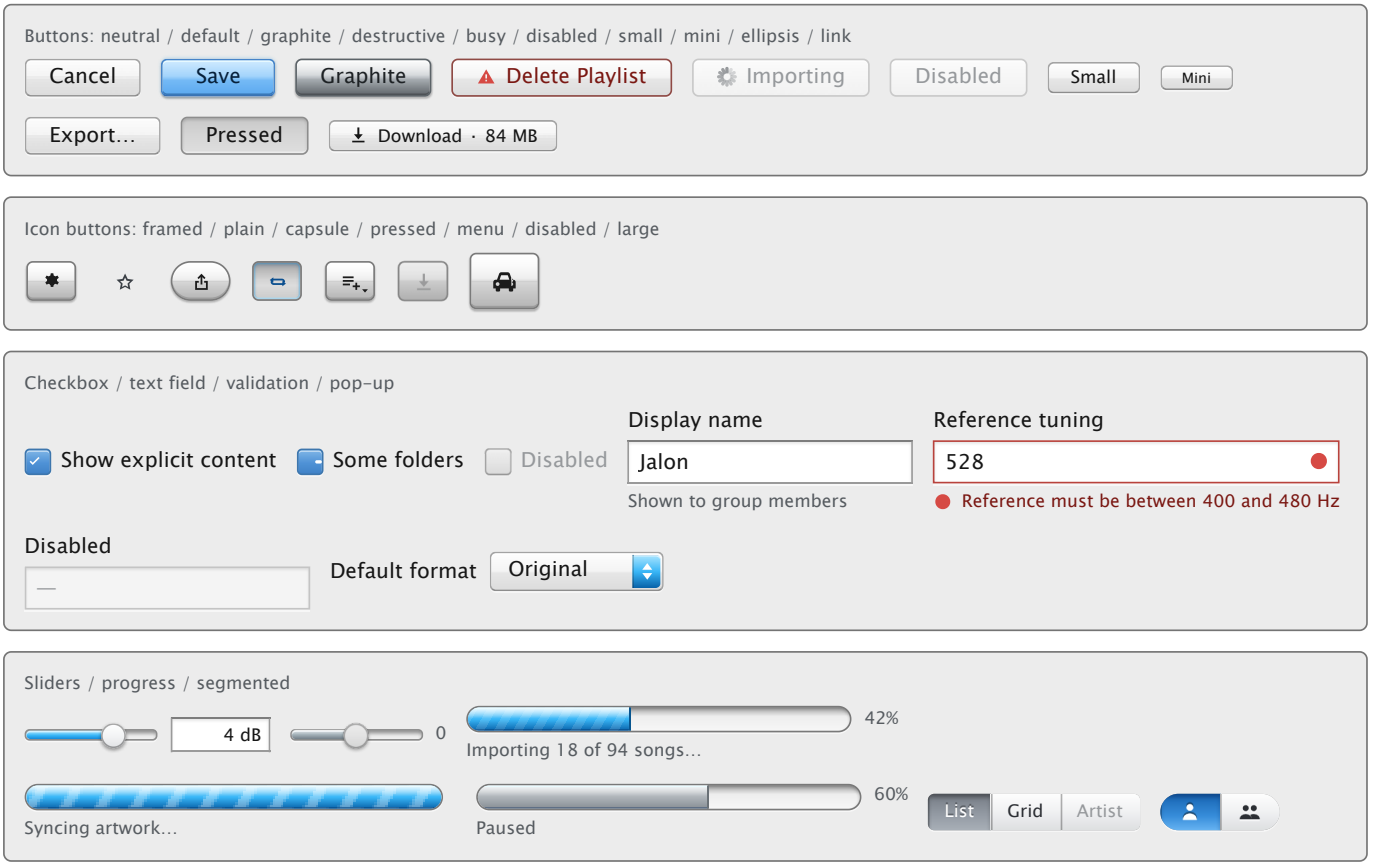
- **The kits are per product.** `docker-container/src/web/ui.tsx` and `windows-companion/src/renderer/ui.tsx` write the same classes — `.push`, `.field`, `.pop`, `.chk`, `.well`, `.rows`, `.sdot` — with different props and, for the sheet, different markup, so they are not one component (DEC-026). By product, below, lists what each exports.
- **Each product's `styles.css` is the remainder.** The hub's holds the layout the design wrote inline, the states it never drew (sign-in, the sheet, loading and failed lists) and the contrast inks. The companion's makes the window fill its frame, makes the chrome the title bar and drag region (DEC-020), and adds the song list and the sheet.

STATES EVERY LIST AND ACTION OWES

- **Loading, empty and failed are one quiet line inside the well** — never an illustration, so nothing jumps when rows arrive (`UX-STATE-001`). A failed list offers Try Again in the same row.
- **A failed action says what to do** in a sentence under the control that failed (`.note--bad`), not in a toast.
- **A disabled button says why** on hover and to assistive technology; a busy one keeps its width.
- **Anything that cannot be undone asks first, in a sheet** that drops over the pane, with Cancel as the safe default (`UX-SAFE-001`, DEC-021). Both sheets are in the mockups.
- **Colour never carries the meaning alone:** every status lamp has its word beside it or in its accessible name.

Library controls

The component library's controls, in both of its skins. The player's React source imports them; the hub and the companion do not — their controls are the Airwave window's, above.



Overlays

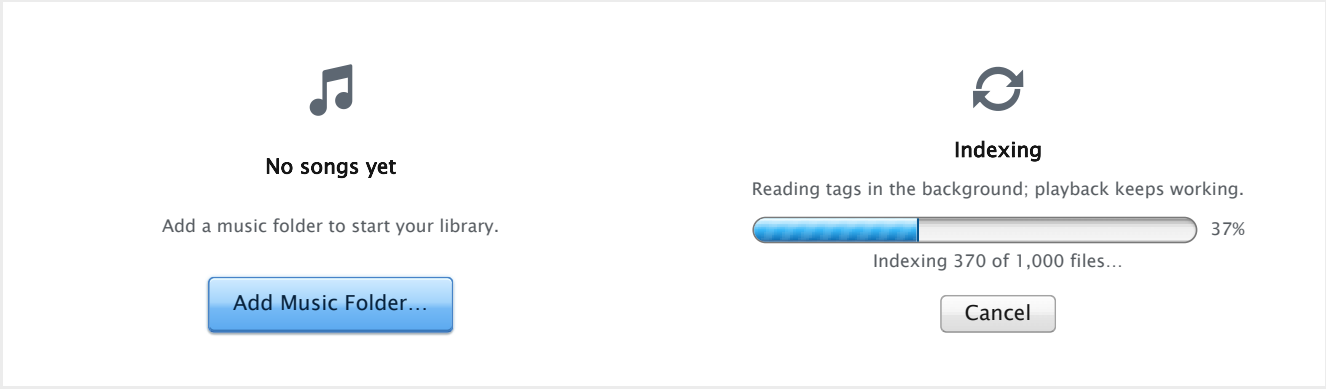
Sheets attach to their window; alerts stand alone; toasts confirm what just happened and get out of the way. Press the buttons. These are the component library's overlays; the Airwave windows' confirmation sheets are drawn in the mockups.



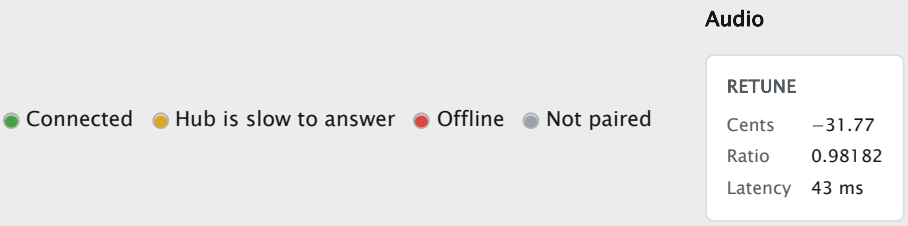
States — the group people skip

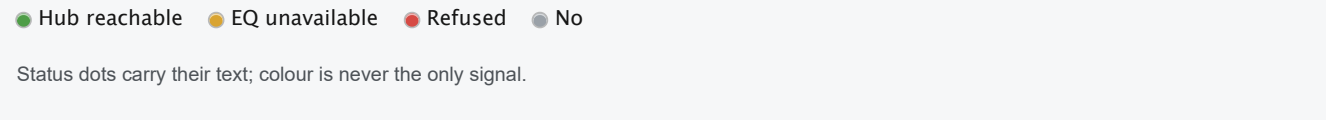
Every screen owes an answer for empty, loading, offline, partial, refused and out-of-date. The component library draws two of them as panels, `EmptyState` and `LoadingState`; the rest are said in the view's own words where the choice is made. In the Airwave windows the same answers are one quiet line inside the list's well (see The Airwave window).

States: empty / loading



Status dots + panel / key-value list



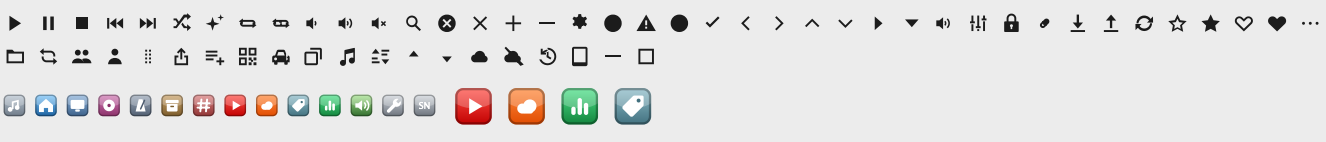


Status dots carry their text; colour is never the only signal.

Icons

Two families in the component library: single-colour glyphs for every control, and the platform marks that say where a track comes from. The Airwave windows draw their own: six 26-pixel tools for the hub (`docker-container/src/web/icons.tsx`) and four tools and three row glyphs for the companion (`windows-companion/src/renderer/icons.tsx`), as their designs drew them; both sets are in the mockups.

Icon families: glyphs (single colour), platform marks



A page-skin glyph, at the transport's size



By product

What each product is built from today, read from its source when this page was built — so a product that moves on cannot leave a tick behind.

THE AIRWAVE WINDOW KITS

PIECE	AIRWAVE HUB	AIRWAVE COMPANION
airwave-window.css (generated)	●	●
airwave-hub.css (generated)	●	–
AquaArt — @now-playing/aqua-ui/airwave-art	●	●
ActionError	●	–
Ago	●	–
Check	●	●
ConfirmProvider	–	●
EmptyCells	●	–
EmptyRow	●	●
Field	●	–
Group	●	–
HubUiProvider	●	–
LoadingRow	–	●
Note	●	–
Option	–	●
Pop	●	●
Progress	–	●
Push	●	●
Remove	–	●
Rows	–	●
Sdot	●	–
Sheet	●	–
Status	–	●
SubHead	●	–

Rows after the first three are what each product's `ui.tsx` exports. A name in both columns is two components that write the same classes, not one shared component: `Push` takes `primary` in the hub and `isDefault` in the companion, `Check` reports a boolean in one and passes input props through in the other, and the confirmation sheet is a `role="alertdialog"` layer in the hub and a modal `<dialog>` in the companion (DEC-026).

THE COMPONENT LIBRARY

`packages/aqua-ui` exports 38 components. The hub and the companion import none of them. All 38 are imported by the player's React source in `music-player/src`, which is kept and tested but is not what the player serves (DEC-019). A component no product imports is removed from the library rather than kept as legacy.

Imported by the player's React source (38)

`AquaTable` `BarClock` `BarSearch` `Button` `ButtonLink` `Checkbox` `EmptyState` `Glyph` `Hero` `HeroArt` `IconButton` `InlineValidation` `JewelStage` `KeyButton` `KeyTransport` `KeyValueList` `LevelSlider` `LoadingState` `ModeSwitch` `MusicList` `PageBar` `Panel` `PanelSection` `PopUpMenu` `ProfileButton` `ProgressBar` `ProviderArtworkProvider` `ProviderMark` `SectionStrip` `SegmentedControl` `Sheet` `Slider` `SourceBadge` `Spinner` `StatusDot` `TextField` `ToastProvider` `TrackScrubber`

The player's shell also draws its search popover, sheets, toast and equalizer window from its own markup rather than through a component.

Interaction and content rules

Every rule has a stable ID. Cite it in a change that touches the behaviour, and change the rule, the owning code and its test together. **Adopted** rules are the standard; **observed** ones describe what the code does today and have not been ratified. The source is `design/ux-rules.json`.

PRINCIPLES THAT OUTRANK THE VISUALS

`UX-PRIN-001` **A control that cannot act is not drawn** — ADOPTED

No button that looks live and does nothing, no toggle whose state is decoration. If an action is impossible in this context the control is omitted or replaced by an explained state.

The equalizer window ships without traffic lights: three circles that close nothing would be decoration.

OWNED BY `music-player/src/views/Equalizer.tsx`

CHECKED BY reviewer Judgement call, checked in review (docs/AQUA_CONFORMANCE.md).
`docs/DEVIATIONS.md`

`UX-PRIN-002` **An unavailable capability says so where the choice is made** — ADOPTED

Not in a tooltip, not in a log, not by disappearing. The control stays visible and, when chosen, reports the reason instead of acting.

The Shared segment of the mode switch stays visible with no hub paired and reports why when pressed.

OWNED BY `packages/aqua-ui/src/components/PageBar.tsx` `packages/aqua-ui/src/components/States.tsx` `windows-companion/src/renderer/views/Library.tsx` `windows-companion/src/renderer/views/Hub.tsx`

CHECKED BY `packages/aqua-ui/tests/dom/page-chrome.test.tsx` — “reports the reason instead of switching when a mode is unavailable”
`music-player/tests/e2e/np/livevol.spec.ts` — “the radio transport: Previous and Next read as unavailable, Play does not, and they say why”
`windows-companion/tests/dom/companion-shell.dom.test.tsx` — “with no hub paired, Send to Hub says where to pair one”

`UX-PRIN-003` **Say what the code does, not what the feature is called** — ADOPTED

Labels describe the implemented behaviour. When a label and the DSP (or the network, or the provider) disagree, the label changes.

Solfeggio presets are described as filters with no claimed physical effect; retuning states that the fallback changes tempo.

OWNED BY `music-player/src/views/Equalizer.tsx` `music-player/src/views/Settings.tsx`

CHECKED BY `music-player/tests/e2e/np/connect.spec.ts` — “the companion, the container and backup”
`music-player/tests/e2e/helper.spec.ts` — “a link the helper will not fetch from is refused, with the reason”

`UX-PRIN-004` **A disabled control is still legible about why** — ADOPTED

Greyed, not gone. The reason is part of the accessible name, not only a tooltip.

“500 Hz band, not used by 528 Hz (MI)”.

OWNED BY `music-player/src/views/Equalizer.tsx`

CHECKED BY `music-player/tests/e2e/np/connect.spec.ts` — “the companion, the container and backup”

KEYBOARD, FOCUS AND TOUCH

UX-KEY-001 **One tab stop per group; arrows move inside it** ● ADOPTED

The section strip, the mode switch, tab strips, tables and menus use a roving tabIndex. Home/End jump, Enter activates, type-ahead selects. Focus travels with the selection. The tools of an Airwave window are such a strip: arrows, Home and End move the selection, and a locked tool is skipped.

OWNED BY
`packages/aqua-ui/src/components/SectionStrip.tsx` `packages/aqua-ui/src/components/AquaTable.tsx` `packages/aqua-ui/src/components/MusicList.tsx` `docker-container/src/web/App.tsx` `windows-companion/src/renderer/App.tsx` `windows-companion/src/renderer/views/Library.tsx`

CHECKED BY
`docker-container/tests/e2e/a11y.spec.ts` — “@a11y the tab strip is navigable with the keyboard alone”
`packages/aqua-ui/tests/dom/table.test.tsx` — “uses a roving tabindex, arrow navigation, Space selection and Enter activation”
`packages/aqua-ui/tests/dom/page-chrome.test.tsx` — “has one tab stop, moves with arrows and Home/End, chooses with Enter, and answers type-ahead”
`docker-container/tests/dom/admin-shell.dom.test.tsx` — “moves between tabs with the arrow keys, Home and End, keeping one tab stop”
`windows-companion/tests/dom/companion-shell.dom.test.tsx` — “moves along the tools with the arrow keys, choosing as it goes”
`windows-companion/tests/dom/companion-controls.dom.test.tsx` — “End reaches the last song, fetching only its stretch; Shift+Home chooses everything between by asking for the ids”
`windows-companion/tests/dom/companion-controls.dom.test.tsx` — “Ctrl+A chooses every song, and Send to Hub sends them in batches”

UX-FOCUS-001 **Focus is always visible** ● ADOPTED

Every focusable element shows --aqua-focus-shadow or a 2px --np-accent outline at 2px offset, including when the ring is drawn on an ancestor via :focus-within.

OWNED BY
`packages/aqua-ui/src/styles/aqua.css` `packages/aqua-ui/src/styles/now-playing.css` `windows-companion/src/renderer/styles.css`

CHECKED BY
`music-player/tests/e2e/a11y.spec.ts` — “@a11y focus is visible wherever it lands”
`windows-companion/tests/dom/companion-controls.dom.test.tsx` — “dims only the pane under the toolbar while a sheet is open, and keeps it modal”

UX-KEY-002 **Every action is reachable with the keyboard alone** ● ADOPTED

Every section and every transport control can be reached and operated without a pointer. Space toggles playback without scrolling the page.

OWNED BY
`music-player/src/App.tsx` `docker-container/src/web/App.tsx` `windows-companion/src/renderer/App.tsx`

CHECKED BY
`music-player/tests/e2e/a11y.spec.ts` — “@a11y the transport is reachable and named from the keyboard alone”
`music-player/tests/e2e/player.spec.ts` — “space toggles playback and does not scroll the page”
`docker-container/tests/e2e/a11y.spec.ts` — “@a11y the tab strip is navigable with the keyboard alone”

UX-TOUCH-001 **Nothing interactive is under 44px on a coarse pointer** ● ADOPTED

Under (pointer: coarse) controls grow; where the visual cannot grow, a transparent ::after takes the taps. Desktop keeps its 18px rows. Hit targets are never shrunk to match a visual; --aqua-hit is 32px on desktop, 44px on touch.

OWNED BY
`packages/aqua-ui/src/styles/aqua.css` `packages/aqua-ui/src/styles/now-playing.css`

CHECKED BY
`music-player/tests/e2e/responsive.spec.ts` — “nothing overflows, no text under 12px, no target under 44px”
`music-player/tests/e2e/responsive.spec.ts` — “the desktop keeps the design it was drawn with”
`packages/aqua-ui/tests/e2e/styleguide.spec.ts` — “every screen of every product fits at every size”

UX-TOUCH-002 **No text under 12px on touch; field text is 16px** ● ADOPTED

The touch layer steps the scale up (15/14/13/12/11px). A field's text is 16px because iOS zooms the page when a smaller field takes focus.

OWNED BY `packages/aqua-ui/src/styles/aqua.css` `packages/aqua-ui/src/styles/now-playing.css`

CHECKED BY `music-player/tests/e2e/responsive.spec.ts` — “settings, where every panel of explanatory text lives”

UX-FORM-001 **A hub settings form sends nothing until Save, and Revert puts back what the hub has** ● ADOPTED

In the hub’s Network, Backup, Recommendations and Discord forms nothing is sent while typing or choosing. Values the hub would refuse are checked first and said under the form; Save sends what changed together and says so; Revert puts back what the hub holds. The companion is different on purpose: its preferences are written as they change and the foot of the pane says so (UX-FEED-001).

OWNED BY `docker-container/src/web/views/Network.tsx` `docker-container/src/web/views/Backup.tsx` `docker-container/src/web/views/Recommendations.tsx` `docker-container/src/web/views/Discord.tsx`

CHECKED BY
- `docker-container/tests/dom/admin-forms.dom.test.tsx` — “sends nothing until Save, checks the address first, and Revert puts it back”
- `docker-container/tests/dom/admin-forms.dom.test.tsx` — “saves the schedule, the parts and Keep together, and Revert puts them back”
- `docker-container/tests/dom/admin-forms.dom.test.tsx` — “saves the edit, and Revert puts back what the hub has”
- `docker-container/tests/dom/admin-forms.dom.test.tsx` — “saves the edited settings, then starts the bot”
- `docker-container/tests/dom/admin-forms.dom.test.tsx` — “refuses a number outside its range before sending it”
- `docker-container/tests/e2e/hub.spec.ts` — “network settings wait for Save, check the address first, and say Saved.”
- `docker-container/tests/e2e/hub.spec.ts` — “backup settings are saved with Save, the schedule says when the next one runs, and an archive downloads”

STATES, FEEDBACK AND MOTION

UX-STATE-001 **Every data view answers empty, loading and error** ● ADOPTED

A view that fetches says it is loading, says what failed and offers a retry, and when it has nothing says what to do next. In the Airwave windows each of those is one quiet line inside the list’s well, so nothing jumps when rows arrive (listState, EmptyRow and EmptyCells in the hub’s kit; LoadingRow and EmptyRow in the companion’s); a section that throws is caught by the hub’s ViewBoundary and the window stays up. In the component library they are LoadingState and EmptyState; a failure is said in the view’s own words with its retry. Blank panes are defects.

OWNED BY `docker-container/src/web/views/common.tsx` `docker-container/src/web/ViewBoundary.tsx` `packages/aqua-ui/src/components/States.tsx` `docker-container/src/web/ui.tsx` `windows-companion/src/renderer/ui.tsx` `docker-container/src/web/lib/overview.tsx`

CHECKED BY
- `music-player/tests/e2e/np/stats.spec.ts` — “before any play: nothing recorded, and no generated source on offer”
- `music-player/tests/e2e/player.spec.ts` — “opens with no music, says so, and raises no errors”
- `docker-container/tests/dom/admin-shell.dom.test.tsx` — “keeps the window alive when a section fails to render”
- `windows-companion/tests/dom/companion-shell.dom.test.tsx` — “shows no sample channels: two empty lists in the design’s words”
- `docker-container/tests/dom/admin-shell.dom.test.tsx` — “draws the badge and the Needs attention list from one reading, so they always agree”
- `windows-companion/tests/dom/companion-controls.dom.test.tsx` — “draws a window of rows, not 50,000, and counts them all”

UX-MOTION-001 **Motion confirms; reduced motion is a contract** ● ADOPTED

Nothing animates that a person did not just cause. Under prefers-reduced-motion the live dot stops, marquees hold their ellipsis, sheets fade without travel, and the 3D stage stops drifting. New animations read --aqua-anim-state or --aqua-sheet-travel.

OWNED BY `packages/aqua-ui/src/styles/aqua.css` `packages/aqua-ui/src/styles/now-playing.css` `packages/aqua-ui/src/components/Marquee.tsx`

CHECKED BY `music-player/tests/e2e/a11y.spec.ts` — “@a11y respects reduced motion: nothing loops that a person did not start”
`packages/aqua-ui/tests/unit/aqua-conformance.test.ts` — “§17.5”

UX-FEED-001 **Progress says what is happening in words** ● ADOPTED

Progress bars carry a name and a value; an indeterminate bar is aria-busy and has text. Live regions are polite and carry one sentence. The Airwave windows have no toast: what the window just did is said in the status strip (hub) or at the foot of the pane and beside the control (companion).

OWNED BY `packages/aqua-ui/src/components/ProgressBar.tsx` `packages/aqua-ui/src/components/Toast.tsx` `docker-container/src/web/ui.tsx` `windows-companion/src/renderer/ui.tsx` `docker-container/src/web/views/Downloads.tsx` `windows-companion/src/renderer/views/Transfers.tsx`

CHECKED BY `packages/aqua-ui/tests/dom/controls.test.tsx` — “progress has a name and value; indeterminate is busy with text”
`windows-companion/tests/dom/companion-shell.dom.test.tsx` — “writes a preference back as it changes, and says so at the foot of the pane”

UX-SETUP-001 **Downloaders set themselves up and say how far along they are** ● ADOPTED

yt-dlp, spotDL and (on Windows) FFmpeg are set up without a prompt, in the background, into the app's own tools folder, and only after the file matches the SHA-256 its project published for that release. Each downloader row reads Setting up… with a named progress bar, Ready (adding "Set up automatically" when the app installed it), or Couldn't set up: <reason> beside a Try Again button; the manual install advice appears only when the tool cannot be set up on this machine. Tools are set up one at a time, and one waiting its turn behind another also reads Setting up… — it is queued, not missing, so it shows no install advice and no badge. Only failed or unsupported tools count as missing — in the companion's Settings badge and on the player's Connections card, where a tool on its way reads "setting up". Where FFmpeg's absence explains an empty Tempo column, the Library says whether it is on its way or could not be set up. The Hub image ships yt-dlp, spotDL and FFmpeg; its Providers details expose each tool's verified readiness and allow separately enabling yt-dlp and spotDL without an API key, while a custom command remains untouched.

Settings ▸ Downloaders: "FFmpeg — Setting up… 42%" with a progress bar; "spotDL — Couldn't set up: GitHub did not answer (503)." and Try Again. Music: "Setting up FFmpeg — tempos appear once it finishes."

OWNED BY

`local-helper/src/provision.ts` `local-helper/src/install.ts` `windows-companion/src/main/helper.ts` `docker-container/src/web/views/Providers.tsx` `windows-companion/src/renderer/views/Settings.tsx` `windows-companion/src/renderer/views/Library.tsx` `windows-companion/src/renderer/App.tsx` `music-player/scripts/make-shell.py`

CHECKED BY

`docker-container/tests/dom/admin-forms.dom.test.tsx` — "shows both hub downloaders and sends only the selected preset to the hub"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "Settings shows setting up with progress, ready and set up automatically, and a failure with Try Again"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "Settings keeps the install advice for a tool this PC cannot set up"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "the Library says tempos are on their way while ffmpeg is being set up"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "the Library points to Settings when ffmpeg could not be set up, and says nothing once it is ready"

`windows-companion/tests/integration/main-process.test.ts` — "sets missing downloaders up by itself, reports how that went, and tries again on request"

`local-helper/tests/unit/install.test.ts` — "refuses bytes that do not match the published digest, and leaves nothing behind"

`local-helper/tests/unit/provision.test.ts` — "records a failure and waits six hours before trying again, unless told to try now"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "a tool setup has not reached yet is on its way, not missing: no badge, no install advice"

`windows-companion/tests/dom/companion-controls.dom.test.tsx` — "Check asks about that tool; one found behind says which version is out, wears the badge, and Update installs it"

`windows-companion/tests/dom/companion-controls.dom.test.tsx` — "a missing tool offers Install, and Check waits for it; a busy helper says why Update must wait"

`windows-companion/tests/dom/companion-controls.dom.test.tsx` — ""Update downloaders automatically" is on by default and writes its preference"

UX-DL-001 **A download carries clean tags and its cover, or says what it lacks** ● ADOPTED

A file the helper or the companion downloads with yt-dlp has the thumbnail embedded as cover art (JPEG) and clean tags: a YouTube upload's title is cleaned ("(Official Video)", "(4K Remaster)", frame rates and a trailing "- Official …" go; "Artist - Song" splits; the channel is the artist when the title names none), featured artists are written into the artist tag as "Artist feat. Guest" (more: "Artist feat. Guest & Other") with the main artist alone as album artist, a source's own album and genre are kept (a site category is not a genre), and the date is the whole release or upload date, YYYY-MM-DD. Each song of a playlist gets its own. The cleaning happens on the link's description in the helper, never on a command line. Without FFmpeg nothing can be embedded: the file is saved as the site sent it and the finished job says so in words. spotDL embeds its own tags and art. The library keeps the whole date beside the year.

ffprobe of a downloaded file: title=Big Buck Bunny, artist=Blender, album_artist=Blender, date=2014-11-10, and an attached picture. Without FFmpeg: "Saved as the site sent it: FFmpeg is not on this PC, so there is no cover art and the title, artist and date were not written into the file."

OWNED BY `local-helper/src/jobs.ts` `packages/domain/src/media-tags.ts` `packages/domain/src/titles.ts` `packages/contracts/src/entities/library.ts` `music-player/src/lib/library.ts` `windows-companion/src/main/library.ts`

CHECKED BY
- `local-helper/tests/unit/download-tags.test.ts` — "embeds the thumbnail as JPEG cover art and writes the tags when FFmpeg is here"
- `local-helper/tests/unit/download-tags.test.ts` — "describes, cleans, then downloads from the cleaned description"
- `local-helper/tests/unit/download-tags.test.ts` — "writes features into the artist tag as "A feat. B", and the album artist alone"
- `local-helper/tests/unit/download-tags.test.ts` — "without FFmpeg, downloads in one go and says what the file lacks"
- `packages/domain/tests/unit/media-tags.test.ts` — "takes the date tag first, then the release date, then the original release date"
- `packages/domain/tests/unit/titles.test.ts` — "takes "4K Remaster", frame rates and a trailing " - Official …" off as noise"

UX-A11Y-001 **Roles are true and colour is never the only signal** ● ADOPTED

A rail that cannot be dragged is role=img, not a slider that refuses changes. Icon buttons have names. The playing row has a glyph as well as a tint; status dots carry text.

OWNED BY `packages/aqua-ui/src/components/Hero.tsx` `packages/aqua-ui/src/components/IconButton.tsx` `packages/aqua-ui/src/components/Badge.tsx`

CHECKED BY
- `packages/aqua-ui/tests/dom/page-chrome.test.tsx` — "goes read-only for a live broadcast and says why"
- `packages/aqua-ui/tests/dom/controls.test.tsx` — "icon buttons require names; menu affordance sets aria-haspopup"

UX-LINK-001 **A download is a link** ● ADOPTED

Downloads and destinations in another tab use ButtonLink (an <a> wearing the button face), never a <button> that assigns location.

OWNED BY `packages/aqua-ui/src/components/Button.tsx`

CHECKED BY
- `music-player/tests/e2e/np/stats.spec.ts` — "after real plays: the page counts them, and export and import round-trip them"

THE AIRWAVE WINDOW

UX-CHROME-001 **One sheet of chrome: the product's name and its tools** ● ADOPTED

An Airwave window has one sheet of chrome carrying the product's name, from BRANDING, and a row of tools — nothing else. The window is the same in every state: signed out and on first run the hub draws its form or its gate inside it, with the tools it cannot open locked and the reason said. In the companion the chrome is the title bar: the operating system's own is hidden but the frame is kept, Windows draws minimise, maximise and close over the top-right corner, the chrome is the drag region and the tools are not, and the application menu is removed (DEC-020).

"Airwave Companion" over Library · Search · Live TV · Remote · Settings; "Airwave Hub" over seven tools (the design's six and Search), six of them locked until a real password is set.

OWNED BY

`docker-container/src/web/App.tsx` `windows-companion/src/renderer/App.tsx` `windows-companion/src/renderer/styles.css` `windows-companion/src/main/index.ts` `windows-companion/src/shared/identity.ts` `packages/aqua-ui/src/styles/airwave-window.css`

CHECKED BY

`windows-companion/tests/contract/app-identity.test.ts` — "hides the OS title bar but never the frame"

`windows-companion/tests/contract/app-identity.test.ts` — "asks Windows to draw its buttons over the chrome, at the height of the design's title strip"

`windows-companion/tests/contract/app-identity.test.ts` — "names the window once, from the shared constant"

`windows-companion/tests/contract/app-identity.test.ts` — "makes the chrome the drag region and leaves the tools clickable"

`windows-companion/tests/contract/app-identity.test.ts` — "wears the design's stylesheet, not a copy of it"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "has one title, five tools and nothing else in its chrome"

`docker-container/tests/dom/admin-shell.dom.test.tsx` — "renders the window with the seven tabs, the status strip and Sign Out"

`docker-container/tests/dom/admin-shell.dom.test.tsx` — "is drawn inside the same window, with every tab locked and the reason said"

TRUST, SAFETY AND DESTRUCTIVE ACTIONS

UX-SAFE-001 **Destructive actions are worded plainly and confirmed** ● ADOPTED

The button names the consequence ("Forget This Hub…", "Remove Folder", "Revoke"), sits apart from the default action, and the hub and the companion ask before doing it — in a sheet over the window it belongs to, never a second window. The safe answer has the keyboard: Cancel takes the focus in the hub, and is the default button for a destructive action in the companion; Escape cancels and focus returns to what asked (DEC-021).

OWNED BY

`packages/aqua-ui/src/components/Sheet.tsx` `docker-container/src/web/views/Devices.tsx` `windows-companion/src/renderer/views/Folders.tsx` `docker-container/src/web/ui.tsx` `windows-companion/src/renderer/ui.tsx` `docker-container/src/web/views/Backup.tsx` `docker-container/src/web/views/Discord.tsx` `docker-container/src/web/views/Downloads.tsx` `docker-container/src/web/views/Groups.tsx` `docker-container/src/web/views/Library.tsx` `docker-container/src/web/views/Profiles.tsx` `docker-container/src/web/views/Shares.tsx` `windows-companion/src/renderer/App.tsx` `windows-companion/src/renderer/views/Backup.tsx` `windows-companion/src/renderer/views/Hub.tsx` `windows-companion/src/renderer/views/Streaming.tsx` `windows-companion/src/renderer/views/About.tsx`

CHECKED BY

`packages/aqua-ui/tests/dom/controls.test.tsx` — "marks the destructive action, separates it from the default, and words it as what it does"

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reviewer Each new destructive action is reviewed for wording and confirmation.

`docker-container/tests/dom/admin-shell.dom.test.tsx` — "asks before revoking a device, and Cancel leaves it alone"

`docker-container/tests/e2e/a11y.spec.ts` — "@a11y the confirmation sheet has no detectable violations, holds the keyboard, and Escape cancels"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "asks before removing one, says the files are not touched, and does nothing on Cancel"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "asks before forgetting the hub, and says the music stays"

`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "asks before restoring the defaults"

`windows-companion/tests/dom/companion-controls.dom.test.tsx` — "shows the cache, asks before clearing it, and clears it through the main process"

UX-SAFE-002 **The bootstrap password must be replaced before anything else** ADOPTED

admin/admin signs in once, into the same window: Overview, with the amber gate asking for a real password and the other five tabs locked with the reason. Every setup-gated route refuses until then, and the hub stays on loopback outside a container. After setup, the signed-in admin can change the password again under System: current password required, two matching new entries, other admin sessions revoked and the current session rotated.

OWNED BY `docker-container/src/web/App.tsx` `docker-container/src/auth/service.ts`

CHECKED BY `docker-container/tests/integration/setup-and-pairing.test.ts` — “refuses every setup-gated route until the password is changed”
`docker-container/tests/e2e/a11y.spec.ts` — “@a11y the sign-in screen has no detectable violations”
`docker-container/tests/dom/admin-shell.dom.test.tsx` — “shows the gate in the window, with only Overview open and the other tabs locked with the reason”
`docker-container/tests/dom/admin-shell.dom.test.tsx` — “explains what stays off until the password is changed”
`docker-container/tests/dom/admin-shell.dom.test.tsx` — “lets a signed-in administrator change the password from System with current-password verification”
`docker-container/tests/integration/setup-and-pairing.test.ts` — “allows a later change with the current password and rotates only admin sessions”

UX-SAFE-003 **Pairing is confirmed by comparing a fingerprint** ADOPTED

A device receives a scoped credential only after the administrator confirms the fingerprint shown on both screens. Codes are never listed again after they are issued.

OWNED BY `docker-container/src/web/views/Devices.tsx` `docker-container/src/pairing/service.ts`

CHECKED BY `docker-container/tests/integration/setup-and-pairing.test.ts` — “issues a scoped credential only after the fingerprint is confirmed”

UX-SAFE-004 **Privacy claims are stated where they can be checked** ADOPTED

Copy that says nothing is uploaded, copied or moved must be true of the code path beside it, and the player loads nothing from outside its own origin.

OWNED BY `music-player/src/views/Settings.tsx` `music-player/src/views/Metrics.tsx`

CHECKED BY `music-player/tests/e2e/helper.spec.ts` — “fetching a link puts a real track in the library, and it plays from this device”
`music-player/tests/e2e/player.spec.ts` — “loads nothing from outside its own origin, including three.js”

UX-TV-001 **A Live TV link is read before it is kept** ● ADOPTED

A pasted M3U or XMLTV link is never kept on the strength of how it is spelled. The companion's main process reads it — http(s) only, public addresses only, redirects re-checked, a deadline, size caps on the wire and after unpacking — and keeps it only when a playlist holds channels or a guide holds programmes. Until that answer arrives the row reads "checking…"; a refusal is a sentence under the field, with what was typed left in it and nothing added. A kept link that stops answering keeps what it last held, says so, and offers Check Again. What is kept is served to the player on this PC through the helper's read-only routes /helper/v1/tv/channels and /helper/v1/tv/guide.

"✓ 112 channels" · "That address is on this PC or your own network. Live TV links need an address on the internet."

OWNED BY

`windows-companion/src/main/live-tv/index.ts` `windows-companion/src/main/live-tv/fetch.ts`
`windows-companion/src/main/live-tv/m3u.ts` `windows-companion/src/main/live-tv/xmltv.ts` `local-helper/src/server.ts`
`windows-companion/src/renderer/views/LiveTv.tsx` `windows-companion/src/main/hub.ts`
`docker-container/src/live-tv/service.ts` `docker-container/src/web/views/LiveTv.tsx`

CHECKED BY

`windows-companion/tests/integration/live-tv.test.ts` — "reads the link, keeps it with how many channels it holds, and remembers it"
`windows-companion/tests/integration/live-tv.test.ts` — "refuses what is not a playlist, in a sentence, and keeps nothing"
`windows-companion/tests/integration/live-tv.test.ts` — "refuses this PC and the home network before anything is connected"
`windows-companion/tests/integration/live-tv.test.ts` — "is refused by the reader itself, not only by the form in front of it"
`windows-companion/tests/integration/live-tv.test.ts` — "stops at the size cap, on the wire and after unpacking"
`windows-companion/tests/integration/live-tv.test.ts` — "keeps what a link last held when it stops answering, says so, and recovers"
`windows-companion/tests/integration/live-tv.test.ts` — "serves both through the helper's routes, in the contract's shape"
`local-helper/tests/integration/helper-tv.test.ts` — "gives a player page the channel list the companion keeps, in the contract's shape, with no token"
`local-helper/tests/integration/helper-tv.test.ts` — "is read-only: nothing but GET is served on these routes"
`windows-companion/tests/dom/companion-shell.dom.test.tsx` — "says why a link was not kept, keeps what was typed, and adds nothing"
`windows-companion/tests/integration/live-tv-to-hub.test.ts` — "sends the channels and their guide while sharing is on, leaving out addresses with a sign-in"
`windows-companion/tests/integration/live-tv-to-hub.test.ts` — "sends nothing while sharing is off, and removes what was sent once it is turned off"
`docker-container/tests/integration/hub-admin-controls.test.ts` — "is accepted only from a companion holding library:share, and read only with library:read"
`docker-container/tests/e2e/hub.spec.ts` — "Live TV from the companion is shown read-only in the Music tab"

UX-NET-001 **The helper reaches the network only when switched on, and then only to be read** ● ADOPTED

The companion's local helper listens on this PC alone by default. Switched on under Settings ▸ Network, other devices on the home network may use only its four token-free read routes; every other route still refuses them, even with the token, and a rebound name or a page from the internet is refused. What the switch exposes is said before it is turned on (DEC-032).

OWNED BY

`local-helper/src/security.ts` `local-helper/src/server.ts` `windows-companion/src/renderer/views/Settings.tsx`

CHECKED BY

`local-helper/tests/security/helper-lan.test.ts` — "cannot be reached from the network at all"
`local-helper/tests/security/helper-lan.test.ts` — "lists only the four token-free reads"
`local-helper/tests/security/helper-lan.test.ts` — "refuses every other route from the network, even with the token"
`local-helper/tests/security/helper-lan.test.ts` — "refuses a rebound name and a page from the internet"
`windows-companion/tests/dom/companion-controls.dom.test.tsx` — "explains what the LAN switch exposes before it is turned on"

UX-PRIV-001 **Asking for a new version needs no key and opens only the release page** ADOPTED

The companion asks the project's public releases for the latest version, with no key, and only while the check is switched on (it is on by default and asks at once when switched on). It says in words when a newer version is out, when there is none and when GitHub does not answer. Download opens the fixed release page through the main process; no address from the answer is ever opened (DEC-033). Logs leave the PC only through Export, with secrets, tokens and folder paths taken out.

OWNED BY `windows-companion/src/main/updates.ts` `windows-companion/src/main/log.ts` `windows-companion/src/renderer/views/Settings.tsx` `windows-companion/src/renderer/views/About.tsx`

CHECKED BY
- `windows-companion/tests/unit/settings-plumbing.test.ts` — "asks the project's releases without a key, says when a newer one is out, and opens only the release page"
- `windows-companion/tests/unit/settings-plumbing.test.ts` — "says plainly when GitHub has nothing or does not answer, and shows nothing when switched off"
- `windows-companion/tests/unit/settings-plumbing.test.ts` — "takes out secrets, tokens and folder paths before anything leaves this PC"
- `windows-companion/tests/dom/companion-controls.dom.test.tsx` — "says a new version is out, and Download opens the release page through the main process"
- `windows-companion/tests/dom/companion-controls.dom.test.tsx` — "asks GitHub at once when the new-version check is turned on, and says what it found"
- `windows-companion/tests/security/companion.test.ts` — "opens the release page, the logs and the data folder without the page naming a place"

SHARED LISTENING AND THE DISCORD BOT

UX-SHARED-001 **The shared queue rejects stale commands and says so** ADOPTED

Every queue command carries the revision it was based on. A stale command is refused with the current revision, never applied over someone else's change.

OWNED BY `docker-container/src/group/service.ts` `packages/domain/src/queue.ts`

CHECKED BY `docker-container/tests/integration/groups-and-queue.test.ts`

UX-DISCORD-001 **Slash and prefix commands behave identically** ADOPTED

Both transports run through one CommandService and render the same templates; the only difference is how the reply is delivered.

OWNED BY `docker-container/src/group/command-service.ts` `docker-container/src/discord/gateway.ts`

CHECKED BY `docker-container/tests/integration/discord-parity.test.ts` — "produces identical outcomes for every command"

UX-DISCORD-002 **Refusals are private and say what was needed** ADOPTED

permissionDenied, noResults, unavailableSource, emptyQueue, wrongChannel and error replies are ephemeral and name the reason or the channel to use.

OWNED BY `packages/domain/src/templates.ts` `docker-container/src/group/command-service.ts`

CHECKED BY `docker-container/tests/integration/discord-parity.test.ts` — "applies the same permission decision on both transports"

UX-DISCORD-003 **The bot never goes silent without a reason** ADOPTED

A track the bot cannot play is skipped with the reason recorded; a failed decode advances as an error; the hub's Discord section (Sharing tab) shows the voice state and last error.

OWNED BY `docker-container/src/discord/voice.ts` `docker-container/src/discord/worker.ts` `docker-container/src/web/views/Discord.tsx`

CHECKED BY
- `docker-container/tests/unit/discord-voice.test.ts` — "skips a track it cannot play instead of going silent"
- `docker-container/tests/integration/discord-worker.test.ts` — "shows the worker's live status in the hub"

UX-STREAM-001 **Streaming from the PC follows what the connection costs, and says why it paused** ADOPTED

Two switches under Remote decide whether the companion streams on Wi-Fi and Ethernet and on metered connections (mobile data, a hotspot); both are on by default (DEC-031). The cost is read from what Windows says about the connection; on a connection that is switched off streaming pauses, the pane says why in a sentence, and it starts again when allowed. The switches are off while streaming itself is off.

OWNED BY
`windows-companion/src/main/network.ts` `windows-companion/src/main/awsp.ts`
`windows-companion/src/renderer/views/Streaming.tsx`

CHECKED BY
`windows-companion/tests/integration/awsp-supervisor.test.ts` — “pauses on a metered connection when that is switched off, says why, and starts again when allowed”
`windows-companion/tests/unit/settings-plumbing.test.ts` — “reads what Windows says a connection costs”
`windows-companion/tests/unit/settings-plumbing.test.ts` — “decides whether streaming may serve, and says why not”
`windows-companion/tests/dom/companion-controls.dom.test.tsx` — “has the two switches, writes each, and says why streaming is paused on this connection”
`windows-companion/tests/dom/companion-controls.dom.test.tsx` — “keeps the switches off while streaming itself is off”

WORDS

VOICE-001 **Empty states say what to do next** OBSERVED

An empty list says what is missing and what to do next, in a few plain words: one sentence inside the list in the Airwave windows, a title and a line in the component library (“Nothing queued”). No apologies, no exclamation marks.

“No folders yet — add the one this PC keeps these in.” · “No guides yet — the guide will list channels without programmes.”

OWNED BY
`music-player/src/views/Queue.tsx` `windows-companion/src/renderer/views/Transfers.tsx`
`windows-companion/src/renderer/views/Folders.tsx` `windows-companion/src/renderer/views/LiveTv.tsx`
`windows-companion/src/renderer/views/Library.tsx`

CHECKED BY
`music-player/src/views/Queue.tsx`
`windows-companion/tests/dom/companion-shell.dom.test.tsx` — “shows no sample channels: two empty lists in the design’s words”

VOICE-002 **Errors explain the problem and the way out** ADOPTED

Name what failed in the person's terms and what fixes it (“Add a bot token first (Admin → Discord → Token)”). Never show a stack trace or an internal identifier; a 500 carries a correlation id instead.

OWNED BY
`docker-container/src/app.ts` `docker-container/src/discord/service-interface.ts`
`docker-container/src/web/views/Discord.tsx`

CHECKED BY
`docker-container/src/discord/service-interface.ts`
`docker-container/tests/dom/admin-forms.dom.test.tsx` — “loads what the hub has and shows its warnings as plain sentences”
`docker-container/tests/dom/admin-forms.dom.test.tsx` — “says what the hub said when it refuses”

VOICE-003 **One product name, set in one place** ADOPTED

The suite is Airwave and its apps are Airwave, Airwave Hub and Airwave Companion; the names come from BRANDING in TypeScript, and “Now Playing” is the name of the player’s home page only (DEC-025). Native resources that cannot import BRANDING (Android strings.xml, the SVG icons' aria-labels, the Electron builder config) repeat it and must be changed together. Machine identifiers keep their spelling.

OWNED BY
`packages/contracts/src/branding.ts` `android/app/src/main/res/values/strings.xml` `music-player/public/icon.svg`
`windows-companion/resources/icon.svg` `windows-companion/src/shared/identity.ts`

CHECKED BY
`packages/contracts/tests/branding.test.ts`
`packages/contracts/src/branding.ts`
`windows-companion/tests/contract/app-identity.test.ts` — “names the window once, from the shared constant”

PLAYER SHELL — PRINCIPLES

NP-PRIN-001 One component, one reference

● ADOPTED

Every component reconstructs one named Apple release (iPod classic, iTunes 10, OS X 10.6, iOS 5…) and is sampled from it, not averaged. Eras mix across components, never inside one. The reference is written beside the component's tokens.

The contextual menu is OS X 10.6 (opaque, 5px radius, 20px rows, 13px text); the action sheet that replaces it on touch is iOS 5/6.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY
- `music-player/tests/e2e/np/menuverify.spec.ts` — “era metrics, keyboard nav intact, escape closes, menu action still fires”
- `music-player/tests/e2e/np/menuverify.spec.ts` — “dark menu opaque”

NP-PRIN-002 Unavailable is shown and explained, never hidden

● ADOPTED

A control that cannot act in this context stays visible and focusable, is marked aria-disabled, looks disabled, and says why when pressed or hovered. It is not removed and it does not silently do nothing.

On a station, previous and next are greyed, titled 'Not available for radio', and pressing them says 'Pick another station from the list'.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY
- `music-player/tests/e2e/np/livevol.spec.ts` — “the radio transport: Previous and Next read as unavailable, Play does not, and they say why”
- `music-player/tests/e2e/np/radio.spec.ts` — “the tab opens on stations, not on a menu, and the panel is the song list’s own rules”
- `music-player/tests/e2e/np/func.spec.ts` — “avatar gradient not black; the transport at rest is a slider with every key reachable”

NP-PRIN-003 Measured, not assumed

● ADOPTED

Numbers shown to the listener come from what actually happened. A track's seconds come from the transport's own clock (a seek is not a listen); stream time is measured from playing to pause/end; anything under five seconds is not a session; a film or song counts as finished only when it played to its end. No invented history, no sample figures — with one labelled exception: Statistics shows a generated demo year while this player has recorded nothing, and says so in the Data menu, the source line and Your data (NPD-023).

Statistics opens on this player's own history once anything has played; before that, on a demo year that says it is one.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY
- `music-player/tests/e2e/np/prefs.spec.ts` — “opening: Settings is a page with its own address, and the toolbar is the navigation”
- `music-player/tests/e2e/np/prefs.spec.ts` — “it is a page at both widths, and a section has its own address”
- `music-player/tests/e2e/np/discover.spec.ts` — “a track’s seconds are measured, not assumed”

NP-PRIN-004 Values live in tokens, once

● ADOPTED

Colour, gradient and metric values are CSS custom properties on :root. Components read variables. The dark list is written twice (Auto and forced Dark) and the two copies must be identical.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY reviewer Carried from airwave-np; its test did not survive the port as a separate assertion. Checked in review against the shell.

PLAYER SHELL — PERIOD FIDELITY

NP-ERA-001 **Menus are opaque** ADOPTED

Menus, sheets and popovers of this period had no vibrancy. Their backgrounds are opaque in both themes; only the iOS HUD is translucent.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/menuverify.spec.ts` — “dark menu opaque”

NP-ERA-002 **Selection is a gradient, not a flat fill** ADOPTED

Selected rows and highlighted menu items use the era's two-stop vertical gradient with white text: iTunes 10 blue in lists (--lib-sel-*), Aqua blue in menus (--ctx-hi-*), iPod blue on the iPod screen (--ipod-sel-*).

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY reviewer Carried from airwave-np; its test did not survive the port as a separate assertion. Checked in review against the shell.

PLAYER SHELL — LIVE AND BROADCAST

NP-LIVE-001 **A live feed says LIVE where the time would be** ADOPTED

On Live TV the video controller replaces the elapsed/duration readout with a red LIVE badge, from the moment the tab opens (not only after a channel plays). The badge is centred on the progress rail horizontally in its box and vertically on the rail's centre line at every width.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/livevol.spec.ts` — “the Live TV tab is live before a channel is chosen, and a channel does not follow you on to TV”

`music-player/tests/e2e/np/devices.spec.ts` — “: the LIVE readout sits right, idle and playing”

`music-player/tests/e2e/np/livevol.spec.ts` — “LIVE on the video bar: the tab is live on arrival and a channel keeps it so”

NP-LIVE-002 **The progress of a broadcast does not move and cannot be seeked** ADOPTED

During a live feed the scrubber is an image (role=img, no range ARIA), the channel loops so LIVE stays true, and pressing it explains instead of seeking.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/livevol.spec.ts` — “LIVE on the video bar: the tab is live on arrival and a channel keeps it so”

`music-player/tests/e2e/np/func.spec.ts` — “24px optical offset preserved; sort headers are buttons and sorting works; page has h1”

`music-player/tests/e2e/np/func.spec.ts` — “avatar gradient not black; the transport at rest is a slider with every key reachable”

PLAYER SHELL — TRANSPORT AND SOUND

NP-TRANS-001 **One source sounds at a time** ADOPTED

Starting a station stops the video; starting a channel or film stops the station. A film does not follow the listener onto the Live TV or TV tab.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/livevol.spec.ts` — “one thing at a time: a channel stops the station, and tuning back stops the channel”

`music-player/tests/e2e/np/livevol.spec.ts` — “the Live TV tab is live before a channel is chosen, and a channel does not follow you on to TV”

NP-TRANS-002 **The volume control owns every sound** ● ADOPTED

The volume slider drives the station and the video as well as music; Home is silence and mutes, End is full, the level survives a change of station and a reload.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/livevol.spec.ts` — “volume reaches the stream: the slider, Home and End, and a new station keeps the level”
`music-player/tests/e2e/np/livevol.spec.ts` — “the level is remembered across a reload”

NP-TRANS-003 **Video starts at the highest quality** ● ADOPTED

A channel, episode or film opens on its highest available rendition; the quality menu lets the viewer step down.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/media.spec.ts` — “Live TV: the guide replaces the song list, the video player takes the top, and a channel plays”

NP-TRANS-004 **The equalizer never silences what it cannot process** ● ADOPTED

Audio is routed through the equalizer only when the browser allows processing it (same-origin, blob:, data:, or the companion). Cross-origin streams play unequalized, and the pane says so.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/prefs.spec.ts` — “opening: Settings is a page with its own address, and the toolbar is the navigation”

PLAYER SHELL — MENUS AND COMMANDS

NP-MENU-001 **Right-click opens the item's commands; keys work inside** ● ADOPTED

Right-clicking a song, station, channel, series or film opens an OS X contextual menu at the pointer with the commands for that kind of item. Arrow keys move, Return chooses, Escape closes and returns focus. Scrolling the page after opening does not close it instantly, and only the user's own scroll (wheel, touch drag or a scrolling key) closes it: a list repainting or bringing its playing row into view as a feed lands leaves it open.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/menuverify.spec.ts` — “era metrics, keyboard nav intact, escape closes, menu action still fires”
`music-player/tests/e2e/np/lists.spec.ts` — “the song menu is untouched”
`music-player/tests/e2e/np/lists.spec.ts` — “Radio: a station is played, kept or filed; a new playlist holds it and can be emptied”
`music-player/tests/e2e/np/menuverify.spec.ts` — “a list scrolling itself leaves the menu open; the user scrolling closes it”

NP-MENU-002 **On touch, a long press raises an action sheet** ● ADOPTED

Touch devices get the same commands as an iOS action sheet from a long press, with Cancel last.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/menuverify.spec.ts` — “long press raises action sheet, listing all commands + playlists + cancel, and its action fires”

NP-MENU-003 **Confirmation is a HUD, not a dialog** ● ADOPTED

A command that completes (added to a playlist, removed) confirms with a brief HUD naming what happened. Dialogs are only for commands that need more input.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/menuverify.spec.ts` — “era metrics, keyboard nav intact, escape closes, menu action still fires”

`music-player/tests/e2e/np/menuverify.spec.ts` — “long press raises action sheet, listing all commands + playlists + cancel, and its action fires”

NP-MENU-004 **One menu for several songs** ● ADOPTED

Right-clicking inside a selection of several songs opens the contextual menu for all of them: its heading says how many, and each command names the count it will act on — Download N Songs (or Remove N Downloads when all are saved), Add N Songs to Group Queue, Add to Playlist ▸ (ticked only when the playlist holds every one), New Playlist from N Songs…, Deselect All. The selection survives the repaint a command causes.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/marquee.spec.ts` — “dragging draws a selection box, and the menu for several acts on all of them”

PLAYER SHELL — LISTS AND PLAYLISTS

NP-LIST-001 **Every tab has playlists, behind the library button** ● ADOPTED

Music, Radio, Live TV, TV and Movies each keep their own playlists. They are reached from the library (menu) button on the left of the scope bar, and filled from the item's contextual menu. They persist across reloads.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/lists.spec.ts` — “Radio: a station is played, kept or filed; a new playlist holds it and can be emptied”

`music-player/tests/e2e/np/lists.spec.ts` — “the playlist is still there after a reload”

NP-LIST-002 **Sorting is a button** ● ADOPTED

Sortable column headers contain buttons and expose aria-sort; the sorted column tints blue (iTunes 10). A set order (a playlist) shows no arrow until a column sort overrides it.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/func.spec.ts` — “24px optical offset preserved; sort headers are buttons and sorting works; page has h1”

`music-player/tests/e2e/np/ipod.spec.ts` — “no column may show a sort arrow while the rows are in set order”

NP-LIST-003 **Drag a box to pick several songs** ● ADOPTED

On a pointer, pressing on a song row's empty space and dragging draws a rubber band; every row it touches is selected. Cmd/Ctrl-click adds or removes one song, Shift-click takes the range, Cmd/Ctrl-A takes the list, Escape clears. Picking is never playing: the click that ends a drag, and a modified click, start nothing. A plain click goes back to one song and plays it.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/marquee.spec.ts` — “dragging draws a selection box, and the menu for several acts on all of them”

PLAYER SHELL — HEADER AND NAVIGATION

NP-CHROME-001 **The header search belongs to Music** ● ADOPTED

The status bar's search pill searches songs and takes pasted song links, so it is shown only on Music. Other tabs search with the list's own field.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/hdr.spec.ts` — “the header field is up on Music and gone on the other tabs, which keep their own field”

NP-CHROME-002 **The profile button opens Settings** ● ADOPTED

The avatar at the right of the status bar opens Settings as its own page (#settings), not a pop-up.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/prefs.spec.ts` — “opening: Settings is a page with its own address, and the toolbar is the navigation”

NP-CHROME-003 **A mini player while Settings is open; leaving video stops it** ● ADOPTED

When Music or Radio is the tab and Settings covers the player, the header's search pill becomes a mini player: title and artist, time and duration, play/pause and skip, driving the real transport. Leaving a video tab — for Settings or another tab — pauses a film or episode, and disconnects Live TV (the stream is closed, not just muted).

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/mini.spec.ts` — “with Settings open, the header shows a mini player that drives the real transport”
`music-player/tests/e2e/np/mini.spec.ts` — “video tabs: Settings from Live TV shows no mini player and disconnects; Music pauses a channel”

FINDING AND HEARING MUSIC BEFORE IT IS YOURS

NP-FIND-001 **A listing can be heard before it is added** ● ADOPTED

In the header search popover, a song with a clip plays it on a click of its artwork tile — the whole clip, up to 30 seconds, with a countdown ring and a fade at the tail — and resting a fine pointer on the row for five seconds arms the same play with a visible fill in the selection blue. Leaving the row, a key, a scroll or another row cancels the hold; touch and reduced motion never arm; the click always works. The main track pauses for the audition and resumes after it. Clips are the catalog's: Apple Music's, then Deezer's (a Deezer clip that has expired is asked for again, once). A row with no clip keeps its tile and says why on hover instead of dressing as a button. Rows are music, never offers: the per-row "find it on YouTube / SoundCloud / Bandcamp" links and the "open it on a platform instead" fallback are gone (owner, 2026-10-06; UX-CAT-003). A hub that predates the catalog still answers with its enriched songs (features, album, genre, tempo), and a match below confidence 0.5 shows the platform's own words.

OWNED BY `music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts`
`music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY `music-player/tests/e2e/np/preview.spec.ts` — “paired, the hub answers first: album, features, genre chip and bpm on the row”
`music-player/tests/e2e/np/preview.spec.ts` — “a click plays up to thirty seconds, and the main track waits its turn”
`music-player/tests/e2e/np/preview.spec.ts` — “resting the pointer on a row arms it, fills the ring, and then plays”
`music-player/tests/e2e/np/preview.spec.ts` — “reduced motion never arms; the click still works”
`music-player/tests/e2e/np/preview.spec.ts` — “a row with no clip says so instead of pretending”
`music-player/tests/e2e/np/catalog-search.spec.ts` — “a search streams songs, artists and albums in sections; rows upsert by id and name their platforms”

NP-FIND-002 **A pasted link of any platform says what it is, keylessly** ● ADOPTED

A music link pasted into the header search — YouTube, YouTube Music, SoundCloud, Bandcamp, Spotify, Deezer, Apple Music, or a store SongLink can read — is not searched but resolved (GET catalog/resolve): by the paired hub, else the companion on this PC (its yt-dlp or spotDL reading the link with no account or key), else this browser (Deezer and Apple links through their public APIs; a YouTube or SoundCloud link named by its oEmbed). The count says "Reading the link…" at once, and that a Spotify link takes spotDL up to a minute. A song becomes one row naming its platform, its details line gaining the whole release date when it has no album; a Spotify song says it plays from YouTube Music, and + files it with that address. An album or playlist becomes one listing (NP-FIND-007). A link nobody can read says why, in the server's words. The companion answers a page it has vetted, or the token; it never answers another device on the network, because reading a link runs a tool.

Pasting https://www.youtube.com/watch?v=dQw4w9WgXcQ: “Never Gonna Give You Up” — “Rick Astley — 2009-10-25”, 3:33, YouTube.

OWNED BY

`music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts` `music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html` `local-helper/src/catalog.ts` `local-helper/src/server.ts` `packages/domain/src/catalog/engine.ts` `packages/contracts/src/api/catalog.ts`

CHECKED BY

`music-player/tests/e2e/np/links.spec.ts` — “a pasted YouTube link: the companion reads it, and the date is on the details line”

`music-player/tests/e2e/np/links.spec.ts` — “with no companion, a pasted link still resolves through oEmbed and nothing asks a local port”

`music-player/tests/e2e/np/links.spec.ts` — “a link nobody can read says why, in words”

`music-player/tests/e2e/np/links.spec.ts` — “a paired hub that reads a Spotify link slowly still lands its answer: Spotify, playing from YouTube Music”

`local-helper/tests/integration/helper-resolve.test.ts` — “reads a YouTube video: clean title and artist, the whole date, the duration and the art”

`local-helper/tests/integration/helper-resolve.test.ts` — “lists a playlist’s songs, each with its own address, and says the count and the cap”

`local-helper/tests/integration/helper-resolve.test.ts` — “reads a Spotify track through spotDL save, in a home of its own that is gone afterwards”

`local-helper/tests/integration/helper-resolve.test.ts` — “answers a vetted page without the token, and no page only with it”

`local-helper/tests/integration/helper-resolve.test.ts` — “refuses a host off the allowlist, a private address and a non-https link before starting anything”

`local-helper/tests/security/helper-lan.test.ts` — “refuses every other route from the network, even with the token”

NP-FIND-003 **Search is music, live: a calm overview of songs, artists, albums and playlists, each naming its platforms** ● ADOPTED

A search in the header (Enter) asks the catalog — the paired hub's /api/v1/catalog/search with the device credential, else the companion's /helper/v1/catalog/search, else the engine run in this browser against the keyless services that answer a page (iTunes, MusicBrainz and LRCLIB over CORS; Deezer over JSONP) — and reads its NDJSON as it arrives. Rows appear as each service answers and upsert by id (a later chunk's merged copy replaces its row in place); they are sorted by rank, ties by arrival. One song is one row: the engine merges its services' copies, and the player applies the engine's own identity (sameRecording) and merger (mergeTrack) to whatever still arrives apart — another chunk, another page — so a copy already shown joins that row with its badge instead of making a second; a live, remix or other version stays its own row. The first view after Enter is a calm overview (owner, 2026-10-07): Songs (five), Artists (three), Albums (three) and Playlists (three), each group ending in “See all N” (N+ while more may exist) that opens that type's own page (NP-FIND-004). The overview has no pager — no dots, no ‹ ›, no page count — and Page Up/Down do nothing there. Every row shows each platform it is on as a small badge (Apple Music, Deezer, YouTube, YouTube Music, SoundCloud, Bandcamp, Spotify, MusicBrainz…); a Spotify song says "plays from YouTube Music". A quiet line under the count names every service in the engine's states — a count, searching…, nothing, failed, timed out, cooling down until a time, needs yt-dlp — with the reason on hover, the platforms that only contributed links (done.linkedOnly: "Spotify: links only"), and who answered; with only this browser it says plainly that YouTube and SoundCloud results need the hub or the companion. A failure says plainly what happened. Arrows move through every row of every section, Enter auditions a song with a clip or opens what it is on, Ctrl/Cmd+Enter adds the song, Shift+Enter queues it to Up Next (NP-FIND-010), Escape goes back a level and then closes; the rows are a listbox of groups with options, and a live region says the counts. A song row shows its cover, title, credit line (the main artist, then “feat. …” from the other credited names), album and year, time, BPM and a small “E” when the store marks it explicit; these arrive with the engine’s last chunk before done (UX-CAT-006: Deezer’s detail by id, ISRC or one exact search, the Cover Art Archive for a MusicBrainz-only cover) and upsert the row in place, so a row may gain “feat.”, its BPM, its E and its cover a moment after it first appeared. The browser’s own Deezer tempo lookup runs only for rows still without a BPM once the search is done.

“harbour”: Results: 12 songs · 2 artists · 2 albums · 4 playlists — Apple Music 6 · Deezer 9 · MusicBrainz cooling down until 12:05 · YouTube 4 · SoundCloud failed — through the companion on this PC; under Songs, “See all 12+ songs”.

OWNED BY

`music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts` `music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html` `packages/domain/src/catalog/view.ts`

CHECKED BY

`music-player/tests/e2e/np/catalog-search.spec.ts` — “a search streams songs, artists and albums in sections; rows upsert by id and name their platforms”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “keys move through every section; Enter opens what it is on, Escape goes back, Ctrl+Enter adds”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “a failure says plainly what happened”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “the overview has no pager, and Page Up/Down do nothing there; a type page has ‹ › and “Page N of M””

`music-player/tests/e2e/np/catalog-search.spec.ts` — “paired, the hub’s catalog answers first, with the device credential”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “with no hub and no companion, this browser searches the keyless services itself and says what it cannot”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “one song from three sources in separate chunks is one row with three badges; a live version stays its own row”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “a later page never repeats a song already shown: its platforms join the row it is”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “the engine’s last chunk fills a row in place: it gains “feat.”, BPM, the E mark and a cover (UX-CAT-006)”

NP-FIND-004 **One type per page: it scrolls on by itself, and ‹ › with “Page N of M” move it a page at a time** ● ADOPTED

“See all” on the overview opens a page for that type alone — Songs, Artists, Albums or Playlists — under Back: a heading with the count (N+ while more may exist) and the words searched, the type's own search field (NP-FIND-005), a segmented control to switch type, the per-service line for it, and the list. The page starts with what the overview already had and reads on from that type's next offset (offset / limit / hasMore, as the engine pages; sections=<type> only), asked at once and again whenever the list is scrolled near its end or the keys reach its last rows, while the server says more may exist (to its offset limit of 1000): infinite scroll. “Loading more…” shows while a page is coming; the end says “That’s all N songs.” A song that a later page brings again — by id or as the same recording — is not shown twice. The one pager lives here, in a footer that stays put while the list scrolls: ‹ › and “Page N of M” (“M+” while more may exist). Pages are 25 rows for songs and 12 for artists, albums and playlists (never past the contract's page bounds). ‹ › scroll the list to the previous or next page's first row and make it the row the keys are on, fetching that page first when it has not arrived; Page Up/Down do the same from the field; the count follows the scroll (at the very end, the last page is the one on show). Nothing is sliced from the first answer: later pages are fetched from the engine.

“See all 25+ songs” → Songs: 25+ for “harbour”, Page 1 of 2+; scrolled to the end: That’s all 57 songs, Page 3 of 3; ‹ lands on song 26, Page 2 of 3.

OWNED BY
`music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts` `music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY
`music-player/tests/e2e/np/catalog-search.spec.ts` — “see all opens one type’s page that scrolls on, page after page, until it is all there”
`music-player/tests/e2e/np/catalog-search.spec.ts` — “the overview has no pager, and Page Up/Down do nothing there; a type page has ‹ › and “Page N of M””
`music-player/tests/e2e/np/search-pages.spec.ts` — “on a type page the list scrolls on by itself and ‹ › move a page at a time, fetching first when needed”
`music-player/tests/e2e/np/search-pages.spec.ts` — “› past what has arrived fetches the next page first, then lands on its first row”

NP-FIND-005 **A filter for sections and services, kept; Track / Artist / Album fields and ISRC; a type page’s own field** ● ADOPTED

Filter in the popover's header opens a sheet with the sections to show (Songs, Artists, Albums, Playlists) and the services to ask (Apple Music, Deezer, MusicBrainz, YouTube, SoundCloud). Saved, it lives in the player's settings store (kv "player:search"), survives a reload, marks the button, and asks a search on screen again; at least one section and one service stay on, said in the sheet. The pill's own switch folds out Track, Artist, Album and ISRC fields, starting from what was typed; while they are out the pill shows them merged, read-only, and folded away they stay merged in it. A search from the fields asks track, artist and album rather than q. An ISRC — typed in the pill or its field — is searched as an ISRC; a malformed one is explained before anything is asked. A type page (NP-FIND-004) has a field of its own, prefilled with the words searched and named for the type (“Search songs”): Enter there searches that type alone (sections=<type>, from offset 0, a page at a time), and the segmented control beside it switches type, searching the new type for the same words.

Track “harbour lamps” + Artist “Lantern Choir” → GET …/catalog/search?track=harbour+lamps&artist=Lantern+Choir; the pill reads “harbour lamps Lantern Choir”. On the Songs page, “lantern” ⏎ → GET …/catalog/search?q=lantern&sections=tracks&offset=0&limit=25.

OWNED BY
`music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts` `music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY
`music-player/tests/e2e/np/catalog-search.spec.ts` — “the filter keeps its sections and services in the settings store, and the search asks only those”
`music-player/tests/e2e/np/catalog-search.spec.ts` — “track, artist and album fields fold out of the pill and back into it; an ISRC is searched as one”
`music-player/tests/e2e/np/search-pages.spec.ts` — “See all opens the type’s page: a scoped field, the segmented control, and a search of that type alone”

NP-FIND-006 **One card, centred under the field: album, artist and song details stack inside it, and go back** ● ADOPTED

The popover is one card that hangs centred under the header's search field at every width — the same centre x as the field, min(560px, 100vw − 20px) wide, moved only as far as it must to stay on screen, with a small caret at the field's centre keeping the joint between the two (the Aqua skin's rounded card); on a phone it spans the width. Everything the search shows opens inside that card, with Back: the overview, a type's page, and the details. An album row opens its details (catalog/album): cover, title, artist, year · label · genre · song count, its platforms, "Open in Music" (NP-FIND-007) and its songs, each auditionable and addable. An artist row opens theirs (catalog/artist): picture (an album cover when the platform has none), genre, fans, top songs and albums; an album there drills into that album. A song row opens the song: its cover, platforms (with any other homes enrichment found), genre, label and year from catalog/enrich ("Looking up…" while MusicBrainz is asked, a sentence if it cannot be), its ISRC, Add to Library, Download… (the fetch sheet with a stated basis, as the Download key: the hub when it may, else the helper, NP-FIND-012; a store-only song says there is nothing the helper can fetch), and its lyrics from catalog/lyrics — synced lines with their times, or plain, or "No lyrics found", or an instrumental said as one. Details stack; Back and Escape walk them back, and the keys come back to the row they left.

At 1280 px the card is 560 wide under the pill's centre; at 390 px it is 370 wide, 10 px from each edge; an album opened inside it is the same card, with Back.

OWNED BY

`music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts`
`music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY

`music-player/tests/e2e/np/catalog-search.spec.ts` — “an artist drills into an album; details stack, and Back walks them back”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “a song’s details: genre, label and year from enrichment, and its lyrics, synced”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “keys move through every section; Enter opens what it is on, Escape goes back, Ctrl+Enter adds”

`music-player/tests/e2e/np/search-pages.spec.ts` — “the card is centred under the field at 390, 820 and 1280 px, and never off screen”

NP-FIND-007 **A pasted album or playlist is one listing that opens in the music list like an album** ● ADOPTED

A pasted album or playlist (catalog/resolve, kind album or playlist) is one listing, not a page of rows: its title, its platform as a badge and in words with its owner and song count, and a cover that is a 2×2 mosaic of the first four songs' covers — or the list's own cover when fewer than four songs have one. Choosing it (or an album's "Open in Music") opens it in the music list the way an album opens: the silver bar names it and its songs are the rows, in the list's order. Every song loads, from any platform, with no cap: pages of up to 200 (offset / limit) are asked until the server says hasMore is false, the bar saying "Playlist · Spotify · Loading 400 of 1,250…" meanwhile and "1,250 songs" when done, and the rows already there can be played (each a visitor: its preview while the hub or the companion fetches it, NP-FIND-011) and fetched while the rest arrive (a server that still caps a list says "(the first N)"; one that fails part way says where it stopped). The songs are on show, not added: + or the Download key (through the helper) makes one the person's. Clearing the bar returns to the library.

“Lantern Mix”, Playlist on Spotify · Ada · 1,250 songs → the bar: Playlist · Spotify · Loading 300 of 1,250… — Lantern Mix ☆ ⓧ, then Playlist · Spotify · 1,250 songs.

OWNED BY

`music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts`
`music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY

`music-player/tests/e2e/np/collections.spec.ts` — “a pasted playlist: its platform, a 2×2 mosaic, and opened, the music list names it and loads every song, page by page”

`music-player/tests/e2e/np/collections.spec.ts` — “the mosaic gives way to the list’s own cover when fewer than four songs have artwork”

`music-player/tests/e2e/np/collections.spec.ts` — “an album from search opens in the list too, and starred, it is kept under Albums”

`music-player/tests/e2e/np/collections.spec.ts` — “a song on show does not play a short clip; Download offers to fetch the full song”

`music-player/tests/e2e/np/links.spec.ts` — “a pasted playlist becomes one listing that names its platform, not a page of rows”

`music-player/tests/e2e/np/links.spec.ts` — “a set whose entries the server looked up shows their covers as a mosaic”

NP-FIND-008 **The silver bar’s star keeps an album or playlist in the library** ● ADOPTED

While the music list shows an album or playlist from the catalog — from search, a pasted link, or reopened from the library — the silver bar has a star beside its name: pressing it saves the list to the library as a SavedCollection (the contract the hub's sync and the companion share: ref, savedAt, artworkUrl, covers, trackCount) in library:state, and pressing it again removes it; the star says which it would do. The library menu lists what is kept under Playlists (after the person's own) and Albums (after the library's), and choosing one reads it again through whichever server answers. Paired with a hub that allows this device `library:sync` (DEC-041), the stars are kept in step with this device's own copy on the hub (`/api/v1/catalog/saved`): on start, GET and merge — a union by {platform, kind, id}, the later savedAt winning when both hold one — then PUT what only this player has (or has newer) and DELETE what it un-starred; every star and un-star is PUT or DELETEd as it happens. An un-star is kept as a tombstone with its time (`player:saved-sync`) until the hub has heard it, so one made while the hub was away wins over the hub's older copy, and a star made since wins over the tombstone. A hub that cannot be reached is asked again later, with backoff, and nothing is said. The hub admin's starred lists come back as `shared` and are listed under Playlists ▸ “Shared on <hub>”; one opens read-only — no star, and the bar says “Shared on <hub> — read-only”. Without the permission, or unpaired, the stars stay this player's alone.

OWNED BY

`music-player/index.html` `music-player/scripts/make-shell.py` `music-player/src/shell/search/client.ts` `music-player/src/shell/search/hub-shelf.ts` `music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts`

CHECKED BY

`music-player/tests/e2e/np/collections.spec.ts` — “the star in the silver bar saves the list to the library, the menu lists it, and it opens again from there”

`music-player/tests/e2e/np/collections.spec.ts` — “an album from search opens in the list too, and starred, it is kept under Albums”

`music-player/tests/e2e/np/hub-shelf.spec.ts` — “starred lists sync with the hub on start: a union, and an un-star made offline wins over the hub’s older copy (NP-FIND-008)”

`music-player/tests/e2e/np/hub-shelf.spec.ts` — “a star and an un-star go to the hub; one made while the hub is away is kept as a tombstone and sent once it is back”

`music-player/tests/e2e/np/hub-shelf.spec.ts` — “the hub admin’s shared lists open read-only under “Shared on TOWER”: no star, and the bar says so”

`music-player/tests/unit/hub-shelf.test.ts` — “is a union by platform, kind and id: this player’s own first, then what only the hub had, oldest first”

`music-player/tests/unit/hub-shelf.test.ts` — “keeps the later copy of a list both hold, and sends this player’s when it is the newer”

`music-player/tests/unit/hub-shelf.test.ts` — “lets an un-star made here (offline) win over the hub’s older copy, and owes the hub a DELETE until it has heard it”

`music-player/tests/unit/hub-shelf.test.ts` — “lets a star made after the un-star win, and drops the tombstone”

NP-FIND-009 **Playlists are a type of their own: public playlists keylessly, the person’s starred lists above them** ● ADOPTED

Search lists playlists as it lists songs, artists and albums (UX-CAT-005): the overview shows three with “See all N”, and the Playlists page lists them all, a page of twelve at a time. A playlist row shows its picture (or a 2×2 mosaic of its first songs' covers when the service gave them), its title, its platform badge, and “Playlist on Deezer · owner · N songs”. Public playlists come from Deezer's public API through whichever server answers — the hub, the companion, or this browser over JSONP — with no key; the other services say they have nothing in the section. Choosing one opens it in the music list the way an album opens (NP-FIND-007): its link is resolved page by page, the silver bar names it, and the star keeps it (NP-FIND-008). On the Playlists page the person's starred playlists sit first under “In your library” (“From the catalog” below), each opening again from there.

Playlists: 30 for “harbour” — In your library: Kept Lanterns (Playlist on Spotify · Ada · 9 songs); From the catalog: Mix 01 (Playlist on Deezer · Playlist Editor · 40 songs), Mix 02… Page 1 of 3.

OWNED BY `music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts` `music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html` `packages/domain/src/catalog/providers/deezer.ts` `packages/contracts/src/api/catalog.ts`

CHECKED BY
- `music-player/tests/e2e/np/search-pages.spec.ts` — “Playlists is a type: Deezer’s public playlists, the starred ones above them, and one opens like an album”
- `music-player/tests/e2e/np/search-pages.spec.ts` — “with only this browser, the Playlists type still answers: Deezer’s playlist search over JSONP”
- `packages/domain/tests/unit/catalog-engine.test.ts` — “lists public playlists from Deezer as a section of their own, keylessly (UX-CAT-005)”

NP-FIND-010 **A song from search goes to Up Next, a playlist or the library through the row’s menu** ● ADOPTED

Every song row in the search — the overview, a type page, an album's or artist's songs — has a “…” at its end, and right-click or a long press on the row does the same: the shell's contextual menu opens over the song with exactly Add to Up Next, Add to Playlist ▸ (the person's playlists, ticked where one holds the song, and New Playlist…; paired, the hub's as well, NP-FIND-013), Add to Library (“In your library”, disabled, once it is), Download… (the fetch sheet: the hub when it may, else the helper, NP-FIND-012; disabled, saying why, for a store-only song) and Audition (disabled when the song has no clip; “Stop Audition” while it plays). Filing a song that is not in the library yet adds it first — quietly, over the library's own wall (library:add), kept in library:state so a playlist that holds it still finds it after a reload; nothing plays — and then queues or files it; the HUD confirms (NP-MENU-003) and the row's + reads ✓. The + badge stays the one-click Add to Library. From the field, Ctrl/Cmd+Enter adds the hot song and Shift+Enter queues it to Up Next. Escape closes the menu before it closes the popover; the menu's first enabled command takes the keys; on a touch screen the commands are 44 px tall and a submenu unfolds in place.

“One” ▸ … ▸ Add to Up Next → HUD “Up Next: “One””; Add to Playlist ▸ New Playlist… → the sheet → “Added to “Harbour Nights””.

OWNED BY `music-player/src/shell/search/index.ts` `music-player/src/shell/search/view.ts` `music-player/src/shell/search/client.ts` `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY
- `music-player/tests/e2e/np/search-pages.spec.ts` — “a song row’s menu: “…”, right-click and long-press open it; it queues to Up Next, files into a playlist and adds to the library”

NP-FIND-011 **A catalog visitor is fetched, not previewed; short clips play only when auditioned from Search** ● ADOPTED

A visitor — a catalog song in the music list that is not on this device: a search result filed as a link row, a song of a catalog list on show, or an online Discover pick — is metadata, not playable audio. Choosing it in the music list never starts a 30-second preview and never marks it playing or heard. If the person has a hub credential that may fetch downloads and receive files (downloads:request, transfers:receive), or a companion helper is available, it opens the fetch sheet; the person states why they may have the file, and the hub/helper fetches a full-length copy. Only after that copy is saved locally does it become the active track and play through the player transport at its full duration. With no route, the UI says what is missing and does not claim playback. 30-second previews are requested and played only by the explicit audition control on Search results; they are never part of Now Playing, Discover, a playlist/album list, or the fetch sheet. A visitor's Download button and right-click remain catalog actions (NP-FIND-010).

Choosing "Night Harbour" from the music list with a hub paired opens the fetch sheet without audio; after an authorized fetch, the kept local copy starts at its full duration. Clicking the Search result's preview button still auditions its 30-second clip.

OWNED BY `music-player/src/shell/search/visit.ts` `music-player/src/shell/bridge.ts` `music-player/src/shell/search/index.ts` `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY
- `music-player/tests/e2e/np/discover-online.spec.ts` — "a visitor row does not play a preview; clips are auditioned in Search and full tracks must be fetched"
- `music-player/tests/e2e/np/discover-online.spec.ts` — "with no preview and nothing to fetch it, it says what is needed instead of pretending"
- `music-player/tests/e2e/np/discover-online.spec.ts` — "with a hub, a visitor fetches the full song without a preview, then plays the full local copy (NP-FIND-012)"
- `music-player/tests/unit/visit.test.ts` — "reads what a fetched file is from its first bytes"
- `music-player/tests/e2e/np/collections.spec.ts` — "a song on show does not play a short clip; Download offers to fetch the full song"

NP-FIND-012 **Download… goes through the hub when it may, and says which source it chose and what it embedded** ● ADOPTED

The player's Download… for a catalog song — the transport's Download key, a visitor row's Download button, the row menu, a song's details — opens one fetch sheet that names who will fetch it. Paired with a hub whose device credential allows downloads:request and transfers:receive, it is the hub: POST /api/v1/catalog/download with the catalog song, the stated basis and destination "player"; the sheet then says, in the hub UI's words, which source the hub chose and what it embedded ("Queued from YouTube Music. Tagged with ISRC, genre and lyrics."), the job's progress is read from /api/v1/downloads, and the finished file is read back from /api/v1/files/<sha256> into the library. Otherwise it is the companion's helper ("fetched by the helper on this PC from <host>"), which cannot fetch a store-only song and says so. With neither, or with a hub that has not allowed this device to download, the sheet says exactly that and Fetch claims nothing. No basis chosen is refused in a sentence.

"Night Harbour" by Alder Quartet — fetched by the hub TOWER from the best source it has for it (YouTube Music, YouTube, SoundCloud, Bandcamp, then Spotify through spotDL), tagged, then played from this device.

OWNED BY `music-player/src/shell/search/visit.ts` `music-player/src/shell/bridge.ts` `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY
- `music-player/tests/e2e/np/discover-online.spec.ts` — "with a hub, a visitor fetches the full song without a preview, then plays the full local copy (NP-FIND-012)"
- `music-player/tests/e2e/np/discover-online.spec.ts` — "paired without downloads allowed, Download… says why and claims nothing"
- `music-player/tests/unit/visit.test.ts` — "names a link row's platform for the hub, and refuses a link it does not know"

NP-FIND-013 **Add to Playlist ▸ files a song into the paired hub's playlists, in a group of their own** ● ADOPTED

A song row's Add to Playlist ▸ — a catalog song in search, a type's page or Discover's catalog set, a song of a catalog list on show, or a song in this device's library (the menu and the touch sheet) — keeps this player's own playlists as they were, under "On this player", and, when a hub is paired, gains a group "On <hub>" (DEC-041). With `playlists:use` it lists the hub's folder playlists, read again for each menu with the song (`GET /api/v1/playlists?catalogId&isrc`), each a menu checkbox ticked from `hasTrack`, then "New Playlist on <hub>…", which asks for the name in a sheet and makes the list with the song in it (`POST /api/v1/playlists`). Choosing an unticked list files the song (`POST /api/v1/playlists/:id/entries`); a ticked one says the song is there already and asks nothing. Filing is additive, so any of the hub's lists may be chosen. A catalog song is sent as its `CatalogTrack`; a library song by its catalog identity — its source link, else its ISRC (as its MusicBrainz ISRC page, which the hub matches to its own library by ISRC). The contract keeps every entry's link (a `CatalogTrack` has at least one source), so a song with neither cannot go to the hub: the group says "Can't file this song on <hub>", and choosing it says why. The HUD confirms ("Added "One" to "Road Songs" on TOWER"), and the toast is a live region. Without the permission the group says "Your hub hasn't allowed this player to use its playlists", and choosing it says how to fix it: pair again and tick "File into the hub's playlists". Those lines are marked disabled but reachable with the arrows, and Enter reads them; the hub's lists take the keys like the menu's other commands.

"One" ▸ … ▸ Add to Playlist ▸ On TOWER ▸ ✓ Harbour Mix · Kitchen Radio · New Playlist on TOWER… → Kitchen Radio → HUD "Added "One" to "Kitchen Radio" on TOWER".

OWNED BY

`music-player/src/shell/search/hub-shelf.ts` `music-player/src/shell/search/index.ts` `music-player/src/shell/bridge.ts` `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY

`music-player/tests/e2e/np/hub-shelf.spec.ts` — "Add to Playlist ▸ lists the hub's playlists in a group of their own, ticked where the song is, and files the catalog song (NP-FIND-013)"

`music-player/tests/e2e/np/hub-shelf.spec.ts` — "New Playlist on TOWER… makes a playlist in the hub's folder with the song in it"

`music-player/tests/e2e/np/hub-shelf.spec.ts` — "without playlists:use the group says the hub hasn't allowed it, and how to fix it; a song with no link or ISRC says why it stays here"

`music-player/tests/unit/hub-shelf.test.ts` — "sends a library song by its source link, with its ISRC"

`music-player/tests/unit/hub-shelf.test.ts` — "sends a library song with only an ISRC as its MusicBrainz ISRC page, so the hub can match it by ISRC"

`music-player/tests/unit/hub-shelf.test.ts` — "says why a song with neither a link nor an ISRC cannot go to the hub (the contract keeps every entry's link)"

NP-FIND-014 **The hub’s playlists open in the music list like an album; only the ones this player made change here** ● ADOPTED

Paired with `playlists:use`, the library menu’s Playlists lists “On <hub>” after the person’s own and the starred lists: the hub’s folder playlists, read when the menu opens (“Checking <hub>…” meanwhile, “No playlists on <hub> yet”, or, without the permission, what is missing and how to fix it). Choosing one opens it in the music list the way an album opens (NP-FIND-007): the silver bar names it, “Playlist · <hub> · N songs”, and its songs are the rows, in its order, read a page of up to 200 at a time until the hub says there are no more. A song the hub’s library holds streams from the hub — a short-lived signed URL from `POST /api/v1/library/stream-urls`, through the same transport, a real play; a song kept by its link is a visitor (NP-FIND-011: its preview, the fetch sheet); one the hub cannot place says “Not found”. There is no star (a hub playlist is not a catalog list). A “…” beside the name opens Rename…, Move Song Up, Move Song Down and Delete Playlist… (asked first, saying the songs stay in the hub’s library and at their sources); Alt+↑/↓ moves the highlighted song. They are enabled only on a list this device made (`createdBy` is its device id) — the hub enforces the same (UX-PL-004) — and on any other the bar adds “Made on another device — read-only” (or “Made outside Airwave — read-only” for a hand-made list), the commands are disabled, and Alt+↑/↓ says why.

Library ▸ Playlists ▸ On TOWER ▸ Harbour Mix → “Harbour Mix” · “Playlist · TOWER · 3 songs”; Quay Lights plays from the hub; Alt+↓ → “Moved “Quay Lights” down to 2 of 3”.

OWNED BY `music-player/src/shell/search/hub-shelf.ts` `music-player/src/shell/bridge.ts` `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY `music-player/tests/e2e/np/hub-shelf.spec.ts` — “a hub playlist opens in the music list like an album; a library entry streams through the hub, and only a list this player made can change (NP-FIND-014)”

RADIO: THE SONG ON THE AIR, AND KEEPING IT

NP-RADIO-001 **The song on the air comes from the station, or is not shown** ● ADOPTED

The tuned station's song is read from what the station itself publishes: a feed it serves with CORS open, or its stream's ICY metadata read for the page by the paired container or the companion app (a browser never sees ICY). Only the tuned station reads its stream; the station list keeps to feeds, so a list never becomes a stream connection per row. With no feed and nothing to read the stream, the line shows the programme format the directory carries, never an invented title. Private and loopback addresses are never read on a page's behalf.

OWNED BY `music-player/scripts/make-shell.py` `docker-container/src/api/routes/providers.ts` `local-helper/src/server.ts` `packages/domain/src/radio-node.ts`

CHECKED BY `music-player/tests/e2e/np/radio-air.spec.ts` — “a station with no feed shows the song its stream carries, read by the companion”

`music-player/tests/e2e/np/radio-air.spec.ts` — “with no companion and no container, a station without a feed invents nothing”

`packages/domain/tests/integration/radio-node.test.ts` — “refuses private and loopback addresses unless a test says otherwise”

NP-RADIO-002 **A song heard on the radio can be kept** ● ADOPTED

Right-clicking a station (a long press on touch) whose song is known adds an “On air” section to its menu: Add Song to Up Next, Add Song to Group Queue, and Add Song to Playlist with New Playlist…. The song becomes a library entry of this player's own — filled in by the same search chain when it finds that song by that artist, labelled Radio when it does not — and it survives a reload. The group item names the group when there is one, lists them when there are several, and is disabled with the reason when the player is unpaired or in no group; the container queues a playable copy by name, the way Discord's /play does, and says why when there is none. A same-titled song by someone else is never taken for it.

OWNED BY `music-player/scripts/make-shell.py` `docker-container/src/api/routes/groups.ts` `docker-container/src/group/command-service.ts`

CHECKED BY
- `music-player/tests/e2e/np/radio-air.spec.ts` — “right-click keeps the song: Up Next, a new playlist, and it survives a reload”
- `music-player/tests/e2e/np/radio-air.spec.ts` — “paired: the container reads the title, and the song goes to the group’s queue by name”
- `music-player/tests/e2e/np/radio-air.spec.ts` — “touch: a long press on a station opens the same menu, and the lift does not retune”
- `docker-container/tests/integration/radio-and-requests.test.ts` — “never queues a same-titled song by someone else”

NP-RADIO-003 **The radio bar reports only a trustworthy broadcast position** ● ADOPTED

When the station publishes a start and a length, the progress bar follows the published millisecond cue clock while visible and catches up when a tab returns; the whole-second stamps never run ahead of it. A new start for the same named song inside its previously published length contradicts the position: keep the song name, show position unavailable and its published length, and do not jump the bar backwards. A legitimate re-cue after the old song ends, or a different song, restores a timed bar. When no length is published, the elapsed stamp can advance but the bar stays empty instead of sitting full. Older overlapping replies cannot replace a newer one; bfcache restoration re-reads the station.

Radio: a timed song advances smoothly; an inconsistent station says —:— and Length 04:17 instead of moving backward or sitting at 100%.

OWNED BY `music-player/scripts/make-shell.py` `music-player/index.html`

CHECKED BY
- `music-player/tests/e2e/np/radio-elapsed.spec.ts` — “radio fill tracks the published cue clock without pausing a second behind it”
- `music-player/tests/e2e/np/radio-elapsed.spec.ts` — “conflicting same-song re-cues stop claiming a broadcast position until a new song”
- `music-player/tests/e2e/np/radio-elapsed.spec.ts` — “a timed cue with no published length advances elapsed but never shows a permanently full bar”
- `music-player/tests/e2e/np/radio-elapsed.spec.ts` — “a late old feed reply cannot replace the new song on the same station”

NP-TV-001 **Live TV channels come from the companion, and the guide says only what a guide said** ● ADOPTED

A player that can reach a companion takes its Live TV channels from the playlists kept in the companion's Live TV tab; nothing has to be loaded by hand in the player, and a playlist the page could not read itself is the companion's to read. A channel with a guide entry shows what is on and what follows; one without says Live and no programme is invented. A guide refresh changes the words and never resets the list or the listener's place in it. A companion with no playlists, or one that has gone, leaves the guide as it was.

OWNED BY `music-player/scripts/make-shell.py` `local-helper/src/server.ts` `windows-companion/src/main/live-tv/index.ts` `docker-container/src/live-tv/service.ts`

CHECKED BY
- `music-player/tests/e2e/np/live-tv.spec.ts` — “the companion’s channels fill the guide, with what is on and what follows”
- `music-player/tests/e2e/np/live-tv.spec.ts` — “a companion with no playlists leaves the guide as it was, and one that is gone breaks nothing”
- `music-player/tests/e2e/np/live-tv.spec.ts` — “with no companion on this machine, a paired hub’s copy of the companion’s Live TV fills the guide”

PLAYER SHELL — PREFERENCES

NP-PREF-001 **Two idioms: iPhone OS on phones, a toolbar on desktops** ● ADOPTED

Below 720px Settings is an iPhone OS drill-down: an index of 44px rows, a back button that names where it returns to. From 720px it is a preferences window with a toolbar of five panes. The browser Back button walks the same history.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/prefs.spec.ts` — “media playlists live behind the library button”
`music-player/tests/e2e/np/prefs.spec.ts` — “opening: Settings is a page with its own address, and the toolbar is the navigation”

NP-PREF-002 **A preference applies when it is changed** ● ADOPTED

No Save and no Apply. Theme, motion, artwork, defaults, group mode and every algorithm setting apply immediately and persist (player:prefs). Revert, in the footer, restores everything to how it was when Settings opened (NPD-016).

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/prefs.spec.ts` — “opening: Settings is a page with its own address, and the toolbar is the navigation”
`music-player/tests/e2e/np/prefs.spec.ts` — “player preferences, Solo-by-default group gating and the equalizer survive reload”
`music-player/tests/e2e/np/discover.spec.ts` — “Revert puts back the weight, the algorithm, and removes the copy”

NP-PREF-003 **The algorithm is editable and cannot be made invalid** ● ADOPTED

Every recommendation setting has a control. Rules the engine enforces (weights summing to 100, ranges) are checked as the user edits and offered a fix, never silently clamped.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/prefs.spec.ts` — “opening: Settings is a page with its own address, and the toolbar is the navigation”

NP-PREF-004 **Built-in algorithms never change** ● ADOPTED

Airwave default, Late night and Crate digger are read-only. The first change to one makes a named copy, selects it and says so; built-ins cannot be renamed or deleted. Delete asks twice.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/discover.spec.ts` — “built-ins, copies, rename, duplicate, delete, export, import, and a reload”

NP-PREF-005 **Algorithms move as files, and imports say what they read** ● ADOPTED

Export writes one algorithm (or all of yours) as an airwave-algorithm JSON file; Import takes those or a bare engine config, reads only keys the engine knows with values of the right type, and reports how many settings it read and how many it ignored.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/discover.spec.ts` — “the shelf records what it shows, and plays from it count”
`music-player/tests/e2e/np/discover.spec.ts` — “built-ins, copies, rename, duplicate, delete, export, import, and a reload”

NP-PREF-006 **Settings controls are Mac OS X controls, drawn over native elements** ● ADOPTED

Push buttons, pop-ups and checkboxes in Settings wear Mac OS X drawings: Snow Leopard (10.6) push buttons — a matte, brushed-metal pill with a graphite hairline, blue when pressed — and 10.4 Aqua pop-ups (white body, blue arrow segment) and gel checkboxes that AquaArt draws from sampled colour profiles and publishes as custom properties. The elements stay native button, select and checkbox, so keyboard, forms and screen readers are unchanged. 22px on desktop, 30px on touch; the drawings and their black labels are the same in both themes (NPD-021).

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/aqua.spec.ts` — “buttons, pop-ups and checkboxes wear the Aqua drawings and still work”
`music-player/tests/e2e/np/aqua.spec.ts` — “the 30px drawings on a phone”

NP-PREF-007 **Connections show what answered and what goes where** ● ADOPTED

Sources ▸ Connections tests the Windows companion app (GET /helper/v1/health; an empty address scans 127.0.0.1:17342–17345) and the Docker container (GET /healthz, /readyz, /api/v1/hub; port 4546), using the AudioWave repo's own endpoints; and Stream from a PC pairs with the companion's AWSP sidecar by its ticket and six-digit code (docs/AWSP.md §2), then shows the PC's name, the connection type (relay-carried in a browser) and how many of its tracks joined the library, with the same lights. Each test shows what came back — version, downloaders, allowed sites, readiness checks, fingerprint, encryption — with a green, amber or red light; every request goes in the connection log with its status and time; and the What's passed table lists each setting, where it goes, and 'Ready' only once that destination has answered.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/connect.spec.ts` — “the companion, the container and backup”
`music-player/tests/e2e/awsp.spec.ts` — “pairs, plays relay-carried, is bit-identical, seeks by range, and resumes across a relay restart”

NP-PREF-008 **Backups say what they need before they run** ● ADOPTED

Sources ▸ Backup chooses what to copy (Music, TV, Movies; playlists, EQ presets and settings) and where (the Docker container, the Windows companion app, or both), and how often. Media moves between the companion app and the container, so it needs both to have answered a test and its folder to be set; settings need only their destination. Back Up Now stays disabled and says which of those is missing; once it runs, the request is written to the connection log and What's passed lists a backup row per destination.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/connect.spec.ts` — “the companion, the container and backup”

NP-PREF-009 **Profiles live on the container, and names are unique there** ● ADOPTED

Once the player is paired with the Docker container (the hub's own /pairing/claim, /status and /complete, confirmed by fingerprint), Settings gains a Profile tab: a username, a picture cropped to 256 × 256, and playlists shared as CSV (title, artist, album, seconds), all saved to the container and never kept in the browser. A username someone else has — compared without case — is caught as you type and refused on save. Search lists people on the container above songs; a profile opens in a sheet with their picture, name and shared playlists, and a song you also have can be played from it. Your picture takes the jewel case's place on the Music page. Until the container has the profile routes, the tab says so.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/profile.spec.ts` — “pair, edit the profile on the container, find someone, open their profile”

NP-PREF-010 **One default volume for every source** ADOPTED

Settings ▸ Equalizer ▸ Default volume sets the level Airwave opens at — where you left it, or a fixed default — and Set Volume Now moves the bar's volume to it. Music, radio, Live TV, TV and films all start at and follow that one volume.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/volume.spec.ts` — “the default volume: Set Volume Now, then each start mode across a reload”

NP-PREF-011 **A backup says how big it is and whether it fits** ADOPTED

Settings ▸ Sources ▸ Backup shows the expected size, part by part, and each destination's free space as a bar (in use · this backup · free after it). Figures the player can't measure itself come from the companion app and the container; until they answer, the row names which figure is waiting and who measures it. Back Up Now is unavailable when the backup doesn't fit. Algorithms can be backed up like the rest. Sizes are decimal (1 GB = 10^9 bytes).

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/connect.spec.ts` — “the companion, the container and backup”

NP-PREF-012 **Groups and invites live on the Profile tab** ADOPTED

Settings ▸ Profile lists your groups on the container with their member count and your role, makes a new one (you own it), and — for an owner or admin whose player was paired with the “Manage groups” permission — makes an invite link that works once and expires after the time chosen. Without that permission no invite link is offered, because the container would refuse it, and the message names the missing permission rather than blaming the role. Invites you have received wait there until you join or decline; ones you have sent show who they were for and when they expire. A code the container can't withdraw is never shown as withdrawn.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/groups.spec.ts` — “the Profile tab: groups, a new group, an invite link, Withdraw, and Leave”

`music-player/tests/e2e/np/groups.spec.ts` — “paired without “Manage groups”, no group offers an invite link — the hub would refuse it”

`tests/journey/cross-app.spec.ts` — “the whole pass: hub set up, player paired, group joined, invite declined, name taken, group owned, someone found”

NP-PREF-013 **An invite link opens an invite page** ADOPTED

Opening an invite link shows a page naming the group, who sent it, the container, the role and when it expires, with Join Group and Decline (Not Now keeps it waiting). What the link says is labelled as the link's until the container confirms it. An unpaired player, or one paired with another container, is asked to pair first and offered Connections. The code is taken out of the address bar as soon as it has been read.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/groups.spec.ts` — “the invite page: from a link, Not Now, Join, a used code, a bad code, Decline”

`music-player/tests/e2e/np/groups.spec.ts` — “not paired: the invite page asks you to pair first”

NP-PREF-014 **How the disc moves is the listener’s choice** ● ADOPTED

Settings ▸ Player ▸ The disc sets whether the disc spins while a song plays (steady, with the song’s tempo, or not at all) and how fast (0.25× to 3×); whether it turns all the way round (steady, with the tempo, held at its angle, or left where it is put) and how fast; its angle (yaw, 0–355°, with presets: the label face on or at an angle, the edge, the playing side) — where it starts and where a held disc stays; and its tilt (pitch, −30° to 80° back). Reset Position puts the disc at that angle and tilt and drops any dragged pose. The jewel-case sleeve and CD label show the current track’s available artwork and update together when the track changes. A live preview at the top shows the disc moving as the settings say, as if a song were playing, and says what it is showing (a song at 120 BPM when tempo is chosen and nothing with a known tempo plays); it draws only while it is on screen. Each choice is saved at once, applies to the playing disc immediately and survives a reload; only the controls that apply are shown. Tempo is the song’s BPM: at 120 BPM the disc moves at the chosen speed, faster songs faster, and one full turn takes 32 beats; a song with no known tempo, and the radio, move at the steady speed. Turn also governs the closed case while nothing plays. The defaults are the motion and pose the player always had, and Reduce animation still slows the spin and stops the turning.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py` `music-player/src/shell/bridge.ts`

CHECKED BY
- `music-player/tests/e2e/np/disc.spec.ts` — “the controls start at today’s behaviour, show only what applies, and are kept across a reload”
- `music-player/tests/e2e/np/disc.spec.ts` — “the stage follows the choices: steady, a speed, off, and with the song’s tempo”
- `music-player/tests/e2e/np/disc.spec.ts` — “held one way, the disc settles on that facing while it plays”
- `music-player/tests/e2e/np/disc.spec.ts` — “Reduce animation still wins: a slow spin and no turning, whatever is chosen”
- `music-player/tests/e2e/np/disc.spec.ts` — “the angle and the tilt are the listener’s, and Reset Position puts the disc back there”
- `music-player/tests/e2e/np/disc.spec.ts` — “the preview in Settings draws the disc, and says what it is showing”
- `music-player/tests/e2e/np/disc.spec.ts` — “the case sleeve and CD label show the current track’s cover, and update when the track changes”

PLAYER SHELL — STATISTICS AND CHARTS

NP-DATA-001 **One hue for amounts, named hues for genres, and every chart as numbers** ● ADOPTED

Amounts use one hue (--viz-1) with a single-hue ramp for the skyline (--viz-sky-0 → --viz-1 → --viz-peak). Genres use seven token hues (--viz-g1…g7, --viz-g0 for the rest), lifted in dark, and a genre's colour never appears without its name beside it. Every chart has its value written beside it or a table under it, and the 3D views are backed by the same numbers in 2D.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/stats.spec.ts` — “before any play: nothing recorded, and no generated source on offer”

NP-DATA-002 **Every figure is set against the period before** ● ADOPTED

For 7, 30, 90 days and the year, each tile compares with the period of the same length just before it, with ▲/▼, the figure, and colour for good or bad news (more skips is up and bad). All time has no period before and shows no comparison. Heard-to-the-end and skips count only plays the transport timed (NPD-018).

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/discover.spec.ts` — “statistics against the period before, and the recommendation funnel”

NP-DATA-003 **Recommendations are counted from the shelf's own record** ● ADOPTED

What Discover shows is logged once per song per day (`recShown: {id, at}` — every page it shows, on opening, on Refresh, on Load more and after an algorithm or lean change, the same record); a play started there is a recommendation taken. The funnel reads recommended → started → heard to the end → replayed → added or favourited from those logs, never from estimates. The ranker reads the same record back as impressions for its overexposure penalty (NP-DISC-001).

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/discover.spec.ts` — “the shelf records what it shows, and plays from it count”
`music-player/tests/e2e/np/discover.spec.ts` — “statistics against the period before, and the recommendation funnel”

NP-DATA-004 **The statistics page works without its 3D views** ● ADOPTED

Statistics loads three.js from the page's import map the first time it opens. If that fails or takes more than eight seconds, each 3D well says it couldn't load and the page carries on; every 3D view has the same numbers in 2D or in a table. A theme change repaints the wells, scenes and labels from the tokens.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/stats.spec.ts` — “before any play: nothing recorded, and no generated source on offer”
`music-player/tests/e2e/np/stats.spec.ts` — “the wells say so, the tables stay”

PLAYER SHELL — LAYOUT ACROSS DEVICES

NP-RESP-001 **The video controller fits the player, not the window** ● ADOPTED

Controller-bar content responds to the player's width (container queries): elapsed only under 560px, download and touch volume hidden, More hidden under 358px; the scrubber keeps at least 60px. Checked at 18 device sizes.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/devices.spec.ts` — “: the LIVE readout sits right, idle and playing”
`music-player/tests/e2e/np/livevol.spec.ts` — “px LIVE is centred in its box, clear of its neighbours, and the scrubber keeps room”

NP-RESP-002 **Picture and controls fit under the status bar** ● ADOPTED

On short screens (landscape phones, short laptops) the video player is capped so it and its controls fit below the status bar without scrolling.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/devices.spec.ts` — “: the LIVE readout sits right, idle and playing”

PLAYER SHELL — ACCESSIBILITY

NP-A11Y-001 **Every control is reachable and named** ● ADOPTED

Icon-only controls carry accessible names; groups of choices are radiogroups moved with arrow keys; scrolling regions are focusable; headings give the page an outline (one h1).

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/func.spec.ts` — “24px optical offset preserved; sort headers are buttons and sorting works; page has h1”
`music-player/tests/e2e/np/prefs.spec.ts` — “opening: Settings is a page with its own address, and the toolbar is the navigation”

NP-A11Y-002 **Motion stops when asked** ADOPTED

With the OS reduced-motion setting, or Preferences > Reduce animation, pulses, barber poles, spinners and transitions stop.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/prefs.spec.ts` — “opening: Settings is a page with its own address, and the toolbar is the navigation”

NP-A11Y-003 **Text meets 4.5:1; exceptions are recorded** ADOPTED

Text/background pairs listed in design/foundations.json meet WCAG 2.2 AA (4.5:1 text, 3:1 marks) in both themes. Period colours that fall short are recorded as decisions, never added silently.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY reviewer Carried from airwave-np; its test did not survive the port as a separate assertion. Checked in review against the shell.

PLAYER SHELL — VOICE AND COPY

NP-VOICE-001 **Commands are Title Case; an ellipsis means a dialog follows** ADOPTED

Menu commands use Title Case verbs ('Go to Album', 'Add to Playlist'). An ellipsis marks exactly the commands that open a dialog for more input ('New Playlist…', 'Change Artist…').

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/menu2.spec.ts` — “six commands in three groups, and an ellipsis marks exactly the dialog commands”

NP-VOICE-002 **The verb matches the thing** ADOPTED

A song or film is Played, a channel is Watched, a series is Opened. Commands that make no sense for a kind of item are absent from its menu, not disabled.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/lists.spec.ts` — “Live TV: a channel is watched, not played, and its playlist still reads as a guide”
`music-player/tests/e2e/np/lists.spec.ts` — “Radio: a station is played, kept or filed; a new playlist holds it and can be emptied”
`music-player/tests/e2e/np/lists.spec.ts` — “the song menu is untouched”

NP-VOICE-003 **Explanations are one plain sentence with a way forward** ADOPTED

When something cannot happen, say what to do instead in one sentence, without apology or error codes: 'Pick another station from the list', 'Couldn’t reach search — open it on a platform instead:'.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY reviewer Carried from airwave-np; its test did not survive the port as a separate assertion. Checked in review against the shell.

MUSIC CATALOG — WHAT THE SEARCH API PROMISES

UX-CAT-001 **Every service a search asks says how it did**

● ADOPTED

A catalog search (hub /api/v1/catalog/search, helper /helper/v1/catalog/search) names every keyless service on every chunk with its state: pending, ok, empty, failed, timeout, cooling-down or skipped, with a sentence for anything but ok and, when resting, the time it may be asked again. The first chunk lists them all as pending (or skipped, or cooling down) before any answers; the last (done) has every final state. A failed service never hides the others' results and never fails the search. Two failures in a row rest a service for 30 s, doubling to at most ten minutes; a Retry-After is honoured at once; iTunes rests within Apple's own budget (about 20 calls a minute) instead of being refused. A hub or PC without yt-dlp says so for YouTube and SoundCloud, and a service switched off in the catalog settings is skipped, saying so.

{"provider":"deezer","state":"cooling-down","error":"api.deezer.com answered 500","retryAt":"2026-01-01T00:00:30.000Z"}

OWNED BY

`packages/domain/src/catalog/engine.ts` `packages/domain/src/catalog/limits.ts` `packages/contracts/src/api/catalog.ts` `docker-container/src/api/routes/catalog.ts` `local-helper/src/server.ts`

CHECKED BY

`packages/domain/tests/unit/catalog-engine.test.ts` — "opens with every service's state, sends a chunk per service, merges, and ends with done"

`packages/domain/tests/unit/catalog-engine.test.ts` — "reports a failing service as failed, and rests it after a second failure (UX-CAT-001)"

`packages/domain/tests/unit/catalog-engine.test.ts` — "says a slow service timed out, and still answers with the rest"

`packages/domain/tests/unit/catalog-engine.test.ts` — "reports a server without yt-dlp honestly, and a switched-off service as skipped"

`packages/domain/tests/unit/catalog-engine.test.ts` — "rests iTunes within Apple's budget instead of being refused"

`packages/domain/tests/unit/catalog-engine.test.ts` — "honours Retry-After at once, and backs off by doubling after repeated failures"

`docker-container/tests/integration/catalog.test.ts` — "streams NDJSON: a chunk per service as it answers, then done"

UX-CAT-002 **One song is one row, however many services have it**

● ADOPTED

Rows from different services are one song when they share an ISRC, or — when either lacks one — when the main artist and the title agree once the noise is set aside (“feat. …”, “Remastered 2011”, “(Official Music Video)”, “[Lyrics]”, “HD”, an uploader’s “Artist - Title”, a “- Topic” or VEVO channel), their version agrees, and the durations are within three seconds. Versions stay apart: live, remix, acoustic, instrumental, demo, radio edit, extended, sped up, slowed, karaoke, cover and reprise are each a row of their own (`versionOf`). A row with no duration joins on names only when it is an official upload — a YouTube Music Topic recording, the artist’s own or a VEVO channel, or a title saying “Official”; a stranger’s re-upload never does. The merged row keeps every source (platform, id, link, preview) and takes each field from the best service that has it: the stores’ titles and credits before MusicBrainz, a YouTube upload’s last; the largest store artwork; the fullest date. A merged row keeps the id it first arrived with, so a later chunk replaces it in place, and a later page’s copy of a song an earlier page sent comes back with that row’s id (the server keeps a query’s rows for 15 minutes), so no page repeats another. The best rows gain their other homes (Spotify, YouTube Music, Tidal…) from MusicBrainz’s links, keylessly, and `done.linkedOnly` names the platforms that only contributed links.

iTunes’s and Deezer’s “Harbour Lights” (214 s), the band’s official video “Cassette Bloom - Harbour Lights (Official Music Video) [HD]” (216 s), its YouTube Music Topic upload (no duration) and SoundCloud’s “Cassette Bloom - Harbour Lights” (215 s) are one row with five badges; “Harbour Lights (Live at the Granary)” and “(Night Shift Remix)” are rows of their own.

OWNED BY

`packages/domain/src/catalog/merge.ts` `packages/domain/src/catalog/engine.ts`

CHECKED BY

`packages/domain/tests/unit/catalog-query-merge.test.ts` — “is one recording by ISRC, whatever the titles say”

`packages/domain/tests/unit/catalog-query-merge.test.ts` — “is one recording by names when the durations are within three seconds, and not otherwise”

`packages/domain/tests/unit/catalog-query-merge.test.ts` — “keeps every source and takes each field from the best one”

`packages/domain/tests/unit/catalog-query-merge.test.ts` — “upserts by the first id in a merger, so a chunk can replace a row it sent before”

`packages/domain/tests/unit/catalog-engine.test.ts` — “folds into one aggregate where Deezer, Apple and YouTube’s copies of one song are one row”

`packages/domain/tests/unit/catalog-engine.test.ts` — “adds the best rows’ other homes (Spotify, Tidal…) from MusicBrainz before done”

`packages/domain/tests/unit/catalog-one-row.test.ts` — “iTunes, Deezer, a YouTube official video, a Topic upload and SoundCloud: one row, five badges”

`packages/domain/tests/unit/catalog-one-row.test.ts` — “keeps versions apart: live, remix, acoustic, sped up and a cover are rows of their own”

`packages/domain/tests/unit/catalog-one-row.test.ts` — “a stranger’s re-upload with no duration does not join on names alone”

`packages/domain/tests/unit/catalog-one-row.test.ts` — “page 2’s copy of a page-1 song comes back with page 1’s id; only new songs are new rows”

`packages/domain/tests/unit/catalog-one-row.test.ts` — “says which platforms only contributed links”

UX-CAT-003 **Search finds music, never a “search on YouTube” offer, and every result says where it is** ● ADOPTED

The catalog answers with songs, artists and albums only. It never returns an entry whose only purpose is to send the person to search another site; YouTube and SoundCloud are searched by the server (yt-dlp) and their results are songs like any other. Every track, artist and album carries at least one source naming its platform, and a song from Spotify that plays from YouTube Music carries both, the second marked as spotDL’s match. A pasted link is not searched: the search says to resolve it, and resolve says what it is. A public Spotify playlist is read from the public embed page first to avoid a slow API crawl; tracks and albums use spotDL first and the embed page only as metadata fallback. Embed previews are never used as audio, and an inaccessible/private link still says why — in words. A Tidal link without a SongLink key also says why. A row shows at most three platform badges and counts the rest (+N), so the title is never cut short by them; every platform is named in the row's tooltip and on the song's page. The facts a row carries beyond its platforms — cover, BPM, featured artists, explicit, date, ISRC, album — and where each comes from are UX-CAT-006.

{"sources":[{"platform":"spotify","matchedBy":"link"},{"platform":"youtube-music","matchedBy":"spotdl"}]} — shown as “Spotify · plays from YouTube Music”.

OWNED BY

`packages/contracts/src/api/catalog.ts` `packages/domain/src/catalog/engine.ts` `packages/domain/src/catalog/links.ts` `packages/domain/src/catalog/providers/ytdlp.ts` `packages/domain/src/catalog/providers/spotify-embed.ts`

CHECKED BY

`packages/domain/tests/unit/catalog-engine.test.ts` — “folds into one aggregate where Deezer, Apple and YouTube’s copies of one song are one row”

`packages/domain/tests/unit/catalog-engine.test.ts` — “yt-dlp: a cleaned YouTube title, a Topic channel as YouTube Music, no channels or sets”

`packages/domain/tests/unit/catalog-engine.test.ts` — “does not search a pasted link: done names it for resolve”

`packages/domain/tests/unit/catalog-engine.test.ts` — “a Spotify song through spotDL, with spotDL’s YouTube Music match as a second source”

`packages/domain/tests/unit/catalog-engine.test.ts` — “a Spotify playlist Spotify will not list says so, with the reason”

`packages/domain/tests/unit/catalog-engine.test.ts` — “an Apple Music playlist whose page changed and a Tidal link without a SongLink key say why they cannot be read”

`packages/domain/tests/unit/catalog-engine.test.ts` — “resolves a public Spotify playlist from its embed page when spotDL cannot list it”

`docker-container/tests/unit/external-tool-runs.test.ts` — “falls back to a public Spotify track embed when spotDL cannot read the track”

`docker-container/tests/integration/catalog.test.ts` — “a Spotify song through spotDL, with its YouTube Music match”

`music-player/tests/e2e/np/catalog-search.spec.ts` — “a song on five platforms keeps its whole title: three badges on the row, the rest counted”

UX-CAT-004 **A loaded list loads every song, from any platform** ● ADOPTED

An album or playlist opened from a link or a result lists every song it has, in order, page by page: `resolve` and `album` serve any `offset` (to 10,000) with a `limit` of at most 200, `total` is the platform's own count when available (otherwise `null`), and `hasMore` holds until the end; `collectAllPages` reads a whole list. Spotify playlists use the public embed listing first and spotDL when the page is unavailable; the embed has no total and at most 100 tracks, so the 100-track ceiling is reported as capped. YouTube and YouTube Music playlists and Bandcamp albums (yt-dlp) are read once, whole; a SoundCloud set's bare ids are described a page at a time; Deezer pages with its own index; Apple Music playlists are read from their public page. A list past 10,000 songs says so (`capped`); a page that carries fewer songs than the list holds says so in words. Nothing is cut silently.

When spotDL can read a 1,200-song Spotify playlist, page 12 is songs 1,101–1,200 with `hasMore: false`; when only the public embed is available, its 100-track ceiling is marked capped.

OWNED BY

`packages/domain/src/catalog/engine.ts` `packages/domain/src/catalog/links.ts`
`packages/domain/src/catalog/providers/applemusic-page.ts` `packages/domain/src/catalog/providers/spotify-embed.ts`
`docker-container/src/catalog/service.ts` `docker-container/src/providers/adapters/external-tool.ts`
`docker-container/src/downloads/service.ts` `local-helper/src/resolve.ts`

CHECKED BY

`packages/domain/tests/unit/catalog-paging.test.ts` — "Deezer: any page of a 450-song playlist, the exact total, hasMore until the end, asked 100 at a time"

`packages/domain/tests/unit/catalog-paging.test.ts` — "Spotify through spotDL: all 1,200 songs of a playlist, read once and served page by page"

`packages/domain/tests/unit/spotify-embed.test.ts` — "reports the embed page limit instead of claiming a 100-track playlist is complete"

`docker-container/tests/integration/downloads-external-tool.test.ts` — "downloads each track from public Spotify playlist metadata without crawling it through spotDL"

`packages/domain/tests/unit/catalog-paging.test.ts` — "stops at CATALOG_COLLECTION_CAP and says so, never silently"

`packages/domain/tests/unit/catalog-paging.test.ts` — "SoundCloud: a set's bare ids are described a page at a time, in one run for that page"

`packages/domain/tests/unit/catalog-paging.test.ts` — "Apple Music: a playlist read from its public page, every song it carries, paged like any other"

`packages/domain/tests/unit/catalog-paging.test.ts` — "collectAllPages pages any resolve-shaped call in order, page size bounded"

`local-helper/tests/unit/resolve-paging.test.ts` — "keeps 1,500 entries of a playlist and 3,000 songs of a Spotify playlist when the cap allows"

UX-CAT-005 **Playlists are a search section, keyless from Deezer**

● ADOPTED

A catalog search can be asked for `playlists` beside `tracks`, `artists` and `albums` (`sections=`, CatalogSearchSection). Deezer's public `/search/playlist` answers it — paged with its own `index`/`limit` — and each row is a CatalogPlaylist: id (`deezer:<id>`), title, owner, trackCount, pictureUrl (the playlist's own picture), covers (the first four songs' artwork only when the service gives them without a call per row; else empty, and the picture stands for it), sources (one, whose url opens the playlist through `resolve`, NP-FIND-007) and rank. A private playlist is never a row. The other services say they have nothing in the section (`skipped`), the merger keeps playlists one row by id, `done.page.playlists` and `done.totals.playlists` page them like the rest, and a client that predates the section still parses the stream (the arrays and page default). The hub and the helper pass `sections` through unchanged.

GET /api/v1/catalog/search?q=harbour&sections=playlists&limit=12 → results{playlists: [{id:"deezer:14632517341", title:"Relaxing Classical Music", owner:"Playlist Editor", trackCount:101, pictureUrl:"https://cdn-images.dzcdn.net/…", covers:[], sources: [{platform:"deezer", url:"https://www.deezer.com/playlist/14632517341"}]}]} … done{page: {playlists:{offset:0, limit:12, hasMore:true}}}

OWNED BY

`packages/contracts/src/api/catalog.ts` `packages/domain/src/catalog/engine.ts` `packages/domain/src/catalog/merge.ts` `packages/domain/src/catalog/providers/deezer.ts` `docker-container/src/web/views/Search.tsx` `windows-companion/src/renderer/views/Search.tsx`

CHECKED BY

`packages/domain/tests/unit/catalog-engine.test.ts` — “lists public playlists from Deezer as a section of their own, keylessly (UX-CAT-005)”

`packages/domain/tests/unit/catalog-engine.test.ts` — “a search of every section asks Deezer for playlists beside the rest, and the overview’s done counts them”

UX-CAT-006 **Every result carries its facts** ● ADOPTED

Every song a search page, an album page or an artist's top songs list carries, on all three apps, has its cover, title, duration, BPM, featured artists, explicit mark, release date and year, ISRC and album when any keyless service knows them — not only the stores' search rows, which carry no tempo and no contributors. After a page is assembled (every service's chunk and the cross-links) the engine fills each row with a bounded, concurrent, cached lookup: a Deezer source → `/track/{id}` (bpm, contributors → `artists`, isrc, explicit_lyrics, release_date, cover_xl, track_position, disk_number, the album's title); else an ISRC → `/track/isrc:{isrc}`; else one exact "artist title" Deezer search, taken only when it is the same recording (`sameRecording`); a MusicBrainz-only row still without a cover takes the Cover Art Archive's front image for the recording's first release (`/release/{mbid}/front-500` answers 307 when it exists, 404 when not; asked with redirects unfollowed). Nothing known is overwritten — only nulls are filled, the fuller date wins, an iTunes row keeps its 600 px artwork — and a video thumbnail gives way to a store's square cover as the merger's rule has it. Budget: at most 8 lookups in flight, 6 s a page (4 s for a detail page), the first chunk never waits; the filled rows are one last `results` chunk with `provider: null` before `done`, upserted by id like the cross-link chunks, and `?stream=0` folds them in. Answers are cached a day by Deezer id, by ISRC and by the name they were found by; Deezer's quota (error code 4) ends the page's lookups and rests the service, and a service switched off, resting or left out of the search is not asked. A row's credit line (`creditLine`) is the main artist, then "feat. A & B" from the other credited names when the artist line does not already name them — repeats dropped, order kept, four at most then "& others" — on the row and the song page of the player, the hub and the companion; a song the store marks explicit shows a small "E" beside its title, read as "explicit". The player's own Deezer tempo lookup is a fallback for rows still without a BPM once the search is done (an older hub's rows), never a race with the engine.

Deezer's search row "Get Lucky (feat. Pharrell Williams and Nile Rodgers)" by Daft Punk arrives with bpm: null and artists: ["Daft Punk"]; the last chunk before done carries the same id with bpm: 116.1, artists: ["Daft Punk", "Pharrell Williams", "Nile Rodgers"], isrc: "USQX91300108", explicit: false, releaseDate: "2013-05-20", trackNumber: 8 — the row reads "Daft Punk feat. Pharrell Williams & Nile Rodgers — Random Access Memories · 2013 · 116 bpm".

OWNED BY

`packages/domain/src/catalog/hydrate.ts` `packages/domain/src/catalog/engine.ts` `packages/domain/src/catalog/view.ts` `packages/domain/src/catalog/providers/musicbrainz.ts` `music-player/src/shell/search/view.ts` `music-player/src/shell/search/index.ts` `docker-container/src/web/views/Search.tsx` `windows-companion/src/renderer/views/Search.tsx`

CHECKED BY

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "fills a search page's rows from Deezer's detail in one last merge-only chunk before done"

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "a MusicBrainz-only row takes the Cover Art Archive's front cover for its first release, never following the hop"

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "fills an ISRC-only row from /track/isrc:, and a store-less row by one exact search taken only for the same recording"

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "never downgrades: an iTunes row keeps its 600 px artwork and its own credit line, and gains what it lacked"

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "asks at most eight at a time, starts nothing past the budget, and never holds done for a slow answer"

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "keeps a detail for a day, under its id and its ISRC: the next page or search asks nothing"

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "respects Deezer's quota: the first code-4 answer ends the page's lookups and rests the service"

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "an album's songs and an artist's top songs are filled in too"

`packages/domain/tests/unit/catalog-hydrate.test.ts` — "names the main artist, then the featured ones a store lists as contributors"

`docker-container/tests/integration/catalog.test.ts` — "a page's rows carry bpm, contributors, ISRC and explicit after the hydration chunk, before done (UX-CAT-006)"

`music-player/tests/e2e/np/catalog-search.spec.ts` — "the engine's last chunk fills a row in place: it gains "feat.", BPM, the E mark and a cover (UX-CAT-006)"

`docker-container/tests/dom/search.dom.test.tsx` — "shows cover, title, artist, album, its platforms, BPM, time, the explicit marker and a preview"

windows-companion/tests/dom/companion-search.dom.test.tsx — “folds the chunks the main process passes on, says every service’s state, and shows rows that are music”

MUSIC SEARCH IN THE HUB AND THE COMPANION

UX-SEARCH-001 **One field finds music; the advanced fields fold back into it; the filter is kept** ● ADOPTED

The Search tool takes free text, an ISRC or a pasted link from any platform. Track, Artist and Album are an expandable set of fields; put away, whatever they hold is folded back into the one line, so nothing typed is lost. Filter… is a sheet choosing the sections shown (Songs, Artists, Albums, Playlists) and the services asked; it will not take a choice that shows nothing, and it is kept — per browser on the hub, in the companion’s settings store — and searching again applies it at once. Search is disabled, with the reason, until there is something to look for.

OWNED BY

docker-container/src/web/views/Search.tsx docker-container/src/web/lib/catalog.ts
windows-companion/src/renderer/views/Search.tsx windows-companion/src/renderer/catalog.ts
packages/domain/src/catalog/view.ts

CHECKED BY

docker-container/tests/dom/search.dom.test.tsx — “puts the Track, Artist and Album fields back into the line when they are closed”

docker-container/tests/dom/search.dom.test.tsx — “is a sheet: it takes the focus, Escape puts it away, and Done keeps the choice and searches again”

docker-container/tests/dom/search.dom.test.tsx — “will not take a filter that shows nothing”

windows-companion/tests/dom/companion-search.dom.test.tsx — “keeps the filter in the companion’s store and searches again”

UX-SEARCH-002 **Results arrive live, and every service’s state is said** ● ADOPTED

Results fold in as each service answers (the NDJSON feed), upserted by id and ordered by rank, ties by arrival. One line under the results — under the overview’s groups, and on a type’s page between its control and its list — names each service with its lamp and its state in words — asking…, N found, nothing found, failed, took too long, resting (with when it is back), not asked — and the platforms that only contributed links; a live region says, once, how far the search has got and what it found (playlists among it when some came).

OWNED BY

docker-container/src/web/views/Search.tsx docker-container/src/web/lib/catalog.ts
windows-companion/src/renderer/views/Search.tsx windows-companion/src/renderer/catalog.ts
packages/domain/src/catalog/view.ts

CHECKED BY

docker-container/tests/dom/search.dom.test.tsx — “folds the live feed by id and says every service’s state, the resting one with when it is back”

packages/domain/tests/unit/catalog-view.test.ts — “upserts rows by id: a later chunk replaces a row in place, a new one is added after”

packages/domain/tests/unit/catalog-view.test.ts — “says each state, with when a resting service is back”

windows-companion/tests/dom/companion-search.dom.test.tsx — “folds the chunks the main process passes on, says every service’s state, and shows rows that are music”

UX-SEARCH-003 **A row is music, and says where it is** ● ADOPTED

Every song row shows its cover, title, credit line (`creditLine`: the main artist, then “feat. …” from the other credited names when the artist line does not already name them), album, time, BPM, an explicit mark, a play control for its 30-second preview (Apple Music or Deezer), its “…” (the row’s menu, UX-SEARCH-012), and one chip per platform it is on; a song that plays only through Spotify says it plays from YouTube Music via spotDL. No row offers to “search on” a site. A song opens to its genre, label and year (MusicBrainz), its lyrics (synced when LRCLIB has them), its preview, and Download: on the hub into the queue with the rights basis, saying which source was chosen and what was written into the tags; on the companion through the helper, as every download on that PC. Those facts come with the search itself: the engine fills every page’s rows before done (UX-CAT-006 — Deezer’s detail, the Cover Art Archive), as one merge-only chunk the window folds in by id, so a row gains its BPM, its featured artists and its cover a moment after it first appeared; the song page says the same credit line.

OWNED BY
`docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/catalog.ts`
`windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/catalog.ts`
`packages/domain/src/catalog/view.ts`

CHECKED BY
`docker-container/tests/dom/search.dom.test.tsx` — “shows cover, title, artist, album, its platforms, BPM, time, the explicit marker and a preview”
`docker-container/tests/dom/search.dom.test.tsx` — “downloads a song to the hub’s queue and says which source was chosen and what was embedded”
`packages/domain/tests/unit/catalog-view.test.ts` — “a Spotify song says it plays from YouTube Music via spotDL; a song that is elsewhere too does not”
`windows-companion/tests/dom/companion-search.dom.test.tsx` — “opens a song with genre, label, year and lyrics, and downloads it through the helper’s path”
`windows-companion/tests/contract/catalog-ipc.test.ts` — “a download takes the helper’s own path from the song’s best source, with the rights basis”
`windows-companion/tests/dom/companion-search.dom.test.tsx` — “folds the chunks the main process passes on, says every service’s state, and shows rows that are music”
`packages/domain/tests/unit/catalog-hydrate.test.ts` — “names the main artist, then the featured ones a store lists as contributors”

UX-SEARCH-004 **A pasted list names its platform, wears a mosaic and opens like an album** ● ADOPTED

A pasted album or playlist link is read, not searched. Its listing names its kind, platform and length, and its cover is a 2×2 mosaic of four songs’ artwork (or its own cover when four do not differ). Opened, it is a page like an album’s: the head, then its songs, every one of them, loaded as you scroll. A link that cannot be read says why, in words.

OWNED BY
`docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/catalog.ts`
`windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/catalog.ts`
`packages/domain/src/catalog/view.ts`

CHECKED BY
`docker-container/tests/dom/search.dom.test.tsx` — “lists a pasted Spotify playlist with its platform and a 2×2 mosaic, opens it like an album, and the star un-stars it”
`docker-container/tests/dom/search.dom.test.tsx` — “says why a link cannot be read, in words”
`packages/domain/tests/unit/catalog-view.test.ts` — “draws four songs’ artwork as a 2×2 mosaic, else the list’s own cover”
`windows-companion/tests/dom/companion-search.dom.test.tsx` — “lists a pasted Spotify playlist with its mosaic and platform, opens it, and the star un-stars it”

UX-SEARCH-005 **A star keeps an album or playlist in the library** ● ADOPTED

An opened album or playlist has a star (a toggle, pressed when kept). It keeps the list in the shared SavedCollection shape: on the hub in the admin's library (`/api/v1/catalog/saved`, one store per admin, for the player to sync later); on the companion in its own store. The status line says what happened.

OWNED BY
`docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/catalog.ts` `windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/catalog.ts` `packages/domain/src/catalog/view.ts` `docker-container/src/api/routes/catalog.ts` `docker-container/src/db/repositories/saved-collections.ts` `windows-companion/src/main/catalog.ts`

CHECKED BY
`docker-container/tests/integration/catalog.test.ts` — "keeps a starred list in the admin's library, in the shared SavedCollection shape, and un-stars it"
`docker-container/tests/integration/catalog.test.ts` — "needs the admin or a device with library:sync (DEC-041), never an anonymous caller, and a change needs the CSRF token"
`docker-container/tests/dom/search.dom.test.tsx` — "opens an album with its label, date and songs, and stars it into the library"
`windows-companion/tests/contract/catalog-ipc.test.ts` — "keeps starred albums and playlists and the filter in the companion's store"

UX-SEARCH-006 **The hub's music search settings: keyless, the key write-only** ● ADOPTED

Music ▸ Music search sets which services the catalog asks, whether a song downloaded from Search carries its lyrics, and the optional SongLink key. Everything works with no key. The key is sealed like the hub's other secrets and never shown again: the field says only whether one is set, and Clear Key removes it. Save sends what changed; Revert puts back what the hub has; at least one service stays on.

OWNED BY
`docker-container/src/web/views/CatalogSettings.tsx` `docker-container/src/catalog/service.ts`

CHECKED BY
`docker-container/tests/dom/search.dom.test.tsx` — "sends only what changed on Save, keeps the SongLink key write-only, and needs one service on"
`docker-container/tests/integration/catalog.test.ts` — "settings: admin only, the SongLink key write-only, a switched-off service skipped"

UX-SEARCH-007 **After a search, a calm overview: a few of each kind, "See all N", and no pager** ● ADOPTED

The owner's words (2026-10-07, given for the player and carried to the hub and the companion): "don't show pagination for the search list until the user clicks See all … it feels too hectic." After a search the results are a calm overview: Songs (five), Artists (three), Albums (three) and Playlists (three) — the kinds the filter shows — each group ending in "See all N" ("See all 9+ songs", N+ while the services say more may exist) that opens that type's own page (UX-SEARCH-009). The services' line sits under the groups (UX-SEARCH-002). The overview has no pager: no page numbers, no ‹ ›, no page count. Rows still fold in live as each service answers, one row per song. This replaces the numbered eight-row pager the overview had.

"harbour": Songs (5 of 9), See all 9+ songs; Artists (3), See all 3 artists; Albums (2), See all 2 albums; Playlists (3), See all 14 playlists; then iTunes 3 found · Deezer 6 found · … SoundCloud resting.

OWNED BY
`docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/catalog.ts` `windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/catalog.ts` `packages/domain/src/catalog/view.ts`

CHECKED BY
`docker-container/tests/dom/search.dom.test.tsx` — "the overview is calm: five songs and three of the rest, each with "See all N", the services under them, and no pager"
`docker-container/tests/e2e/search.spec.ts` — "searches, opens a song and comes back with the keyboard alone, on stock answers"
`windows-companion/tests/dom/companion-search.dom.test.tsx` — "the overview has no pager; See all opens the Songs page, which reads on and moves a page at a time (UX-SEARCH-007, UX-SEARCH-009)"
`packages/domain/tests/unit/catalog-view.test.ts` — "shows a few of each kind on the overview and pages a type in 25 songs or 12 of the rest"

UX-SEARCH-008 **Search is keyboard-first and says where it is** ● ADOPTED

Each list is a list box with one tab stop: the arrows, Home and End move the highlight, Page Up/Down move eight rows (on a type's page they turn its pages, UX-SEARCH-009), Enter opens, Space plays a song's preview, and Shift+F10 or the Menu key opens a song's menu (UX-SEARCH-012). A type's page has a tab list for its kinds (the arrows switch it) and a tab panel for its list; turning a page moves the keys to that page's first row. Opening a row or a type stacks a page with Back, and the focus moves to its heading; Escape walks back a page, and on the overview puts the results away and gives the keys back to the field — never from a field being typed in, and a menu's Escape closes only the menu. The filter is a modal sheet: the focus starts in it, stays in it, and returns to Filter… when it closes. A live region carries the search's progress, and another the window's confirmations.

OWNED BY

`docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/catalog.ts`
`windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/catalog.ts`
`packages/domain/src/catalog/view.ts`

CHECKED BY

`docker-container/tests/dom/search.dom.test.tsx` — "is a list box: one tab stop, arrows move, Enter opens the song with genre, label, year and synced lyrics"

`docker-container/tests/dom/search.dom.test.tsx` — "is a sheet: it takes the focus, Escape puts it away, and Done keeps the choice and searches again"

`docker-container/tests/dom/search.dom.test.tsx` — "Escape walks back a page, then puts the results away"

`docker-container/tests/dom/search.dom.test.tsx` — "the segmented control switches type, searching the new type alone for the same words"

`docker-container/tests/e2e/search.spec.ts` — "searches, opens a song and comes back with the keyboard alone, on stock answers"

`docker-container/tests/e2e/search.spec.ts` — "@a11y the search results, the filter sheet, a type page, a row's menu and a song have no detectable violations"

UX-SEARCH-009 **One type per page: it scrolls on by itself, and ‹ › with “Page N of M” move it a page at a time** ● ADOPTED

The owner’s words (2026-10-07): “The search list pagination should have arrows and let the user infinite scroll the search bar list, go to the next or previous page … these should have a separate search list page where the user can search for specifically that type of music media.” “See all” on the overview opens a page for that type alone — Songs, Artists, Albums or Playlists — in the same column, under Back: a heading, the count with the words searched (“31 songs for “harbour””, N+ while more may exist), a field of its own (“Search songs”, prefilled; Enter searches that type alone, `sections=<type>`, from offset 0), a segmented control (Songs · Artists · Albums · Playlists, a tab list) that searches the new type for the same words, the services’ line, and the list. From “See all” the page starts with the overview’s rows and asks that type’s next offset at once; afterwards it asks again whenever the end of the list comes near or the keys reach its last rows, while the services say more may exist (to their offset bound of 1000) — infinite scroll, “Loading more…” while a page is coming, and “That’s all N songs.” at the end. A song a later page brings again is not shown twice (by id and by `sameRecording`). The pager lives here alone, in a footer that stays put at the foot of the pane while the list scrolls: ‹ › and “Page N of M” (“M+” while more may exist). Pages are 25 songs, or 12 artists, albums or playlists; each row carries its page (`data-page`). ‹ › and Page Up/Down (from the list or the field) go to the previous or next page’s first row — fetching that page first when it has not arrived — and move the keys there; the count follows the scroll, and at the very end the last page is the one on show. Nothing is sliced from the first answer: later pages are fetched from the engine.

“See all 9+ songs” → Songs: 31 songs for “harbour”, Page 1 of 2 (the next offset asked at once); › lands on song 26, Page 2 of 2; the Songs field with “harbour lights” ⏎ → GET …/catalog/search?q=harbour+lights&sections=tracks&limit=25, Page 1 of 1+; › asks offset 25 first, then lands on its first row.

OWNED BY

`docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/catalog.ts`
`windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/catalog.ts`
`packages/domain/src/catalog/view.ts`

CHECKED BY

`docker-container/tests/dom/search.dom.test.tsx` — “See all opens the Songs page: it reads on at once, ‹ › and Page Up/Down move a page and land on its first row”

`docker-container/tests/dom/search.dom.test.tsx` — “› past what has arrived fetches that page first, then lands on its first row”

`docker-container/tests/dom/search.dom.test.tsx` — “the segmented control switches type, searching the new type alone for the same words”

`docker-container/tests/e2e/search.spec.ts` — “searches, opens a song and comes back with the keyboard alone, on stock answers”

`windows-companion/tests/dom/companion-search.dom.test.tsx` — “the overview has no pager; See all opens the Songs page, which reads on and moves a page at a time (UX-SEARCH-007, UX-SEARCH-009)”

`packages/domain/tests/unit/catalog-view.test.ts` — “counts the pages loaded, says when there are more, and asks for the next offset only at the end”

UX-SEARCH-010 **Playlists are a type: Deezer’s public playlists, the starred ones first** ● ADOPTED

Search lists playlists as it lists songs, artists and albums (UX-CAT-005): the filter has Playlists, the overview shows three with “See all N”, and the Playlists page lists them a page of twelve at a time. A playlist row shows its picture (or a 2×2 mosaic when the service gave four covers), its title, its platform chip and “Playlist on Deezer · owner · N songs”. Public playlists come from Deezer’s public API through the hub or the companion’s helper, keylessly. On the Playlists page the starred playlists whose title or owner holds the words searched sit first under “In your library” (the hub admin’s `/api/v1/catalog/saved`, the companion’s own store), “From the catalog” below. Choosing one opens it the way an album opens (UX-SEARCH-004): its link resolved page by page, every song, with the star (UX-SEARCH-005).

Playlists: 14 playlists for “harbour” — In your library: ★ Harbour Mix (Playlist · Spotify · 6 songs); From the catalog: Harbour Evenings (Playlist on Deezer · Playlist Editor · 40 songs) … Page 1 of 2.

OWNED BY
`docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/catalog.ts`
`windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/catalog.ts`
`packages/domain/src/catalog/view.ts` `packages/domain/src/catalog/providers/deezer.ts`

CHECKED BY
`docker-container/tests/dom/search.dom.test.tsx` — “Deezer’s public playlists, the starred ones first under “In your library”, and one opens like an album”
`windows-companion/tests/dom/companion-search.dom.test.tsx` — “Playlists is a type: the starred ones first, then Deezer’s, twelve a page (UX-SEARCH-010)”
`windows-companion/tests/contract/catalog-ipc.test.ts` — “a search can ask for playlists alone, and they reach the window in the chunks (UX-SEARCH-010)”
`packages/domain/tests/unit/catalog-view.test.ts` — “says what a playlist is, and finds the starred ones for the words searched”
`packages/domain/tests/unit/catalog-view.test.ts` — “folds playlists by id like every other kind”

UX-SEARCH-011 **One column, centred under the field: results, type pages and details open in it** ● ADOPTED

The owner’s words (2026-10-07): “Update the search modal UI and UX to be centered under the search bar.” In the hub’s Search tab and the companion’s Search tool the search is one column, at most 720px wide and centred in the pane, with the field at its top: the overview, a type’s page with its sticky footer, and every detail — an album, an artist, a song, a playlist — open in that same column under the field, with Back. A narrow window narrows the column with it.

Hub at 1280 px: the pane is 740 px wide, the column 720 px, its centre the pane’s; the Playlists page’s footer stays inside it at the foot of the pane.

OWNED BY
`docker-container/src/web/views/Search.tsx` `docker-container/src/web/styles.css`
`windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/styles.css`

CHECKED BY
`docker-container/tests/e2e/search.spec.ts` — “the results are one column centred under the field, and a type page and its footer stay in it”

UX-SEARCH-012 **A song from search goes to Up Next, a playlist or the library through the row’s menu** ● ADOPTED

The owner’s words (2026-10-07): “ensure the user can add songs from search to their Now Playing queue, playlist, or music library.” Every song row — the overview, a type’s page, an album’s, a playlist’s or an artist’s songs — has a “…” at its end; a right-click, or Shift+F10 or the Menu key on the highlighted row, opens the same menu: the window kit’s contextual menu (`.menu` in airwave-window.css), keyboard-operable (the arrows, Home and End, Right and Left for a submenu, Enter and Space, Escape back to the list, Tab or a click away closes it; on a touch screen its commands are 44px tall). It offers Add to Up Next, Add to Playlist ▸, Add to Library…, Download…, Audition (Stop Audition while it plays; disabled, saying why, without a clip) and Open Details. Add to Up Next is a group’s queue — on the hub, `POST /api/v1/groups/:id/requests` with the song’s best fetchable link (YouTube Music, YouTube, SoundCloud, Bandcamp, spotDL), a submenu when there are several active groups; the companion has no queue of its own, so the paired hub’s groups it is in (`hub:groups`, `hub:request`), or a sentence saying queueing needs a group (no hub, no permission to join groups, not in one). A song only a store has says there is nothing to queue or fetch. Add to Library… is a download with a stated rights basis — on the hub into its download queue (`POST /api/v1/catalog/download`, Music ▸ Downloads, what the hub’s library is made of), on the companion through the helper’s own path into this PC’s library. Add to Playlist ▸ writes (amended 2026-10-10, DEC-041): each app keeps a playlist folder, so the submenu lists its playlists, ticked where the song is already, and New Playlist…; filing keeps the song’s link, and “Also add to Library”, on by default, queues the download too (UX-PL-005, CMP-PL-004). The hub’s players’ shared copies stay read-only, shown apart. Download… is the same with a choice of format. Every action confirms in the window’s status line, which is a live region.

Hub: “Harbour Lights” ▸ … ▸ Add to Up Next ▸ Studio → POST /groups/…/requests {query: https://music.youtube.com/watch?v=…} → “Up Next in Studio: “Harbour Lights”, number 3.” Companion with no hub: “Queueing needs a group on a hub: this companion has no queue of its own. Pair a hub under Remote first.”

OWNED BY

`docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/catalog.ts` `windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/catalog.ts` `packages/domain/src/catalog/view.ts` `docker-container/src/web/lib/menu.tsx` `windows-companion/src/renderer/menu.tsx` `packages/aqua-ui/scripts/make-window-css.py` `packages/aqua-ui/src/styles/airwave-window.css` `windows-companion/src/main/hub.ts` `windows-companion/src/shared/ipc.ts` `docker-container/src/web/lib/playlists.ts` `windows-companion/src/renderer/playlists.ts`

CHECKED BY

`docker-container/tests/dom/search.dom.test.tsx` — ““…” opens it; Add to Up Next ▸ a group queues the song’s fetchable source through the group’s requests, and says so”

`docker-container/tests/dom/search.dom.test.tsx` — “right-click and Shift+F10 open it too; the keys walk it, and Escape gives the keys back to the list”

`docker-container/tests/dom/search.dom.test.tsx` — “Add to Library and Download… go to the hub’s queue with a rights basis”

`docker-container/tests/dom/search.dom.test.tsx` — “a song only a store has says there is nothing to fetch; with no group, queueing says it needs one”

`docker-container/tests/dom/search.dom.test.tsx` — “Audition plays the clip and then reads Stop Audition”

`docker-container/tests/e2e/search.spec.ts` — “a song row’s “…” and right-click open its menu; the keys walk it and Add to Library asks for the rights basis”

`windows-companion/tests/dom/companion-search.dom.test.tsx` — “a song row’s “…”: Add to Up Next goes to the paired hub’s group, Add to Library through the helper with a basis (UX-SEARCH-012)”

`windows-companion/tests/dom/companion-search.dom.test.tsx` — “with no hub paired, Up Next says plainly that queueing needs a group (UX-SEARCH-012)”

`windows-companion/tests/contract/catalog-ipc.test.ts` — “a group is an id and a name; a request names the group by id and the song by its link or its name”

`windows-companion/tests/integration/companion-and-hub.test.ts` — “lists the groups this companion is in, and a request comes back with the hub’s own answer”

`windows-companion/tests/integration/companion-and-hub.test.ts` — “says why there is no group to offer: no hub, or no permission to join groups”

`docker-container/tests/dom/search.dom.test.tsx` — “Add to Playlist ▸ lists the hub’s playlists, ticked where the song is, and files it — with the library too, by default (UX-PL-005)”

`windows-companion/tests/dom/companion-search.dom.test.tsx` — “Add to Playlist ▸ lists this PC’s playlists, ticked where the song is, and files it — with the library too, by default (CMP-PL-004)”

NP-DISC-001 **Discover is ranked by the chosen algorithm, through one ranker** ● ADOPTED

Discover (library menu ▸ Discover) is the recommendation engine's ranking of the eligible songs — owned, not starred, not waiting in Up Next — under the current algorithm (Settings ▸ Recommendations, `cfg.algoCur`) and its mode, using every attribute a song carries: plays and skips, tempo, genre, release and added dates, what Discover has shown before. The ranker is a pure function (music-player/src/shell/recommend/rank.ts, `window.NP_RECOMMEND`): the weighted factors less the engine's penalties, then the familiarity mix (each tier's share), the artist cap and exploration — deterministic for a seed, so the same library and settings give the same list. The preview beside the sliders ranks through the same call (seed 0), and a change there re-ranks an open Discover. A factor this library cannot feed scores zero and the panel says so beside its slider, read from the rows themselves. The mode's `excludeOwned` is not applied to a library that is all owned (NPD-032).

Eight seeded songs, Alder Quartet played, Airwave default in For you: the strong tier (Alder) opens the list, Birch Ensemble follows, Gantry (played this week) is left out. Crate digger in Deep cuts leaves out every artist you know and ranks what you have heard least first.

OWNED BY `music-player/src/shell/recommend/rank.ts` `music-player/src/shell/bridge.ts` `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY
- `music-player/tests/unit/recommend-rank.test.ts` — “gives the same order for the same library, log, config and seed, and another for another seed”
- `music-player/tests/unit/recommend-rank.test.ts` — “Airwave default leans on the artists you play; Crate digger puts the least-played first”
- `music-player/tests/unit/recommend-rank.test.ts` — “leaves out starred songs, queued songs and anything played inside the repeat window”
- `music-player/tests/unit/recommend-rank.test.ts` — “reads tempo, genre, release and added dates, and says which factors this library can feed”
- `music-player/tests/e2e/np/discover.spec.ts` — “opens ranked by the chosen algorithm, with the chip and Refresh in the silver bar”
- `music-player/tests/e2e/np/prefs.spec.ts` — “the recommender ranks, and its validation is enforced here”

NP-DISC-002 **Refresh gives new songs; Load more extends the list** ● ADOPTED

While Discover is on show the silver bar has a Refresh button ("New songs", ↻, a 44px target on touch). It re-seeds and re-ranks, leaves out what this Discover session has already shown until everything eligible has been shown once — then starts over and the HUD and the live region say so — scrolls the list to its top and announces "N new songs" in the list's live region. Shift-click, or the menu's Start Discover Over, clears the shown record first. Discover shows fifty at a time; a Load more row under the rows says how many are left and extends the same ranking without moving what is already shown, and announces how many more. Opening Discover again shows the list as it stands; the menu's Discover row counts what is eligible.

56 songs: a page of 50 and "Load more — 6 more songs". Refresh: "6 new songs". Refresh again: "50 new songs — everything eligible has been shown once, so Discover started over".

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY
- `music-player/tests/e2e/np/discover.spec.ts` — “Refresh re-seeds, leaves out what was shown until everything has been, pages with Load more, and speaks”
- `music-player/tests/unit/recommend-rank.test.ts` — “a refresh excludes what was already shown, and says how many were eligible”

NP-DISC-003 **The algorithm chip: see and change the algorithm from the bar** ● ADOPTED

While Discover is on show the silver bar carries a chip with the current algorithm's colour dot, name and mode ("Crate digger · Deep cuts", plus any lean as a suffix); the scope chip wears a thin tint of the colour. Clicking it opens the shell's own menu (NP-MENU-001 behaviour: arrows, Return, Escape back to the chip) under it: every algorithm, built in and yours, each with its colour dot, mode and one line on what it favours (from its two largest effective weights and its exploration rate), a tick on the current one; Lean ▸; New Songs; Start Discover Over; Edit Algorithms…, which opens Settings ▸ Recommendations. Choosing one sets `cfg.algoCur` and its mode, saves, re-ranks Discover at once from the top and the HUD says which algorithm is on. Settings and the chip share one source of truth: a choice in either shows in the other.

Chip: "Airwave default · For you". Menu row: "Crate digger · Deep cuts — Favours what you have heard least and what sounds like what you play, wide exploration". HUD: "Discover ranked by “Crate digger” · Deep cuts".

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py` `music-player/src/shell/recommend/rank.ts`

CHECKED BY `music-player/tests/e2e/np/discover.spec.ts` — “the chip’s menu switches the algorithm: the list re-ranks, the HUD says so, and Settings shows the same choice”
`music-player/tests/unit/recommend-rank.test.ts` — “describes an algorithm from its largest weights and its exploration”
`music-player/tests/e2e/a11y.spec.ts` — “@a11y Discover, with the algorithm chip, Refresh and the open algorithm menu, has no detectable violations”

NP-DISC-004 **Every algorithm has a colour of its own** ● ADOPTED

An algorithm carries `color` (`#rrggbb`), saved with `cfg.algos`, exported and imported in `.airwave-algorithm.json` as `color` (optional on import: a file without one, or with one another algorithm here already wears, gets the next free hue). The twelve hues are tokens.json `algorithm` (--np-algo-*), thirty degrees apart; the built-ins are fixed — Airwave default blue, Late night indigo, Crate digger orange. A new or duplicated algorithm takes the next hue no other algorithm uses, in palette order; with all twelve taken, the least-used hue farthest from the others. Settings ▸ Recommendations shows the swatch beside the name and a legend of every algorithm in its colour; the swatch opens a keyboard-operable menu of the twelve, naming who has each, and a colour another algorithm uses is refused in a plain sentence. The colour is on the bar chip, the menu's dots, the Settings legend and the scope chip's tint.

Duplicate Airwave default twice: the copies are red, then gold (orange is Crate digger's). Pick blue for a copy: "“Airwave default” already uses blue — pick another, or recolour that one first."

OWNED BY `packages/aqua-ui/src/styles/tokens.json` `packages/aqua-ui/src/styles/now-playing.css` `music-player/src/shell/recommend/rank.ts` `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY `music-player/tests/e2e/np/discover.spec.ts` — “every algorithm has a colour of its own: new ones take the next free hue, a taken one is refused, and they survive a reload and an import without one”
`music-player/tests/unit/recommend-rank.test.ts` — “hands out the next unused hue, never one another algorithm has”
`music-player/tests/unit/recommend-rank.test.ts` — “with all twelve taken, picks the least crowded hue, farthest from the others”

NP-DISC-005 **Lean: a different taste or genre without editing the algorithm** ● ADOPTED

The chip's menu has Lean ▸ Familiar / Balanced / Adventurous and, when the library carries genres, the genres to lean toward. A lean biases the ranking for this player without touching the saved algorithm: Familiar lowers exploration and raises the artists and genres you play (the mix leans on the strong tier); Adventurous widens exploration and raises what you have heard least (the mix leans on the far tiers); a genre lean scores that genre's songs as fully in your taste, treats the genre as known and triples the genre weight. The lean is kept in the player's settings (`cfg.lean`), shown on the chip as a suffix ("· Adventurous", "· Jazz"), and Reset lean clears it; the HUD says what changed.

"Airwave default · For you · Adventurous · Folk": Birch Ensemble's folk songs open the list.

OWNED BY `music-player/index.html` `music-player/scripts/make-shell.py` `music-player/src/shell/recommend/rank.ts`

CHECKED BY `music-player/tests/e2e/np/discover.spec.ts` — “a lean bends the ranking without editing the algorithm: Adventurous, then a genre; Reset clears it”

`music-player/tests/unit/recommend-rank.test.ts` — “a genre lean moves that genre up without touching the saved weights”

`music-player/tests/unit/recommend-rank.test.ts` — “Familiar and Adventurous bend exploration and discovery, Balanced leaves the algorithm alone”

NP-DISC-006 **From the catalog: songs this device does not have, ranked by the same algorithm** ● ADOPTED

Under the library's ranked songs, Discover shows a set from the catalog, under a heading row "From the catalog". The queries come from the listener's own profile, as the ranker builds it from the play log: the artists played (most played first) and the genres played, a session lean's genre first; a mode that leaves out the artists you know (Deep cuts) asks for genres only, and one that leaves out your top artist does not ask for it. They are asked through whichever client answers the search — the paired hub, the companion on this PC, or, with neither, this browser's own engine against the keyless services (keyless-first) — two at a time. The answers are ranked by the same rankSongs, under the current algorithm, mode and lean, with the library supplying the taste and never offered again: what the device already has (the same recording, however spelled) and what this Discover session already showed are left out, and one recording is one pick. Twelve picks at most; each row carries its platforms as badges after the title, the catalog's artist credit, and why the ranker put it there (the strongest factor, or "Picked to explore") in its tooltip. The heading says how many, by which algorithm and mode, through whom, and that each pick is not on this device. Choosing one never starts a preview; it opens the authorized full-song fetch where a route exists, or says what is needed (NP-FIND-011). Refresh asks the catalog again and shows what the session has not shown; a new algorithm or lean asks again from the top. Offline, with nothing played, with nothing answering, or with nothing new, the heading says which in a sentence. The rows' menu is the catalog song's (Add to Up Next, Add to Playlist ▸, Add to Library, Download…); short auditions remain in Search results. The chip's menu has From the Catalog, a tick while it is on, kept in the player's settings; off, Discover shows only the library and asks nothing.

Discover, Airwave default · For you: eight library songs, then two catalog picks ranked by the same algorithm; choosing a visitor opens a fetch sheet without a preview, and a saved full-length copy plays locally.

OWNED BY

`music-player/src/shell/recommend/online.ts` `music-player/src/shell/recommend/rank.ts`
`music-player/src/shell/search/discover.ts` `music-player/src/shell/search/index.ts` `music-player/index.html`
`music-player/scripts/make-shell.py`

CHECKED BY

`music-player/tests/unit/discover-online.test.ts` — “asks for the artists and genres actually played, most played first”
`music-player/tests/unit/discover-online.test.ts` — “follows a mode that leaves out the artists you know: it asks for genres instead”
`music-player/tests/unit/discover-online.test.ts` — “drops what this device already has, however the two spell it”
`music-player/tests/unit/discover-online.test.ts` — “puts an artist you play above one you have never played”
`music-player/tests/unit/discover-online.test.ts` — “ranks under the mode chosen, never another: Deep cuts keeps only artists you have never played”
`music-player/tests/unit/discover-online.test.ts` — “names the factor that put it there and says it is not on this device yet”
`music-player/tests/e2e/np/discover-online.spec.ts` — “an online pick appears under the library’s songs, ranked by the current algorithm, with its platforms and why”
`music-player/tests/e2e/np/discover-online.spec.ts` — “its row menu adds it to the library, Up Next or a playlist; Download… opens the fetch sheet”
`music-player/tests/e2e/np/discover-online.spec.ts` — “Refresh asks the catalog again and shows what this session has not shown”
`music-player/tests/e2e/np/discover-online.spec.ts` — “offline it says so plainly, and so it does when nothing answers”
`music-player/tests/e2e/np/discover-online.spec.ts` — “with no hub and no companion, this browser’s own engine answers, keylessly”

NP-DISC-007 **Discover never preloads or plays preview clips** ● ADOPTED

Discover's From the catalog rows are metadata, not audio. When they land or a row is chosen, no short clip is prefetched or played; only an explicit Search result audition requests a 30-second preview. Choosing a visitor in Discover offers the authorized full-song fetch path; once its full copy is saved locally, the player plays that copy at full duration. The catalog rows and controls still answer while search is pending; the UI never suggests a clip is ready or playing in Discover.

Five catalog picks include preview URLs; Discover requests none until the person explicitly auditions a result in Search.

OWNED BY `music-player/src/shell/recommend/prefetch.ts` `music-player/src/shell/search/discover.ts` `music-player/index.html` `music-player/scripts/make-shell.py`

CHECKED BY
- `music-player/tests/unit/discover-prefetch.test.ts` — "asks for the best pick first, one request at a time"
- `music-player/tests/unit/discover-prefetch.test.ts` — "keeps only the depth it promises ahead, and says the ring is full"
- `music-player/tests/unit/discover-prefetch.test.ts` — "stops at the byte budget, measured from what arrived, and explains itself"
- `music-player/tests/unit/discover-prefetch.test.ts` — "asks for nothing at all while it is off, and lets go of what it held"
- `music-player/tests/unit/discover-prefetch.test.ts` — "lets go of a played clip and of one no longer ahead"
- `music-player/tests/e2e/np/discover-online.spec.ts` — "catalog clips are neither fetched nor played here; only an explicit Search audition requests one"

PLAYLIST FOLDERS IN THE HUB AND THE COMPANION

UX-PL-001 **The hub keeps its playlists in a folder inside its data volume, which the admin can change** ● ADOPTED

The owner's words (2026-10-10): "Give the hub and companion app a folder to store or write playlists to." The hub's playlist folder is `<data>/playlists` by default (the data volume, `/data` in the container). Music ▸ Playlists ▸ Playlist folder shows it with Change…, a sheet with a path field: a folder under `playlists/` or `library/` in the data volume, typed relative ("library/Playlists") or written out ("/data/library/Playlists"). The server decides (`PUT /api/v1/playlists/folder`, admin only): each segment is plain (letters, digits, spaces, dots, dashes, underscores, brackets), no drive, no root, no `..`, no reserved name, and once made its real path must still be inside the data volume, so a link inside the volume cannot lead out. When the folder holds playlists the sheet offers "Move the N playlists there too" (on): they are rewritten in the new folder, relative paths worked out again, and removed from the old one only once written; what could not be moved is named. A refused folder leaves the old one in place and says why under the field.

"library/Playlists" with Move on → "Playlists are kept in /data/library/Playlists now. Moved 2 playlists there." "../etc" → "Playlists can be kept in a folder inside the hub's data volume, under /data/playlists or /data/library …"

OWNED BY `docker-container/src/playlists/service.ts` `docker-container/src/api/routes/playlists.ts` `docker-container/src/web/views/Playlists.tsx` `docker-container/src/web/lib/playlists.ts`

CHECKED BY
- `docker-container/tests/integration/playlists.test.ts` — "moves the playlists to a new folder inside the data volume when asked"
- `docker-container/tests/security/playlist-folder.test.ts` — "refuses %j and keeps the folder it had"
- `docker-container/tests/security/playlist-folder.test.ts` — "refuses a symbolic link inside the volume that leads out of it"
- `docker-container/tests/dom/playlists.dom.test.tsx` — "changes the folder inside the data volume and offers to move the playlists there"
- `docker-container/tests/dom/playlists.dom.test.tsx` — "says why when the server refuses a folder"

UX-PL-002 **A playlist is an .m3u8 any player can open, with an .airwave.json sidecar beside it** ● ADOPTED

One `<name>.m3u8` per playlist: `#EXTM3U`, `#PLAYLIST:<name>`, then for each song `#EXTINF:<seconds>,<artist> - <title>` (−1 when the length is unknown) and its location — a path relative to the folder, with forward slashes, when the library has the file (matched by ISRC, else by artist and title within 3 s), otherwise the song's best source URL (the best fetchable one, else its first). Beside it `<name>.airwave.json` (format `airwave-playlist`, version 1) carries what M3U cannot: the playlist's id, created and updated times, who made it, its description and its 2×2 cover mosaic, and each entry's id, catalog id, ISRC, artists, album, artwork, platforms, sources, added time and who added it. The M3U decides which songs are in the list and in what order; the sidecar is matched back to it by location. File names are the playlist's name made safe for Windows and Linux (`sanitizeFilename`: no separators, control characters, reserved names or trailing dots, 120 characters at most) and unique in the folder, case-insensitively ("Mix (2)"). Line breaks never reach a line. Every file is written to a temporary name beside it and renamed over the old one, sidecar first, one write at a time. Shared by the hub and the companion (`@now-playing/domain/playlist-folder`).

#EXTM3U / #PLAYLIST:Road Trip / #EXTINF:215,Daft Punk - Digital Love / ../library/music/Daft Punk/Discovery/03 Digital Love.flac / #EXTINF:220,Röyksopp - Eple / https://www.deezer.com/track/3135556

OWNED BY

`packages/domain/src/playlist-folder/format.ts` `packages/domain/src/playlist-folder/store.ts` `packages/contracts/src/api/playlist-folder.ts`

CHECKED BY

`packages/domain/tests/unit/playlist-folder.test.ts` — "writes #EXTM3U, #PLAYLIST and #EXTINF:<secs>,<artist> - <title> before each location"

`packages/domain/tests/unit/playlist-folder.test.ts` — "round-trips what it writes"

`packages/domain/tests/unit/playlist-folder.test.ts` — "cannot be made to write an extra line: names, artists and titles lose their line breaks"

`packages/domain/tests/unit/playlist-folder.test.ts` — "creates a playlist as an .m3u8 and a sidecar: a library song as a relative path, another by its source URL"

`packages/domain/tests/unit/playlist-folder.test.ts` — "round-trips through a fresh store (nothing kept in memory)"

`packages/domain/tests/unit/playlist-folder.test.ts` — "keeps names unique and safe, and never writes outside the folder"

`packages/domain/tests/unit/playlist-folder.test.ts` — "writes atomically: a temporary file renamed over the old one, nothing left behind"

`docker-container/tests/integration/playlists.test.ts` — "files a song the library has as a path relative to the folder, and a player can stream it by id"

UX-PL-003 **A list dropped into the folder by hand is shown, read-only until it is changed here; outside edits show up** ● ADOPTED

Every read rescans the folder (a file is parsed again only when its size or time, or its sidecar's, changed), so a list edited in another player, or an `.m3u`/`.m3u8` put there by hand, appears the next time anything is opened; the companion also watches the folder. A list without a sidecar is "made outside Airwave": listed with "Hand-made", opened with a note, and never rewritten by reading it. The first change made to it here adopts it — it is written as `.m3u8` with a sidecar, keeping its id — and the person's own lines are kept as they wrote them. An absolute path in a hand-made list (`C:\…`, `/…`, `file:`) is neither shown nor followed; a relative one is looked at only when it resolves inside the playlist folder or the library's folders; a symbolic link in the folder is not followed. A list reordered or trimmed elsewhere keeps the metadata of the songs it still holds.

"Old Favourites.m3u" → "Old Favourites · 1 song · made outside Airwave"; after Add to Playlist ▸ Old Favourites → "Old Favourites.m3u8" and "Old Favourites.airwave.json".

OWNED BY
`packages/domain/src/playlist-folder/format.ts` `packages/domain/src/playlist-folder/store.ts`
`packages/contracts/src/api/playlist-folder.ts` `windows-companion/src/main/playlists.ts`

CHECKED BY
`packages/domain/tests/unit/playlist-folder.test.ts` — "reads a hand-made list: a BOM, CRLF, blank lines, no #EXTINF, comments"
`packages/domain/tests/unit/playlist-folder.test.ts` — "matches the sidecar back by location after the list was reordered and trimmed elsewhere"
`packages/domain/tests/unit/playlist-folder.test.ts` — "shows a hand-made .m3u read-only, adopts it on the first change, and keeps its id"
`packages/domain/tests/unit/playlist-folder.test.ts` — "picks up a list edited outside Airwave the next time it is read"
`packages/domain/tests/unit/playlist-folder.test.ts` — "does not follow a symbolic link in the folder"
`docker-container/tests/integration/playlists.test.ts` — "shows a hand-made .m3u8 read-only, with its absolute paths hidden"

UX-PL-004 **The hub's playlists are an API: the admin may do everything, a device with playlists:use files into any list and changes only its own** ● ADOPTED

`/api/v1/playlists` lists (with `catalogId`/`isrc`, whether each list holds that song), creates, reads a page of songs (≤ 200), renames, deletes, adds (≤ 500 a request; a song already there is skipped), removes, moves and exports (`audio/x-mpegurl`, the file as it is). The admin session may do everything, with the CSRF token on every change; the folder itself is the admin's alone. A paired device needs `playlists:use`: it reads every list, makes new ones and files songs into any — filing is additive and the hub is the household's shelf — but renames, reorders and deletes only lists it made, and removes only songs it added or songs in a list it made. A playlist holds at most 10,000 songs and a folder 1,000 playlists; past either the hub says so in words. Ids are opaque (`[A-Za-z0-9_-]`), never paths. Every change is audited.

A player: POST /api/v1/playlists/{id}/entries {tracks:[…]} → {added: 1, skipped: 0}; DELETE /api/v1/playlists/{admin's id} → 403 "Only the hub admin, or the device that made it, may change …".

OWNED BY
`docker-container/src/playlists/service.ts` `docker-container/src/api/routes/playlists.ts`
`packages/contracts/src/api/routes.ts` `packages/contracts/src/api/playlist-folder.ts`

CHECKED BY
`docker-container/tests/integration/playlists.test.ts` — "creates, files, ticks, pages, moves, removes, renames, exports and deletes"
`docker-container/tests/integration/playlists.test.ts` — "lets a device read every list, make its own, and file into any"
`docker-container/tests/security/playlist-folder.test.ts` — "needs a signed-in admin or a device with playlists:use"
`docker-container/tests/security/playlist-folder.test.ts` — "asks the admin for the CSRF token on every change"
`docker-container/tests/security/playlist-folder.test.ts` — "lets a device change only what it made, and remove only what it added"
`docker-container/tests/security/playlist-folder.test.ts` — "holds a playlist to 10,000 songs"
`docker-container/tests/security/playlist-folder.test.ts` — "holds a request to 500 songs and a folder to 1,000 playlists"
`docker-container/tests/security/playlist-folder.test.ts` — "turns a name into a safe file inside the folder, whatever it says"
`docker-container/tests/security/playlist-folder.test.ts` — "refuses an id that is a path"
`packages/domain/tests/unit/playlist-folder.test.ts` — "enforces the caps: songs per playlist and playlists per folder"

UX-PL-005 **Add to Playlist ▸ in the hub's Search lists the folder's playlists, ticked where the song is, and writes** ● ADOPTED

A song row's menu (UX-SEARCH-012) has Add to Playlist ▸: the folder's playlists, read again for each menu with the song, each a menu checkbox ticked when the song is already in it, then New Playlist…. A ticked one says the song is there already. Choosing one opens a sheet: the playlist keeps the song's link, so it plays before it is in the library; "Also add to Library", on by default, queues the download as well, with the rights basis every download states (DEC-036); a song only a store has is filed by its store link with nothing to fetch. New Playlist… asks for the name in the same sheet. The status line says what was written, and what the library did.

"Harbour Lights" ▸ … ▸ Add to Playlist ▸ Road Trip ▸ Add to Playlist → "Harbour Lights" is in "Road Trip" now. It is joining the hub's library too. Queued from YouTube Music. …

OWNED BY `docker-container/src/web/views/Search.tsx` `docker-container/src/web/lib/playlists.ts` `docker-container/src/web/lib/menu.tsx`

CHECKED BY `docker-container/tests/dom/search.dom.test.tsx` — "Add to Playlist ▸ lists the hub's playlists, ticked where the song is, and files it — with the library too, by default (UX-PL-005)"
`docker-container/tests/e2e/search.spec.ts` — "Add to Playlist ▸ New Playlist… files a song into the hub's playlist folder, and Music ▸ Playlists opens it like an album (UX-PL-005, UX-PL-007)"

UX-PL-006 **Music ▸ Playlists lists the hub's playlists with their mosaics, and the players' copies apart** ● ADOPTED

A list box (UX-KEY-001) of the folder's playlists: the 2×2 mosaic of its first four songs' artwork, its name, "N songs · length", and "Hand-made" for a list made outside Airwave; Enter or a click opens it. New Playlist… asks for a name. The folder's path is said in the group's hint and in Playlist folder (UX-PL-001). The playlists paired players sync to the hub stay theirs, read-only, under "Shared from players".

Road Trip · 3 songs · 11 min; Old Favourites · 1 song · made outside Airwave · Hand-made. Shared from players: Kitchen Mix, 12 songs.

OWNED BY `docker-container/src/web/views/Playlists.tsx` `docker-container/src/web/lib/playlists.ts` `packages/domain/src/catalog/folder-playlists.ts`

CHECKED BY `docker-container/tests/dom/playlists.dom.test.tsx` — "lists the folder's playlists with their mosaics and lengths, a hand-made one marked, the folder and the players' copies apart"
`docker-container/tests/dom/playlists.dom.test.tsx` — "makes a new playlist from New Playlist…"

UX-PL-007 **A playlist opens like an album, and its songs move, leave and export from the keyboard as from the pointer** ● ADOPTED

Opened (UX-SEARCH-004 style): Back, its name as the page heading (focused), the big mosaic, "Playlist · N songs · length", its file name, and Rename…, Export .m3u8 (the file, named as it is) and Delete… (asked in a sheet first, saying the songs stay in the library and at their sources). Its songs are a list box: number, cover, title, artist and album, where it plays from ("In the library", "Plays from Deezer", or "Not found"), and length. A song moves by dragging it, or with Alt+↑/↓ on the highlighted row; Delete or Backspace takes it out; its "…", a right-click, Shift+F10 or the Menu key open its menu (Move Up, Move Down, Open Its Link, Remove from Playlist). More Songs reads the next 200.

Alt+↓ on "Harbour Lights" → "Moved "Harbour Lights" to number 2."

OWNED BY `docker-container/src/web/views/Playlists.tsx` `docker-container/src/web/lib/playlists.ts`

CHECKED BY `docker-container/tests/dom/playlists.dom.test.tsx` — "opens like an album: the mosaic, the name, the length, and where each song plays from"
`docker-container/tests/dom/playlists.dom.test.tsx` — "moves a song with Alt+↓, by drag, and takes one out with Delete"
`docker-container/tests/dom/playlists.dom.test.tsx` — "its menu moves, opens the link and removes; Shift+F10 opens it"
`docker-container/tests/dom/playlists.dom.test.tsx` — "renames, and deletes only once asked"
`docker-container/tests/e2e/search.spec.ts` — "Add to Playlist ▸ New Playlist… files a song into the hub's playlist folder, and Music ▸ Playlists opens it like an album (UX-PL-005, UX-PL-007)"

UX-PL-008 **A device syncs its starred albums and playlists with the hub through library:sync** ● ADOPTED

`/api/v1/catalog/saved` answers the admin (its own starred lists) and a device with `library:sync`: the device's own (read and written, kept apart from every other device's) and the admin's as `shared`, read-only. A device never writes or removes anyone else's.

GET /api/v1/catalog/saved (device) → {items: [its own], shared: [the admin's]}.

OWNED BY `docker-container/src/api/routes/catalog.ts` `docker-container/src/db/repositories/saved-collections.ts` `packages/contracts/src/api/routes.ts`

CHECKED BY `docker-container/tests/security/playlist-folder.test.ts` — "needs library:sync, keeps each device's own, and shows the admin's as shared, read-only"
`docker-container/tests/integration/catalog.test.ts` — "needs the admin or a device with library:sync (DEC-041), never an anonymous caller, and a change needs the CSRF token"

CMP-PL-001 **The companion keeps its playlists in Music\Airwave Playlists, changed only through the system's folder picker** ● ADOPTED

The default is `%USERPROFILE%\Music\Airwave Playlists` — beside the library, not inside a library folder, which may be read-only or a network share and would be scanned (DEC-041). Settings ▸ Playlists shows it with Change… (the system's folder picker: the window never names a path, and `playlistDir` is not a preference the window can set), Open Folder, and "Move my playlists when the folder changes" (on). Moving rewrites the lists in the new folder with their relative paths worked out again, and keeps the old folder when the new one cannot be written. Restore Defaults and restoring a backup leave the folder where it is.

Change… ▸ D:\Music\Playlists → "Playlists are kept in D:\Music\Playlists now. Moved 1 playlist there."

OWNED BY `windows-companion/src/main/playlists.ts` `windows-companion/src/shared/ipc.ts` `windows-companion/src/main/index.ts` `windows-companion/src/renderer/views/Playlists.tsx` `windows-companion/src/renderer/views/Settings.tsx`

CHECKED BY `windows-companion/tests/integration/playlist-folder.test.ts` — "starts at Airwave Playlists in the Music folder, the default, and answers by the IPC contract"
`windows-companion/tests/integration/playlist-folder.test.ts` — "shows a hand-made list, adopts it on the first change, exports it, and moves everything to a newly chosen folder"
`windows-companion/tests/contract/playlists-ipc.test.ts` — "never take a folder from the window: the picker and Open Folder take no path, and preferences refuse one"
`windows-companion/tests/dom/companion-playlists.dom.test.tsx` — "shows the folder, changes it only through the system picker with the playlists moved along, and opens it"

CMP-PL-002 **The companion's playlists are the same files and shapes as the hub's, over playlists:* IPC** ● ADOPTED

Twelve channels in `windows-companion/src/shared` (list, get, create, update, delete, add, remove, move, export, folder, pick-dir, open-folder), validated both ways, answering the shapes `/api/v1/playlists` serves; a change answers with its result or null and the reason in words. File access lives in the main process beside the library's: a song a music folder has is filed as a path relative to the playlist folder (and played by the library's id), any other by its source URL.

playlists:get {playlistId: "not-there"} → {result: null, reason: "That playlist isn't in the playlist folder any more."}

OWNED BY `windows-companion/src/main/playlists.ts` `windows-companion/src/shared/ipc.ts` `windows-companion/src/shared/channels.ts`

CHECKED BY `windows-companion/tests/contract/playlists-ipc.test.ts` — "are all allowlisted and registered, with the change event"
`windows-companion/tests/contract/playlists-ipc.test.ts` — "take ids, not paths, and refuse what they do not know"
`windows-companion/tests/contract/playlists-ipc.test.ts` — "answer in the shapes the hub serves"
`windows-companion/tests/integration/playlist-folder.test.ts` — "files a library song as a relative path the library resolves, and any other song by its source"
`windows-companion/tests/integration/playlist-folder.test.ts` — "answers a failure with its reason instead of throwing"

CMP-PL-003 **A list changed in the companion's folder by another player shows up without a click** ● ADOPTED

The main process watches the playlist folder while the app runs and tells the window (`event:playlists-changed`, debounced); Library ▸ Playlists reads the folder again, and an open playlist reads itself again.

A .m3u8 saved into the folder by another player appears in the list.

OWNED BY `windows-companion/src/main/playlists.ts` `windows-companion/src/renderer/views/Playlists.tsx`

CHECKED BY `windows-companion/tests/integration/playlist-folder.test.ts` — "notices a list changed on disk by another player"
`windows-companion/tests/dom/companion-playlists.dom.test.tsx` — "reads the folder again when it changes on disk"

CMP-PL-004 **Add to Playlist ▸ in the companion's Search writes into this PC's playlists** ● ADOPTED

As UX-PL-005, through `playlists:list` (with the song, for the ticks), `playlists:add` and `playlists:create`; "Also add to Library" (on by default) takes the helper's own download path into this PC's library (`catalog:download`) with the rights basis.

"Harbour Lights" is in "Road Trip" now. It is joining this PC's library too. Downloading from YouTube Music.

OWNED BY `windows-companion/src/renderer/views/Search.tsx` `windows-companion/src/renderer/playlists.ts`

CHECKED BY `windows-companion/tests/dom/companion-search.dom.test.tsx` — "Add to Playlist ▸ lists this PC's playlists, ticked where the song is, and files it — with the library too, by default (CMP-PL-004)"

CMP-PL-005 **Library ▸ Playlists in the companion: the mosaics, a list opened like an album, its songs moved and taken out** ● ADOPTED

As UX-PL-006 and UX-PL-007, in the companion's kit: the list box of playlists with their mosaics, opened with Rename…, Export… (the save dialog) and Delete… (asked first); songs moved by drag or Alt+↑/↓, taken out with Delete or from the row's menu.

Road Trip · 2 songs · 7 min.

OWNED BY `windows-companion/src/renderer/views/Playlists.tsx` `windows-companion/src/renderer/playlists.ts`

CHECKED BY `windows-companion/tests/dom/companion-playlists.dom.test.tsx` — "lists the folder's playlists with their mosaics, and opens one like an album"
`windows-companion/tests/dom/companion-playlists.dom.test.tsx` — "moves a song with Alt+↓, takes one out with Delete, exports, and deletes once asked"

CMP-PL-006 **On touch, a long press on a search or playlist row opens the same menu as a right-click** ● ADOPTED

A finger or pen held still on a row for 500 ms (moving more than 10 px first is a scroll, lifting early is a tap) opens the row's menu where it was pressed; the click it leaves behind does not also open the row. A mouse is never timed. The rows do not start a text selection on a long press.

Hold "Harbour Wall" half a second → its menu.

OWNED BY `windows-companion/src/renderer/long-press.ts` `windows-companion/src/renderer/views/Search.tsx`
`windows-companion/src/renderer/views/Playlists.tsx`

CHECKED BY `windows-companion/tests/dom/companion-search.dom.test.tsx` — "on touch, a long press on a row opens the same menu; a short tap or a moved finger does not (CMP-PL-006)"
`windows-companion/tests/dom/companion-playlists.dom.test.tsx` — "opens a song's menu with a long press on touch (CMP-PL-006)"

Accessibility

Not bolted on afterwards. These are parts of the design.

- **One tab stop per group, arrows to move.** The section strip, the mode switch, the music list, the context menu and the tools of an Airwave window all use a roving `tabIndex`, and the focus ring travels with the selection.
- **Focus is always visible.** An end-to-end test tabs the whole page and fails on any element that shows no indicator.

- **Roles must be true.** A rail that cannot be dragged is `role="img"` with no value pair, not a slider that refuses every change. A button that opens a section does not claim `aria-haspopup="menu"`.
- **Names say what will happen**, including the refusal.
- **Colour is never the only signal.** The playing row has a glyph as well as a tint; status dots carry text.
- **Text meets 4.5:1.** Where the Airwave design's status colours fall short as text on white, the hub reads sentences in darker inks of the same hues (DEC-022); an axe pass runs on every hub tab, the sign-in window and the confirmation sheet.

Every screen

Every surface found in the products' own navigation, with the states it has to answer and the rules that govern it. `pnpm styleguide:check` rebuilds this list from the `ViewId` declarations, the sheet files and the Discord command table, and fails when something appears that is not recorded in `design/coverage.json`.

Seen running: opened in the running product for this revision (see Coverage). **Mockup only**: rendered at every size in the mockups with fixture data — the hub and the companion from their own stylesheets and kits, the player from the component library's page skin. **Not run here**: the product could not be started where this guide was built. **Source only**: checked against the code, not looked at.

AIRWAVE — THE PLAYER

The shell generated from design/frontends/origin/airwave-now-playing.html: the PWA, the single-file build, the Android app and the local helper all show it. Its surfaces are the player-shell rows (adopted). The rows marked proposed are the React interface in music-player/src, which is kept and tested but no longer served (DEC-019).

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Status bar, mode switch, hero and section strip
(React specimen, not served — DEC-019) every section
`music-player/src/App.tsx`
`packages/aqua-ui/src/components/PageBar.tsx`
	• nothing playing
	• playing
	• paused
	• live stream (read-only scrubber)
	• unknown duration
	• shared unavailable (reason shown)
	• narrow (strip scrolls)
	• transport: volume beneath when it cannot flank the keys (DEC-016)
	• transport: keys wrapped, previous/play/next alone on the first line (DEC-016)
		`UX-PRIN-002`
		`UX-KEY-001`
		`UX-KEY-002`
		`UX-TOUCH-001`
			○ mockup only

Music Library
(React specimen, not served — DEC-019)
section strip → Music Library
`music-player/src/views/Library.tsx`
`packages/aqua-ui/src/components/MusicList.tsx`
	• empty (how to add music)
	• scanning
	• loaded
	• sorted
	• selected row
	• playing row
	• unplayable format
		`UX-KEY-001`
		`UX-STATE-001`
		`UX-A11Y-001`
		`VOICE-001`
			○ mockup only

Now playing
(React specimen, not served — DEC-019)
section strip → Now playing
`music-player/src/views/NowPlaying.tsx`
`packages/domain/src/visualisers/spectrum.ts`
	• nothing playing
	• playing (spectrum moving)
	• paused (spectrum stopped)
	• reduced motion (spectrum at 4 per second)
	• equalizer not applied (reason shown)
	• shared (group queue)
		`UX-STATE-001`
		`UX-MOTION-001`
			○ mockup only

Queue
(React specimen, not served — DEC-019)
section strip → Queue
`music-player/src/views/Queue.tsx`
	• nothing queued
	• queued
	• shuffled
	• shared (hub revision)
		`UX-STATE-001`
		`UX-SHARED-001`
		`VOICE-001`
			○ source only

Playlists (React specimen, not served — DEC-019) section strip → Playlists `music-player/src/views/Playlists.tsx`	• no playlists • list • playlist open • empty playlist	`UX-STATE-001` `VOICE-001`	○ source only

Search
(React specimen, not served — DEC-019)
status-bar search → Enter
`music-player/src/views/Search.tsx`
`music-player/src/components/SearchPopover.tsx`
	• prompt (device only / with hub)
	• searching hub
	• no local matches
	• results
	• hub unreachable
		`UX-STATE-001`
		`UX-PRIN-002`
			○ source only

Listening history (React specimen, not served — DEC-019) section strip → Listening history `music-player/src/views/Metrics.tsx`	• nothing yet • history	`UX-STATE-001` `UX-SAFE-004`	○ source only

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Constellation
(React specimen, not served — DEC-019)
section strip → Constellation
`music-player/src/views/Constellation.tsx`
`packages/domain/src/visualisers/constellation-layout.ts`
	• nothing to map
	• field
	• table (equivalent view)
	• reduced motion (table)
	• WebGL unavailable
		`UX-MOTION-001`
		`UX-STATE-001`
			○ mockup only

Settings and equalizer
(React specimen, not served — DEC-019)
profile button in the status bar
`music-player/src/views/Settings.tsx`
`music-player/src/views/Equalizer.tsx`
	• no folders
	• folders
	• hub paired
	• hub not paired
	• equalizer on/off
	• solfeggio preset (dimmed bands)
	• crossfade
	• car/lock-screen report
		`UX-PRIN-001`
		`UX-PRIN-003`
		`UX-PRIN-004`
		`UX-LINK-001`
		`UX-TOUCH-002`
		`UX-SAFE-004`
			○ mockup only

Add to playlist (React specimen, not served — DEC-019) ✚ beside the transport, or row menu `music-player/src/components/AddToPlaylistSheet.tsx`	• open • no playlists	`UX-KEY-001`	○ source only

New playlist
(React specimen, not served — DEC-019)
Playlists → New, or row menu
`music-player/src/components/NewPlaylistSheet.tsx`
`packages/aqua-ui/src/components/music-list-behaviours.tsx`
	• empty name (validation)
	• valid
		`VOICE-002`	○ source only

Download from the hub (React specimen, not served — DEC-019) row menu → Download `music-player/src/components/DownloadSheet.tsx`	• choose format • basis required • in progress • failed with reason	`UX-FEED-001` `UX-PRIN-002`	○ source only

Fetch with a local tool
(React specimen, not served — DEC-019)
Library → Fetch (when a helper or the Android tools are present)
`music-player/src/components/FetchSheet.tsx`
`music-player/src/lib/fetch-helper.ts`
	• no tool (control hidden)
	• url entry
	• running
	• finished (files)
	• failed (tool's own reason)
		`UX-PRIN-001`
		`UX-FEED-001`
		`VOICE-002`
			○ source only

Share (React specimen, not served — DEC-019) share button beside the transport, or a playlist `music-player/src/components/ShareSheet.tsx`	• no hub (explains why) • link created • copied	`UX-PRIN-002`	○ source only

Search popover
(React specimen, not served — DEC-019)
typing in the status-bar search
`music-player/src/components/SearchPopover.tsx`
	• counting
	• results
	• none
	• dismissed (arrow brings back)
	• a row filled in by the engine's last chunk: "feat. …", BPM, the E mark and a cover arrive after the row (UX-CAT-006)
		`UX-FEED-001`
		`UX-KEY-001`
		`UX-CAT-006`
			○ source only

Notice bar and toasts
(React specimen, not served — DEC-019) any warning the store raises
`music-player/src/components/NoticeBar.tsx`
`packages/aqua-ui/src/components/Toast.tsx`
	• info
	• warning
	• dismissed
		`UX-FEED-001`
		`VOICE-002`
			○ source only

Row context menu (React specimen, not served — DEC-019) right-click or the row's menu key `packages/aqua-ui/src/components/music-list-behaviours.tsx`	• open • disabled items with reason	`UX-KEY-001` `UX-PRIN-004`	○ source only

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Library
the list under the player; Music view
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• empty (nothing indexed)
	• rows from this device
	• link rows from search
	• sorted
	• narrowed by scope chip
	• filtered by the bar
	• an album or playlist from the catalog (search-collection-list)
	• Discover: ranked by the chosen algorithm, the chip and Refresh in the silver bar (player-shell-discover)
	• visitor rows: catalog metadata; no player preview, fetch the full song before it plays
		`NP-LIST-001`
		`UX-KEY-001`
		`UX-STATE-001`
		`NP-FIND-007`
		`NP-FIND-008`
		`NP-DISC-001`
		`NP-FIND-011`
			● seen running

Music list ▸ Discover
library menu ▸ Discover
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• ranked by the chosen algorithm (the chip names it, in its colour; the scope chip tinted)
	• refreshed: a new seed, what this session showed left out, N new songs spoken
	• everything eligible shown once: started over, and said so
	• capped at fifty with Load more
	• leaned: Familiar / Adventurous, or toward a genre (the chip's suffix)
	• nothing eligible (every song starred or queued)
	• From the catalog: asking (the heading says so; the library rows answer meanwhile)
	• From the catalog: picks ranked by the same algorithm, each with its platforms and why, through the hub / the companion / this browser
	• From the catalog: offline / nothing played / nothing answered / nothing new (a sentence in the heading)
	• From the catalog: no preview clips fetched or played; explicit auditions stay in Search
		`NP-DISC-001`
		`NP-DISC-002`
		`NP-DISC-003`
		`NP-DISC-005`
		`NP-DATA-003`
		`NP-DISC-006`
		`NP-DISC-007`
			○ source only

Discover ▸ the algorithm menu
the chip in the silver bar while Discover is on show
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• every algorithm, built in and yours, with its colour dot, mode and what it favours; a tick on the current one
	• Lean ▸ Familiar / Balanced / Adventurous, the library's genres, Reset lean
	• New Songs, Start Discover Over, Edit Algorithms…
	• touch: 44px rows
	• From the Catalog (ticked while on)
		`NP-DISC-003`
		`NP-DISC-004`
		`NP-DISC-005`
		`NP-MENU-001`
		`NP-DISC-006`
			○ source only

Now Playing
the jewel case and transport at the top of the page
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• nothing playing
	• a track from this device playing (the bar follows the element; case sleeve and CD label use its available cover art)
	• paused
	• a link row chosen (does not pretend to play)
	• a station (LIVE)
	• a channel (video bar, LIVE)
	• a catalog visitor selected with no preview or playback; after fetch, its full local copy starts
		`NP-TRANS-001`
		`UX-KEY-002`
		`NP-FIND-011`
			● seen running

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Search popover
the header search field
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• empty
	• searching (rows arriving, upserted by id)
	• the overview: songs (five), artists, albums and playlists (three each) in sections, each with “See all N”, no pager
	• a Spotify song: “plays from YouTube Music”
	• the per-service line: counts, searching, failed, timed out, cooling down until, needs yt-dlp; who answered
	• no matches for the words searched
	• failed (said plainly, per service)
	• Track / Artist / Album / ISRC fields folded out
	• a stale query (dimmed, “Press ⏎ to search”)
	• an older hub’s enriched songs (features, album, genre, bpm)
	• auditioning (click, or a five-second hold that fills the ring)
	• no clip (says why)
	• a song row’s “…” (and right-click, long-press): the contextual menu
	• pasted link being read (Spotify: spotDL takes up to a minute)
	• pasted song: one row with its platform and date
	• pasted album or playlist: one listing with its platform and a 2×2 mosaic
	• pasted link nobody can read: the reason
	• people on the hub
	• centred under the field at 390, 820 and 1280 px (phone: spans the width)
		`NP-PRIN-002`
		`NP-FIND-001`
		`NP-FIND-002`
		`NP-FIND-003`
		`NP-FIND-005`
		`NP-FIND-006`
		`NP-FIND-007`
		`NP-FIND-009`
		`NP-FIND-010`
			○ source only

Row menu
right-click or long-press a row
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• one row
	• several rows (marquee)
	• a station with a known song: the On air section (Up Next, Group Queue, Playlist)
	• touch: a long press on a station opens the same menu
	• a visitor (catalog song not on this device): Add to Up Next, Add to Playlist ▸, Add to Library, Download…, Audition
	• Add to Playlist ▸ On <hub>: the hub’s playlists, ticked from hasTrack, and New Playlist on <hub>…
	• paired without playlists:use: “Your hub hasn’t allowed this player to use its playlists”, and how to fix it
	• a song with neither a link nor an ISRC: “Can’t file this song on <hub>”, and why
		`NP-MENU-001`
		`NP-RADIO-002`
		`NP-FIND-010`
		`NP-FIND-011`
		`NP-FIND-013`
			○ source only

New playlist sheet
row menu ▸ Add to Playlist ▸ New Playlist…
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• empty name refused
	• created
		`NP-MENU-002`	○ source only

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Fetch sheet
transport ▸ Download, on a link row
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• no basis chosen (refused, said why)
	• helper found / not found
	• fetching
	• refused by the helper (its reason)
	• fetched and playing
	• the date beside the title, when the link row has one
	• the hub fetches it: the source it chose and what it embedded, then “Fetching… N%”
	• opened for a visitor without audio; the kept full copy starts after fetch
	• nothing can fetch it (the hub has not allowed downloads, no helper): said, and Fetch claims nothing
		`UX-SAFE-001`
		`UX-FEED-001`
		`NP-FIND-002`
		`NP-FIND-011`
		`NP-FIND-012`
			○ source only

Mini player
the header while Settings is open
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• hidden
	• shown for music and radio
		`NP-CHROME-001`	○ source only

Radio
toolbar ▸ Radio
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• directory unreachable (offline shelf)
	• a station playing
	• station list
	• the song on the air, read from the stream (paired or with the companion)
	• no feed and nothing to read the stream: the programme format, no invented title
	• station directory from the paired Hub (metadata only, not a stream proxy); the public directory when unpaired, on an older Hub, or after a named refusal
	• timed radio cue advancing smoothly from its published start and length, including tab return
	• conflicting start times for one song: song named, length shown, broadcast position unavailable
	• published start with no length: elapsed stamp advances, progress bar empty
		`NP-LIVE-001`
		`NP-RADIO-001`
		`NP-RADIO-002`
		`NP-RADIO-003`
			● seen running

Live TV
toolbar ▸ Live TV
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• no playlist loaded (empty guide)
	• channels from a loaded M3U
	• a channel playing (LIVE)
	• channels and guide from the companion (now and next where a guide has them; Live where it has none)
	• no companion, or one with no playlists: the guide is as it was
	• no companion on this machine: a paired hub’s copy of the companion’s Live TV fills the guide
	• Until: when the programme on now ends; blank where a channel has no guide
		`NP-LIVE-002`
		`NP-TV-001`
			● seen running

TV
toolbar ▸ TV
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• no catalogue (nothing invented)	`NP-PRIN-001`	○ source only

Movies
toolbar ▸ Movies
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• no catalogue (nothing invented)	`NP-PRIN-001`	○ source only

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Settings ▸ Statistics
profile button ▸ Statistics, or #settings/stats
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• nothing recorded yet
	• this player's history
	• imported file
	• 3D views unavailable (tables still carry the numbers)
		`NP-DATA-001`	● seen running

Settings ▸ Recommendations
#settings/rec
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• defaults
	• edited
	• saved algorithm
	• reverted
	• the swatch and the legend: every algorithm in its colour; a built-in's is fixed
	• a colour picked from the twelve; one another algorithm uses, refused in a sentence
	• the preview beside the sliders ranks through the same ranker as Discover
		`NP-PREF-001`
		`NP-DISC-001`
		`NP-DISC-004`
			○ source only

Settings ▸ Sources
#settings/src
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• music on this device (folders/files)
	• companion found / missing downloaders
	• companion downloader setting up (with %), not counted as missing
	• hub tested / paired
	• backup measured (nothing sent)
	• backup will not fit (reason)
	• Hub tools — authenticated: unpaired / loaded / credential refused (401) / missing permission naming its scope (403) / older Hub (unsupported) / unavailable (503) / unreachable, kept apart from the Companion cards (UX-SETUP-001)
	• each Hub runtime tool (yt-dlp, spotDL, FFmpeg, MusicSearch) shown as missing / checking / installing / ready / failed, or status unknown; refreshing never installs
		`NP-PREF-002`
		`UX-FEED-001`
		`UX-SETUP-001`
			○ source only

Stream from a PC
#settings/src ▸ Connections ▸ Stream from a PC; the indicator beside the transport while a PC track plays
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• not paired
	• not a ticket / not six digits (refused before sending, said why)
	• pairing
	• wrong code / code expired or used (the PC's answer, in words)
	• paired and connected: PC name, relay-carried, track count
	• reconnecting (the PC or its relay is away)
	• refused (revoked on the PC: pair again)
	• a PC track playing: indicator reads relay-carried
	• single-file copy: unavailable, and says so
		`NP-PREF-007`
		`NP-PRIN-002`
		`UX-FEED-001`
			○ source only

Settings ▸ Player
#settings/player
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• appearance
	• listening
	• defaults
	• disc: spin steady / with the tempo / off, with its speed
	• disc: turn steady / with the tempo / held one way (facing) / left where it is
	• disc: no known tempo (steady speed)
	• disc: angle (presets or exact) and tilt; Reset Position
	• disc: live preview (as if playing; 120 BPM when no tempo is known)
		`NP-PREF-003`
		`NP-PREF-014`
			○ source only

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Settings ▸ Equalizer
#settings/eq
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• preset
	• custom
	• default volume
		`NP-PREF-004`	○ source only

Settings ▸ Profile
#settings/profile (shown when paired)
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• name available / taken
	• picture
	• shared playlists
	• groups
	• invites received / sent
	• declined
		`NP-PREF-005`
		`UX-SAFE-001`
			○ source only

Profile viewer
search ▸ a person, or a group member
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• their name, picture and shared playlists
	• a playlist opened
		`NP-PREF-005`	○ source only

Invite page
a #invite/<code> link
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• previewed (group, sender, role, expiry)
	• unknown or used code
	• joined
	• declined
		`NP-PREF-005`	○ source only

Search ▸ a type's page (See all)
a type's "See all" on the overview, or the segmented control on another type's page
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• the overview's rows, then the next page at once; "Page 1 of M+"
	• loading more as it scrolls; the count following the scroll
	• ‹ › (or Page Up/Down) landing on a page's first row, fetched first when needed
	• the type's own field: a fresh search of that type
	• another type chosen in the control
	• that's all N; "Page M of M"
	• nothing for the words (said)
	• a page that failed (said, and no more asked)
		`NP-FIND-004`
		`NP-FIND-005`
		`NP-FIND-003`
			○ source only

Search ▸ Playlists
"See all playlists" on the overview, or Playlists in a type page's control
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• In your library: the person's starred playlists, first
	• From the catalog: Deezer's public playlists, twelve a page, with their pictures, owners and counts
	• the pager: ‹ › and "Page N of M"
	• with only this browser: answered over JSONP
	• one chosen: opened in the music list, with the star
	• none for the words (said)
		`NP-FIND-009`
		`NP-FIND-007`
		`NP-FIND-008`
		`NP-FIND-004`
		`UX-CAT-005`
			○ source only

Search ▸ a song's menu
a song row's "…", right-click, or a long press, in the search
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• Add to Up Next / In Up Next
	• Add to Playlist ▸ the playlists (ticked where one holds the song), New Playlist…
	• Add to Library / In your library
	• Download… (disabled with the reason for a store-only song)
	• Audition / Stop Audition (disabled with no clip)
	• touch: 44 px commands, the submenu unfolding in place
	• the HUD confirming (NP-MENU-003)
	• Add to Playlist ▸ On <hub>: the hub's playlists, ticked from hasTrack, and New Playlist on <hub>…
	• paired without playlists:use: "Your hub hasn't allowed this player to use its playlists", and how to fix it
	• a song with neither a link nor an ISRC: "Can't file this song on <hub>", and why
		`NP-FIND-010`
		`NP-MENU-003`
		`NP-FIND-013`
			○ source only

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Search ▸ Album
an album row in the search popover
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• opening
	• cover, facts, platforms and songs
	• could not be opened (the reason)
		`NP-FIND-006`
		`NP-FIND-007`
			○ source only

Search ▸ Artist
an artist row in the search popover
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• opening
	• picture, facts, top songs and albums
	• an album drilled into (Back returns)
		`NP-FIND-006`	○ source only

Search ▸ Song
a song row in the search popover
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• genre, label and year being looked up
	• enriched
	• lyrics synced / plain / none / instrumental
	• Download… through the hub (its chosen source and tags)
	• Download… (the hub, else the helper) / only in stores (says so)
	• Hub Metadata: explicit MusicSearch metadata-only results as plain text (no play, preview, download or path action); refused while MusicSearch is not configured, installing or failed; cleared when the paired Hub changes (NP-FIND-012)
		`NP-FIND-006`
		`NP-FIND-001`
		`NP-FIND-012`
			○ source only

Search filter sheet
Filter in the search popover's header
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• every section and service on
	• some switched off (the button says so)
	• nothing left on (refused, in words)
		`NP-FIND-005`	○ source only

Music list ▸ an album or playlist from the catalog
a pasted playlist or album, an album's "Open in Music", or the library menu's kept lists
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• loading ("Loading 400 of 1,250…", the rows already there usable)
	• every song, page by page, no cap
	• a server that still caps it (the first N)
	• stopped part way or could not be read (the reason)
	• star off / on
	• a song played: its preview while the hub or the companion fetches it
	• star synced with the hub (library:sync)
	• shared by the hub admin: read-only, no star ("Shared on <hub> — read-only")
		`NP-FIND-007`
		`NP-FIND-008`
		`NP-FIND-011`
			○ source only

Music list ▸ a playlist on the hub
library menu ▸ Playlists ▸ On <hub>
`music-player/index.html`
`music-player/scripts/make-shell.py`
	• the hub's playlists listed / checking / none yet / not allowed (how to fix it)
	• opened: rows page by page, "Playlist · <hub> · N songs"
	• a song the hub's library holds, streaming from the hub
	• a song kept by its link: a visitor (fetch sheet; no audio until its full copy is saved)
	• a song the hub cannot place ("Not found")
	• made here: "…" ▸ Rename…, Move Song Up/Down, Delete Playlist…; Alt+↑/↓
	• made on another device or by hand: read-only, said in the bar
	• Shared on <hub>: the admin's starred lists, read-only
		`NP-FIND-014`
		`NP-FIND-007`
		`NP-FIND-011`
		`NP-FIND-008`
			○ source only

AIRWAVE HUB — ADMIN WINDOW AND PUBLIC PAGES

The Airwave window of design/frontends/origin/airwave-hub.html, served by the container on port 4546. Sign-in and the first-run gate are drawn in the same window.

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Sign in
any admin URL without a session — drawn in the same window, every tab locked
`docker-container/src/web/App.tsx`
DEC-017
	• connecting to the hub
	• hub not answering (Try Again)
	• first run (admin / admin hint)
	• set up (no hint)
	• username or password missing
	• wrong password (the hub's sentence)
	• rate limited
		`UX-SAFE-002`
		`VOICE-002`
		`UX-CHROME-001`
			● seen running

First-run gate: choose a real password first sign-in — the amber gate at the top of Overview, the other five tabs locked `docker-container/src/web/App.tsx` DEC-017	• new password asked twice (the current one is remembered from sign-in) • current password asked for too, after a reload • under twelve characters refused • the two entries differ • refused by the hub (its sentence) • saved: every tab unlocks	`UX-SAFE-002` `UX-CHROME-001`	○ not run here
Section error boundary any section that throws while rendering `docker-container/src/web/ViewBoundary.tsx`	• one sentence where the section would have been, with Try Again; the window, tools and status strip stay up	`UX-STATE-001`	○ not run here

The window: title, seven tools, pane and status strip
every admin URL, in every state
`docker-container/src/web/App.tsx`
`docker-container/src/web/icons.tsx`
DEC-017
	• signed out (every tool locked)
	• first run (only Overview open; the rest locked with the reason)
	• signed in (seven tools: the design's six and Search; arrows, Home and End move between them)
	• a section opened by its #id in the address
	• attention badge on Overview
	• status strip: hub name, bind address and reach; what the window just did replaces it for a few seconds
	• below 640px: tools scroll sideways, tiles two-up, tables drop secondary columns
		`UX-CHROME-001`
		`UX-KEY-001`
		`UX-KEY-002`
		`UX-FEED-001`
			● seen running

Confirmation sheet
any destructive action: revoke, remove, restore, import, withdraw, archive, cancel a download, remove the bot token
`docker-container/src/web/ui.tsx`
`docker-container/src/web/styles.css`
DEC-021
	• open: the question, what will happen, Cancel and the verb
	• Cancel has the focus
	• Tab moves between the two buttons only
	• Escape cancels
	• focus returns to the button that asked
		`UX-SAFE-001`
		`UX-FOCUS-001`
			○ not run here

Overview
Overview tab
`docker-container/src/web/views/Overview.tsx`
`docker-container/src/web/views/common.tsx`
	• loading (tiles read — and Loading…)
	• loaded (tiles: hub, devices, groups, library, providers, storage)
	• first run (Devices, Groups and Library read Off, with the reason)
	• error (tiles read Not available; Try Again in Needs attention)
	• alerts (count on the Overview tool)
	• nothing needs you
	• signed-in line and Sign Out at the foot
	• the badge on the Overview tool and the Needs attention list come from one reading, so they always agree
		`UX-STATE-001`
		`UX-KEY-001`
		`UX-CHROME-001`
			○ not run here

Devices and pairing
Devices tab → Devices
`docker-container/src/web/views/Devices.tsx`
	• no devices
	• permissions ticked before a code is made
	• pairing code shown on the LCD plate (with QR)
	• code expired
	• claim waiting for the verification code
	• pending pairing refused
	• revoke (confirm sheet)
		`UX-SAFE-001`
		`UX-SAFE-003`
		`UX-STATE-001`
			○ not run here

SURFACE	STATES IT MUST ANSWER	RULES	SEEN
Groups Groups tab `docker-container/src/web/views/Groups.tsx` DEC-017	• no groups • group with queue and members (drift per device) • archived • new group • invite made (code shown once) • invite listed by who it is for, its role and state (never its code) • invite withdrawn (confirm sheet) • archive (confirm sheet) • a new invite opens under its own row with its code, folded away with its button or Escape	`UX-SHARED-001` `UX-STATE-001`	● seen running

Providers
Music tab → Providers
`docker-container/src/web/views/Providers.tsx`
	• working
	• limited
	• needs setting up
	• off
	• down
	• details open (capabilities, limits, credit)
	• set-up form (secrets write-only; blank keeps what is stored)
	• saved and tested (result beside the button)
	• external tool on without setup (DEC-024)
	• external tool row names the tools on this hub: (yt-dlp, spotDL) (DEC-036)
	• external tool details show independent yt-dlp, spotDL and FFmpeg readiness, allow each downloader to be enabled without credentials, and do not replace a custom command
	• requests and limits
	• details open under their own row; Escape closes them and puts the caret back on the button
	• Requests and limits lists the providers in use, the rest behind a disclosure
		`UX-PRIN-002`
		`UX-A11Y-001`
		`UX-SETUP-001`
			○ not run here

Library Music tab → Library `docker-container/src/web/views/Library.tsx`	• no folders • folder scanning • folder scan failed • tracks • no matches • remove folder (confirm sheet) • Scan on a folder’s row scans that folder alone • Scan Now scans every folder	`UX-STATE-001` `UX-SAFE-001`	● seen running

Downloads
Music tab → Downloads
`docker-container/src/web/views/Downloads.tsx`
	• none (who asks for one)
	• waiting
	• running (stage as words or a bar)
	• paused
	• failed with reason
	• cancelled
	• done
	• cancel (confirm sheet)
	• playlist entry: one row each, From names the list and the position (DEC-036)
	• formats this hub cannot write struck through, with why
	• storage line
		`UX-FEED-001`
		`UX-PRIN-002`
		`UX-SAFE-001`
			○ not run here

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Live TV from the companion
Music tab → Live TV from the companion (drawn after Downloads, inside the downloads section)
`docker-container/src/web/views/LiveTv.tsx`
`docker-container/src/live-tv/service.ts`
	• loading
	• nothing yet: a sharing companion sends its channels when it syncs
	• kept: which companion sent it, how many channels, the guide, when it was updated
	• remove (confirm sheet); the companion sends them again when it next syncs
	• failed to load (Try Again)
	• read-only: channels are added and removed in the companion
		`UX-TV-001`
		`UX-STATE-001`
		`UX-SAFE-001`
			○ not run here

Profiles Devices tab → Profiles `docker-container/src/web/views/Profiles.tsx`	• none • listed • renaming • name taken • remove picture (confirm sheet)	`UX-SAFE-001` `UX-FEED-001`	● seen running
Shared links Sharing tab → Shared links `docker-container/src/web/views/Shares.tsx` DEC-028	• none • active (expiry in words) • revoked • revoke (confirm sheet) • Create Link: a hub playlist or album from the picker and how long the link works; the hub builds the item list (DEC-028) • link made: the full link shown once, under the form, with Copy • nothing to share yet: the picker says so	`UX-SAFE-001`	● seen running
Recommendations Music tab → Recommendations `docker-container/src/web/views/Recommendations.tsx`	• defaults • edited • nothing sent until Save; Revert puts back what the hub has • a number outside its range refused before sending • refused by the hub: its sentence under the form	`UX-PRIN-003` `UX-FORM-001`	● seen running

Discord
Sharing tab → Discord
`docker-container/src/web/views/Discord.tsx`
`docker-container/src/discord/service-interface.ts`
	• no token
	• has a token, switched off
	• worker not running
	• connected
	• voice playing
	• last error
	• template preview
	• remove token (confirm sheet)
	• warnings and errors as plain sentences
	• server IDs checked before anything is sent
	• Save and Start sends the lot; Revert puts back what the hub has
	• Invite the Bot disabled with its reason until a token is saved
		`UX-PRIN-002`
		`UX-DISCORD-003`
		`VOICE-002`
		`UX-SAFE-001`
		`UX-FORM-001`
			● seen running

Network System tab → Network `docker-container/src/web/views/Network.tsx`	• localhost • lan • remote (restart required) • edited: nothing sent until Save; the public address checked first; Revert puts back what the hub has • saved: Saved. in the status strip • what works where: this machine, your network, the internet	`UX-SAFE-002` `UX-FORM-001`	● seen running

SURFACE	STATES IT MUST ANSWER	RULES	SEEN
Change admin password System tab → Change admin password `docker-container/src/web/views/Security.tsx`	• signed-in admin only, current/new/confirmation required • mismatch refused before sending • weak or incorrect password refused by the server • password changed, admin sessions rotated and sensitive inputs cleared	`UX-SAFE-002` `UX-FORM-001`	○ not run here

Backup
System tab → Backup
`docker-container/src/web/views/Backup.tsx`
DEC-029
	• no backups
	• list
	• Back Up Now (said in the status strip)
	• restore (confirm sheet)
	• restore failed with reason
	• import: checked first, then confirmed in a sheet
	• space measured
	• space unknown
	• next backup will not fit
	• settings: the folder inside the data volume, the parts included, how often (every day or week, at a time) and how many to keep — sent together with Save; Revert puts them back
	• a folder outside the data volume refused before sending
	• the next scheduled backup said in words, on the hub's clock; a slot missed while the hub was off is taken when it starts (DEC-029)
	• Download on each archive (an attachment, admin only)
		`UX-SAFE-001`
		`UX-STATE-001`
		`UX-FORM-001`
			● seen running

Diagnostics System tab → Diagnostics `docker-container/src/web/views/Diagnostics.tsx`	• loaded • log level chosen (the list follows it) • support bundle saved as a file, with what it leaves out said beside the button	`UX-STATE-001`	● seen running
Public share page /s/<token> `docker-container/src/api/routes/shares.ts` DEC-008	• track or playlist • stream allowed • link unavailable	`UX-SAFE-004`	○ not run here

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Search
Search tab
`docker-container/src/web/views/Search.tsx`
`docker-container/src/web/lib/catalog.ts`
DEC-039
	• nothing asked yet: the field, Track/Artist/Album, Filter…
	• searching: every service “asking…”, rows folding in as each answers
	• the calm overview, centred under the field: Songs (five), Artists, Albums and Playlists (three each), each with “See all N”, no pager, the services’ line under them (UX-SEARCH-007, UX-SEARCH-011)
	• done: each service’s state in words (one resting, with when it is back) and the platforms that only gave links
	• a section with nothing (“No songs.”) and a filtered search (“Showing … from …”)
	• Filter… (sheet): sections (Songs, Artists, Albums, Playlists) and services; refuses a choice that shows nothing
	• a type’s own page (See all): heading, Back, its own field, the Songs · Artists · Albums · Playlists control, the services’ line and the list (UX-SEARCH-009)
	• a type’s page reading on: the next offset at once, “Loading more…” as it scrolls, “That’s all N songs.” at the end
	• a type’s page footer: ‹ › and “Page N of M” (“M+” while more may exist), landing on a page’s first row, fetched first when needed
	• another type chosen in the control, or the type’s field searched: that type alone, from the start
	• Playlists: the starred ones first under “In your library”, then Deezer’s public playlists, twelve a page; one opened like an album (UX-SEARCH-010)
	• a song row’s menu (“…”, right-click, Shift+F10): Add to Up Next (▸ groups), Add to Playlist ▸, Add to Library…, Download…, Audition, Open Details (UX-SEARCH-012)
	• Add to Up Next: queued in a group (“Up Next in Studio: …, number 3.”), the hub’s reason when it could not, “Queueing needs a group” with none
	• a song only a store has: nothing to queue or fetch, said in words
	• Add to Library… / Download… (sheet): the rights basis (and format), into the hub’s download queue
	• Add to Playlist ▸: the hub’s playlists, ticked where the song is, and New Playlist…; the sheet keeps the song’s link and “Also add to Library” (on) queues the download with its basis (UX-PL-005)
	• a song: cover, platforms, preview, genre/label/year (looking it up…), synced or plain lyrics (or none, or instrumental), Download with the rights basis and what happened
	• an album: cover, label, date, songs, the star
	• an artist: picture, top songs, albums (more as you scroll)
	• a pasted playlist: its 2×2 mosaic, platform and length; opened like an album with every song; starred
	• a link that cannot be read: the reason in words
	• a failure: the reason in words under the field
	• a row filled in by the engine’s last chunk: the credit line (“feat. …”), BPM, the explicit mark and a cover arrive after the row (UX-CAT-006)
		`UX-SEARCH-001`
		`UX-SEARCH-002`
		`UX-SEARCH-003`
		`UX-SEARCH-004`
		`UX-SEARCH-005`
		`UX-SEARCH-007`
		`UX-SEARCH-008`
		`UX-CAT-001`
		`UX-CAT-002`
		`UX-CAT-003`
		`UX-CAT-004`
		`UX-KEY-001`
		`UX-CAT-006`
		`UX-SEARCH-009`
		`UX-SEARCH-010`
		`UX-SEARCH-011`
		`UX-SEARCH-012`
		`UX-CAT-005`
		`UX-PL-005`
			○ not run here

SURFACE	STATES IT MUST ANSWER	RULES	SEEN
Music search settings Music tab → Music search `docker-container/src/web/views/CatalogSettings.tsx` DEC-039	• loading • every service on (the default) • a service switched off • none on (refused, in words) • no SongLink key (optional) • a key set (never shown; Clear Key) • changed (Save, Revert) • refused by the hub (its sentence)	`UX-SEARCH-006` `UX-FORM-001`	○ not run here

Playlists
Music tab → Playlists
`docker-container/src/web/views/Playlists.tsx`
`docker-container/src/web/lib/playlists.ts`
DEC-041
	• no playlists yet: the sentence says where they come from (here, or Search ▸ … ▸ Add to Playlist)
	• the folder's playlists: mosaic, name, "N songs · length"; a hand-made one marked (UX-PL-006)
	• New Playlist… (sheet): the name
	• a playlist opened like an album: Back, heading, big mosaic, length, file name, Rename…, Export .m3u8, Delete… (UX-PL-007)
	• its songs: where each plays from (In the library, Plays from Deezer, Not found)
	• a song moved by drag or Alt+↑/↓; taken out with Delete or its menu (Move Up, Move Down, Open Its Link, Remove from Playlist)
	• More Songs past 200
	• a hand-made list: "made outside Airwave", kept as it is until changed (UX-PL-003)
	• Rename… (sheet) and Delete… (confirmation sheet)
	• Playlist folder: the path in the data volume (the default), how many it holds, Change… (sheet: the path, "Move the N playlists there too") (UX-PL-001)
	• a folder refused: the server's reason under the field
	• the folder unreadable, or holding more than 1,000 playlists: said in words
	• Shared from players: the paired players' synced copies, read-only, or the sentence saying none yet
		`UX-PL-001`
		`UX-PL-002`
		`UX-PL-003`
		`UX-PL-004`
		`UX-PL-006`
		`UX-PL-007`
		`UX-KEY-001`
		`UX-SAFE-001`
		`UX-SEARCH-004`
			○ not run here

AIRWAVE COMPANION — WINDOWS

The Airwave window of design/frontends/origin/airwave-companion.html in an Electron window whose chrome is its title bar.

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Folders (Saved Music, Saved TV, Saved Movies)
Library tab → Saved Music · Saved TV · Saved Movies
`windows-companion/src/renderer/views/Folders.tsx`
	• no folders (one line per kind)
	• loading (the list holds its place)
	• watched / kept / N songs
	• scanning (n of m)
	• drive not connected (kept in the list, said under it)
	• couldn't scan (reason)
	• added (said beside the button)
	• remove (confirm sheet; files untouched)
		`UX-SAFE-001`
		`VOICE-001`
		`UX-STATE-001`
			○ not run here

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Music on This PC
Library tab → Music on This PC
`windows-companion/src/renderer/views/Library.tsx`
	• nothing indexed: no music folder yet
	• nothing indexed: the scan is still reading
	• no matches
	• tracks
	• selection: click, Ctrl/Shift-click, arrows, Space, Ctrl+A; Enter or double-click shows the file
	• Scan Now disabled with reason (no music folder)
	• Send to Hub disabled with reason (no hub paired / nothing chosen)
	• Send to Hub: queued, progress under Remote ▸ Transfers
	• a tempo read from a tag is plain; a measured one wears ≈
	• tempo waiting: Setting up FFmpeg
	• tempo unavailable: FFmpeg could not be set up (Try again in Settings)
	• tempo unavailable: no FFmpeg (install it and scan again)
	• a windowed list: only the rows in view are drawn and every song is counted; End, Shift+Home and Ctrl+A reach every song, fetching only what they need
		`UX-STATE-001`
		`VOICE-001`
		`NP-PRIN-002`
		`UX-SETUP-001`
		`UX-KEY-001`
		`UX-PRIN-002`
			○ not run here

Hub connection and what is shared
Remote tab → Hub connection
`windows-companion/src/renderer/views/Hub.tsx`
DEC-023
	• not paired (address and pairing code)
	• pairing (verification code on the LCD plate; waiting for the hub to confirm)
	• pairing refused (the hub's sentence)
	• connected
	• paired but not reachable
	• sharing on from pairing when the hub grants it (DEC-023)
	• sharing as it was left, not "off" each time the window opens
	• sharing off: Sync Now disabled with reason
	• plain-http warning (notice bar)
	• forget (confirm sheet; turns sharing off for the next hub)
		`UX-SAFE-001`
		`UX-SAFE-003`
		`UX-PRIN-002`
			○ not run here

Transfers Remote tab → Transfers `windows-companion/src/renderer/views/Transfers.tsx`	• nothing moving (paired / not paired) • sending (bar and bytes) • waiting • sent • cancelled • didn't send (reason under the list)	`UX-FEED-001` `VOICE-001`	○ not run here
Backup Settings tab → Backup `windows-companion/src/renderer/views/Backup.tsx`	• no folder chosen (Back Up Now disabled, reason shown) • measuring • fits / does not fit (space bar: used · this backup · free) • running with progress • archives listed • restore (confirm sheet) • delete (confirm sheet) • what goes in: music, TV, movies, playlists, EQ presets, recommendation algorithms (from the paired hub, named; the reason when there is none) and these settings with the Live TV links • how often: by hand, every day or every week while the companion is open; a run that is due is taken when the companion next opens • how many to keep • a scheduled run that could not happen is reported, not dropped	`UX-SAFE-001` `VOICE-002`	● seen running

SURFACE	STATES IT MUST ANSWER	RULES	SEEN
About Settings tab → About `windows-companion/src/renderer/views/About.tsx`	• where its data is kept, with Open • cache by part, Clear Cache (confirm sheet), then what was cleared • logs: Open, Export (secrets, tokens and folder paths taken out), detailed logs switch • build, signing, and how updates are checked	`VOICE-003` `UX-PRIV-001`	● seen running

Live TV
Live TV tab → Channel playlists (M3U) · Programme guides (EPG)
`windows-companion/src/renderer/views/LiveTv.tsx`
DEC-018
	• no links (two empty lists, in the design's words)
	• checking…
	• kept: ✓ N channels / ✓ N-day guide
	• refused (the reason under the field; what was typed is kept; nothing added)
	• stopped answering (keeps what it last held; Check Again)
	• remove (the round minus)
	• Add disabled with reason until a link is pasted
	• while sharing is on, its channels and guide are kept on the hub too (links with a sign-in left out); turning sharing off removes them
		`UX-TV-001`
		`UX-STATE-001`
		`VOICE-001`
		`UX-PRIN-001`
			○ not run here

Settings
Settings tab → Settings
`windows-companion/src/renderer/views/Settings.tsx`
`windows-companion/src/renderer/views/Playlists.tsx`
DEC-018
	• downloader setting up (progress bar and %)
	• downloader waiting its turn (Setting up…)
	• downloader ready (set up automatically, or found on this PC)
	• downloader couldn't be set up (amber, reason, Try Again; badge on the Settings tool)
	• downloader cannot be set up on this PC (amber, with what to install)
	• downloaders not looked at yet: found / missing (helper not running)
	• Check All (last checked …)
	• a preference written back as it changes, said at the foot of the pane
	• helper running / not running with reason
	• port invalid
	• token shown / hidden / copied
	• restore defaults (confirm)
	• a tool: Check asks about that tool; Update when a newer version is out (badge on the tool); Install when it is missing; Update waits, with the reason, while the helper is busy
	• Update downloaders automatically: on by default
	• Downloads: the folder from the system's picker, format, how many at once and a speed limit
	• new versions of the companion: checked by default, asked at once when switched on, said when one is out; Download opens the release page
	• notifications when a download finishes: one preference with "When one finishes"
	• helper on your network: off by default; what it exposes said before it is switched on (DEC-032)
	• Playlists: the folder (Airwave Playlists in Music by default), Change… (the system picker), Open Folder, and "Move my playlists when the folder changes" (CMP-PL-001)
		`UX-PRIN-001`
		`UX-FEED-001`
		`UX-SETUP-001`
		`UX-SAFE-001`
		`UX-NET-001`
		`UX-PRIV-001`
		`CMP-PL-001`
			● seen running

Notice bar any main-process notice `windows-companion/src/renderer/views/NoticeBar.tsx`	• info • warning • error (announced at once) • dismiss	`VOICE-002`	○ not run here

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

The window: title bar chrome, five tools, pane and status line
the app window
`windows-companion/src/renderer/App.tsx`
`windows-companion/src/renderer/icons.tsx`
DEC-020
	• five tools — Library, Search, Live TV, Remote, Settings (arrows, Home and End move between them, choosing as they go)
	• a pane is built the first time its tool is chosen, then kept
	• attention badge on Settings (downloaders that need the person)
	• status line: connected and last synced / can't reach the hub / no hub paired; folders, playlists, guides and devices counted
	• opened in a browser, without the bridge: says so and offers nothing
	• Windows draws minimise, maximise and close over the chrome's top-right corner
		`UX-CHROME-001`
		`UX-KEY-001`
		`VOICE-003`
			● seen running

Confirmation sheet
removing a folder, forgetting the hub, revoking a device, restoring or deleting a backup, restoring the defaults
`windows-companion/src/renderer/ui.tsx`
`windows-companion/src/renderer/styles.css`
DEC-021
	• destructive: the action, then Cancel as the default button
	• not destructive: Cancel, then the action as the default button
	• Escape cancels
	• a second question answers the first with no
	• focus returns to where it was
	• the sheet dims only the pane under the toolbar, and stays modal
		`UX-SAFE-001`
		`UX-FOCUS-001`
			○ not run here

Stream to your devices (Pair a device)
Remote tab → Stream to your devices
`windows-companion/src/renderer/views/Streaming.tsx`
`windows-companion/src/main/awsp.ts`
DEC-018
	• off (default; the pairing controls say so)
	• starting
	• running: pairing code on the LCD plate with its time left, ticket and QR
	• code expired
	• sidecar missing or crashed (reason, restarting)
	• devices paired, each direct / through a relay / not connected, with its quality cap
	• revoke (confirm sheet)
	• port invalid
	• Wi-Fi and Ethernet / metered connections: two switches, both on by default; paused with the reason on a connection that is switched off (DEC-031)
	• the connection switches are off while streaming itself is off
	• each paired device says when it was last seen, and how it is connected while it is
		`UX-SAFE-001`
		`UX-FEED-001`
		`UX-PRIN-002`
		`UX-STREAM-001`
			○ not run here

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Search
Search tool
`windows-companion/src/renderer/views/Search.tsx`
`windows-companion/src/renderer/catalog.ts`
DEC-039
	• nothing asked yet: the field, Track/Artist/Album, Filter…
	• searching: every service “asking…”, rows folding in as each answers
	• the calm overview, centred under the field: Songs (five), Artists, Albums and Playlists (three each), each with “See all N”, no pager, the services’ line under them (UX-SEARCH-007, UX-SEARCH-011)
	• done: each service’s state in words (one resting, with when it is back) and the platforms that only gave links
	• a section with nothing (“No songs.”) and a filtered search (“Showing … from …”)
	• Filter… (sheet): sections (Songs, Artists, Albums, Playlists) and services; refuses a choice that shows nothing
	• a type’s own page (See all): heading, Back, its own field, the Songs · Artists · Albums · Playlists control, the services’ line and the list (UX-SEARCH-009)
	• a type’s page reading on: the next offset at once, “Loading more…” as it scrolls, “That’s all N songs.” at the end
	• a type’s page footer: ‹ › and “Page N of M” (“M+” while more may exist), landing on a page’s first row, fetched first when needed
	• another type chosen in the control, or the type’s field searched: that type alone, from the start
	• Playlists: the starred ones first under “In your library”, then Deezer’s public playlists, twelve a page; one opened like an album (UX-SEARCH-010)
	• a song row’s menu (“…”, right-click, Shift+F10): Add to Up Next (▸ groups), Add to Playlist ▸, Add to Library…, Download…, Audition, Open Details (UX-SEARCH-012)
	• Add to Up Next: the paired hub’s group (the companion has no queue), its reason when it could not, or why there is no group (no hub, no permission, not in one)
	• a song only a store has: nothing to queue or fetch, said in words
	• Add to Library… / Download… (sheet): the rights basis (and format), through the helper into this PC’s library
	• Add to Playlist ▸: this PC’s playlists, ticked where the song is, and New Playlist…; “Also add to Library” (on) takes the helper’s path (CMP-PL-004)
	• a song: cover, platforms, preview, genre/label/year (looking it up…), synced or plain lyrics (or none, or instrumental), Download to this PC through the helper
	• an album: cover, label, date, songs, the star
	• an artist: picture, top songs, albums (more as you scroll)
	• a pasted playlist: its 2×2 mosaic, platform and length; opened like an album with every song; starred
	• a link that cannot be read: the reason in words
	• a failure: the reason in words under the field
	• the helper not running: the reason in words
	• a row filled in by the engine’s last chunk: the credit line (“feat. …”), BPM, the explicit mark and
		`UX-SEARCH-001`
		`UX-SEARCH-002`
		`UX-SEARCH-003`
		`UX-SEARCH-004`
		`UX-SEARCH-005`
		`UX-SEARCH-007`
		`UX-SEARCH-008`
		`UX-CAT-001`
		`UX-CAT-002`
		`UX-CAT-003`
		`UX-CAT-004`
		`UX-KEY-001`
		`UX-CAT-006`
		`UX-SEARCH-009`
		`UX-SEARCH-010`
		`UX-SEARCH-011`
		`UX-SEARCH-012`
		`UX-CAT-005`
		`CMP-PL-004`
		`CMP-PL-006`
			○ not run here

SURFACE	STATES IT MUST ANSWER	RULES	SEEN
	a cover arrive after the row (UX-CAT-006) • on touch: a long press (500 ms, within 10 px) on a row opens its menu (CMP-PL-006)		

Playlists
Library tool → Playlists
`windows-companion/src/renderer/views/Playlists.tsx`
`windows-companion/src/renderer/playlists.ts`
DEC-041
	• no playlists yet: the sentence says where they come from
	• the folder's playlists with their mosaics; a hand-made one marked (CMP-PL-005)
	• New Playlist… (sheet)
	• a playlist opened like an album: Rename…, Export… (the save dialog), Delete… (asked first)
	• its songs: where each plays from; moved by drag or Alt+↑/↓; taken out with Delete or its menu
	• on touch: a long press on a song opens its menu (CMP-PL-006)
	• a list changed on disk by another player: read again by itself (CMP-PL-003)
	• the folder unreadable: the reason in words
		`CMP-PL-001`
		`CMP-PL-002`
		`CMP-PL-003`
		`CMP-PL-005`
		`CMP-PL-006`
		`UX-PL-002`
		`UX-PL-003`
		`UX-KEY-001`
		`UX-SAFE-001`
			○ not run here

DISCORD BOT

Messages rendered from templates the operator can edit.

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Bot replies
/play, /queue, /skip … or the prefix form
`docker-container/src/discord/gateway.ts`
`packages/domain/src/templates.ts`
DEC-005
	• queued
	• now playing
	• skipped / vote counted
	• refused (ephemeral)
	• no results
	• cannot be played here
	• joined / left voice
	• not in a voice channel
	• missing Connect/Speak
		`UX-DISCORD-001`
		`UX-DISCORD-002`
		`UX-DISCORD-003`
			○ not run here

ANDROID SHELL

A WebView around the player, plus system UI.

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

WebView shell
launcher icon
`android/app/src/main/java/com/nowplaying/player/MainActivity.kt`
`android/app/src/main/res/values/strings.xml`
DEC-012
	• player loaded from assets
	• renderer crashed (reloads)
	• back = history, else background
	• hub over http refused
		`UX-SAFE-004`	○ not run here

Notification, picker and save dialog
a fetch starts; a file input opens; an export is saved
`android/app/src/main/java/com/nowplaying/player/FetchService.kt`
`android/app/src/main/java/com/nowplaying/player/BlobDownloads.kt`
DEC-010
	• fetching (foreground notification)
	• permission prompt (API 33+)
	• saved
	• could not save
		`VOICE-002`	○ not run here

LOCAL HELPER

A console program that serves the player and runs download tools.

SURFACE	STATES IT MUST ANSWER	RULES	SEEN

Local helper console
now-playing-helper
`local-helper/src/cli.ts`
`local-helper/src/options.ts`
	• tools found / missing with install hint
	• missing tools setting up (verified download)
	• --no-auto-tools: found / missing only
	• token shown
	• stopping
		`VOICE-002`
		`UX-SETUP-001`
		`UX-DL-001`
			○ not run here

Journeys

The 17 things people come to do, from where they start to where they can fail. Each step list is what the code does today; each failure is one the product answers in words.

First run of the hub

A fresh data volume; the container is up on 127.0.0.1:4546.

1. Open the admin page: the window, every tool locked, the sign-in form in the pane
2. Sign in with admin / admin
3. Overview opens with the amber gate; the other five tools stay locked
4. Choose a real password
5. Every tool unlocks

WHEN IT GOES WRONG

- Fewer than twelve characters, or two entries that differ: refused in a sentence
- Login rate-limited after repeated failures
- A deep link to a locked tab is ignored

`UX-SAFE-002` `UX-CHROME-001`

`docker-container/tests/integration/setup-and-pairing.test.ts`
`docker-container/tests/e2e/first-run.setup.ts`
`docker-container/tests/dom/admin-shell.dom.test.tsx`

Pair a player or the companion

Setup complete; the device can reach the hub.

1. Admin ticks the permissions and starts pairing (Devices): a code on the LCD plate
2. Device enters the hub's address and the code
3. Both screens show the same verification code
4. Admin types it and confirms
5. Device receives a scoped credential; a companion granted library:share starts sharing (DEC-023)

WHEN IT GOES WRONG

- Code expired
- Fingerprint differs: do not confirm
- Wrong codes lock out that address for a while

`UX-SAFE-003`

`docker-container/tests/integration/setup-and-pairing.test.ts`
`windows-companion/tests/integration/companion-and-hub.test.ts`
`windows-companion/tests/dom/companion-shell.dom.test.tsx`

Pair with a PC and play its music, relay-carried

The companion app runs its AWSP sidecar on a PC; the player is the served build (a service worker is the bridge).

1. On the PC, Settings ▸ Remote shows a ticket and a six-digit code
2. In the player, Sources ▸ Connections ▸ Stream from a PC: paste the ticket, type the code, Pair
3. The card shows the PC's name, relay-carried, and its track count; the tracks join the library
4. Play one; the bar follows the element and the indicator beside the transport reads relay-carried
5. The relay drops mid-track; the player reconnects and carries on from the last byte it held

WHEN IT GOES WRONG

- A wrong or expired code is refused in words, and nothing is stored
- A revoked player is told to pair again, and stops reconnecting
- The single-file copy has no service worker, and says streaming needs the served player

`NP-PREF-007` `NP-PRIN-002`

`music-player/tests/e2e/awsp.spec.ts`
`windows-companion/awsp-server/tests/awsp.rs`

Add music on this device and play it

The player is open (installed or not).

1. Library shows how to add music
2. Choose a folder or files
3. Tracks appear; unplayable formats say so
4. Play; the hero and the list mark the playing row

WHEN IT GOES WRONG

- Browser without folder access falls back to file choice
- A file that disappeared says it cannot be found

`UX-STATE-001` `UX-A11Y-001`

`music-player/tests/e2e/local-file.spec.ts`
`music-player/tests/e2e/player.spec.ts`

Listen together

Player paired with a hub; a group exists.

1. Switch the mode to Shared
2. Join or create a group
3. Anyone queues; everyone hears it at the same point
4. Skips and pauses apply to the group

WHEN IT GOES WRONG

- No hub: the switch explains why
- Stale command: refused with the current revision
- Member without the DJ role: vote-skip instead of skip

`UX-PRIN-002` `UX-SHARED-001`

`docker-container/tests/integration/groups-and-queue.test.ts`

Play music into a Discord voice channel

Bot token installed, bot enabled, default group chosen, worker container running.

1. Join a voice channel
2. /play <search or link>
3. The bot joins your channel and replies Queued
4. Audio follows the group timeline; pause, skip and seek from any device apply

WHEN IT GOES WRONG

- Not in a voice channel: the reply says so
- Missing Connect/Speak: the reply names them
- Source that cannot stream in Discord: skipped with the reason
- Worker not running: the admin panel says so

`UX-DISCORD-001` `UX-DISCORD-002` `UX-DISCORD-003`

`docker-container/tests/integration/discord-worker.test.ts`
`docker-container/tests/unit/discord-voice.test.ts`
`docker-container/tests/integration/discord-parity.test.ts`

Download a track the hub may give you

Paired player; the provider allows it.

1. Row menu → Download
2. Choose format and the basis (e.g. content I own)
3. Progress in words
4. The file saves as a link download

WHEN IT GOES WRONG

- Format unavailable without FFmpeg
- Provider refuses the basis
- Cancelled: the partial file is removed

`UX-FEED-001` `UX-LINK-001`

`docker-container/tests/security/ssrf-and-downloads.test.ts`
`docker-container/tests/integration/downloads-queue.test.ts`

Share a track or playlist

Paired player.

1. Share
2. Choose whether streaming is allowed
3. Copy the link
4. The recipient opens the public page

WHEN IT GOES WRONG

- No hub: the sheet explains why
- Link revoked or used up: the page says it is unavailable

`UX-PRIN-002`

`docker-container/tests/integration/shares-and-sync.test.ts`
`docker-container/tests/security/share-limits.test.ts`

Send files from Windows to the hub

Companion paired; a folder indexed.

1. Add a music folder (Library)
2. Choose songs in Music on This PC
3. Send to Hub
4. Remote ▸ Transfers shows progress; Cancel stops it

WHEN IT GOES WRONG

- Drive not connected
- No hub paired, or nothing chosen: Send to Hub is disabled and says why
- Hub unreachable: the transfer fails with the reason under the list

`UX-FEED-001`

`windows-companion/tests/integration/companion-and-hub.test.ts`

Downloaders set themselves up

The companion (or the standalone helper) starts on a machine without yt-dlp, spotDL or FFmpeg.

1. The helper starts; the window opens without waiting
2. Each missing tool is fetched from its project's GitHub release, one at a time, and checked against the published SHA-256
3. Settings ▸ Downloaders shows Setting up… with progress, then Ready — Set up automatically
4. When FFmpeg lands, tempos are measured for the library without a rescan

WHEN IT GOES WRONG

- No published SHA-256, or a mismatch: nothing is installed; Couldn't set up with the reason and Try Again
- Offline: retried on the next start or after six hours; Try Again retries at once
- No build for this platform (FFmpeg on macOS/Linux): the install advice is shown instead

`UX-SETUP-001` `UX-FEED-001`

`local-helper/tests/unit/install.test.ts`

`local-helper/tests/unit/provision.test.ts`

`local-helper/tests/integration/helper-server.test.ts`

`windows-companion/tests/integration/main-process.test.ts`

`windows-companion/tests/dom/companion-shell.dom.test.tsx`

Add a Live TV playlist and watch the guide fill in

The companion is running on the PC the player is opened on.

1. Companion ▸ Live TV: paste an M3U link and choose Add
2. The row reads checking… while the main process reads the link
3. Kept: the row reads ✓ and how many channels it holds
4. Paste an XMLTV link the same way for now and next
5. Open the player's Live TV: the guide lists the companion's channels, with what is on and what follows

WHEN IT GOES WRONG

- Not a playlist, not a guide, too large or on this PC's own network: refused in a sentence under the field; what was typed stays; nothing is kept
- A kept link stops answering: it keeps what it last held, says so, and offers Check Again
- No companion, or one with no playlists: the player's guide is as it was

`UX-TV-001` `NP-TV-001`

`music-player/tests/e2e/np/live-tv.spec.ts`

`windows-companion/tests/integration/live-tv.test.ts`

`local-helper/tests/integration/helper-tv.test.ts`

`windows-companion/tests/dom/companion-shell.dom.test.tsx`

Back up and restore

Setup complete.

1. Hub: System ▸ Backup → Back Up Now
2. Restore → confirm in the sheet
3. The hub reports success or the reason it failed
4. Companion: Settings ▸ Backup → choose a folder, Back Up Now; restore and delete ask first

WHEN IT GOES WRONG

- Invalid backup refused
- Swap failed: previous database put back

`UX-SAFE-001`

`docker-container/tests/integration/backup-and-search-cursor.test.ts`

`windows-companion/tests/integration/backup.test.ts`

Fetch with yt-dlp or spotDL

The local helper is running, or the Android app has unpacked its tools.

1. Library → Fetch
2. Paste a link and pick a format
3. Progress
4. Files are added to the library

WHEN IT GOES WRONG

- Tool missing: the control is not shown
- Tool error: its own message, without paths
- FFmpeg missing: the file is saved as the site sent it, and the job says it has no cover art or tags

`UX-PRIN-001` `VOICE-002` `UX-DL-001` `NP-FIND-002`

`music-player/tests/e2e/helper.spec.ts`
`local-helper/tests/integration/helper-server.test.ts`
`local-helper/tests/unit/download-tags.test.ts`
`local-helper/tests/integration/helper-resolve.test.ts`
`music-player/tests/e2e/np/links.spec.ts`

Find music, open a type's page, file a song, keep a playlist

The player on Music. A paired hub, the companion on this PC, or neither (this browser asks the keyless services itself).

1. Type a few words in the header search and press Enter: a calm overview arrives as each service answers — five songs, three artists, albums and playlists, each with "See all N" — every row naming its platforms, the quiet line saying how each service did; the card hangs centred under the field
2. Narrow with Filter (sections, services) or the Track / Artist / Album / ISRC fields
3. See all songs: the Songs page, scrolling on by itself; ‹ › or Page Up/Down move a page at a time (fetched first when needed); the page's own field searches songs alone; the control switches to Artists, Albums or Playlists
4. Playlists: the starred ones first, then Deezer's public playlists; open one in the music list, as an album opens
5. A song's "…" (or right-click, long-press): Add to Up Next, Add to Playlist ▸, Add to Library, Download…, Audition — the HUD confirms
6. Open an artist, then one of their albums; open a song for its genre, label, year and lyrics; Back walks it back
7. Paste a playlist link: one listing with its platform and a 2×2 mosaic
8. Open it: the music list names it and loads every song, page by page, saying how far it has got
9. Star it in the silver bar: it is in the library menu under Playlists, under "In your library" on the Playlists page, and opens again from there

WHEN IT GOES WRONG

- A service fails, times out or is resting: the line says so; the rest still answer
- Nothing reachable: the search says what happened, in words
- A link nobody can read (a private or Spotify-made playlist, an Apple Music playlist, a Tidal link without a SongLink key): the reason
- A list that stops loading part way: the songs so far stay, and it says why

`NP-FIND-001` `NP-FIND-002` `NP-FIND-003` `NP-FIND-004` `NP-FIND-005`
`NP-FIND-006` `NP-FIND-007` `NP-FIND-008` `NP-FIND-009` `NP-FIND-010`
`UX-CAT-001` `UX-CAT-003` `UX-CAT-005`

`music-player/tests/e2e/np/catalog-search.spec.ts`
`music-player/tests/e2e/np/search-pages.spec.ts`
`music-player/tests/e2e/np/collections.spec.ts`
`music-player/tests/e2e/np/links.spec.ts`
`music-player/tests/e2e/np/preview.spec.ts`

Find music, open a type's page, queue or keep a song, keep a playlist (hub and companion)

The hub (signed in) or the companion with its helper running; no account or key anywhere.

1. Search ▸ type a song, an artist, an album, an ISRC or paste a link
2. A calm overview arrives as each service answers, in one column under the field: five songs, three artists, albums and playlists, each with "See all N"; the services' line under them
3. See all songs: the Songs page reads on as it scrolls; ‹ › or Page Up/Down move a page at a time; its own field searches songs alone; the control switches to Artists, Albums or Playlists
4. Playlists: the starred ones first, then Deezer's public ones; open one like an album
5. A song's "…" (or right-click, Shift+F10): Add to Up Next (a group's queue — on the companion, the paired hub's), Add to Playlist ▸ (a playlist of the app's own folder, or New Playlist…; Also add to Library on), Add to Library… (with the rights basis), Download…, Audition — the status line confirms
6. Open a song: preview, genre, label, year, lyrics; download it (hub: into the queue with the rights basis; companion: through the helper)
7. Or open an album or a pasted playlist and star it into the library

WHEN IT GOES WRONG

- A service resting or failing: said on the status line, the others still answer
- A link that cannot be read: the reason in words
- A song only in a store: nothing to queue or fetch, said in words
- No group to queue to (or, on the companion, no hub paired): said in a sentence
- The companion's helper not running: the reason, with where to fix it

`UX-SEARCH-001` `UX-SEARCH-002` `UX-SEARCH-003` `UX-SEARCH-004` `UX-SEARCH-005` `UX-SEARCH-007` `UX-CAT-002` `UX-CAT-004` `UX-SEARCH-009` `UX-SEARCH-010` `UX-SEARCH-011` `UX-SEARCH-012` `UX-PL-005` `CMP-PL-004`

`docker-container/tests/dom/search.dom.test.tsx`
`windows-companion/tests/dom/companion-search.dom.test.tsx`
`windows-companion/tests/contract/catalog-ipc.test.ts`
`docker-container/tests/integration/catalog.test.ts`
`docker-container/tests/e2e/search.spec.ts`
`packages/domain/tests/unit/catalog-one-row.test.ts`
`packages/domain/tests/unit/catalog-paging.test.ts`
`windows-companion/tests/integration/companion-and-hub.test.ts`
`packages/domain/tests/unit/catalog-view.test.ts`

Keep playlists as files: file a song from search, open the playlist, move the folder (hub and companion)

The hub signed in, or the companion; no account or key anywhere.

1. Search ▸ a song's … ▸ Add to Playlist ▸ New Playlist… ▸ a name ▸ Create (Also add to Library on: the download is queued with its basis)
2. The playlist is an .m3u8 and an .airwave.json in the playlist folder: a library song as a relative path, any other by its link
3. Hub: Music ▸ Playlists; companion: Library ▸ Playlists — open it like an album, move a song with Alt+↓ or by drag, take one out, rename, export
4. Add the same song again from search: the menu ticks the playlist, and says it is there already
5. Hub: Playlist folder ▸ Change… ▸ library/Playlists with Move on; companion: Settings ▸ Playlists ▸ Change… (the system picker) — the playlists go with it
6. A player (playlists:use) files into the hub's playlists through /api/v1/playlists

WHEN IT GOES WRONG

- A folder outside the data volume (hub): refused, the old folder kept, the reason under the field
- A folder the companion cannot write: the old one kept, the reason said
- A playlist at 10,000 songs, or a folder at 1,000 playlists: said in words
- A list made outside Airwave: shown read-only until it is changed here
- A device changing a list it did not make: refused with the reason

`UX-PL-001` `UX-PL-002` `UX-PL-003` `UX-PL-004` `UX-PL-005` `UX-PL-006` `UX-PL-007` `CMP-PL-001` `CMP-PL-002` `CMP-PL-003` `CMP-PL-004` `CMP-PL-005` `CMP-PL-006`

`packages/domain/tests/unit/playlist-folder.test.ts`
`docker-container/tests/integration/playlists.test.ts`
`docker-container/tests/security/playlist-folder.test.ts`
`docker-container/tests/dom/playlists.dom.test.tsx`
`docker-container/tests/dom/search.dom.test.tsx`
`docker-container/tests/e2e/search.spec.ts`
`windows-companion/tests/integration/playlist-folder.test.ts`
`windows-companion/tests/contract/playlists-ipc.test.ts`
`windows-companion/tests/dom/companion-playlists.dom.test.tsx`
`windows-companion/tests/dom/companion-search.dom.test.tsx`

File a song into the hub’s playlists, play one, and keep starred lists in step

The player paired with a hub that allows it “File into the hub’s playlists” (playlists:use) and “Sync starred albums and playlists” (library:sync).

1. A song’s “…” (or right-click, long-press) ▸ Add to Playlist: this player’s playlists under “On this player”, the hub’s under “On <hub>”, ticked where the song is
2. Choose one of the hub’s: the HUD says “Added … on <hub>”; or New Playlist on <hub>…, named in a sheet
3. Library ▸ Playlists ▸ On <hub> ▸ a playlist: it opens in the music list like an album
4. Play a song the hub’s library holds: it streams from the hub; a song kept by its link plays its preview
5. On a list this player made: “…” ▸ Rename…, Delete…, or Alt+↑/↓ to move a song
6. Star an album or playlist: the hub has it too; un-star it while the hub is away, and it is removed there once the hub is back
7. Library ▸ Playlists ▸ Shared on <hub>: the admin’s starred lists, read-only

WHEN IT GOES WRONG

- The hub has not allowed playlists: the group says so, and how to fix it (pair again with the permission)
- A song with neither a link nor an ISRC: it stays in this player’s playlists, and the group says why
- A list made on another device or by hand: read-only, said in the bar
- The hub away: stars are kept here, tombstones remembered, and sent later, quietly

`NP-FIND-008` `NP-FIND-013` `NP-FIND-014` `UX-PL-004` `UX-PL-008`

`music-player/tests/e2e/np/hub-shelf.spec.ts`

`music-player/tests/unit/hub-shelf.test.ts`

The Discord bot

The bot is the one surface Aqua does not draw: Discord does. What the suite controls is the words, the embed colour, whether a reply is private, and when the bot moves in and out of voice. Replies come from templates an operator can edit in the hub (Admin → Discord); these are the defaults, rendered with sample values by the same function the bot uses.

Commands: `/play` `/nowplaying` `/queue` `/skip` `/pause` `/resume` `/stop` `/shuffle` `/clear` `/join` `/leave` `/settings`. Every one also works as a prefix command (`!play`) once the Message Content intent is granted, and the two forms are identical by construction (`UX-DISCORD-001`). Embed colours are the system's own: selection blue, success, warning and danger.

Illustration — Discord renders its own interface. This shows content and colour, not Discord's layout.

`success` A command with nothing more specific to say	Done.
`queued` /play found something playable	**Queued #3** **Signal Fade** — Cassette Bloom Requested by alex · 0:04 · Public domain
`nowPlaying` /nowplaying	**Now Playing** **Signal Fade** — Cassette Bloom Live from Pier 9 Requested by alex in Kitchen
`skipped` /skip by a DJ or the requester	Skipped **Signal Fade** (skipped).
`permissionDenied` The caller lacks the DJ or admin role only the caller sees this	You don't have permission to do that (the DJ role is required).
`noResults` /play found nothing only the caller sees this	No results for “No results”.
`unavailableSource` The match cannot stream in Discord only the caller sees this	**Signal Fade** cannot be played here: YouTube only plays in its own player.
`emptyQueue` /queue or /nowplaying with nothing queued only the caller sees this	The queue is empty. Use /play to add something.
`joined` /join, or /play when the bot was not in voice	Joined #music.
`left` /leave, or the idle timeout	Left the voice channel.
`error` Anything unexpected; the reason is the error message only the caller sees this	Something went wrong: Queue is at revision 12
`wrongChannel` A command outside the designated channel only the caller sees this	Music commands work in #music.

paused /pause	Paused.
resumed /resume	Resumed.
stopped /stop	Stopped and cleared the queue.
shuffled /shuffle	Shuffled 12 tracks.
cleared /clear	Cleared 12 tracks from the queue.

VOICE-CHANNEL REPLIES

Added to the reply when the bot cannot move into voice. They are sentences in the gateway, not templates.

- “Join a voice channel first (or use join <channel>), so I know where to play.”
- “I need View Channel, Connect and Speak in <channel> to play there.”
- “There is no voice channel called “<name>” that I can see.”
- “Could not connect to the voice channel. Check that the bot has Connect and Speak there, and that outbound UDP is not blocked.”

> **Known gap (DEC-005).** The default `noResults` template quotes `{{reason}}`, but the command service passes the search text as `title`, so a failed search currently reads “No results for “No results”.”

Platforms

WEB AND PWA — PLAYER AND HUB ADMIN

- **Breakpoints follow content, not a device list.** The Airwave window changes at 640px — the tools scroll sideways, tiles go two-up, the key–value list stacks and tables drop their secondary columns — and at 560px, where side-by-side groups stack. The library's page skin changes at 480, 560, 620, 720 and 900px. Check at the mockup sizes: 320, 390, 768, 1280 and 1680px.
- **The pointer decides density, not the width.** `(pointer: coarse)` switches the touch layer on at any width; a narrow desktop window keeps desktop density (`UX-TOUCH-001`). The Airwave window's touch layer is small — list rows and the round minus grow, buttons do not — which is an open question for the hub on a phone (DEC-027).
- **Installable and offline.** The player ships a manifest, icons and a service worker, opens with the network off once it has loaded online, and loads nothing from another origin. The single-file build, `now-playing.html`, runs straight from disk.
- **Themes.** The player follows the system colour scheme. The Airwave window is light only, in both OS appearances (DEC-003).
- **Keyboard.** In the player, Space toggles playback and Escape closes a popover without losing what was typed. In the Airwave window the tools are one tab stop: arrows, Home and End move the selection, and locked tools are skipped (`UX-KEY-001`).

ANDROID

- The app is the player inside a WebView, loaded from its own assets, so every web rule applies unchanged. The player is built with `NP_BASE_PATH=/assets/app/`.
- Back goes back in the page's history; with nowhere to go it moves the app to the background, so music keeps playing.
- System UI carries its own copy (`strings.xml`): the "Fetching music" notification while a download runs, the document picker for file inputs, and the save dialog for exports ("Saved …" / "Could not save …"). Notification permission is asked for the first time a fetch starts (DEC-010).
- Hubs are reached over https only (DEC-012).

WINDOWS COMPANION — ELECTRON

- One window with four tools, 640 × 760 as it opens and never smaller than 520 × 440. Its chrome is its title bar: the operating system's own is hidden, the page draws the title, Windows draws minimise, maximise and close over the top-right corner, and the chrome is the drag region while the tools stay clickable (DEC-020, `UX-CHROME-001`).
- A tray icon (`resources/tray.svg`) whose menu has Open, Scan Library Now and Quit. With "Keep running in the notification area when the window closes" on, closing the window keeps the companion running.
- It is the only product that reads the filesystem; its copy says so and never implies files are copied or uploaded in the background.
- Pairing shows the hub's fingerprint, and a notice warns when the hub address is plain http off the local network.
- Live TV links are read by the main process, never by the page: public addresses only, with a size cap and a deadline, and kept only when they hold channels or programmes (`UX-TV-001`).

TERMINAL — LOCAL HELPER

- Takes the first free port in a small range, prints the token to paste into Settings → Platforms, and names each missing tool with the command that installs it.
- Ctrl+C removes only the helper's own working folder, never the folder it was pointed at.

ENGINEERING APPENDIX — DOCKER DELIVERY

Docker is how the hub is delivered, not a design surface. What it means for a design change:

- The image (`docker-container/Dockerfile`) builds the admin window from `docker-container/src/web` with the generated Airwave stylesheets and AquaArt from `@now-playing/aqua-ui`, so a design change reaches the hub only when the image is rebuilt: `docker compose up -d --build` or `./nowplaying update`.
- The hub listens on port 4546, published to `127.0.0.1` by default, with `/healthz` for the health check. Host names other than IP addresses, localhost and `.local`/`.lan` names must be listed in `NP_ALLOWED_HOSTS` or set as `NP_PUBLIC_ENDPOINT`.
- The Discord worker is the same image under `--profile discord`. Its default replies change when `packages/domain/src/templates.ts` changes and the image is rebuilt, or immediately when an operator edits a template in the hub.
- The player, the Android app and the companion are built separately; each needs its own rebuild to pick up a design change.

Every product, every size — and editable

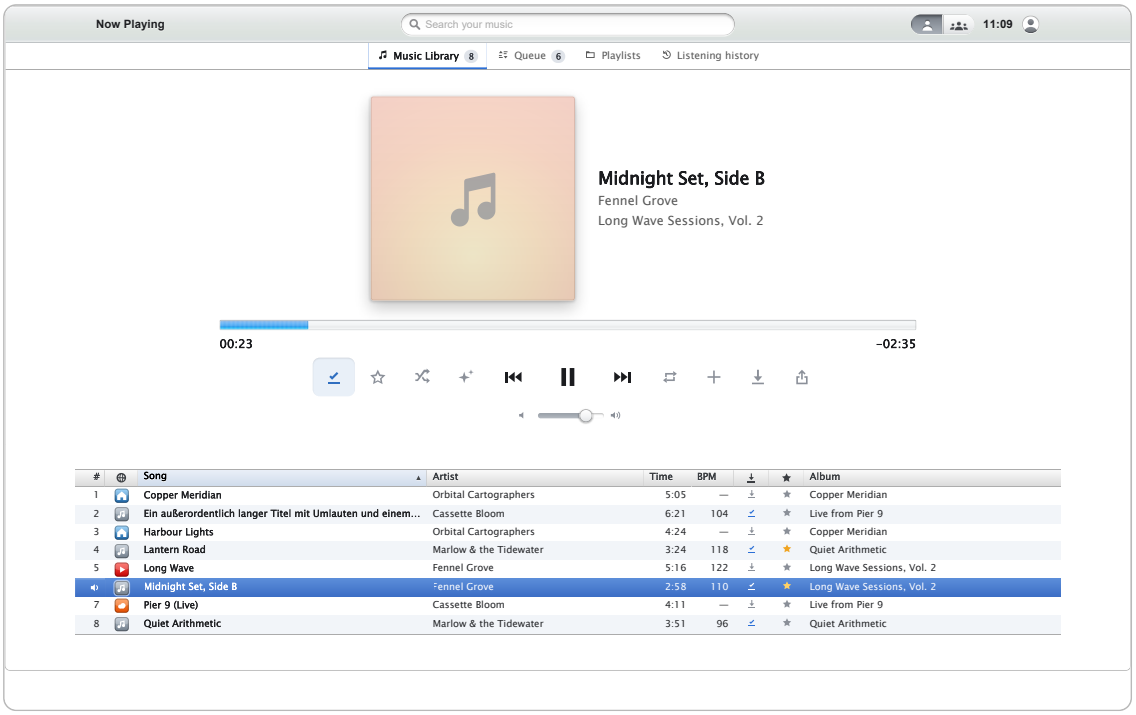
Each frame below is an iframe with its own viewport, so a 320px column is a real 320px viewport and the media queries behave exactly as they do on a phone. The hub's and the companion's frames hold only what those products load — the generated Airwave stylesheets, their own `styles.css`, AquaArt and their own kits — and the player's hold the component library's page skin. Under each one is a measurement taken *inside* that frame: what runs off the side, the smallest text, the smallest thing you could tap.

A frame marked *touch layer emulated* has its `(pointer: coarse)` rules re-applied inside it, read back out of the real stylesheets — an iframe inherits the desktop's pointer, so without that a phone frame would quietly show desktop sizes. The Airwave window has almost no touch layer (list rows and the round minus grow; buttons do not), so the hub's phone frames report their target sizes without being failed on them (DEC-027). At the bottom you can edit any custom property the stylesheets define and watch every frame repaint at once; what you copy out says which file each change belongs in.

In print, every screen is shown at two of its product’s sizes with the fit measured inside each frame. The interactive version — every size, the product switch and the token editor — is `docs/design/styleguide.html`.

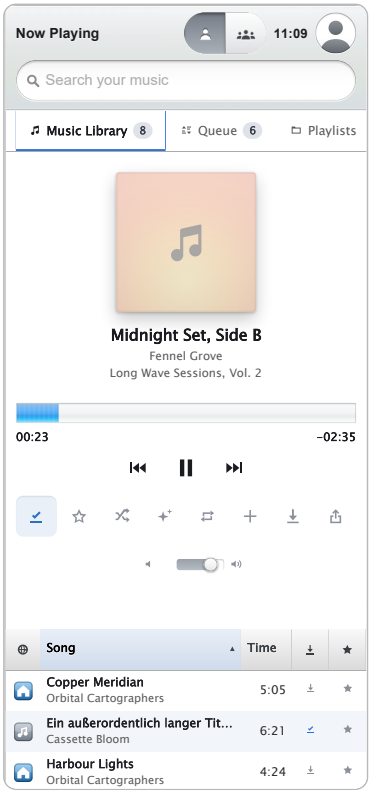
Now playing and library

The whole page skin at once. Narrow it and watch the list drop its number and artist columns and put the artist under the title instead.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 10px

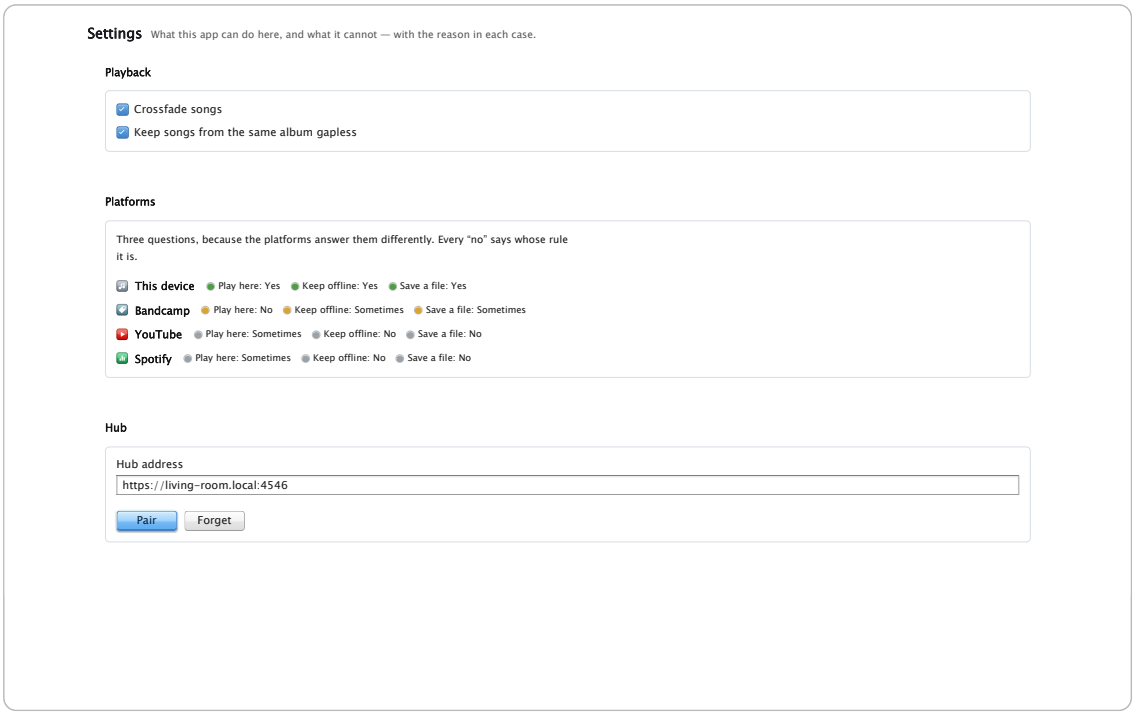


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 12px · smallest target 44px

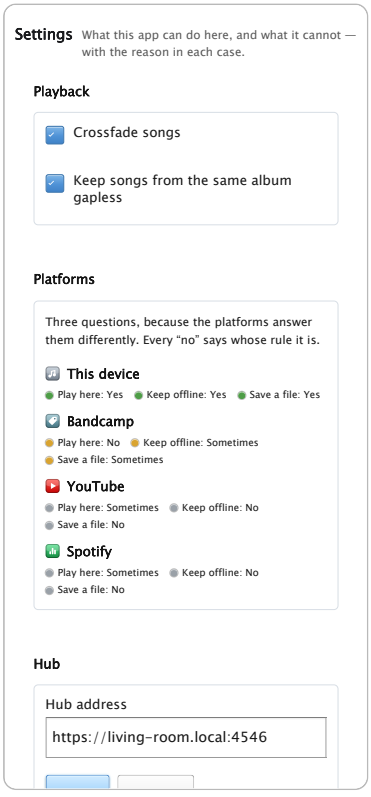
Settings

Where most of the app's explaining happens, so it is the screen where text size and wrapping matter most.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 12px

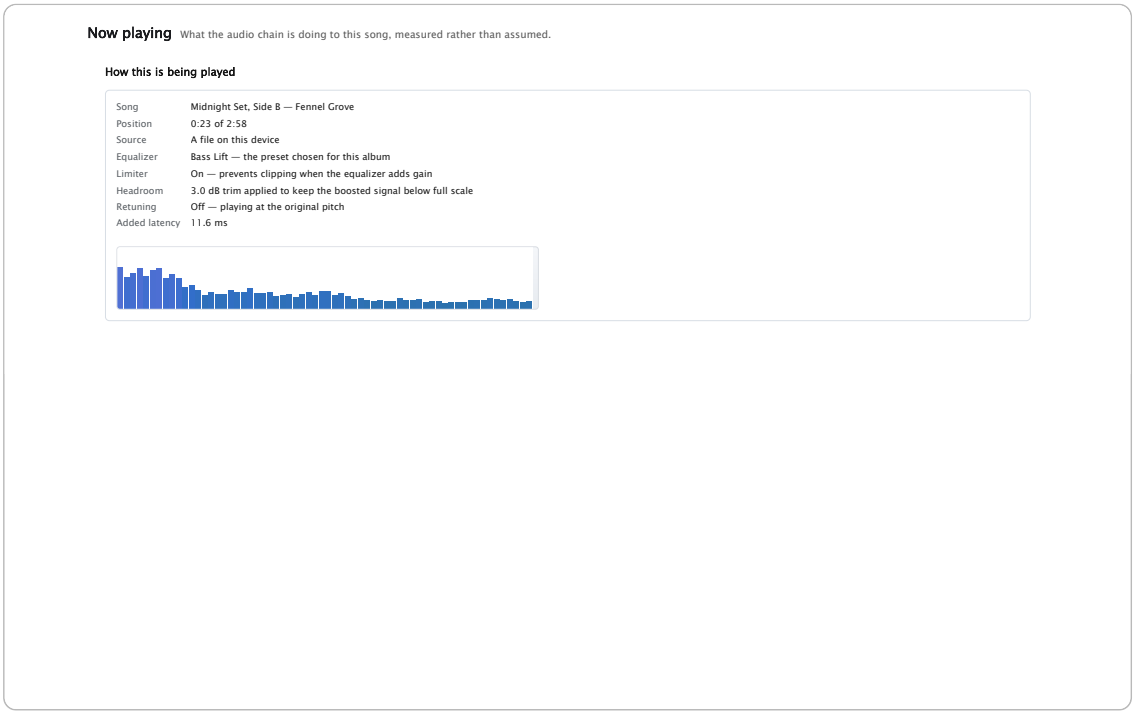


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 13px · smallest target 44px

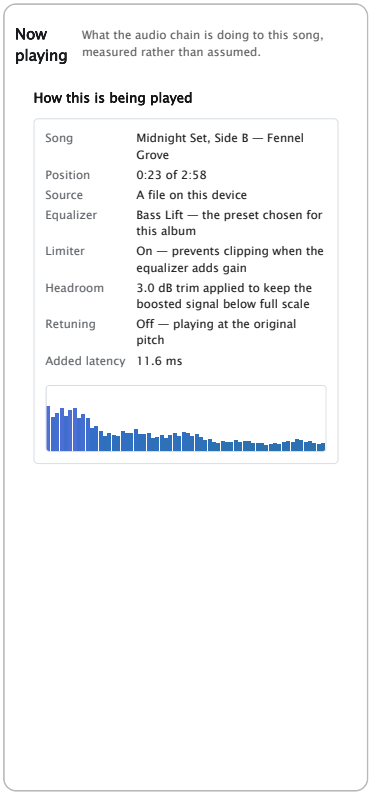
Now playing (audio chain)

What the audio chain is doing to the song, and the spectrum visualiser under it — drawn by the player’s own spectrum code from a generated analyser frame.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

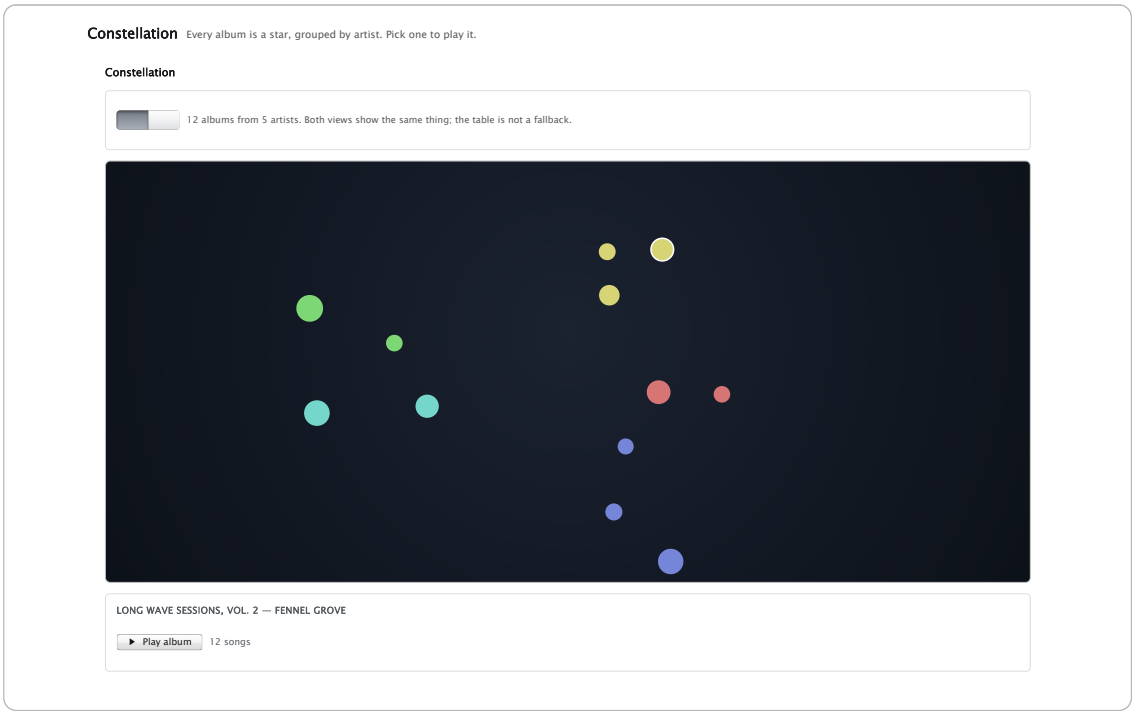


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 13px · smallest target —

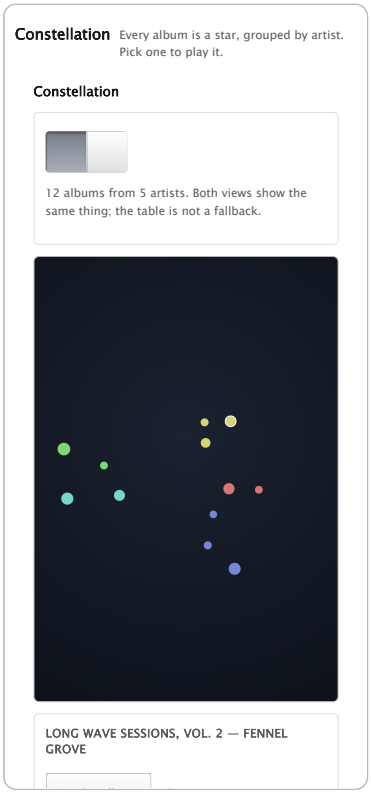
Constellation

The library as a star field, placed by the player's own layout code. The table beside it is the same information, not a fallback.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

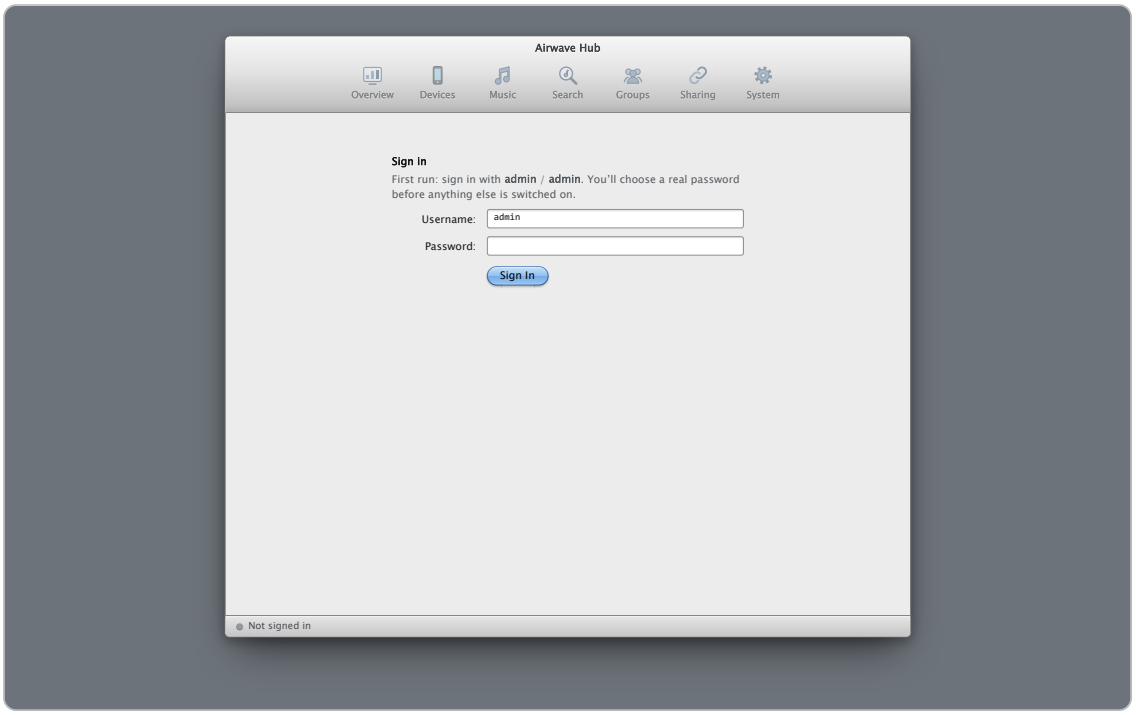


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 13px · smallest target 44px

Sign in

Not signed in: the same window with every tab locked and the form in the pane. There is no separate sign-in page.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

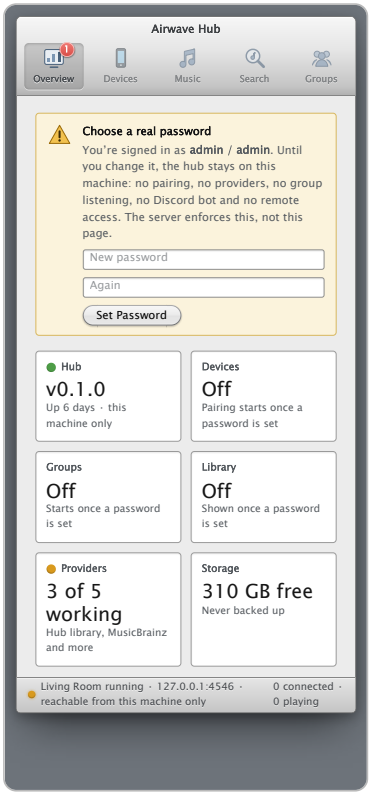
First run

Signed in with the first-run password: the amber gate on Overview, the other five tabs locked. The tiles that depend on gated routes say they are off instead of showing a number.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

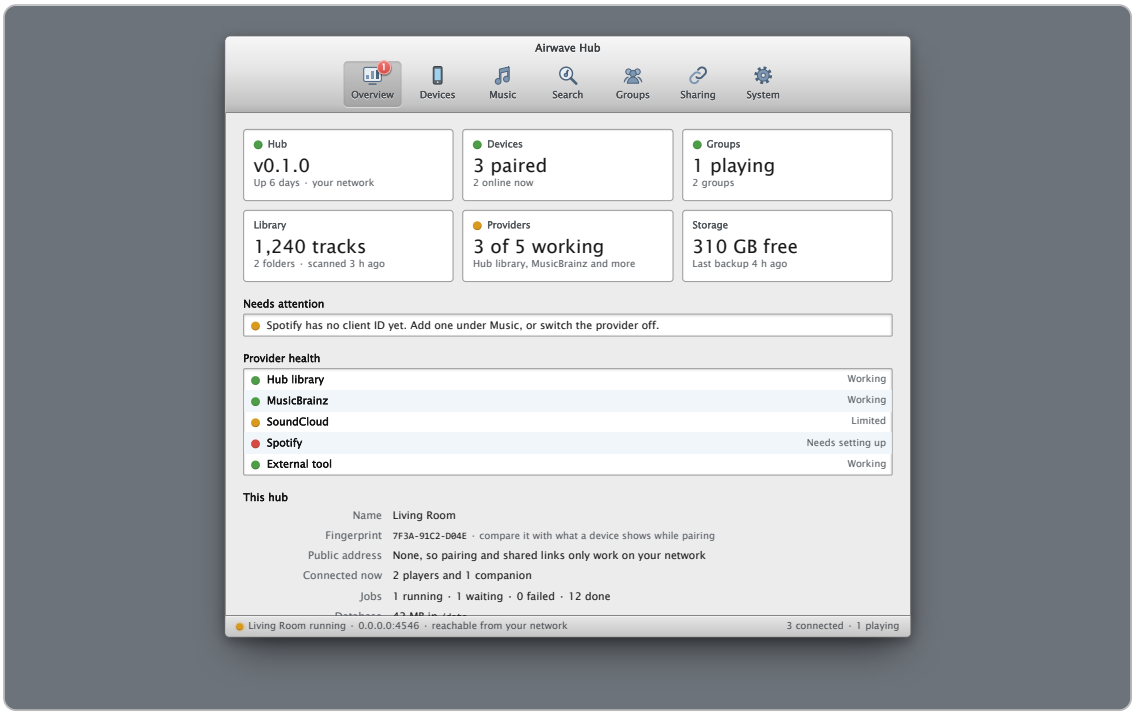


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)

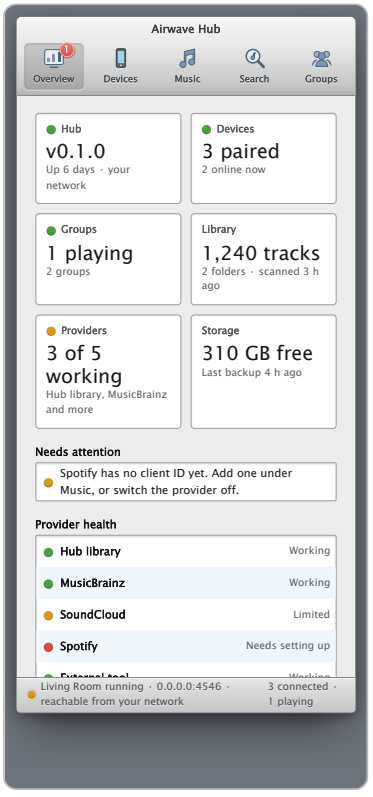
Overview

Six tiles, what needs attention, each provider's health as a word, and the hub's own facts. Below 640px the tiles go two-up and the key–value list stacks.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)

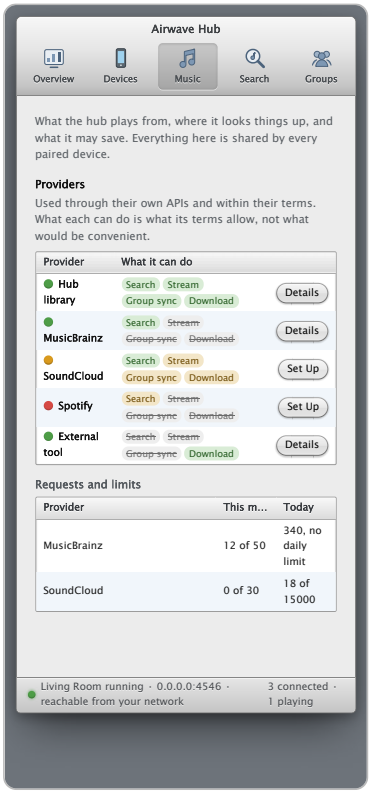
Music ▸ Providers

A table inside a well. Below 640px the Role column and the two request counters are dropped rather than scrolled sideways.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 10px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)

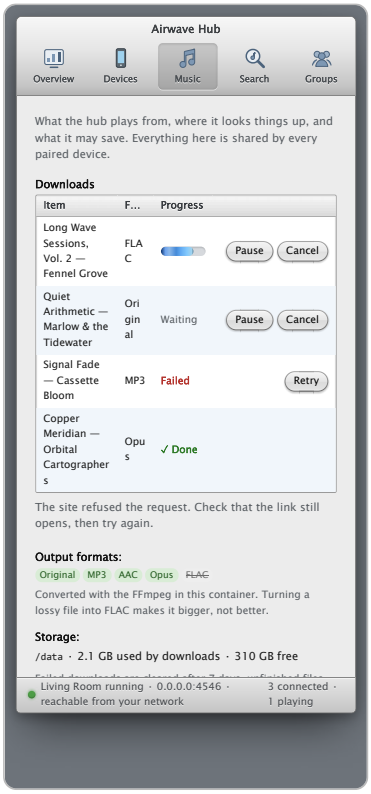
Music ▸ Downloads

Every state a job can be in, as words or a bar; the formats this hub can write, struck through where it cannot; and the storage line.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 10px

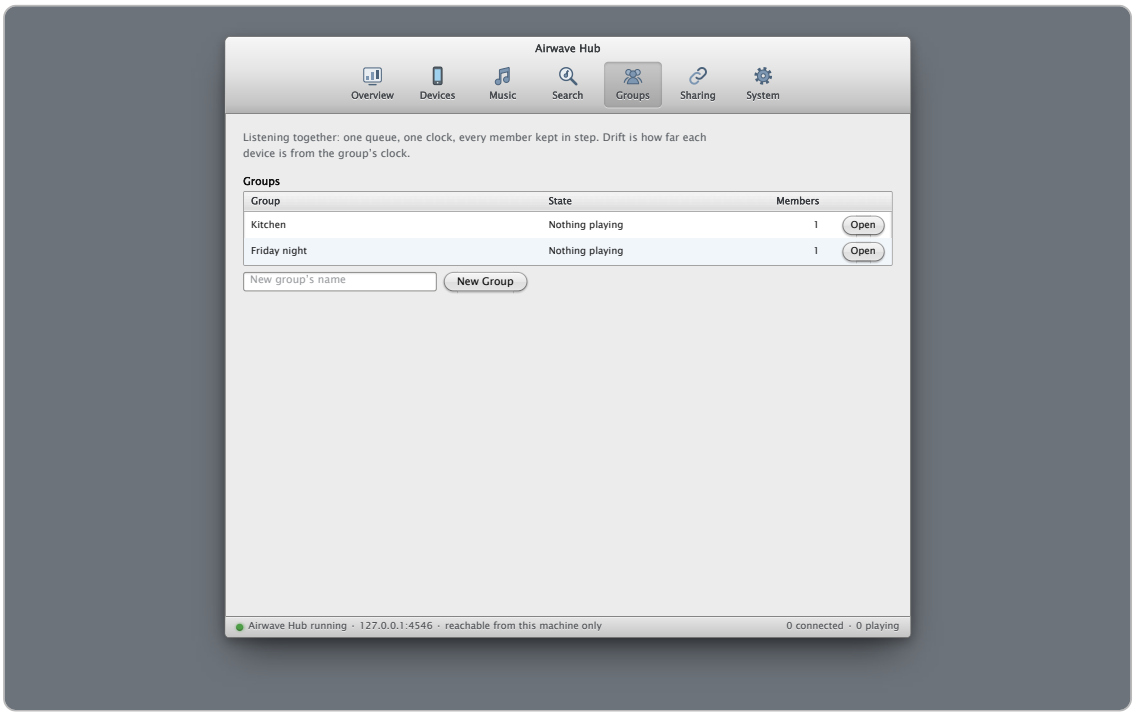


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)

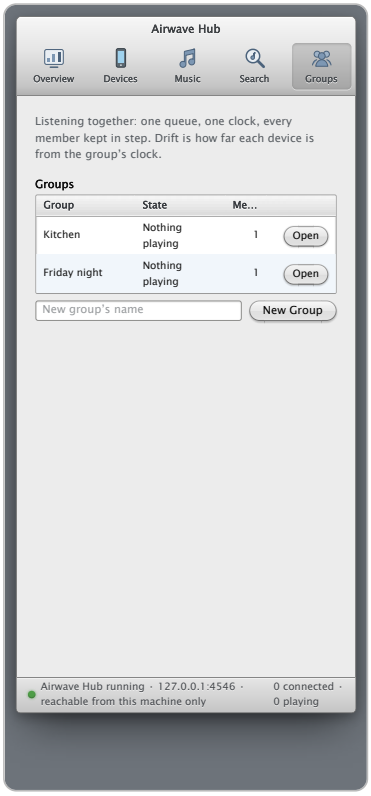
Groups

The hub's own Groups view, answered by a recorded hub: two groups, nothing playing, and the New Group row. Open one to see its transport, queue and invites.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

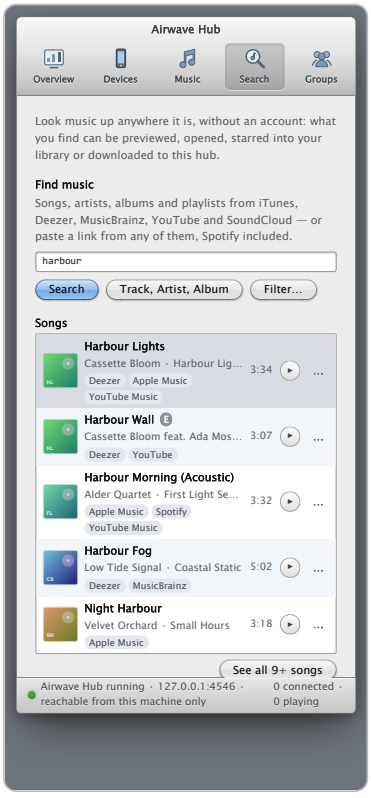
Search

The hub's own Search view, answering "harbour" from the stock catalog: the calm overview in one column under the field — five songs, three artists, albums and playlists, each with "See all N", no pager — every row with its platforms, BPM, time, the explicit mark, a preview and its "…" menu (Up Next, the library, Download…), and every service's state under them (one resting). Open a row for a song, an album or an artist; paste a link for a playlist with its mosaic and star.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 10px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 10px · smallest target 22px · desktop density, not held to 44px (DEC-027)

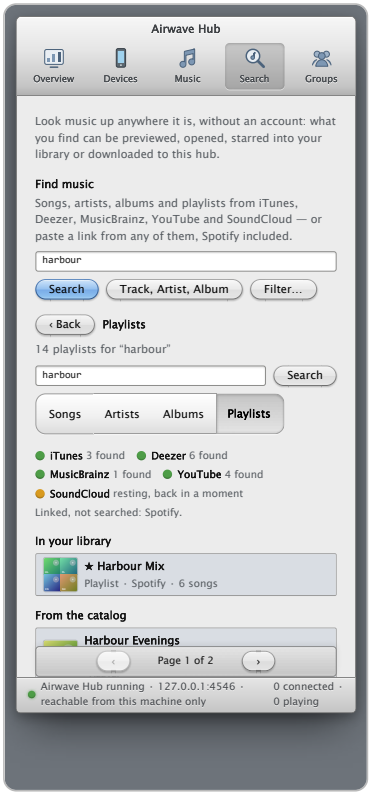
Search ▸ Playlists

A type's own page ("See all 14 playlists"): its field, the Songs · Artists · Albums · Playlists control, the services' line, the starred playlist first under "In your library", then Deezer's public ones, twelve a page, with the sticky ‹ › "Page 1 of 2" footer.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

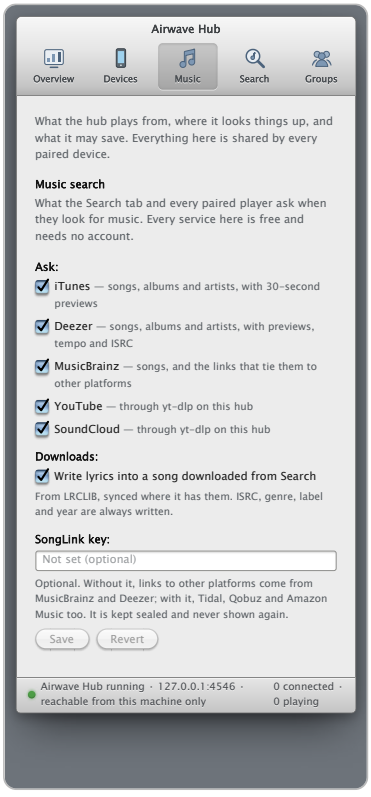
Music ▸ Music search

The services the catalog asks, lyrics in downloads, and the optional SongLink key — write-only, never shown again.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

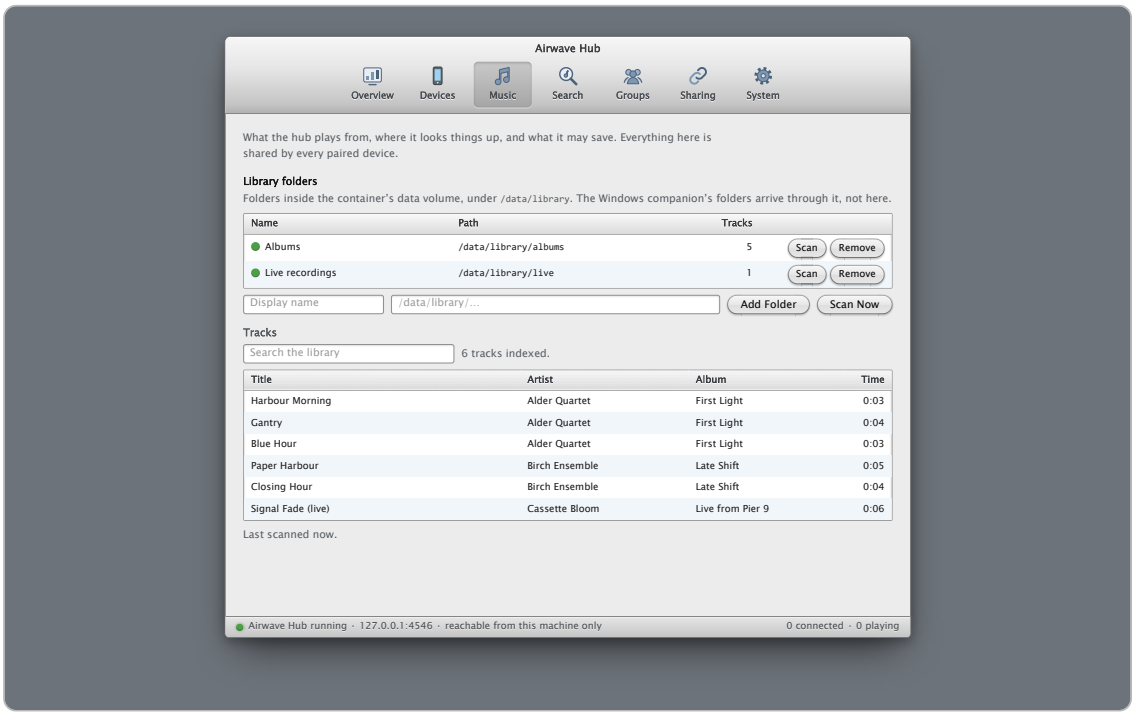


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 18px · desktop density, not held to 44px (DEC-027)

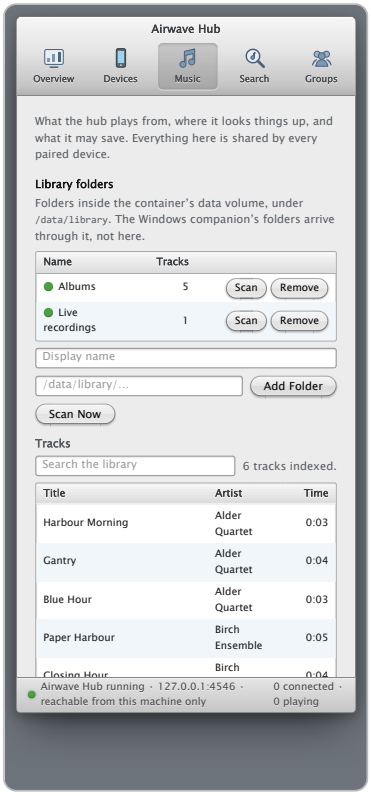
Music ▸ Library

The hub's own Library view: two folders in the data volume, each with its own Scan (that folder alone) and Remove, the fields to add a folder beside Scan Now, and the songs found.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

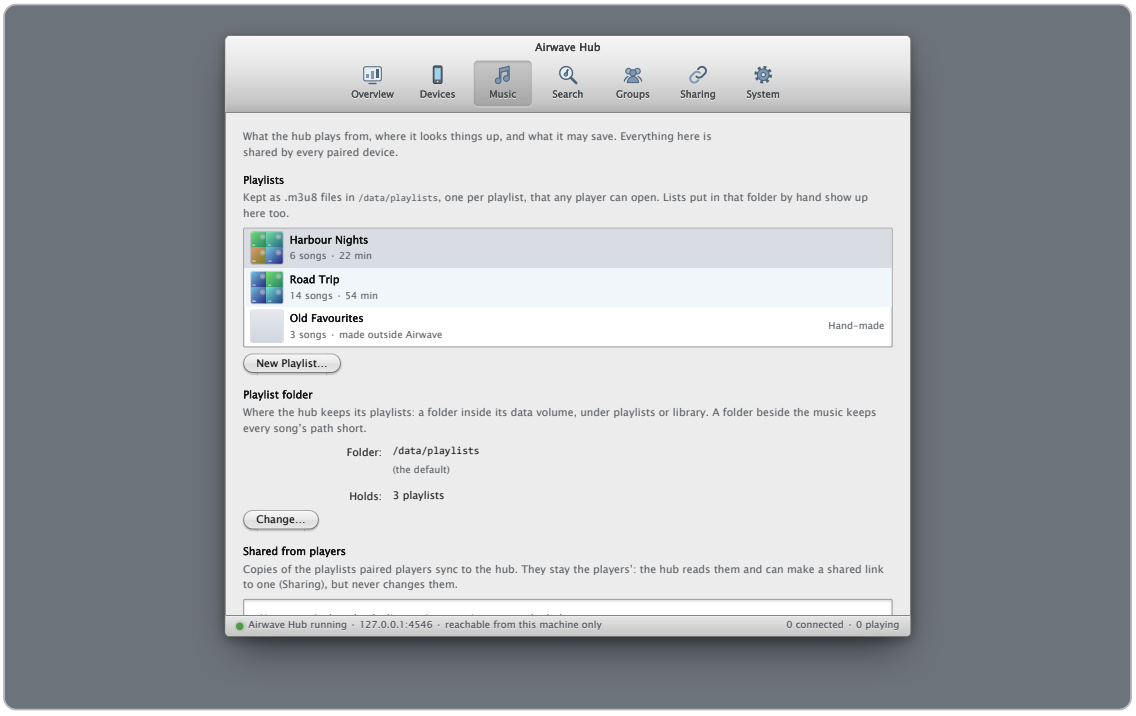


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

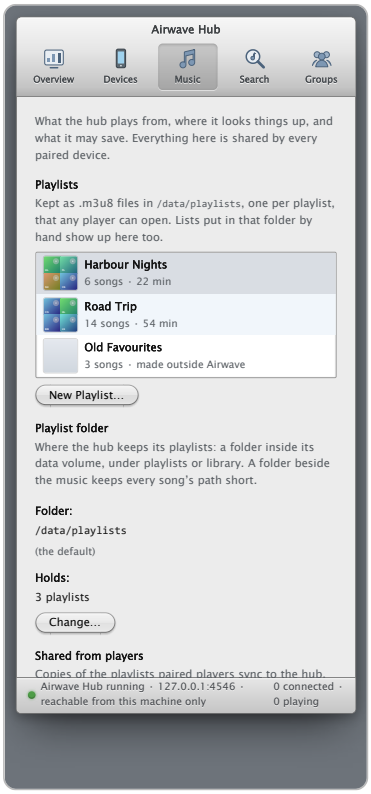
Music ▸ Playlists

The hub's own Playlists view (DEC-041): the playlists kept as .m3u8 files in /data/playlists, each with its 2×2 mosaic and length — one made outside Airwave marked Hand-made — New Playlist…, the Playlist folder with Change…, and the players' synced copies apart under Shared from players. Open a list for its songs, where each plays from, and Rename…, Export .m3u8 and Delete….



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

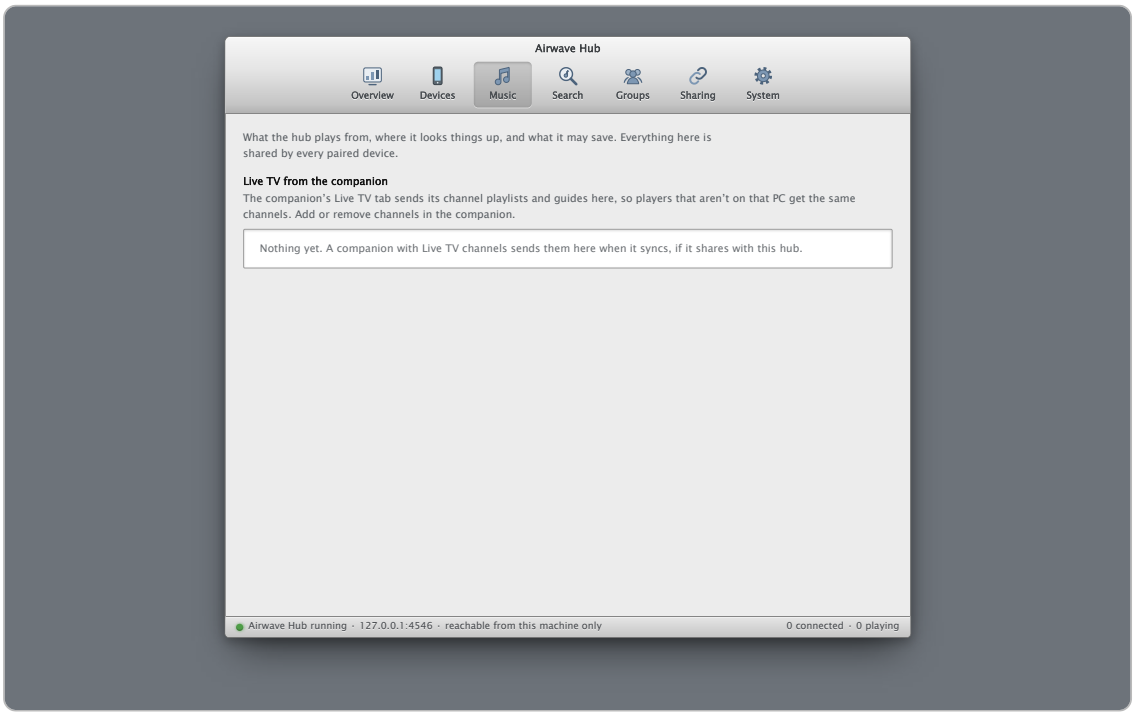


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

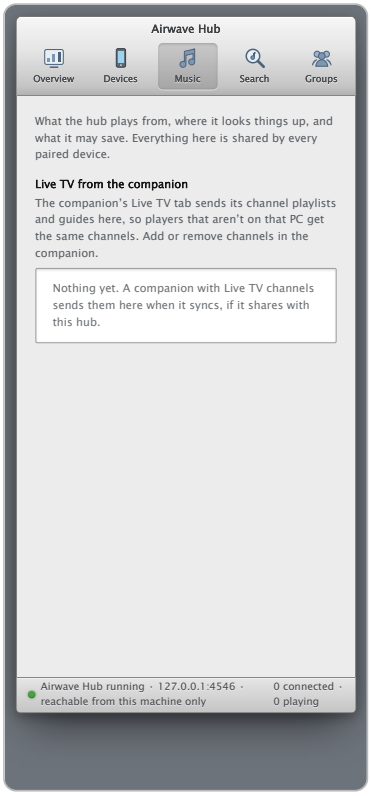
Music ▸ Live TV from the companion

The hub's own Live TV row, read-only: what a sharing companion sent, or — as recorded here — the sentence saying nothing has arrived yet. Channels are added and removed in the companion.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

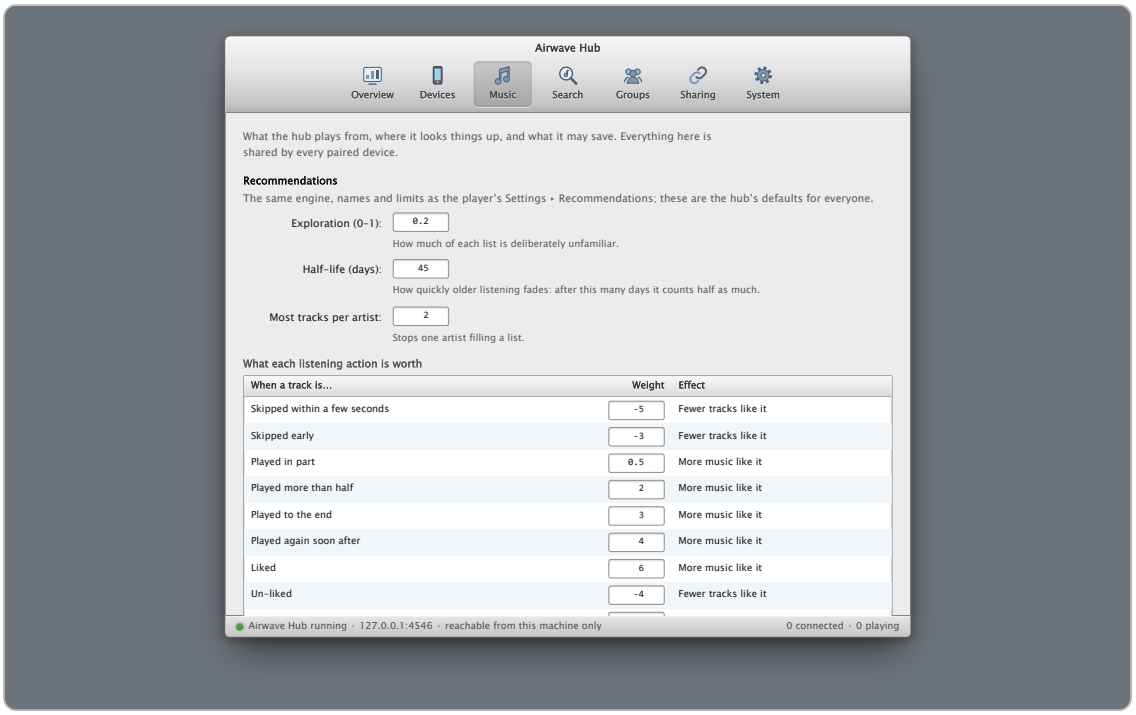


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 50px · desktop density, not held to 44px (DEC-027)

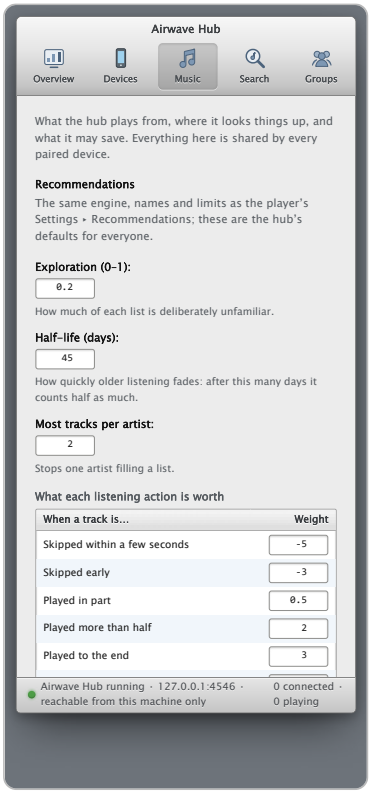
Music ▸ Recommendations

The hub's own Recommendations view: the defaults every player's engine starts from, with Save and Revert for the whole form.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

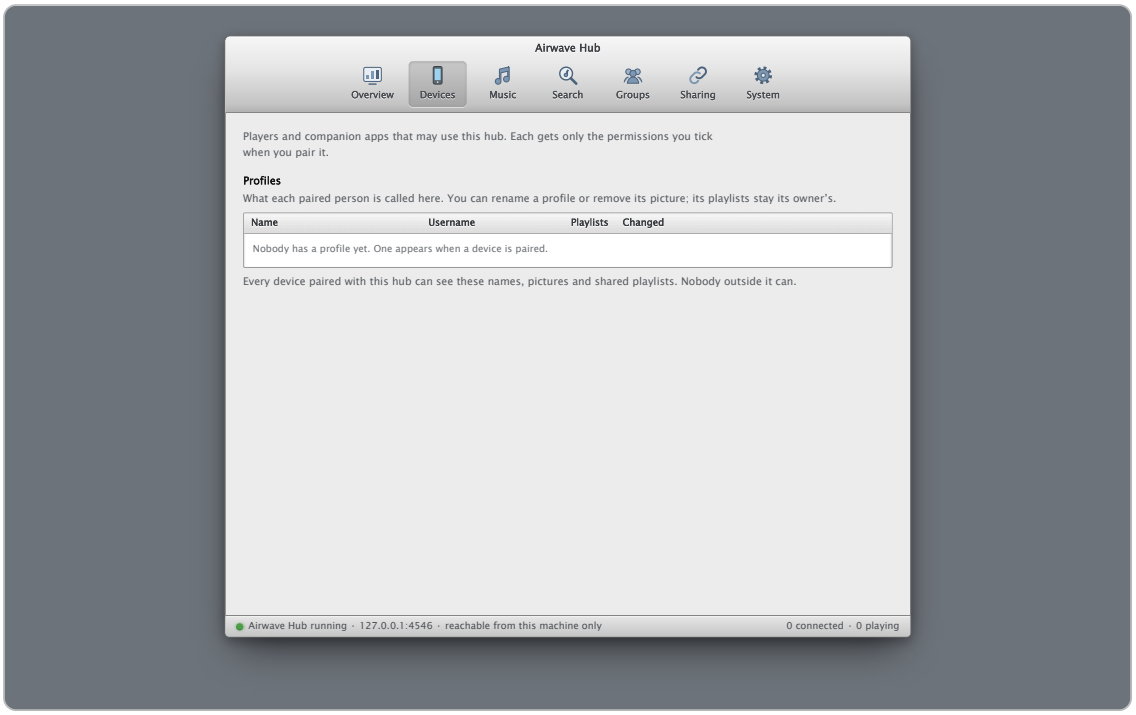


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

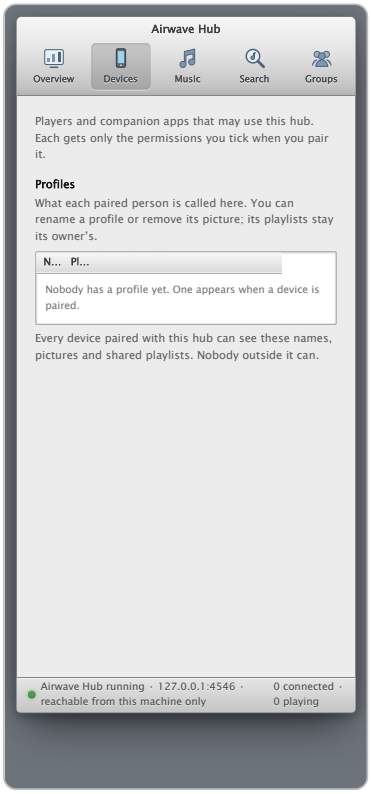
Devices ▸ Profiles

The hub's own Profiles view on a hub with no paired players yet: the empty list says what fills it.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 50px · desktop density, not held to 44px (DEC-027)

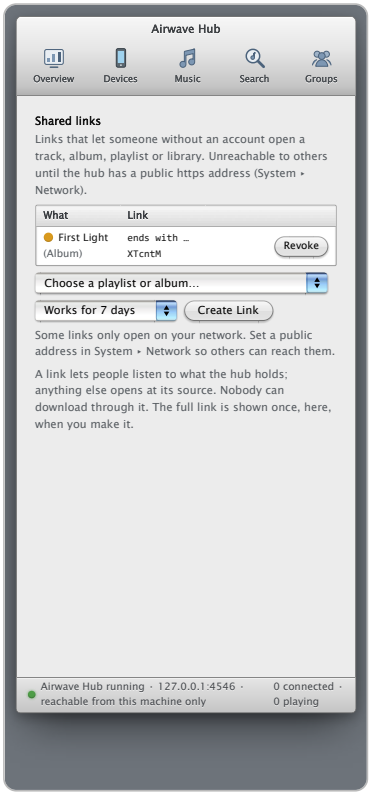
Sharing ▸ Shared links

The hub's own Shared links view: one album link with its expiry and counts, and Create Link with the picker of hub playlists and albums and how long the link lasts.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

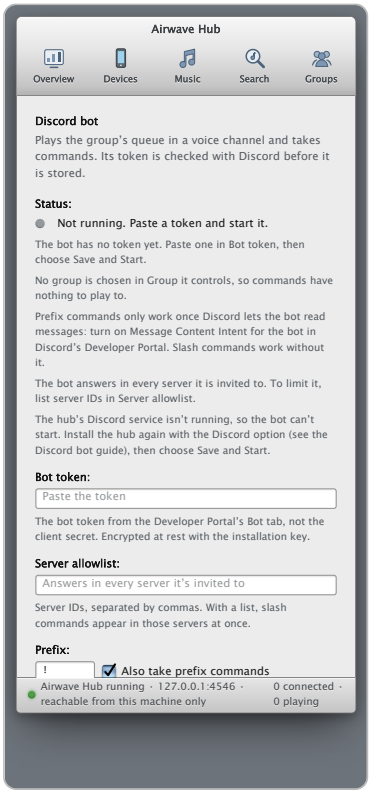
Sharing ▸ Discord

The hub's own Discord view with no bot token yet: the form, Save and Start with Revert, and Invite the Bot disabled with its reason until a token is saved.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

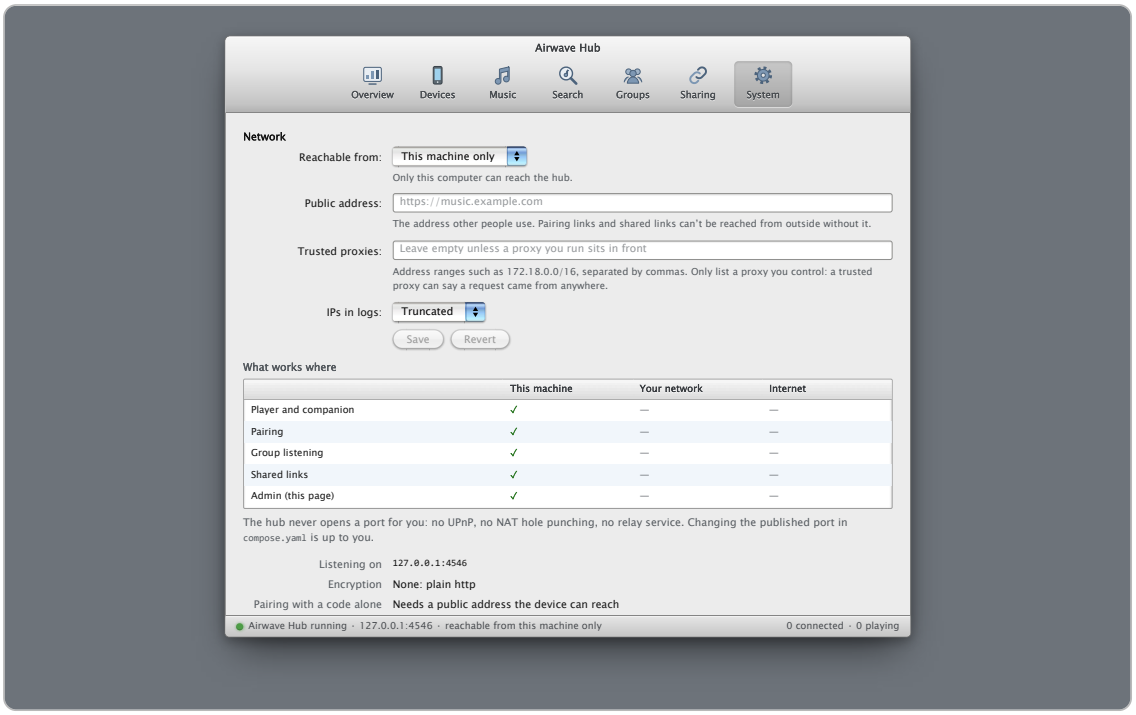


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 18px · desktop density, not held to 44px (DEC-027)

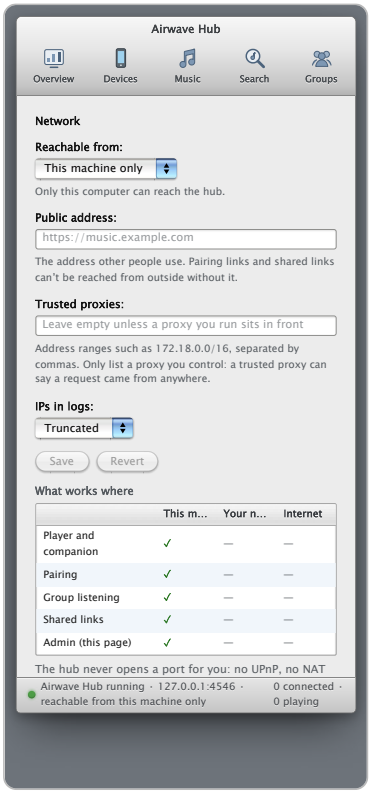
System ▸ Network

The hub's own Network view: how it listens and where it is reached from, with an explicit Save and Revert — nothing applies as you type.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px



Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

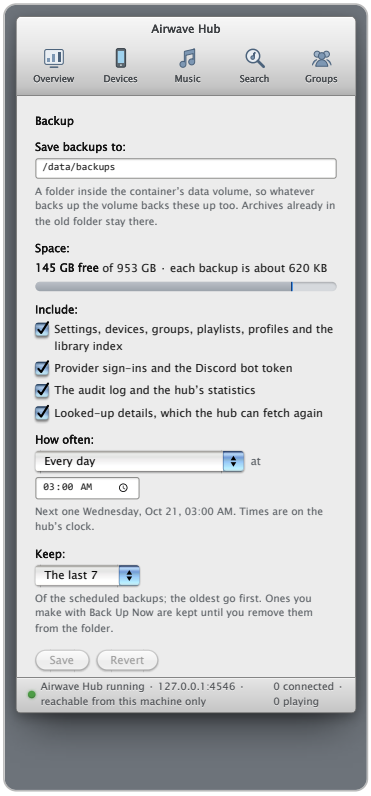
System ▸ Backup

The hub's own Backup view: one archive with its own Download, the space it has, and the settings — where, what goes in, the daily schedule and how many to keep.



Laptop 1280 × 800, shown at 53%

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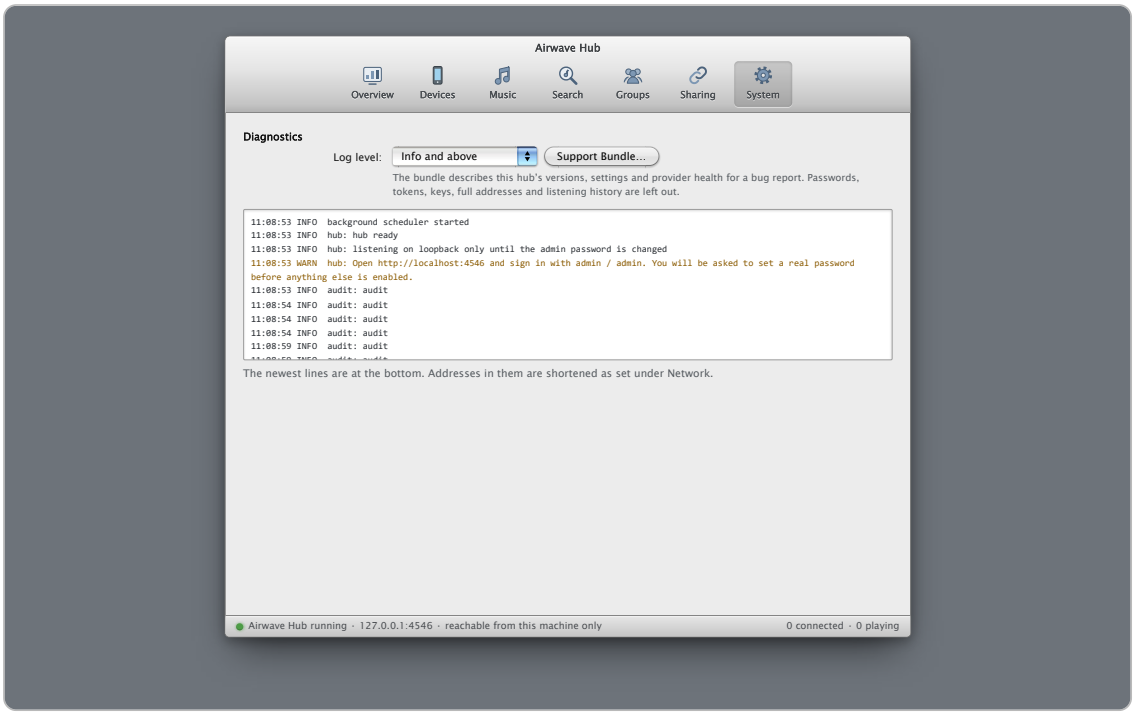


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 18px · desktop density, not held to 44px (DEC-027)

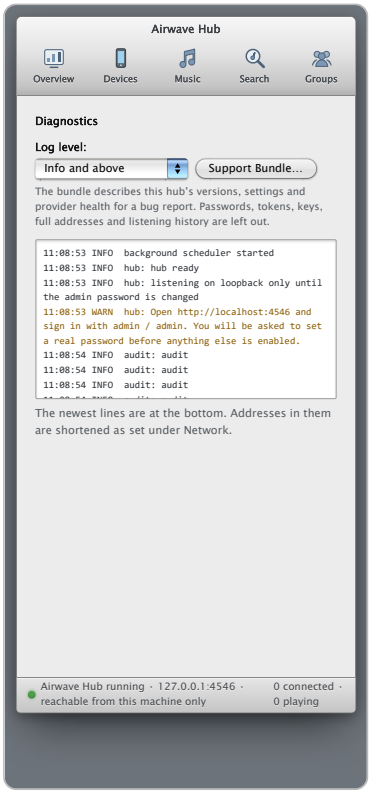
System ▸ Diagnostics

The hub's own Diagnostics view: the recent log at a chosen level and the support bundle, which says what it leaves out.



Laptop 1280 × 800, shown at 53%

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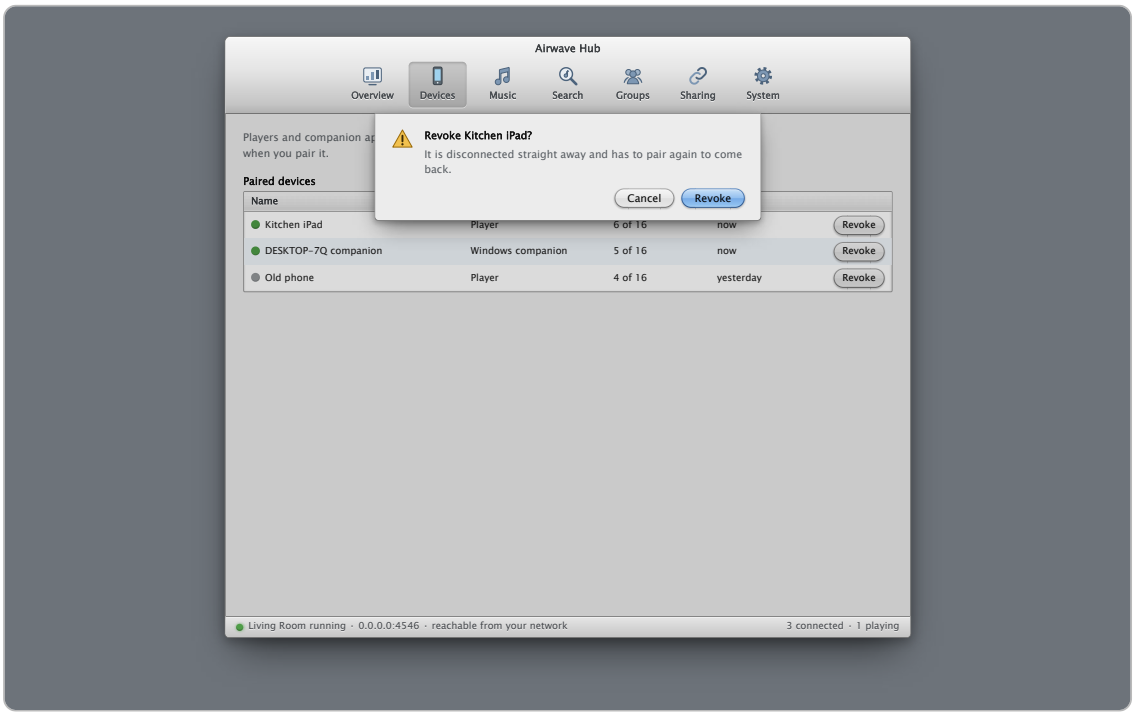


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

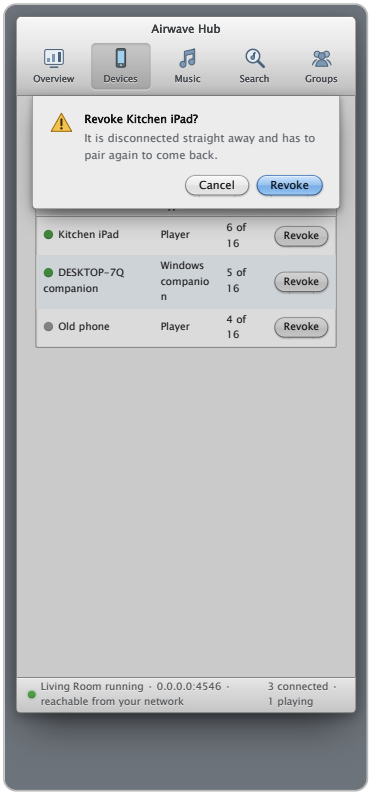
Confirmation sheet

A destructive action asks in a sheet that drops over the pane. Cancel has the focus, Escape cancels, and the window behind it is inert. This is the hub kit's own sheet, asked for real.



Laptop 1280 × 800, shown at 53%

Fits nothing off the side · smallest text 11px

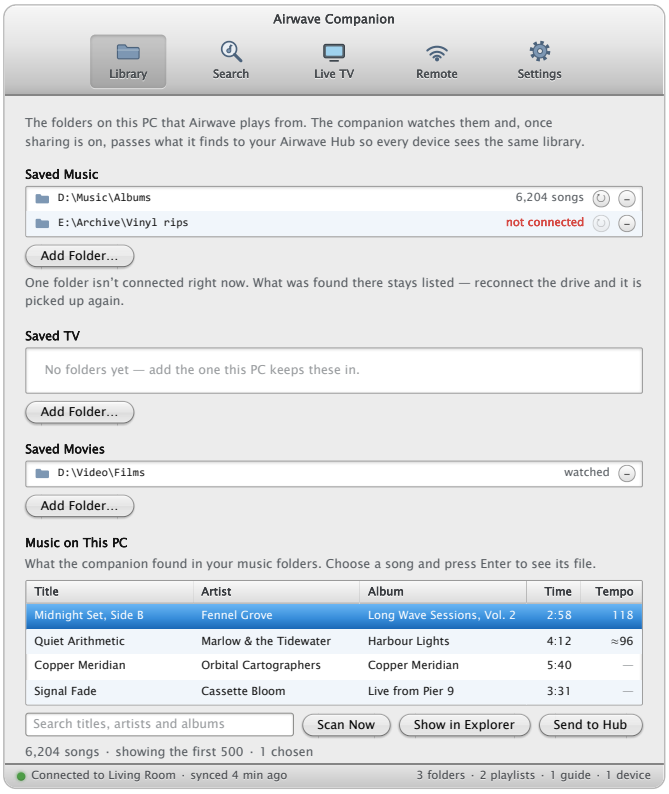


Phone 390 × 844, shown at 56% touch layer emulated

Fits nothing off the side · smallest text 11px · smallest target 22px · desktop density, not held to 44px (DEC-027)

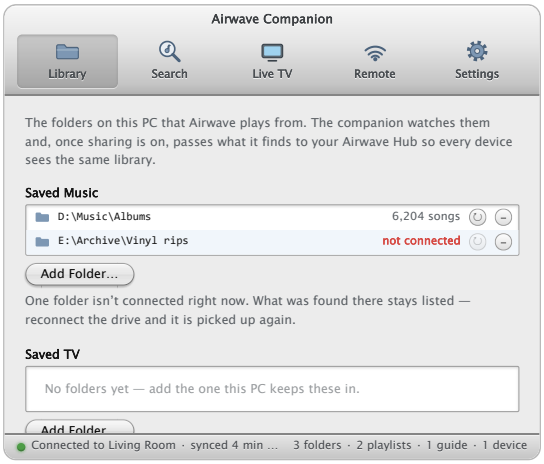
Library

Saved Music, Saved TV and Saved Movies, then the songs found. A folder whose drive is away is said so in its row and under the list.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px

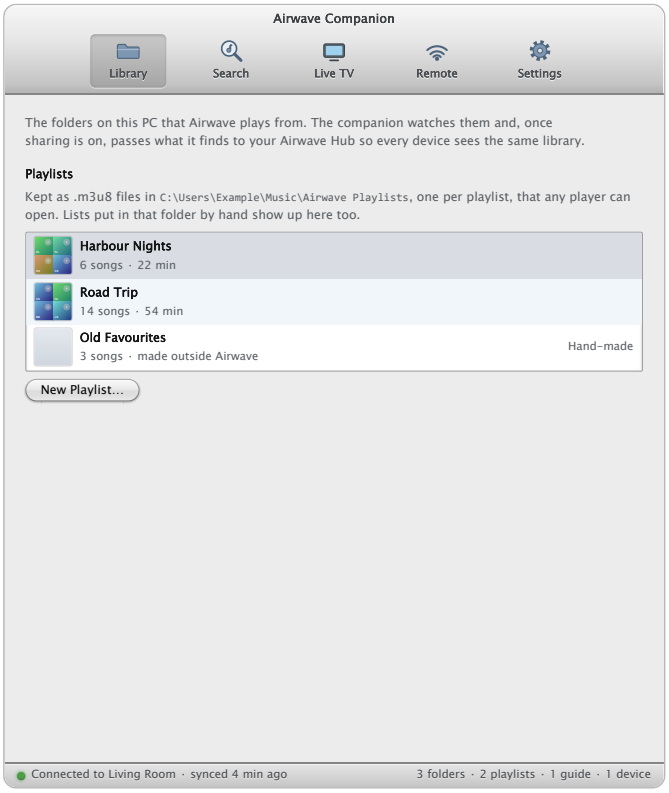


Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 11px

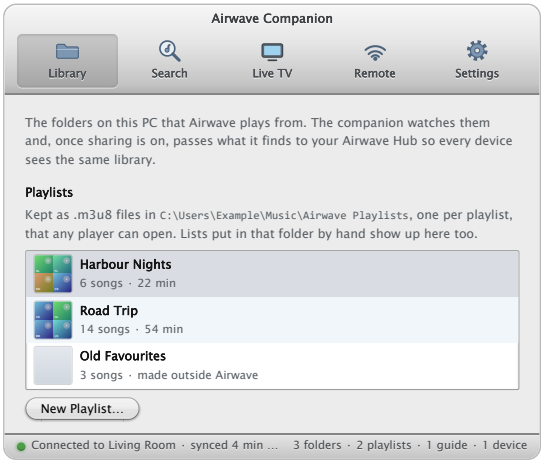
Library ▸ Playlists

The companion's own Playlists view (DEC-041): this PC's playlists, kept as .m3u8 files in Airwave Playlists (in the Music folder), with their mosaics. Its folder, Change… and Open Folder are in Settings ▸ Playlists. On touch, a long press on a song opens its menu.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px



Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 11px

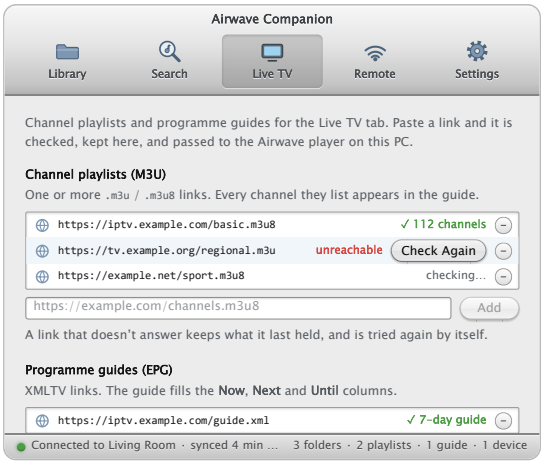
Live TV

A kept link, one that stopped answering (Check Again), one being checked, and a refused one: the reason under the field, the typed text left in it.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px

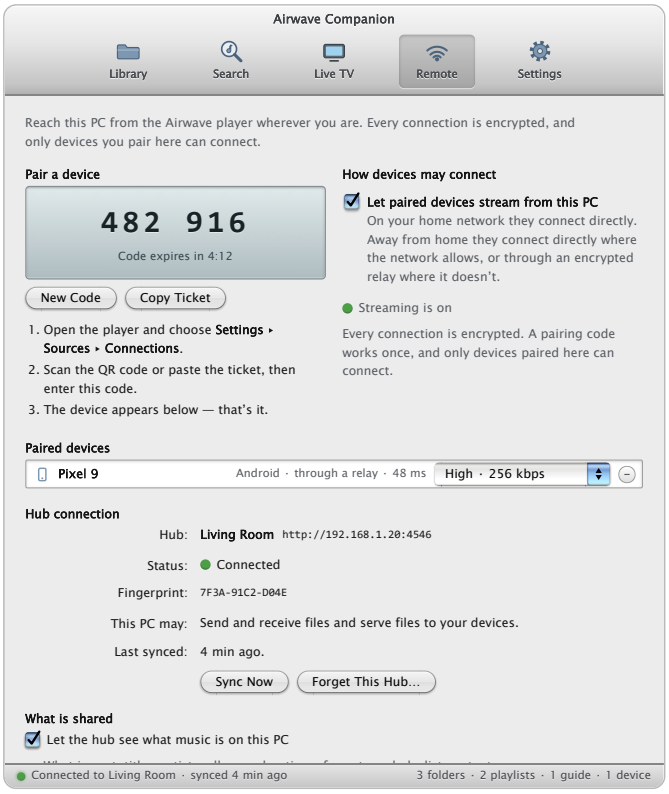


Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 11px

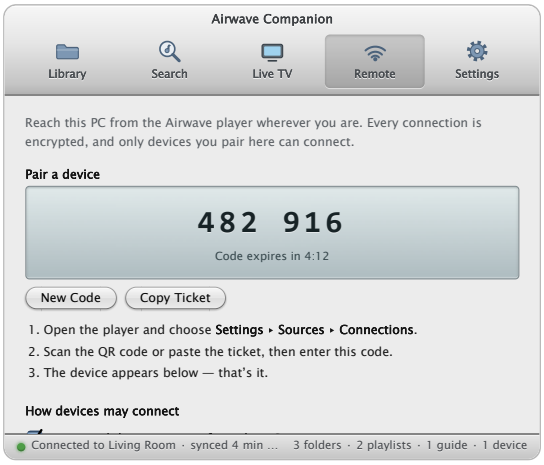
Remote

Pairing a device, the hub connection and what is shared, and transfers. Untick sharing and Sync Now is disabled with its reason.



As it opens 640 × 760, shown at 62%

- Fits nothing off the side · smallest text 11px

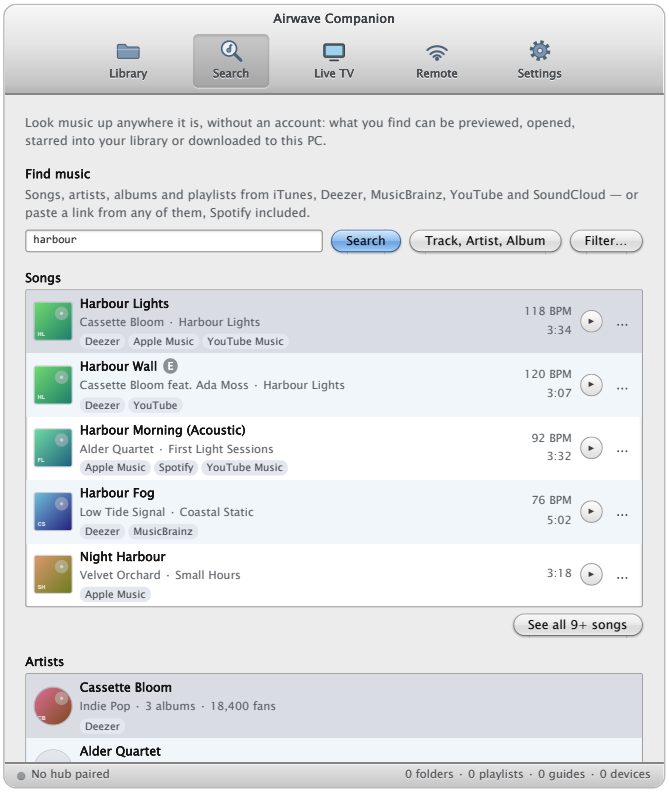


Smallest window 520 × 440, shown at 62%

- Fits nothing off the side · smallest text 11px

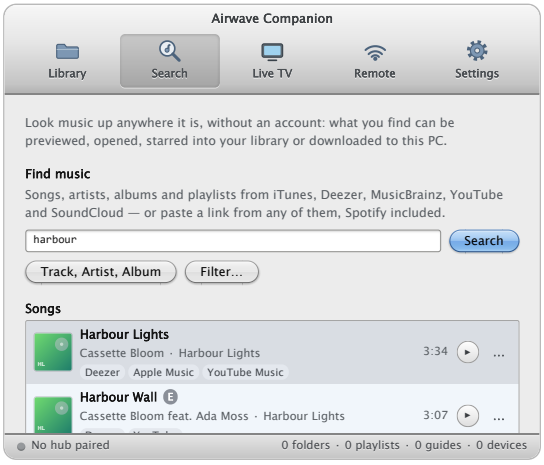
Search

The companion's own Search view, the hub's search through this PC's helper, answering "harbour" from the stock catalog through the stand-in bridge: the calm overview, each row with its "…" menu (Up Next goes to a paired hub's group: the companion has no queue of its own).



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 10px

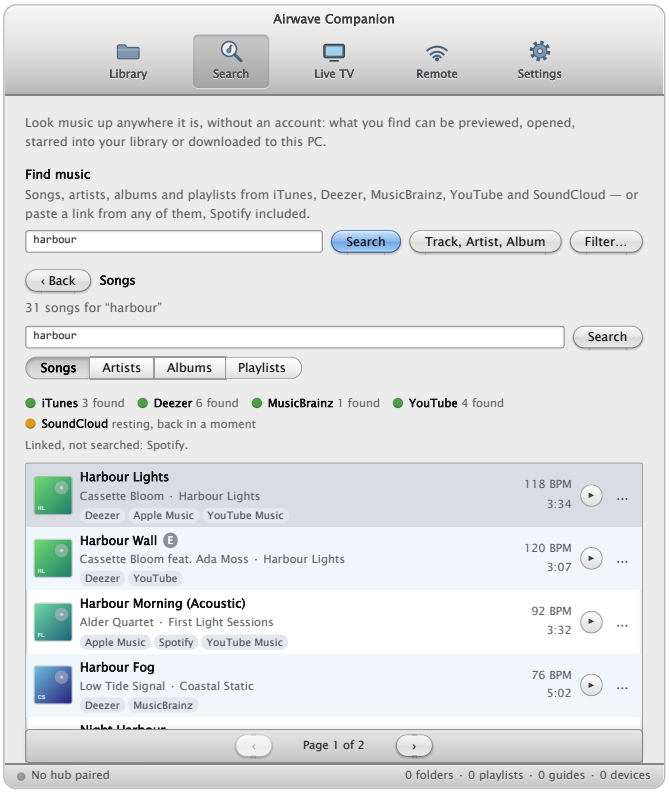


Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 10px

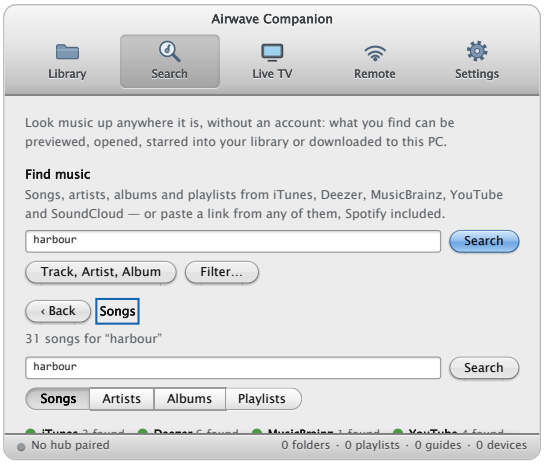
Search ▸ Songs

A type's own page ("See all 9+ songs"): the next page read at once (31 songs), its field and control, and the sticky ‹ › "Page 1 of 2" footer.



As it opens 640 × 760, shown at 62%

- Fits nothing off the side · smallest text 10px



Smallest window 520 × 440, shown at 62%

- Fits nothing off the side · smallest text 10px

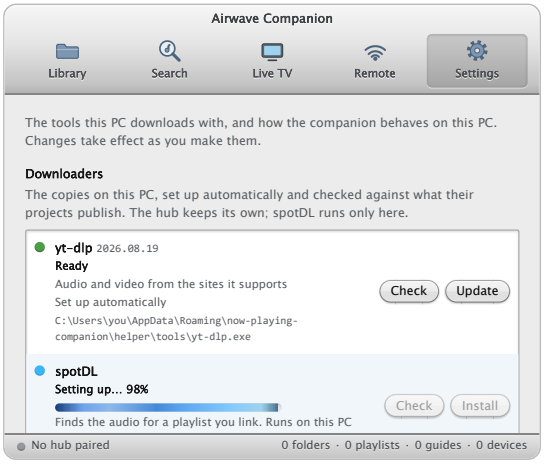
Settings

The companion's own Settings view, answered by a recorded companion: one downloader ready, one still being set up (a bar, and the badge on the tool), FFmpeg found on the PATH; then General with update checks and notifications, Downloads, and Network with the helper on this PC only.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px

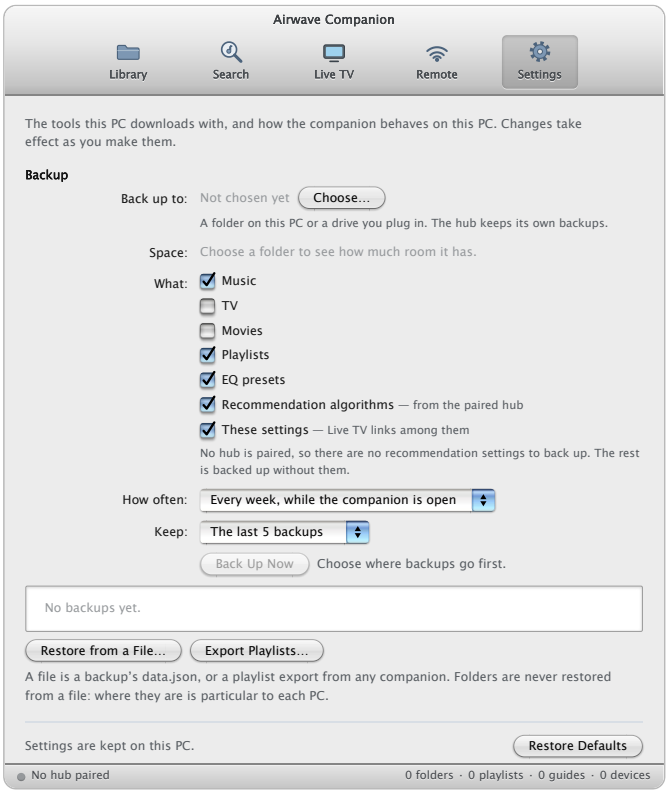


Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 11px

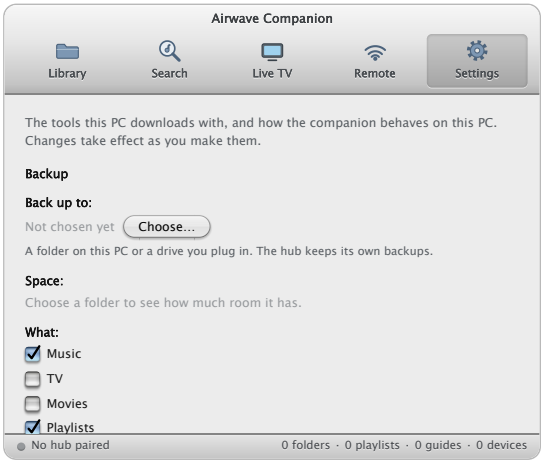
Settings ▸ Backup

The companion's own Backup view, recorded on a new install: a weekly schedule, five kept, what goes in (recommendation settings only once a hub is paired), and Back Up Now waiting for a folder to be chosen.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px

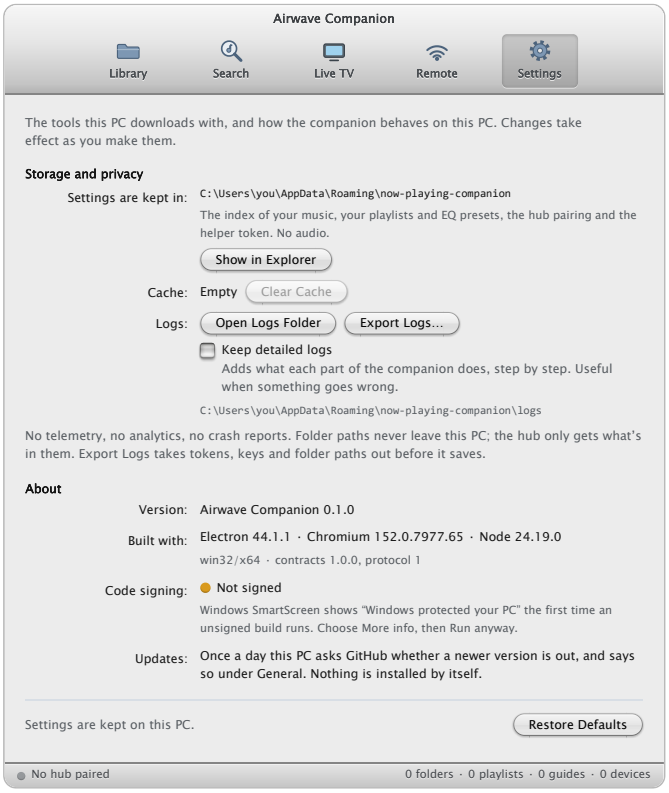


Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 11px

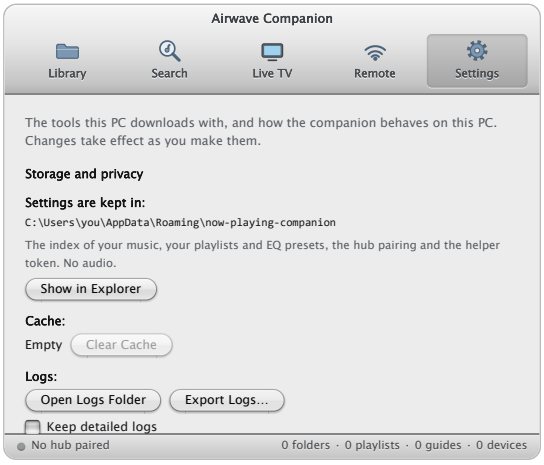
Settings ▸ Storage and About

The companion's own About view: where its data lives, the cache and logs with their controls, and the build, with how updates are checked. The paths are the installed ones.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px

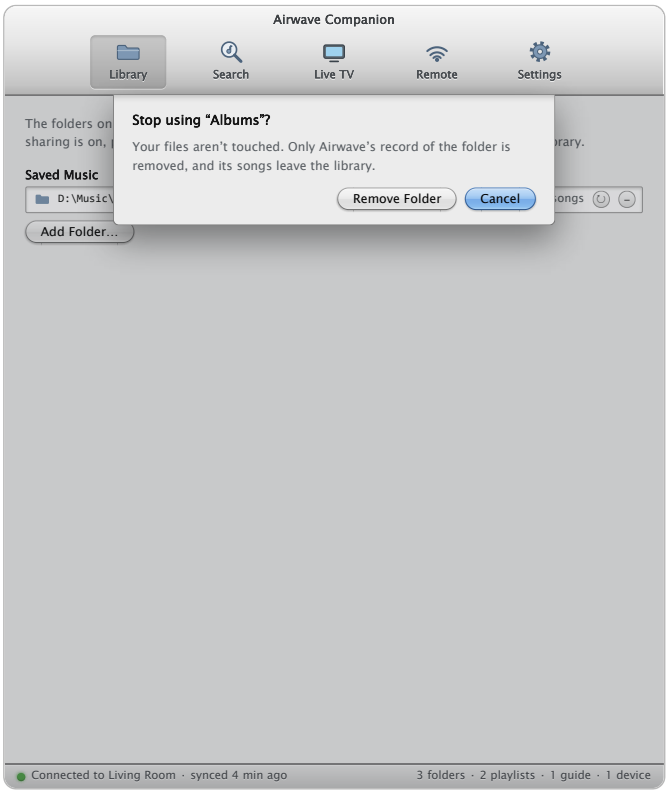


Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 11px

Confirmation sheet

The companion’s sheet is a modal dialog. For a destructive action Cancel is the default button, so Return never removes anything. This is the companion kit’s own sheet, asked for real.



As it opens 640 × 760, shown at 62%

Fits nothing off the side · smallest text 11px



Smallest window 520 × 440, shown at 62%

Fits nothing off the side · smallest text 11px

Sources and commands

This page is generated. The sources below are the design authority: change them, then rebuild. The exports — this page and its PDF — are views, so an edit or annotation on either is a request to change a source, not a change in itself.

CONCERN	SOURCE	OWNS

designs	`design/frontends/origin/airwave-now-playing.html`
	`design/frontends/origin/airwave-hub.html`
	`design/frontends/origin/airwave-companion.html`
		Where each app's look began: the owner's three hand-made designs, frozen as history (DEC-038). They are still the generators' inputs — music-player/scripts/make-shell.py writes music-player/index.html from the player's, and packages/aqua-ui/scripts/make-window-css.py writes the window stylesheets from the other two — but they are not edited: a change is a new asserted step in the generator, or a change to the product's own source.

windowStylesheets	`packages/aqua-ui/src/styles/airwave-window.css` `packages/aqua-ui/src/styles/airwave-hub.css`	What the hub and the companion wear: the window designs' own CSS, copied out by packages/aqua-ui/scripts/make-window-css.py (pnpm build:window-css). Generated — never edited by hand; a change to the window kit is an asserted edit in make-window-css.py (DEC-038).
windowArt	`packages/aqua-ui/src/airwave/aqua-art.ts`	AquaArt: the drawn push button, pop-up and checkbox both windows wear, published as --aq-* custom properties. One module, imported as @now-playing/aqua-ui/airwave-art.

windowKits	`docker-container/src/web/ui.tsx`
	`docker-container/src/web/icons.tsx`
	`docker-container/src/web/styles.css`
	`windows-companion/src/renderer/ui.tsx`
	`windows-companion/src/renderer/icons.tsx`
	`windows-companion/src/renderer/styles.css`
	`windows-companion/src/renderer/search-kit.tsx`
		Each window's own kit, icons and remaining styles: the design's markup as components, the states the designs never drew, and (hub) the contrast inks. The companion's search-kit.tsx adds the parts its Search tool needs (a titled group, notes, the <dialog> form sheet). Per product on purpose (DEC-026).

tokens	`packages/aqua-ui/src/styles/tokens.json`	Documented values: colour, type, space, size, shadow, motion and the page skin for the component library; the `airwave` group records the Airwave window's palette, which the design files own.
stylesheets	`packages/aqua-ui/src/styles/aqua.css` `packages/aqua-ui/src/styles/now-playing.css`	The component library's custom properties (light scheme on :root, dark and touch layers after it). Mirror tokens.json by hand; see tokenMap. Loaded by the player's React source, the gallery and this guide — not by the hub or the companion.
tokenMap	`design/token-map.json`	Which custom property carries each token, and recorded exceptions.
overrides	`packages/aqua-ui/src/styles/overrides.css`	Temporary trials written by the styleguide's token editor, for the component library only. Empty in the repository.
components	`packages/aqua-ui/src`	The component library: every shared React control, shell, overlay and state. Imported by the player's React source in music-player/src (kept, not served — DEC-019), the gallery and this guide; it exports only what the player imports, and the styleguide's usage scan lists any component no product imports so it can be removed. The hub and the companion import only the Airwave stylesheets and AquaArt from this package (DEC-026).
rules	`design/ux-rules.json`	Interaction and content rules with stable IDs, their owning code and their tests.
coverage	`design/coverage.json`	Every surface, state and journey, with discovery sources.
decisions	`design/decisions.md`	Adopted, proposed and unresolved decisions and exceptions (DEC-###).
prose	`docs/design/STYLEGUIDE.md`	The reasoning behind the visuals, in long form.
conformance	`docs/AQUA_CONFORMANCE.md`	Spec §17/§18 MUSTs mapped to tests, reviewer checks or deviations.
specification	`docs/design/APPLE_AQUA_2009_2010_UI_DESIGN_SPEC.md`	The reconstruction target (Snow Leopard / iTunes 9).

CHANGING SOMETHING

1. Find the owner: a token, a component, a rule by ID, or a surface in the ledger.
2. Change it there — and, for a rule, the code that implements it and the test that proves it.
3. Run `pnpm styleguide:check` and the tests the rule names; then `pnpm styleguide:build` and `pnpm styleguide:pdf`.
4. Look at the result in the mockups at every size, and at every page of the PDF.
5. A deliberate exception goes in `design/decisions.md` with its reason. Never widen an exception to make a check pass.

COMMANDS

`BUILD:WINDOW-CSS`	regenerates airwave-window.css and airwave-hub.css from design/frontends/origin/airwave-companion.html and airwave-hub.html, then applies the asserted edits make-window-css.py records.
`STYLEGUIDE:BUILD`	rebuilds docs/design/styleguide.html from the sources above (same as pnpm build:styleguide).
`STYLEGUIDE:CHECK`	sources and evidence exist, rule IDs are unique, every discovered screen is in the ledger, tokens.json matches the stylesheets (the library's and the Airwave window's), and the committed HTML was built from the current sources.
`STYLEGUIDE:SHELL`	after pnpm build:player, retakes the served player's screenshots for The served shell (packages/aqua-ui/styleguide/shell) and records which shell they show.
`STYLEGUIDE:PDF`	prints the committed HTML to docs/design/styleguide.pdf with Playwright's Chromium (set NP_STYLEGUIDE_BROWSER to use an installed Chrome or Edge).
`E2E`	measures every mockup at every size.
`CONFORMANCE`	includes packages/aqua-ui/tests/unit/aqua-conformance.test.ts and design-sources.test.ts.
`MOCKUPS:BUILD`	renders the living mockups in design/frontends from the apps (installed Chrome or Playwright Chromium; NP_MOCKUP_BROWSER names one).
`MOCKUPS:DIFF`	renders the apps into a scratch folder and prints what an edited mockup changes, by section; --check exits 1 on any difference (the mockups-up-to-date gate).
`MOCKUPS:EXPORT`	copies the three mockups to a folder.
`MOCKUPS:RECORD-HUB`	re-records the hub answers the hub mockup is drawn with (scripts/mockups/fixtures/hub.json).

Changing any of this

- **Tokens first.** In the component library a new colour or size goes in `tokens.json`, then into the stylesheet as a custom property. In the Airwave window it goes in the design file's `:root`, then in the `airwave` group of `tokens.json`, which the check compares with the generated stylesheet. A literal hex in a component is a bug.
- **Both schemes.** Light on bare `:root`; redefine only what changes inside the dark guard.
- **The Airwave window changes through its generator.** The designs in `design/frontends/origin/` are frozen (DEC-038): a change to the window kit is an asserted edit in `make-window-css.py`, then `pnpm build:window-css`; AquaArt changes in `src/airwave/aqua-art.ts`. The player's shell changes through a new step in `make-shell.py`. Then `pnpm mockups:build` regenerates the living mockups in `design/frontends/`.
- **Both skins, if it is a library control.** Button, Slider and friends are drawn in the window skin and the page skin.
- **Say why in the CSS.** Every unobvious value carries a comment explaining what it reconstructs.
- **The gates.** `AQUA_CONFORMANCE.md` maps every §17/§18 MUST to a test, a named reviewer check, or a recorded deviation, and `pnpm verify` runs the lot. A deviation is legitimate; an undocumented one is not.
- **Rebuild this page.** `pnpm styleguide:build` regenerates `docs/design/styleguide.html` from the same sources the products use, `pnpm styleguide:check` fails if it no longer matches its sources, and `pnpm styleguide:pdf` prints it.

The correction log in `DEVIATIONS.md` is worth reading before a large change. It records the mistakes this system has already made — buttons a decade too glossy, a list redesigned instead of expanded, a picker that indexed music it could never play — and every one was found by building the thing and looking at it.

Coverage and decisions

What this guide covers and how each part was checked, for source revision `41724ca7c225392f`. The denominator is every surface in `design/coverage.json`, which the check keeps in step with the products' navigation.

PRODUCT	SURFACES	IN MOCKUPS	SEEN RUNNING	NOT RUN BECAUSE
Airwave — the player	48	5	5	—
Airwave Hub — admin window and public pages	24	21	11	Seen at runtime on 2026-10-03 (see runtime). Overview, Devices, Providers and Downloads were not captured in that run; they are covered by the hub's DOM, e2e, a11y and journey suites.
Airwave Companion — Windows	14	13	4	Electron-only; its renderer needs the preload bridge. Its Settings tab was seen in the packaged app on 2026-10-03 (see runtime); the other tabs are covered by its DOM and contract tests and the mockups.
Discord bot	1	0	0	Needs a bot token and a Discord server; covered by the voice and parity tests.
Android shell	2	0	0	No Android SDK or JDK 17 on the machine that built this revision.
Local helper	1	0	0	Console output was not captured for this revision; covered by its integration tests.

Automatic checks when this page was built: 138 rules with evidence, 70 surfaces discovered from navigation, 12 Discord commands, 249 tokens compared with the stylesheets (1 recorded exceptions). No open problems.

recorded exception: font.family ↔ --aqua-font: Lucida Grande, Lucida Sans Unicode, Helvetica Neue, Arial, sans-serif vs "Lucida Grande", "Aqua Fallback", "Lucida Sans Unicode", "Helvetica Neue", Arial, sans-serif (DEC-002)

SEEN IN A RUNNING PRODUCT

WHEN	SCENARIO	VIEWPORTS · THEME	SURFACES
2026-10-03	A scratch hub (pnpm build:hub, then dist/server.js on port 4564 with a fresh data folder, stopped by its PID) seeded through its API the way an administrator would: sign in with admin/admin, set a real password, two library folders with six tagged songs, two groups and an invite, an album link valid for a week, a daily backup schedule keeping 7, one backup. The admin window was signed into in Chrome and each section opened by its #id and captured (outerHTML and a 1280×900 screenshot each), and the same hub's answers to every route those sections read were recorded (packages/aqua-ui/styleguide/fixtures/hub-api.json). Seen: the sign-in in the window, the six tools, and each section with its new controls — Create Link with the playlist-or-album picker and how long it works, the backup folder, parts, schedule and Keep with Save and Revert and a Download per archive, a Scan per library folder, Network and Recommendations with Save and Revert, Discord's plain-sentence warnings with Invite the Bot withheld, and Diagnostics' log and support bundle. Playwright driving installed Chrome (DOM capture and screenshots); recorded API answers	1280×900 · light	11
2026-10-03	The packaged companion (win-unpacked, built 2026-10-03 17:37) started with PORTABLE_EXECUTABLE_DIR in a scratch folder, so no person's data was read or written, attached over CDP on port 9335 and stopped by its PID. The backup schedule was set to weekly keeping 5 through the bridge; the Settings tab was opened and the Backup, Storage and privacy and About groups captured (outerHTML and screenshots), and the answers to every channel the Settings tab reads were recorded (packages/aqua-ui/styleguide/fixtures/companion-ipc.json). Seen: yt-dlp ready with Check and Update, spotDL setting itself up, FFmpeg found on the PATH; Update downloaders automatically on; the update check on; the helper on this PC only; Backup with no folder yet (Back Up Now disabled with the reason), recommendation algorithms unavailable with the reason (no hub paired), Live TV links among these settings; the cache by part, the logs, and the build with how updates are checked. Playwright over CDP against the packaged app	640×760 · light	4

WHEN	SCENARIO	VIEWPORTS · THEME	SURFACES
2026-10-04	pnpm build:player, then pnpm styleguide:shell: the production build served by vite preview on port 4183 (stopped by its PID), every outside host refused, eight tagged songs added through the shell's own import path, Gantry played, Radio opened with the bundled stations, Live TV filled from a stubbed companion helper (three channels, two with a guide), and Settings opened — at 1280×860 and 390×844. The screenshots are the guide's The served shell (packages/aqua-ui/styleguide/shell). scripts/styleguide-shell-shots.mjs (Playwright, installed Chrome)	1280×860, 390×844 · light	5

EARLIER RECORDS, SUPERSEDED

What was seen running before an interface was replaced. Kept as history; these surfaces are not counted as seen.

- **2026-09-16** · Player dev server, empty library, no hub: each section, Settings, search with no match, the Shared segment (toast by pointer and keyboard), and one token change (--np-accent) seen in the player and the rebuilt guide, then reverted. These are the React interface's surfaces, served then and not now: the player serves the Airwave shell (DEC-019).
- **2026-09-16** · Hub admin dev server against a fresh data folder: the first-run sign-in screen. Signing in was not attempted; it needs a person to enter the password. The sign-in screen seen here was replaced by the Airwave Hub window, which draws sign-in inside the window (DEC-017).

DECISIONS AND EXCEPTIONS

ID	STATUS	DECISION
DEC-001	● ADOPTED	Lucida Grande is not shipped (not redistributable). The window skin names it first and falls back through a tuned chain.
DEC-002	● ADOPTED	`--aqua-font` inserts the bundled `"Aqua Fallback"` face after Lucida Grande; `tokens.json font.family` does not list it. The CSS is authoritative.
DEC-003	● ADOPTED	The windows (the Airwave window of the hub and the companion, and the library's window-skin tokens) have no dark palette, because Snow Leopard had none: `airwave-window.css` declares its values for both OS appearances and renders light in each. The player and the styleguide follow the system colour scheme.
DEC-005	● UNRESOLVED	Discord `noResults` renders `No results for "{{reason}}"`, but the command service passes the search text as `title` and `No results` as `reason`, so the bot says "No results for "No results"." Proposed fix: the template uses `{{title}}`. Changing a default template changes what existing servers see.
DEC-006	● ADOPTED	Cover Flow is not implemented; the 3D jewel-case stage stands in for it.
DEC-007	● ADOPTED	The player replaces the inset LCD display with the hero, which states the same information at reading size.
DEC-008	● UNRESOLVED	The public share page (`docker-container/src/api/routes/shares.ts`) is standalone HTML under a strict CSP with its own inline stylesheet. It uses a 12px pill button and weight 600, which the system otherwise forbids. Proposed: restyle it with the page-skin values (4px push button, weights 400/700).
DEC-009	● ADOPTED	The mockups are not the product view files executed verbatim: those are wired to a database, an API or Electron's bridge. The hub's and the companion's frames hold the stylesheets those products load (the generated Airwave sheets and each `styles.css`), AquaArt, and the products' own kits and icons, arranged as the views arrange them, with fixture data. The player's frames are the component library's page skin, because the served shell is one document with its own scripts and cannot be portalled into a frame. Runtime coverage says which surfaces were seen in the running product and which only in a mockup. Since DEC-030 most hub and companion sections are the products' own views; this still holds for the screens arranged in `airwave-screens.tsx` and for the player's page-skin frames.
DEC-010	● ADOPTED	Android notifications, the document picker and the save dialog are system UI. The guide documents their copy (`strings.xml`) and when they appear, not their appearance.
DEC-011	● PROPOSED	`tokens.json` is mirrored into the stylesheets by hand; `pnpm styleguide:check` compares the two through `design/token-map.json`. Proposed: generate the custom properties from `tokens.json` so the map becomes unnecessary.
DEC-012	● UNRESOLVED	The Android shell only reaches hubs over https (mixed content is never allowed and the manifest forbids cleartext), but the player's pairing form still accepts an `http://` address, so the refusal happens at connection time. Proposed: say so in the pairing form on Android.
DEC-013	● ADOPTED	The Windows companion is Electron-only; its renderer cannot run in a browser without the preload bridge (opened in one, it says so and offers nothing), so its runtime evidence comes from its DOM tests and the mockups.
DEC-014	● ADOPTED	The equalizer window (`.eqw`) has an intrinsic width: `max-width: 600px`, which holds its rails about 36 px apart the way the iTunes window does. It had been stretching to whatever container held it — 958 px in the styleguide — and read as a strip rather than a window.

ID	STATUS	DECISION
DEC-015	● ADOPTED	The note in every app icon is placed by centring the path's measured box (x 4.4–19.8, y 3–21.7 of its 24-unit grid) on the icon's optical target: the tile centre for the player and maskable icons, the folder mouth (240, 308) for the companion, the 32-square for the tray. The player icon had shipped 27 px left of centre; `tests/unit/icon-glyphs.test.ts` now pins all four placements.
DEC-016	● ADOPTED	Play sits on the progress rail's centre line at every width, with previous and next on its line either side. `KeyTransport` measures its row: the volume line stays beside the keys only while it fits on both sides (otherwise it goes beneath, centred), and the keys wrap only when they do not fit on one line, with previous, play and next alone on the first line. The player's eleven keys never leave room for the volume beside them inside the 820 px hero, so the player shows it beneath at every width. Covered by `tests/dom/page-chrome.test.tsx`.
DEC-017	● ADOPTED	The hub admin GUI is the six-tab window of `design/frontends/origin/airwave-hub.html` (Overview, Devices, Music, Groups, Sharing, System), drawn with the stylesheets generated from that file (`airwave-window.css`, `airwave-hub.css`), AquaArt, and the hub's own kit (`docker-container/src/web/ui.tsx`); the thirteen sections keep their data flows and ledger entries and stack inside those tabs, and `#section-id` in the address still opens the right one. It is one window in every state. Not signed in, the sign-in form is drawn in the same window with every tab locked. On first run the design's amber gate sits at the top of Overview and the other five tabs are locked with the reason: the server refuses every gated route until the password is changed, so the tiles that depend on those routes say they are off rather than showing a number. What the design draws and the hub did not have when the window was first built — a form that creates a shared link, Live TV on the hub, and the Backup tab's path, schedule and include controls — it has now: see DEC-028, DEC-029 and the hub's Live TV row (`hub-live-tv`). The invite table shows who an invite is for, its role and its state, not its code, because the hub keeps only a code's hash; the code appears once, when made. The header and gate colours that joined `tokens.json` as `header*` and `warningBg`/`warningEdge` for the earlier window skin stay in the library; the hub now reads the design's own `--head-*` and `--warn-*`.
DEC-018	● ADOPTED	The Windows companion is the four-tab window of `design/frontends/origin/airwave-companion.html` (Library, Live TV, Remote, Settings), drawn with `airwave-window.css`, AquaArt and its own kit (`windows-companion/src/renderer/ui.tsx`). Live TV is real: a pasted M3U or XMLTV link is read in the main process, kept only when it holds channels or programmes, and served to the player on this PC through the helper routes `/helper/v1/tv/channels` and `/helper/v1/tv/guide` (UX-TV-001, NP-TV-001). Remote ▸ Pair a device is the AWSP streaming server's pairing; the hub connection and transfers sit under it. Design controls not adopted, because the app cannot do what they promise: separate Wi-Fi and mobile-data toggles (one switch: paired devices may stream); per-tool Check, Update and Install and an auto-update toggle (tools are set up automatically and re-checked together with Check All); the Downloads fieldset; update checks and update notifications; running the helper without pairing as a choice (it always runs); cache and log controls; and "last seen" on a paired device (the row says how it is connected now). A backup is a dated folder (`data.json` + copied folders), measured with `local-helper/src/measure.ts` so the helper route and this window report one number; sizes are decimal units as the player shows them.
DEC-019	● ADOPTED	The player's interface is the Airwave frontend served as the shell (shape (a) of the master prompt): `music-player/index.html` is generated from `design/frontends/origin/airwave-now-playing.html` by `music-player/scripts/make-shell.py`, every edit asserted against the text it replaces, and `src/shell/bridge.ts` stands behind the file's seams with `db.ts`, `library.ts`, `PlaybackEngine`, `HubClient` and bundled three.js. Chosen because the edits that make playback real touch one transport block and one boot block, and the file's twenty-one suites come with it. What the shell no longer contains: the sixteen invented rows, the generated Statistics year, the demo history, the invented TV catalogue, the base64 clips, the two invented albums (now "Nothing playing" / "No group session"), and the direct call to api.anthropic.com (a third-party account). The React interface under `music-player/src/{App,views,components}` is no longer served; its logic layer (`src/lib`, `src/state`) is what the bridge uses. Back Up Now in the player measures and says nothing was sent: no route lets one machine start a backup on another.
DEC-020	● ADOPTED	The companion's chrome is its title bar. The operating system's title bar is hidden (the frame is kept, so the window still snaps, resizes and maximises), the page draws the title, and Windows draws its own minimise, maximise and close over the top-right corner in a clear 26 px overlay. The chrome is the drag region; the tools are not, so they stay clickable. The application menu is removed. The window is named once, from `PRODUCT_NAME`. Rule UX-CHROME-001; pinned by `windows-companion/tests/contract/app-identity.test.ts`.
DEC-021	● ADOPTED	In the hub and the companion a confirmation is a sheet that drops from under the toolbar over the window it belongs to, never a second window or the browser's own dialog. The safe answer has the keyboard: in the hub Cancel takes the focus; in the companion Cancel is the default button for a destructive action, so Return never removes anything. Escape cancels, focus is held in the sheet and returns to what asked. The two are built differently — the hub's is a `role="alertdialog"` layer over an inert window, the companion's a modal `<dialog>` — and each product keeps its own (DEC-026). Rule UX-SAFE-001.

ID	STATUS	DECISION
DEC-022	● ADOPTED	The hub web adds four inks to the design's palette, in `docker-container/src/web/styles.css`: `--ink-quiet` #6b6f73, `--success-ink` #2f7a28, `--danger-ink` #b3322c and `--link` #1764b2, and sets a struck-through capability chip's text to #6b6b6b. The design's `--ink-3` (#96999c), `--success` (#4e9d47) and `--danger` (#d64a44) measure 2.9:1, 3.4:1 and 3.9:1 as text on white, under WCAG AA's 4.5:1; the hues are kept and only their lightness moves. The design's values still colour lamps, bars and placeholders. All five are recorded in `tokens.json` (`airwave.inkQuiet`, `successInk`, `dangerInk`, `link`, `capOffInk`) and checked against the stylesheet.
DEC-023	● ADOPTED	In the companion, pairing with a hub that grants `library:share` is the decision to share: sharing is on from pairing until it is switched off under Remote ▸ What is shared, which is how the app behaved before the checkbox was wired. The checkbox shows what was chosen, Sync Now is disabled with the reason while sharing is off, and forgetting a hub turns sharing off for the next one.
DEC-024	● ADOPTED	The hub's downloader is ready without setup (owner decision, 2026-10-03; see docs/DOWNLOADS_AND_LEGAL.md). The external-tool provider is enabled by default and is yt-dlp when nothing is configured; a hub outside the image finds the tool in the image path, its own tools folder or PATH, and fetches a verified copy when there is none. Allowlisted hosts, no cookies and a rights basis on every request are unchanged, and the Providers table still shows on or off as the hub reports it.
DEC-025	● ADOPTED	The suite is named Airwave, as its three designs name it: Airwave (the player), Airwave Hub and Airwave Companion, from `packages/contracts/src/branding.ts`. The player's shell keeps the design's name; "Now Playing" is the player's home page and nothing else. Machine identifiers (the slug, URL scheme, storage namespace, User-Agent, the Windows appId, the package names and the companion's data folder) keep their spelling, so no library, pairing or taskbar pin is orphaned. This guide is the Airwave Style Guide.
DEC-026	● ADOPTED	One style guide covers the three apps, and the guide says what each is built from. AquaArt — the drawn push button, pop-up and checkbox — is one module, `packages/aqua-ui/src/airwave/aqua-art.ts`, imported by the hub and the companion as `@now-playing/aqua-ui/airwave-art`; the two copies it replaces drew identical bytes. The two window kits stay per product: they write the same classes, but with different props and, for the sheet and the checkbox, different markup, so one shared component would change what one of the products renders. The `packages/aqua-ui` React components and the stylesheets `aqua.css` and `now-playing.css` are kept and tested for the player's React source; the styleguide reads from the products' imports which components it uses. The components no product imported, and `aqua-window.css` and `aqua-media.css`, which only they wore, were removed on 2026-10-03; the guide names any component that falls out of use so it can go the same way.
DEC-027	● UNRESOLVED	The hub's admin window is often opened on a phone, and there it keeps the design's desktop density: 22 px push buttons and 11 px text, under the 44 px and 12 px floors the page skin holds (UX-TOUCH-001). The design's touch layer covers only list rows and the round minus. The styleguide measures and prints those sizes in the hub's phone frames but does not fail the hub on them. Proposed: add a touch layer to `design/frontends/origin/airwave-hub.html` (taller buttons, fields and tools under `pointer: coarse`) and hold the hub to the same floors. The companion is a desktop window and is not affected.
DEC-028	● ADOPTED	The admin window makes shared links from what the hub holds. A device shares from its own library and uploads the item list itself; the admin has no library of its own, so Sharing ▸ Create Link offers the playlists devices have synced to the hub and the albums in the hub's library, and for a playlist the hub builds the item list from its synced copy (`docker-container/src/shares/sources.ts`) — the same metadata a device would have sent. What a link grants does not change: items stream only where the hub holds the same file, and nothing can be downloaded through it. The full link is shown once, when made.
DEC-029	● ADOPTED	Hub backups run on a schedule on the hub's own clock: every day, or one day a week, at a time; off is allowed. The next run is the first slot after the schedule was last saved or last ran, so a hub that was off at its slot takes the backup when it next starts — once, not once per missed slot — and a changed schedule counts from the moment it is saved (changing weekly to daily at 02:59 does not mean a backup at 03:00 for every day that was missed). A failed run waits for its next slot rather than retrying every minute. Only scheduled backups count against "keep the last N"; ones made with Back Up Now stay until removed from the folder. The companion's schedule is the same idea while it is open: a day or a week since the last backup makes a run due, and one that could not happen is reported.
DEC-030	● ADOPTED	The style guide runs the hub's and the companion's own view components where it can. The views reach their product through one module each — the hub's API client (`docker-container/src/web/lib/api.ts`) and the companion's bridge (`windows-companion/src/renderer/bridge.ts`) — and the styleguide build, only, swaps those two for `packages/aqua-ui/styleguide/fixtures/fake-hub-api.ts` and `fake-companion-bridge.ts`, which answer reads from recordings of a real hub and a real companion (`record-hub-api.mts`, `record-companion-ipc.mjs`; machine paths written as `/data` and the installed data folder) and refuse every change with a sentence. Screens that need a state a recording cannot reach stay arranged by hand (DEC-009). The served player cannot be put in a frame, so it is shown as screenshots of the production build (`pnpm styleguide:shell`); `shots.json` records the hash of the shell they show and `pnpm styleguide:check` fails when the shell changes without new ones.

ID	STATUS	DECISION
DEC-031	● ADOPTED	The companion streams to paired devices on metered connections by default, as it does on Wi-Fi and Ethernet: the person who switched streaming on asked for it to work away from home, where a phone hotspot is common, and a default that silently refused would read as broken. Each connection kind has its own switch under Remote; on one that is switched off streaming pauses and says why, reading the cost from what Windows reports (UX-STREAM-001).
DEC-032	● ADOPTED	The local helper can be opened to the home network, and only to be read. It is off by default. Switched on, other devices may use the four token-free read routes in `LAN_READ_ROUTES` (health, the radio's now-playing, and the Live TV channels and guide) by this PC's own address only; every other route still refuses them, even with the token, and the radio and TV routes keep their own rule that no page and no token is not enough. A rebound name or a page from the internet is refused. Downloads, files and settings stay on this PC (UX-NET-001).
DEC-033	● ADOPTED	The companion's check for a new version is keyless and opens nothing it was told to. It asks the project's public releases on GitHub (no key, no account), compares versions as numbers, and says what it found; Download opens the fixed release page through the main process. No URL from the API's answer is ever opened, so a tampered answer cannot send the person anywhere. The check is on by default, can be switched off, and asks at once when switched on (UX-PRIV-001).
DEC-034	● ADOPTED	The companion backs up the paired hub's recommendation settings as seeds, not as the hub's internal profile. The backup keeps the artists and genres the taste profile weighs highest (`seedsFromProfile`), named with the hub they came from; restoring hands them back to a hub as starting points, which that hub weighs with its own data. A raw profile would tie a backup to one hub's engine version and could not be offered to another hub. With no hub paired, or one that cannot give them, the rest is backed up and the reason is said.
DEC-035	● ADOPTED	The component library keeps only what a product imports. The 57 components no product imported — the earlier window skin (window frame, toolbar, transport, LCD, scrubber, search field, results popover, source list and their icons), tabs, menus, avatars and the unused state panels — were removed with their tests, gallery entries, specimens and the two stylesheets only they wore (`aqua-window.css`, `aqua-media.css`), at the owner's request. What remains is imported by the player's React source; the parts MusicList uses (Marquee, its row menu and scroller) stay private to it.
DEC-036	● ADOPTED	The hub's downloader matches the companion's (owner decision, 2026-10-04): the image ships spotDL beside yt-dlp (verified through the hub's own installer; the image moved from Debian bookworm to trixie because spotDL's only Linux build needs glibc 2.38), downloads carry full tags and the cover, playlist, set and album links are supported, and a link has its real title without any API key. A playlist link is one job per entry — at most 200, duplicates skipped, one rights basis checked against every entry — and Music ▸ Downloads shows it as one row per entry whose From cell names the list and the position ("Soulhack, 2 of 11"), in the existing table markup with no new chrome. The provider's row names the tools that are on the hub ("External media tool (yt-dlp, spotDL)"). Unchanged: allowlisted hosts (open.spotify.com added for spotDL only), no cookies or credentials, a rights basis on every request, and Spotify's own audio is never touched — spotDL fetches a matching recording from YouTube Music. Supersedes the TOOL_PRESETS note that spotDL belonged only to the local helper.
DEC-037	● ADOPTED	Spacing follows each design's own values, measured, and three places depart from a design on purpose. (1) Player: a slider row in Settings (`.prefs__row > .weight.vol__row`, the design's Default volume and the disc's four) has no bottom margin. The design writes `.vol__row { margin: 0 }` but its later `.weight { margin-bottom: 11px }` wins the cascade, which left the label 5px below the slider; `make-shell.py` restores the design's stated 0, so the player's Default volume row is 11px shorter than the design file draws it. (2) Hub: `.kv` (a read-only list) uses the 158px label column of `.pref` above 640px instead of the hub design's 150px, so a list under a preference grid (System ▸ Network) lines its values up with the controls; the design never draws a `.kv`. (3) Hub and companion: neither design draws a sheet, so both apps' sheets share one spacing — 16px 18px 14px round it, 12px between the question and its buttons, 8px between buttons (the window's `.barrow` gap). Checked by `docker-container/tests/e2e/hub.spec.ts` and `music-player/tests/e2e/np/prefs.spec.ts`.

ID	STATUS	DECISION
DEC-038	● ADOPTED	**Living mockups; the hand-made designs are history.** `design/frontends/airwave-now-playing.html`, `airwave-hub.html` and `airwave-companion.html` are generated from the apps by `pnpm mockups:build` (`scripts/mockups/`). The owner's hand-made designs they replaced moved, untouched, to `design/frontends/origin/` and are frozen: they stay the inputs of `make-shell.py` and `make-window-css.py` (whose outputs the move left byte-identical, but for the window sheets' two-line header, which now names the folder) and are never edited — a change to the shell is a new asserted step in `make-shell.py`, a change to the window kit an asserted entry in `make-window-css.py`'s `EDITS`, anything else the product's own source. **How they are made:** each app is built by its own Vite config, run in Chrome on sample data (the player on the e2e suites' tagged songs and a stubbed companion; the hub on what a real hub answered, `fixtures/hub.json`, re-recorded by `pnpm mockups:record-hub`; the companion on IPC answers checked against its contract), and walked through its screens and states. Each state is the live DOM written out as readable, indented markup in a `<template>` — typed values, ticked boxes, open dialogs and scroll positions written back; hidden subtrees emptied (they are shown in the state where they appear); line breaks only where whitespace cannot render — beside verbatim copies of the stylesheets the app imports (`<style data-mock-source>`, so a line there is a line in the source), with a small navigator that shows one state at a time and follows the clicks the walk-through made (tabs, menus, sheets). **Why this and not a single-file build:** the owner edits these by hand, and a bundled app keeps its structure in minified JavaScript; rendering the real DOM keeps all of the UI in HTML and CSS a person can change, and `pnpm mockups:diff` can name the state and the source line a change belongs to. What is given up: behaviour beyond the recorded states (sound, video, live search, dragging, animation), and canvases, which are photographs. **Deterministic:** Playwright's installed clock (moved only in fixed steps, after outstanding requests are answered), seeded `Math.random`/`getRandomValues`, fixed locale, time zone and viewport, the media position held at 0:47, UUIDv7 ids renumbered; WebGL pixels may differ by a level or two between runs, so a picture within that noise of the committed one keeps the committed bytes, and the gate compares markup and styles, not pictures. **Kept in step:** `pnpm verify` renders them again (`mockups-up-to-date`); `pnpm styleguide:check` compares each file's inputs stamp (`design/manifest.json mockups`); an owner's edit reaches the app only through the workflow in `AGENTS.md`, "Living mockups".

ID	STATUS	DECISION
DEC-039	● ADOPTED	**The music catalog: MusicSearch's search, reimplemented, keyless (owner decision, 2026-10-06).** Airwave's search gains what [Okami6997/MusicSearch](https://github.com/Okami6997/MusicSearch) does for *finding* music, on all three apps. That repository has **no licence**, so none of its code is used: it was read for behaviour (`README.md`; `app.py` `/api/search`, `/api/search/expand`, `/api/resolve*`, `/api/lyrics`, `/api/musicbrainz`; `backend/{deezer,applemusic,soundcloud,youtube,songlink,musicbrainz,lyrics}.py`; `static/js/app.js`), and the engine was written fresh in this codebase's style: `packages/domain/src/catalog` (pure, network and processes injected), `packages/contracts/src/api/catalog.ts`, hub `/api/v1/catalog/*`, helper `/helper/v1/catalog/*`. **Adopted:** free text, the expandable Track/Artist/Album fields, ISRC and pasted music links (Spotify, Apple Music, Deezer, YouTube/YouTube Music, SoundCloud incl. `on.soundcloud.com`, Bandcamp, Tidal, Qobuz, Amazon Music); a concurrent fan-out to every enabled service with a chunk as each answers (the live feed, NDJSON); a merge across services by ISRC, else cleaned artist + title within ±3 s, every source kept and the richest fields taken (UX-CAT-002); per-service status with cooldown and backoff (UX-CAT-001); tracks, artists and albums each paged for infinite scroll; album and artist details and an artist's album list; 30-second previews from iTunes and Deezer; lyrics from LRCLIB; genre, label and year from MusicBrainz; album and playlist expansion with collection refs, four mosaic covers and saved-collection shapes; a catalog download through the existing queue from the best downloadable source, tagged with ISRC, genre, label, year and (setting, default on) lyrics. **Keyless sources:** iTunes Search/Lookup, Deezer's public API, MusicBrainz ws/2 (1 request/s, named User-Agent), LRCLIB, and yt-dlp (`ytsearch`/`scsearch`) for YouTube and SoundCloud. **Spotify is first-class through spotDL**, which the hub and helper already run: a pasted track, album, playlist or artist link is read with `spotdl save` (a track with `--preload`, so spotDL's own YouTube Music match comes back as a second source, `matchedBy: spotdl`, shown as "Spotify · plays from YouTube Music"); a playlist Spotify will not list (private, or one Spotify made) is reported with the reason. There is **no Spotify free-text search**: no scraped web-player token, no reuse of spotDL's bundled client credentials, and a spotDL query (seconds per song) is not fit for live search; songs found elsewhere show Spotify when MusicBrainz (or SongLink, with a key) links them. **Measured 2026-10-06: SongLink/Odesli closed keyless access** (`401 PUBLIC_API_ACCESS_DEPRECATED`), so cross-platform links are found keylessly from MusicBrainz's recording links and Deezer's ISRC lookup, and SongLink is an optional, write-only admin key that adds Tidal, Qobuz and Amazon. Deezer's `artist:` field filter matched nothing that day, so the advanced fields fall back to plain words when the strict syntax finds nothing. Apple Music playlists are not in Apple's public API; **amended the same day (owner requirement: every platform's playlists load whole)**, they are read from the playlist's public page — one GET of `music.apple.com`, the songs the page embeds as data (`serialized-server-data`), no token, no account, written fresh from the page's shape (MusicSearch's `applemusic.py` reads the same page; none of its code is used) — and a page that carries fewer songs than the playlist holds, or whose shape changed, says so in words. **Lists load whole:** the 200-song cap is gone; `catalog/resolve` and `catalog/album` serve any page (`limit` ≤ `CATALOG_PAGE_MAX` 200, `offset` ≤ `CATALOG_COLLECTION_CAP` 10,000), `total` is the platform's count, `hasMore` holds to the end, `collectAllPages`/`resolveAll` read a list in order, and a list past 10,000 is reported (`capped`), never cut silently; tool-read lists (spotDL, yt-dlp) are read once, whole, and a SoundCloud set's bare ids are described a page at a time (`--playlist-items`). **Excluded, and why:** MusicSearch's lossless downloads from Tidal, Qobuz, Amazon Music, Deezer and Spotify through third-party proxy pools (the SpotiFLAC registry) or scraped private tokens (Spotify's TOTP web token, Amazon page scraping), DRM decryption, the resampling to 192 kHz/24-bit, the file manager and the operation history. These circumvent DRM or the services' access controls, or need accounts (docs/SECURITY.md, "What is deliberately not done"); they are not search. **Checked against the real services, 2026-10-10** (yt-dlp 2026.08.19, spotDL 4.5.2, FFmpeg 9.0, from the engine with the hub's own presets; the image itself could not be run that day): every keyless source answered as designed, and five behaviours changed. A SoundCloud song SoundCloud only lets others preview (Go+, listed at exactly 30 s, `*_preview` formats only) is not a row, since all it can play or save is 30 seconds. A tool-read list entry is the page the tool names, never the stream address a full entry carries (`--playlist-items` puts a signed, expiring CDN URL in `url`), and a SoundCloud set's bare `api-v2` address gives way to that page. A Spotify song whose `--preload` match fails (spotDL wrote `[null]` after its YouTube Music search found no song) or takes over 60 s is read again without it: the row then has its Spotify source only, and spotDL matches it when it is downloaded. `spotdl save` runs with `--lyrics` and no provider (it fetched Genius/AzLyrics/MusixMatch lyrics nobody read: 32 → 18 s for a song, 184 → 106 s for a 50-song playlist), and a playlist's owner and picture are left unknown rather than taken from its first song. An MP3's lyrics are a real USLT frame, written by the hub after FFmpeg's pass, and its ISRC a TSRC frame: FFmpeg has no USLT writer and files both `lyrics` and `isrc` as TXXX frames. M4A carries the lyrics (©lyr) but no ISRC, which FFmpeg's MP4 writer cannot write. Spotify no longer gives spotDL an ISRC (`"isrc": ""`), and spotDL listed Spotify's own "Today's Top Hits" that day. **Sources:** the MusicSearch repository as of 2026-10-06 (read only); Apple's iTunes Search API documentation; Deezer's API reference (developers.deezer.com/api); MusicBrainz's API and rate-limiting documentation; LRCLIB's API; SongLink's public API page; yt-dlp's search extractors; spotDL's `save`/`--preload`. Recorded fixtures: `packages/domain/tests/fixtures/catalog/`. Plan and API notes for the UI: `.agents/plans/2026-10-06-catalog-engine.md`.

ID	STATUS	DECISION
DEC-040	● ADOPTED	**Search in the hub and the companion takes the player's third-round shape (owner, 2026-10-07; UX-SEARCH-007…012).** A calm overview without a pager, one type per page with infinite scroll and a sticky ‹ › footer, Playlists as a type, one centred column, and a song row's menu. Three things are decided here because the two apps differ from the player. (1) **Up Next is a group's queue.** The hub admin has no player of its own, so Add to Up Next is `POST /api/v1/groups/:id/requests` (the Discord /play path) with the song's best fetchable link (`pickDownloadSource`), a submenu when several groups are active; the companion has no Now Playing queue either, so it asks the paired hub (`hub:groups`, `hub:request`, both new IPC channels answered by `HubClient` with the device credential and its `group:member` scope) and otherwise says, in a sentence, that queueing needs a group. A store-only song is never searched for elsewhere: it says there is nothing to queue. (2) **No playlist store is invented.** The hub holds only players' shared playlist copies (admin-readable as share sources, never writable) and the companion holds synced copies it never changes, so Add to Playlist… opens the library sheet and says why, adding the song to the library instead. *(2) is superseded by DEC-041 (2026-10-10): both apps now keep a playlist folder.* (3) **Add to Library is a download with a stated rights basis** (DEC-036): on the hub into its download queue (`POST /api/v1/catalog/download`, Music ▸ Downloads, which is what the hub's library holds); on the companion through the helper's own path (`catalog:download`). The basis is asked in a sheet every time; nothing is fetched on a single click. The window kit gains a contextual menu (`.menu`) and a segmented control (`.seg`) through asserted entries in `make-window-css.py` (DEC-038), drawn by each app's own component (DEC-026: `docker-container/src/web/lib/menu.tsx`, `windows-companion/src/renderer/menu.tsx`).
DEC-041	● ADOPTED	**The hub and the companion keep playlists as files in a folder (owner, 2026-10-10: "Give the hub and companion app a folder to store or write playlists to."; UX-PL-001…008, CMP-PL-001…006). Format: an `.m3u8` per playlist, with an `.airwave.json` sidecar.** M3U8 is what every player and every other music app opens, and a person can read it: `#EXTM3U`, `#PLAYLIST:<name>`, then `#EXTINF:<secs>,<artist> - <title>` and the location per song — a path relative to the folder when the library has the file, otherwise the song's best source URL (`pickDownloadSource`, else its first) — so a list filed from search plays before the song is downloaded, and plays in another player at all. What M3U cannot hold goes beside it in `<name>.airwave.json` (`PlaylistSidecar`, `packages/contracts/src/api/playlist-folder.ts`): the id, created/updated, who made it, the description, the 2×2 cover mosaic, and each entry's id, catalog id, ISRC, artists, album, artwork, platforms, sources, added time and who added it. The M3U decides membership and order; the sidecar is matched back to it by location, so a list reordered or trimmed elsewhere keeps the rest. Rejected: JSON only (no other player opens it), XSPF (few players, no relative-path convention on Windows), and a database index (the files would stop being the truth, and an outside edit would need a sync). **One engine:** `@now-playing/domain/playlist-folder` (Node only, never in the player's bundle) reads and writes a folder for both apps: atomic writes (a temporary file renamed over the old one, sidecar first, one write at a time), names made safe and unique case-insensitively, symbolic links not followed, a relative location looked at only inside the allowed roots, absolute paths never written by Airwave and never shown, a hand-made `.m3u`/`.m3u8` shown read-only until the first change adopts it (keeping its id), a rescan on every read (by size and time), caps of 10,000 songs a playlist, 1,000 playlists a folder and 500 songs a request. **Hub:** `<data>/playlists` by default — the data volume is what the hub may write and what its backups and Docker volume already carry; changeable to a folder under `playlists/` or `library/` (beside the music keeps paths short), checked as text and again by real path, the way library roots and the backup folder are; changing offers to move the playlists. No DB index and so no migration: the folder path is a setting. Routes `/api/v1/playlists/*`. **Device scopes:** `playlists:use` lets a player read every hub playlist, make its own and file songs into any — filing is additive and the hub is the household's shared shelf — while renaming, reordering and deleting are kept to lists it made and removing to songs it added (or any song in a list it made), so no device can empty someone else's playlist; `library:sync` lets a device read and write its own starred albums and playlists at `/api/v1/catalog/saved` and read the admin's as `shared`. The players' synced copies stay read-only, shown apart as "Shared from players". **Companion:** `%USERPROFILE%\Music\Airwave Playlists` by default, chosen over a `Playlists` subfolder of the first library folder: a library folder may be a read-only share, a removable drive or absent, and files written inside it would be scanned as part of the library; the companion's library folders are a list, not one root. File access lives in the main process, beside the library's (`windows-companion/src/main/playlists.ts`), over twelve contract-validated `playlists:*` channels; the folder changes only through the system picker (`playlists:pick-dir`, with the move offer) and is kept across Restore Defaults and restores; the folder is watched (`event:playlists-changed`). A long press (500 ms, 10 px) opens a search or playlist row's menu on touch. **The player (2026-10-10; NP-FIND-008, NP-FIND-013, NP-FIND-014):** it files into the hub's folder and opens its lists in the music list, but keeps no playlist folder of its own; its playlists stay in its library state, shown apart as "On this player". The contract keeps every entry's link (`CatalogTrack.sources` has at least one), so a library song is filed by its source link, else its ISRC as its MusicBrainz ISRC page (`https://musicbrainz.org/isrc/<ISRC>`, which the hub matches to its own file by ISRC); a song with neither is not filed and the menu says why — no link-less entry is invented. Starred lists merge as a union by `{platform, kind, id}`, the later `savedAt` winning, with a local un-star's tombstone winning over an older hub copy.

NOT COVERED, ON PURPOSE

- `docker-deployment` — Docker is delivery, not a visual surface. Its ports, health check and how rebuilt assets reach users are in the engineering appendix of the styleguide.
- `styleguide` — The styleguide documents the products; it is not one of them.
- `aqua-ui-gallery` — Developer specimen page (pnpm --filter @now-playing/aqua-ui dev); its specimens are rendered in the styleguide.